NKG DAG Retrieval Debug Report

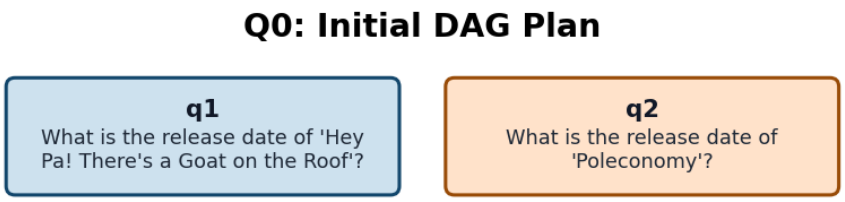
dataset: hotpotqa
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run-dir: /home/taluka/discourse_hyper_graph/sample_exp/nkg/hotpotqa/run_v1
questions in report: 100
retrieval args: {'top_k_seeds': 9, 'max_depth': 1, 'beam_width': 6, 'final_top_k': 10, 'mode': 'all', 'beam_mmr_lambda': 0.6, 'final_mmr_lambda': 0.6, 'search_mode': 'beam_search', 'deduplicate_paths': True, 'neutral_layout': True, 'max_fork_branches': 3}
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aggregate over this subset: EM=0.5000 F1=0.6298 n=100

Q0 [comparison / hard] 5a7555215542996c70cfaee1

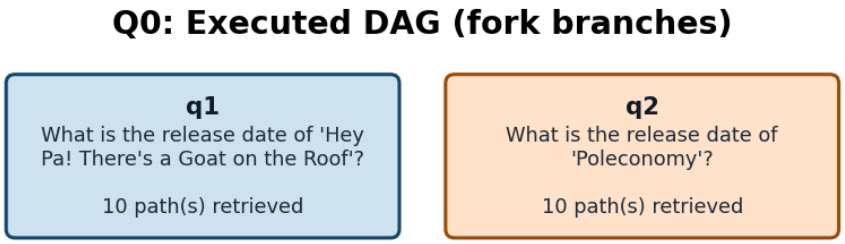
Which came to market first, "Hey Pa! There's a Goat on the Roof" or "Poleconomy"

Gold answer	Hey Pa! There's a Goat on the Roof
LLM answer	Hey Pa! There's a Goat on the Roof came to market first.
EM / F1	**0.00 / 0.75**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the release date of 'Hey Pa! There's a Goat on the Roof'?	10 path(s)
q2	0	—	What is the release date of 'Poleconomy'?	10 path(s)

Final context fed to generation (1310 tokens)

```
Query plan:
- q1: What is the release date of 'Hey Pa! There's a Goat on the Roof'?
- q2: What is the release date of 'Poleconomy'?

Level 0:

Subquery: What is the release date of 'Hey Pa! There's a Goat on the Roof'?
"Hey Pa! There's a Goat on the Roof was issued in 1966." -> (has ) -> [hey pa! there's a goat on the roof
(Children's board game from 1966)]
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"Hey Pa! There's a Goat on the Roof was issued in 1966." -> YEAR_OF -> "Hey Pa! There's a Goat on the Roof was issued by Parker Brothers."
Logical Conclusions:
The board game 'Hey Pa! There's a Goat on the Roof' was issued by Parker Brothers in 1966.

"Hey Pa! There's a Goat on the Roof was issued in 1966." -> YEAR_OF -> "Hey Pa! There's a Goat on the Roof was a children's board game."
Logical Conclusions:
The children's board game 'Hey Pa! There's a Goat on the Roof' was issued in 1966.

[hey pa! there's a goat on the roof (Children's board game from 1966)] -> (participates in event: Hey Pa! There's a Goat on the Roof was issued by Parker Brothers.) -> "Hey Pa! There's a Goat on the Roof was issued by Parker Brothers."

[hey pa! there's a goat on the roof (Children's board game from 1966)] -> (participates in event: Hey Pa! There's a Goat on the Roof was a children's board game.) -> "Hey Pa! There's a Goat on the Roof was a children's board game."

"He Will Have His Way was released on 12 November 2010." -> (has) -> [Finn Brothers Compilation Albums (Compilation albums of Finn Brothers songs)]

"He Will Have His Way was released on 12 November 2010." -> RELEASE_DATE_OF -> "He Will Have His Way is a compilation album featuring male Australian and New Zealand musicians performing songs written by Neil Finn and Tim Finn (The Finn Brothers)."
Logical Conclusions:
'He Will Have His Way' was released on 12 November 2010.

"He Will Have His Way was released on 12 November 2010." -> FOLLOWS -> "He Will Have His Way is a sequel to 'She Will Have Her Way'."
Logical Conclusions:
'He Will Have His Way' was released in 2010, while its predecessor 'She Will Have Her Way' was released in 2005.

"He Will Have His Way was released on 12 November 2010." -> PART_OF -> "The two albums were later released together with extra tracks as the double CD 'They Will Have Their Way'."
Logical Conclusions:
'He Will Have His Way' was later released together with 'She Will Have Her Way' as the double CD 'They Will Have Their Way'.

Subquery: What is the release date of 'Poleconomy'?
"Poleconomy was first published in 1980." -> (has) -> [Monopoly-like board games (Economic board games from 1980)]

"Poleconomy was first published in 1980." -> YEAR_OF -> "Poleconomy is a board game."
Logical Conclusions:
The board game Poleconomy was first published in 1980.

"Poleconomy was first published in 1980." -> HAS_EDITIONS -> "Different editions of Poleconomy were published for several different countries."
Logical Conclusions:
The game Poleconomy, first published in 1980, had different editions for several countries.

"Poleconomy was first published in 1980." -> INVENTED_BY -> "Poleconomy was invented by New Zealander Bruce E. Hatherley."
Logical Conclusions:
Poleconomy was invented by Bruce E. Hatherley and first published in 1980.

"He Will Have His Way was released on 12 November 2010." -> RELEASE_DATE_OF -> "He Will Have His Way is a compilation album featuring male Australian and New Zealand musicians performing songs written by Neil Finn and Tim Finn (The Finn Brothers)."
Logical Conclusions:
'He Will Have His Way' was released on 12 November 2010.

"He Will Have His Way was released on 12 November 2010." -> FOLLOWS -> "He Will Have His Way is a sequel to 'She Will Have Her Way'."
Logical Conclusions:
'He Will Have His Way' was released in 2010, while its predecessor 'She Will Have Her Way' was released in 2005.

"He Will Have His Way was released on 12 November 2010." -> (has) -> [Finn Brothers Compilation Albums (Compilation albums of Finn Brothers songs)]

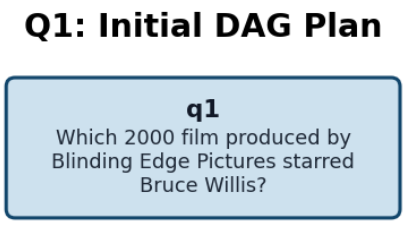
"He Will Have His Way was released on 12 Nov …[truncated for report]

Q1 [bridge / hard] 5a8b3af455429971feec467b

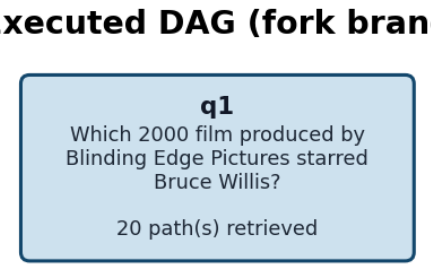
Blinding Edge Pictures produced which 2000 film starring Bruce Willis?

Gold answer	Unbreakable
LLM answer	Unbreakable
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which 2000 film produced by Blinding Edge Pictures starred Bruce Willis?	20 path(s)

Final context fed to generation (1006 tokens)

```
"Blinding Edge Pictures produced 'Unbreakable' (2000)." -> (has ) -> [M. Night Shyamalan and Related Films (Films including Unbreakable and Lay the Favorite)] -> (participates in event: 'Unbreakable' stars Bruce Willis.) -> "'Unbreakable' stars Bruce Willis." -> FEATURES -> "'Unbreakable' is a 2000 American superhero thriller film."
Logical Conclusions:
Bruce Willis stars in the film 'Unbreakable'.

[blinding edge pictures (Production company for Signs)] -> (participates in event: Blinding Edge Pictures produced 'Unbreakable' (2000).) -> "Blinding Edge Pictures produced 'Unbreakable' (2000)."

"'Unbreakable' stars Bruce Willis." -> CO_STARS -> "'Unbreakable' stars Robin Wright."
```

Logical Conclusions:
Both facts list stars of the film 'Unbreakable'.

"'Unbreakable' stars Bruce Willis." -> (has) -> [bruce willis (Actor starring in multiple films)] -> (participates in event: Mercury Rising stars Bruce Willis and Alec Baldwin.) -> "Mercury Rising stars Bruce Willis and Alec Baldwin."

[bruce willis (Actor starring in multiple films)] -> (participates in event: The two veteran NYPD partners are played by Willis and Morgan.) -> "The two veteran NYPD partners are played by Willis and Morgan."

[bruce willis (Actor starring in multiple films)] -> (participates in event: In Country stars Bruce Willis and Emily Lloyd.) -> "In Country stars Bruce Willis and Emily Lloyd."

[bruce willis (Actor starring in multiple films)] -> (participates in event: Striking Distance stars Bruce Willis.) -> "Striking Distance stars Bruce Willis."

[bruce willis (Actor starring in multiple films)] -> (participates in event: Bruce Willis earned a best supporting actor Golden Globe nomination for his role) -> "Bruce Willis earned a best supporting actor Golden Globe nomination for his role in In Country."

"'Unbreakable' stars Bruce Willis." -> PLOT_SUMMARY -> "In 'Unbreakable', Dunn explores and reluctantly confronts his powers while trying to navigate a difficult family life."
Logical Conclusions:
In Unbreakable, Dunn explores and reluctantly confronts his powers while trying to navigate a difficult family life.

[bruce willis (Actor starring in multiple films)] -> (participates in event: Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Josh) -> "Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson."

"'Unbreakable' stars Bruce Willis." -> APPEARS_IN -> "In 'Unbreakable', a security guard named David Dunn survives a horrific train crash."
Logical Conclusions:
Fact 1 identifies a star of 'Unbreakable', while fact 10 describes the plot involving that star's character.

"'Unbreakable' stars Bruce Willis." -> PLOT_SUMMARY -> "After the train crash in 'Unbreakable', with the help of a manipulative disabled comic book shop owner named Elijah Price, David Dunn learns that he possesses superhuman powers."
Logical Conclusions:
After the train crash in Unbreakable, with the help of a manipulative disabled comic book shop owner named Elijah Price, David Dunn learns that he possesses superhuman powers.

[M. Night Shyamalan and Related Films (Films including Unbreakable and Lay the Favorite)] -> (participates in event: Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Josh) -> "Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson." -> SAME_ACTOR -> "In Mercury Rising, Bruce Willis plays Art Jeffries."
Logical Conclusions:
Both facts involve Bruce Willis starring in different films.

"Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson." -> SAME_ACTOR -> "In Country stars Bruce Willis and Emily Lloyd."
Logical Conclusions:
Both facts involve Bruce Willis starring in different films.

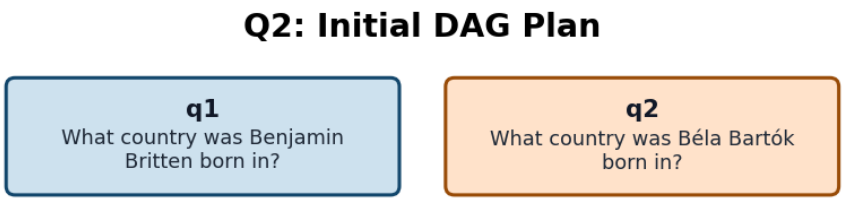
"Lay the Favorite stars Bruce Willis, Rebecca Hall, Catherine Zeta-Jones and Joshua Jackson." -> SAME_ACTOR -> "Mercury Rising stars Bruce Willis and Alec Baldwin."
Logical Conclusions:
Both facts involve Bruce Willis starring …[truncated for report]

Q2 [comparison / hard] 5ae12bcc5542997b2ef7d11a

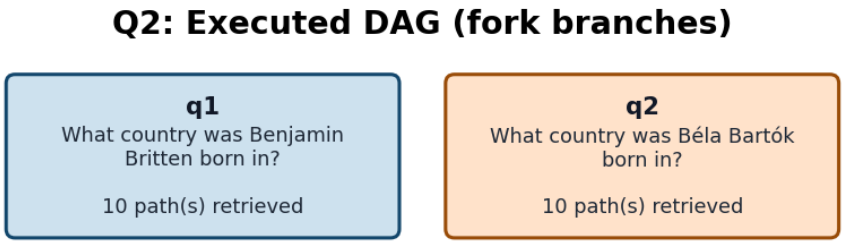
Are Benjamin Britten and Béla Bartók from the same country?

Gold answer	no
LLM answer	No
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What country was Benjamin Britten born in?	10 path(s)
q2	0	—	What country was Béla Bartók born in?	10 path(s)

Final context fed to generation (1107 tokens)

```
Query plan:
- q1: What country was Benjamin Britten born in?
- q2: What country was Béla Bartók born in?

Level 0:

Subquery: What country was Benjamin Britten born in?
"Benjamin Britten was born on 22 November 1913." -> BIRTH_DATE -> "Edward Benjamin Britten, Baron Britten
was an English composer, conductor and pianist."
Logical Conclusions:
Benjamin Britten was born on 22 November 1913.
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"Benjamin Britten was born on 22 November 1913." -> (has) -> [Benjamin Britten (English composer and conductor)]

"Benjamin Britten was born on 22 November 1913." -> PRECEDES -> "Benjamin Britten died on 4 December 1976."
Logical Conclusions:
Britten's birth date precedes his death date, establishing his lifespan.

[Benjamin Britten (English composer and conductor)] -> (participates in event: Edward Benjamin Britten, Baron Britten was an English composer, conductor and pi) -> "Edward Benjamin Britten, Baron Britten was an English composer, conductor and pianist."

[Benjamin Britten (English composer and conductor)] -> (participates in event: Benjamin Britten was a central figure of 20th-century British classical music.) -> "Benjamin Britten was a central figure of 20th-century British classical music."

[Benjamin Britten (English composer and conductor)] -> (participates in event: The English composers whose works were premiered and recorded by the Zorian Quar) -> "The English composers whose works were premiered and recorded by the Zorian Quartet include Benjamin Britten and Michael Tippett."

[Benjamin Britten (English composer and conductor)] -> (participates in event: Benjamin Britten's best-known works include the orchestral piece.) -> "Benjamin Britten's best-known works include the orchestral piece."

[Benjamin Britten (English composer and conductor)] -> (participates in event: Notable composers who have been influenced by Bartók's string quartets include B) -> "Notable composers who have been influenced by Bartók's string quartets include Benjamin Britten."

"He lived in Berlin-Lankwitz, Germany." -> (has) -> [Germany (Country in Europe)]

Subquery: What country was Béla Bartók born in?
"Béla Bartók was born on 25 March 1881." -> BIRTH_DATE -> "Béla Viktor János Bartók was a Hungarian composer, pianist and an ethnomusicologist."
Logical Conclusions:
Béla Bartók was a Hungarian composer born on 25 March 1881.

"Béla Bartók was born on 25 March 1881." -> (has) -> [béla bartók (Composer of the string quartets)]

"Béla Bartók was born on 25 March 1881." -> PRECEDES -> "Béla Bartók died on 26 September 1945."
Logical Conclusions:
Bartók's death date follows his birth date, completing his lifespan.

"After his mother's death, Bartók decided to leave with his family for the United States." -> (has) -> [béla bartók (Composer of the string quartets)]

"Béla Bartók Music High School is situated in the Palace of Music in Bartók square Miskolc, Hungary." -> (has) -> [béla bartók music high school (Music school in Miskolc, Hungary)]

"Béla Bartók Music High School is situated in the Palace of Music in Bartók square Miskolc, Hungary." -> NAMED_AFTER -> "Béla Bartók Music High School is named after the famous Hungarian composer Béla Bartók."
Logical Conclusions:
Béla Bartók Music High School is named after the composer Béla Bartók.

"Béla Bartók Music High School is situated in the Palace of Music in Bartók square Miskolc, Hungary." -> FOUNDED_IN -> "Béla Bartók Music High School was founded in 1966."
Logical Conclusions:
Béla Bartók Music High School was founded in 1966.

"Béla Bartók Music High School is situated in the Palace of Music in Bartók square Miskolc, Hungary." -> HOUSED_IN -> "The construction of the Palace of Music was financed from USA loans."
Logical Conclusions:
The Palace of Music, which houses the Béla Bartók Music High School, was financed from USA loans.

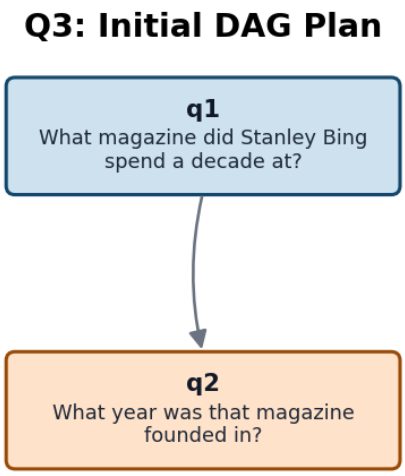
"Béla Bartók Music High School is situated in the Palace of Music in Bartók square Miskolc, Hungary." -> LOCATED_IN -> "The Palace of Music is a building in Bartók squar …[truncated for report]

Q3 [bridge / hard] 5adfecd755429942ec259b90

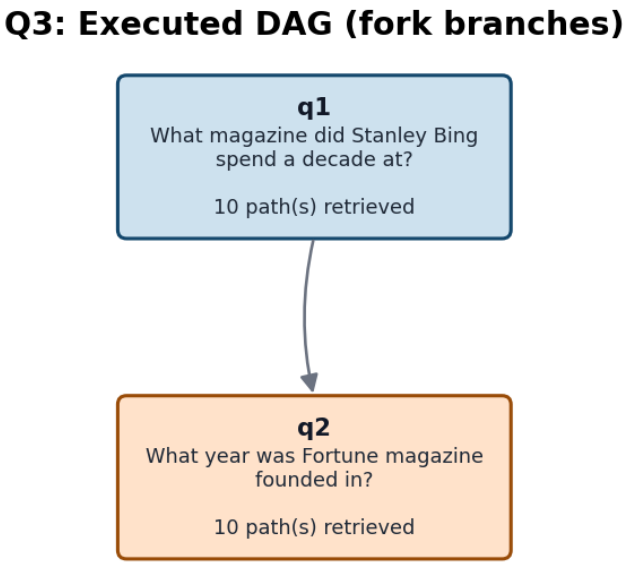
What year was the magazine that Stanley Bing spent a decade at founded in?

Gold answer	1933
LLM answer	1933
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What magazine did Stanley Bing spend a decade at?	10 path(s)
q2	1	q1	What year was Fortune magazine founded in?	10 path(s)

Final context fed to generation (1254 tokens)

```
Query plan:
- q1: What magazine did Stanley Bing spend a decade at?
- q2: What year was that magazine founded in? (depends on: q1)

Level 1:

Subquery: What year was Fortune magazine founded in?
"Esquire was founded in 1933." -> PRECEDES -> "Gil Schwartz spent a decade at 'Esquire' before writing for
'Fortune'."
Logical Conclusions:
Gil Schwartz spent a decade at Esquire, which was founded in 1933, before writing for Fortune.

"Esquire was founded in 1933." -> FOUNDED -> "Esquire is an American men's magazine."
Logical Conclusions:
Esquire is an American men's magazine founded in 1933.

"Esquire was founded in 1933." -> (has ) -> [Esquire (men's magazine and watch brand)]

"Esquire was founded in 1933." -> FOUNDING_YEAR -> "Esquire was founded under the guidance of founders
Arnold Gingrich, David A. Smart and Henry L. Jackson."
Logical Conclusions:
Esquire was founded in 1933 under the guidance of Arnold Gingrich, David A. Smart and Henry L. Jackson.

"Esquire was founded in 1933." -> CO_OCCURS -> "Esquire flourished during the Great Depression."
Logical Conclusions:
Esquire was founded in 1933 and flourished during the Great Depression.

[fortune (Magazine where Stanley Bing wrote column)] -> (participates in event: Gil Schwartz has written a
column for 'Fortune' magazine for more than twenty ye) -> "Gil Schwartz has written a column for 'Fortune'
magazine for more than twenty years."

[Palace Entertainment (Current owner and operator of Castle Park)] -> (participates in event: The Palace of
Music was built between 1926 and 1927.) -> "The Palace of Music was built between 1926 and 1927."

[Billboard (Music industry publication)] -> (participates in event: Nielsen SoundScan data was incorporated
from 1991.) -> "Nielsen SoundScan data was incorporated from 1991."

[fortune brands, inc. (company where Mackay served as director)] -> (participates in event: In addition to
his executive positions, A. D. David Mackay is also a director of) -> "In addition to his executive
positions, A. D. David Mackay is also a director of Fortune Brands, Inc."

[Jane Rhodes (French opera singer known for Carmen role)] -> (participates in event: Jane Marie Andrée
Rhodes was born on March 13, 1929.) -> "Jane Marie Andrée Rhodes was born on March 13, 1929."

Level 0:

Subquery: What magazine did Stanley Bing spend a decade at?
[fortune (Magazine where Stanley Bing wrote column)] -> (participates in event: Gil Schwartz has written a
column for 'Fortune' magazine for more than twenty ye) -> "Gil Schwartz has written a column for 'Fortune'
magazine for more than twenty years."

[stanley bing (Pen name of business humorist Gil Schwartz)] -> (participates in event: Stanley Bing is the
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pen name of Gil Schwartz.) -> "Stanley Bing is the pen name of Gil Schwartz."

[James Garner (Actor in films and TV series)] -> (participates in event: In 'Shady Deal at Sunny Acres', gambler Bret Maverick (James Garner) is swindled) -> "In 'Shady Deal at Sunny Acres', gambler Bret Maverick (James Garner) is swindled by a crooked banker (John Dehner)."

[James Garner (Actor in films and TV series)] -> (participates in event: James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAda) -> "James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams."

[Hank Williams (American country music singer-songwriter)] -> (participates in event: 'Leave Them Boys Alone' was written by Hank Williams Jr., Dean Dillon, Gary Stew) -> "'Leave Them Boys Alone' was written by Hank Williams Jr., Dean Dillon, Gary Stewart, and Tanya Tucker."

[Randy W. Schekman (Co-recipient of 2013 prize)] -> (participates in event: Ekman and Thomas C. Südhof were recognised after discovering how cells precisely) -> "Ekman and Thomas C. Südhof were recognised after discovering how cells precisely transport material."

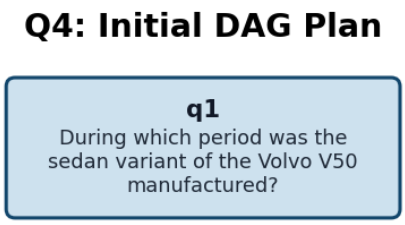
[Palace Entertainment (Current owner and operator of Castle Park)] -> (participates in event: The Béla Bartók Music Institute is a faculty of the Univers …[truncated for report]

Q4 [bridge / hard] 5ab465095542991751b4d75a

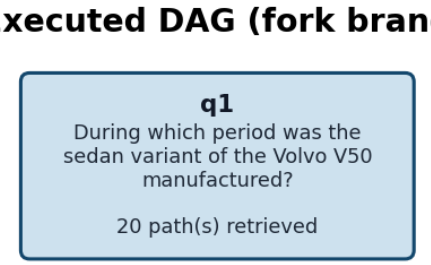
During which period the sedan variant of the Volvo V50 car was manufactured?

Gold answer	1995 to 2012
LLM answer	1995 to 2012
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	During which period was the sedan variant of the Volvo V50 manufactured?	20 path(s)

Final context fed to generation (1224 tokens)

```
[volvo v50 (Volvo car model with sedan variant)] -> (participates in event: The sedan variant of the Volvo
V50 is the Volvo S40.) -> "The sedan variant of the Volvo V50 is the Volvo S40." -> SUPPORTS -> "The Volvo
S40 was produced from 1995 to 2012." -> SPANS -> "The Volvo S40 was produced across two generations."
Logical Conclusions:
The Volvo S40 was produced from 1995 to 2012 and the sedan variant of the V50 is the S40.
The S40's production from 1995 to 2012 spanned two generations.

"The Volvo S40 was produced from 1995 to 2012." -> (has ) -> [volvo s40 (compact luxury car marketed from
1995 to 2012)] -> (participates in event: The Volvo S40 was produced across two generations.) -> "The Volvo
S40 was produced across two generations."
```

"The sedan variant of the Volvo V50 is the Volvo S40." -> (has) -> [volvo s40 (compact luxury car marketed from 1995 to 2012)]

"The sedan variant of the Volvo V50 is the Volvo S40." -> RELATES_TO -> "The Volvo S40 was produced across two generations."
Logical Conclusions:
The Volvo S40 is the sedan variant of the Volvo V50 and was produced across two generations.

"The Volvo S40 was produced from 1995 to 2012." -> PRODUCTION_PERIOD_OF -> "The Volvo S40 is a compact luxury car."
Logical Conclusions:
The Volvo S40, a compact luxury car, was produced from 1995 to 2012.

"The Volvo S40 was produced from 1995 to 2012." -> PRODUCTION_PERIOD_OF -> "The Volvo S40 was marketed and produced by the Swedish manufacturer Volvo Cars."
Logical Conclusions:
Volvo Cars produced the S40 from 1995 to 2012.

"The Volvo S40 was produced from 1995 to 2012." -> SUPPORTS -> "The Volvo C30 is a variant of the Volvo S40/V50/C70 range."
Logical Conclusions:
The Volvo S40 was produced from 1995 to 2012 and the C30 is a variant of the S40/V50/C70 range.

[volvo s40 (compact luxury car marketed from 1995 to 2012)] -> (participates in event: The Volvo S40 is a compact luxury car.) -> "The Volvo S40 is a compact luxury car."

"The Volvo S40 was produced from 1995 to 2012." -> SUPPORTS -> "The Volvo C30 shares the same Ford C1/Volvo P1 platform as the S40/V50/C70 range."
Logical Conclusions:
The Volvo S40 was produced from 1995 to 2012 and shares the Ford C1/Volvo P1 platform with the C30.

[volvo s40 (compact luxury car marketed from 1995 to 2012)] -> (participates in event: The Volvo C30 is a variant of the Volvo S40/V50/C70 range.) -> "The Volvo C30 is a variant of the Volvo S40/V50/C70 range."

[volvo s40 (compact luxury car marketed from 1995 to 2012)] -> (participates in event: The Volvo C30 shares the same Ford C1/Volvo P1 platform as the S40/V50/C70 range) -> "The Volvo C30 shares the same Ford C1/Volvo P1 platform as the S40/V50/C70 range."

[volvo s40 (compact luxury car marketed from 1995 to 2012)] -> (participates in event: The Volvo S40 was marketed and produced by the Swedish manufacturer Volvo Cars.) -> "The Volvo S40 was marketed and produced by the Swedish manufacturer Volvo Cars."

[volvo v50 (Volvo car model with sedan variant)] -> (participates in event: The Volvo V50 is a car model.) -> "The Volvo V50 is a car model." -> HAS_VARIANT -> "The sedan variant of the Volvo V50 is the Volvo S40."
Logical Conclusions:
The Volvo V50 is a car model and its sedan variant is the Volvo S40.

"The sedan variant of the Volvo V50 is the Volvo S40." -> RELATES_TO -> "The Volvo C30 shares the same Ford C1/Volvo P1 platform as the S40/V50/C70 range."
Logical Conclusions:
The Volvo S40 is the sedan variant of the V50 and shares the Ford C1/Volvo P1 platform with the C30.

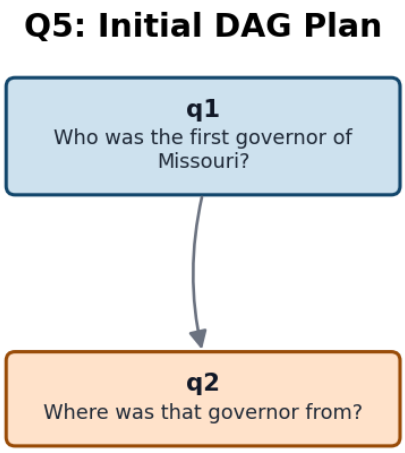
"The sedan variant of the Volvo V50 is the Volvo S40." -> RELATES_TO -> "The Volvo C30 is a variant of the Volvo S40/V50/C70 range."
Logical Conclusions:
The Volvo S40 is the sedan variant of the V50, and the C30 is a variant of the S40/V50/C70 range.

Q5 [bridge / hard] 5adfdef9554299025d62a36b

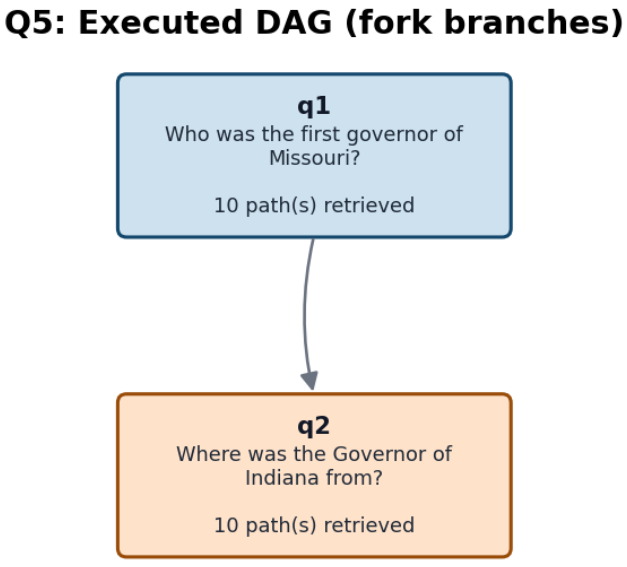
Where was the first governor after the The Missouri Compromise from?

Gold answer	Bath, Maine
LLM answer	The first governor of Missouri was Alexander McNair, and he was from Virginia.
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was the first governor of Missouri?	10 path(s)
q2	1	q1	Where was the Governor of Indiana from?	10 path(s)

Final context fed to generation (1222 tokens)

```
Query plan:
- q1: Who was the first governor of Missouri?
- q2: Where was that governor from? (depends on: q1)

Level 1:

Subquery: Where was the Governor of Indiana from?
"The Governor of Indiana became an early leader in the Republican Party starting in 1856." -> (has ) ->
[governor of indiana (Shortest-serving governor who resigned for Senate)]

"The Governor of Indiana became an early leader in the Republican Party starting in 1856." -> FOLLOWS_FROM
-> "The Governor of Indiana was known for his opposition to slavery."
Logical Conclusions:
The Governor's opposition to slavery led him to become an early leader in the Republican Party.

"The Governor of Indiana became an early leader in the Republican Party starting in 1856." -> ENABLES ->
"The Governor of Indiana served as the president of the first party convention."
Logical Conclusions:
As an early Republican leader, the Governor served as president of the first party convention.

"The Governor of Indiana became an early leader in the Republican Party starting in 1856." ->
CONTRIBUTES_TO -> "The Governor of Indiana was influential in the nomination of Abraham Lincoln."
Logical Conclusions:
The Governor's leadership in the Republican Party was influential in the nomination of Abraham Lincoln.

[governor of indiana (Shortest-serving governor who resigned for Senate)] -> (participates in event: The
Governor of Indiana was known for his opposition to slavery.) -> "The Governor of Indiana was known for his
opposition to slavery."

"The Governor of Indiana supported compromise with the south." -> INFLUENCES -> "The Governor of Indiana
became an early leader in the Republican Party starting in 1856."
Logical Conclusions:
The Governor's support for compromise with the south influenced his role in the Republican Party.

"The Governor of Indiana became an early leader in the Republican Party starting in 1856." -> FOLLOWS ->
"The Governor of Indiana was a Whig until the party collapsed."
Logical Conclusions:
After the Whig party collapsed, the Governor became an early leader in the Republican Party.

"The Governor of Indiana became an early leader in the Republican Party starting in 1856." -> ENABLES ->
"The Governor of Indiana delivered the keynote address at the first party convention."
Logical Conclusions:
As an early Republican leader, the Governor delivered the keynote address at the first party convention.

[governor of indiana (Shortest-serving governor who resigned for Senate)] -> (participates in event: The
Governor of Indiana was influential in the nomination of Abraham Lincoln.) -> "The Governor of Indiana was
influential in the nomination of Abraham Lincoln."

Level 0:

Subquery: Who was the first governor of Missouri?
"With the repeal of the Missouri Compromise, the Governor of Indiana became a full-fledged something." ->
(has ) -> [governor of indiana (Shortest-serving governor who resigned for Senate)]
```

"With the repeal of the Missouri Compromise, the Governor of Indiana became a full-fledged something." -> (has) -> [Missouri Compromise (Legislation dividing slave and free states)]

"With the repeal of the Missouri Compromise, the Governor of Indiana became a full-fledged something." -> CAUSES -> "Bleeding Kansas was largely brought about by the Kansas–Nebraska Act."
Logical Conclusions:
The repeal of the Missouri Compromise by the Kansas-Nebraska Act caused the Governor to become a full-fledged something.

"With the repeal of the Missouri Compromise, the Governor of Indiana became a full-fledged something." -> CONTEXTUALIZED_BY -> "The Governor of Indiana supported compromise with the south."
Logical Conclusions:
The Governor of Indiana supported compromise with the south and became a full-fledged something with the repeal of the Missouri Compromise.

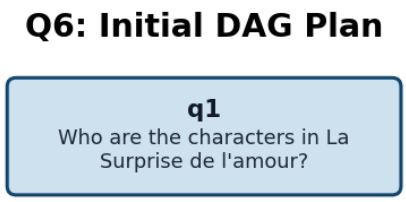
[alexander hamilton (won decision for national government to pay state debts)] -> (participates in event: The Compromise of 1790 was a compromise between Alexander Hamilton on the one ha) -> "The Compromise of 1790 was …[truncated for report]

Q6 [bridge / hard] 5a7f53ff5542992097ad2f17

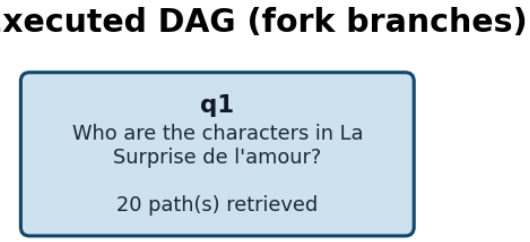
In addition to the best known comic servant from Commedia dell'arte, who else is featured in La Surprise de l'amour?

Gold answer	Columbine
LLM answer	Columbine
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who are the characters in La Surprise de l'amour?	20 path(s)

Final context fed to generation (534 tokens)

```
"In 'La Surprise de l'amour', Arlequin is featured." -> APPEARS_IN -> "Marivaux's play 'La Surprise de
l'amour' makes use of stock characters from the Commedia dell'arte." -> EXEMPLIFIES -> "Columbina is a
stock character in the 'Commedia dell'Arte'."
Logical Conclusions:
Marivaux's play 'La Surprise de l'amour' makes use of stock characters, including Arlequin.
Marivaux's play uses stock characters from Commedia dell'arte, such as Columbina.

[marivaux (Playwright using commedia dell'arte characters)] -> (participates in event: In 'La Surprise de
l'amour', Arlequin is featured.) -> "In 'La Surprise de l'amour', Arlequin is featured."

"In 'La Surprise de l'amour', Columbine is featured." -> APPEARS_IN -> "Marivaux's play 'La Surprise de
l'amour' makes use of stock characters from the Commedia dell'arte."
Logical Conclusions:
Marivaux's play 'La Surprise de l'amour' makes use of stock characters, including Columbine.
```

"In 'La Surprise de l'amour', Columbine is featured." -> CO_OCCURS_WITH -> "In 'La Surprise de l'amour', Arlequin is featured."
Logical Conclusions:
In 'La Surprise de l'amour', both Arlequin and Columbine are featured.

"Marivaux's play 'La Surprise de l'amour' makes use of stock characters from the Commedia dell'arte." -> (has) -> [marivaux (Playwright using commedia dell'arte characters)] -> (participates in event: In 'La Surprise de l'amour', Columbine is featured.) -> "In 'La Surprise de l'amour', Columbine is featured."

"In 'La Surprise de l'amour', Arlequin is featured." -> (has) -> [arlequin (Stock character in La Surprise de l'amour)]

"In 'La Surprise de l'amour', Columbine is featured." -> (has) -> [columbine (Stock character in La Surprise de l'amour)]

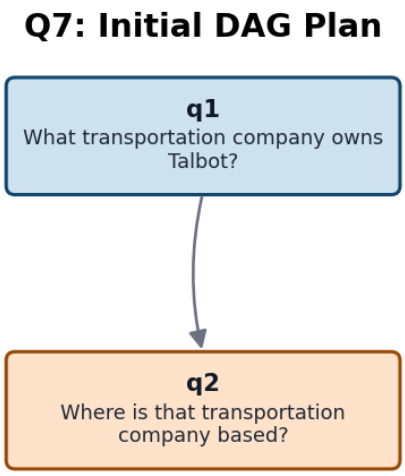
"In 'La Surprise de l'amour', Columbine is featured." -> EXEMPLIFIES -> "Columbina is Harlequin's mistress."
Logical Conclusions:
Fact 10 identifies Columbina as Harlequin's mistress, and fact 6 mentions Columbine appears in 'La Surprise de l'amour'.

Q7 [bridge / hard] 5abe535055429976d4830aea

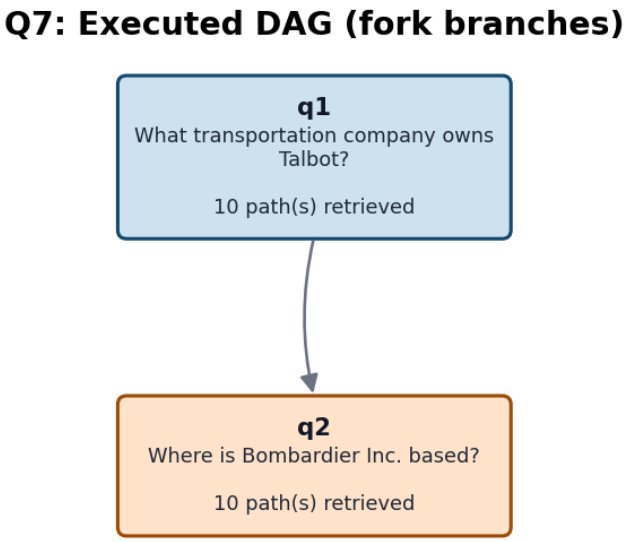
Where is the transportation company that owns Talbot based in?

Gold answer	Montreal
LLM answer	Montreal, Quebec, Canada
EM / F1	0.00 / 0.50
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What transportation company owns Talbot?	10 path(s)
q2	1	q1	Where is Bombardier Inc. based?	10 path(s)

Final context fed to generation (990 tokens)

```
Query plan:
- q1: What transportation company owns Talbot?
- q2: Where is that transportation company based? (depends on: q1)

Level 1:

Subquery: Where is Bombardier Inc. based?
"Bombardier Inc. is based in Montreal, Quebec, Canada." -> HAS_BASE -> "Bombardier Inc. is a multinational
aerospace and transportation company."
Logical Conclusions:
Bombardier Inc., a multinational aerospace and transportation company, is based in Montreal, Quebec,
Canada.

"Bombardier Inc. is based in Montreal, Quebec, Canada." -> (has ) -> [bombardier inc. (multinational
aerospace and transportation company)]

"Bombardier Inc. is based in Montreal, Quebec, Canada." -> LOCATED_IN -> "Over the years, Bombardier Inc.
has grown into a large manufacturer of mass transportation equipment."
Logical Conclusions:
Bombardier Inc., based in Montreal, manufactures mass transportation equipment.

"Bombardier Inc. is based in Montreal, Quebec, Canada." -> LOCATED_IN -> "Over the years, Bombardier Inc.
has grown into a large manufacturer of business jets."
Logical Conclusions:
Bombardier Inc., based in Montreal, manufactures business jets.

"Bombardier Inc. is based in Montreal, Quebec, Canada." -> LOCATED_IN -> "Bombardier Inc. started as a
maker of snow machines or snowmobiles."
Logical Conclusions:
Bombardier Inc., based in Montreal, started as a snowmobile maker.

"Bombardier Inc. is based in Montreal, Quebec, Canada." -> CONTAINS -> "Talbot is part of Bombardier Inc."
Logical Conclusions:
Talbot is part of Bombardier Inc., which is based in Montreal, Quebec, Canada.

[bombardier inc. (multinational aerospace and transportation company)] -> (participates in event:
Bombardier Inc. is a multinational aerospace and transportation company.) -> "Bombardier Inc. is a
multinational aerospace and transportation company."

[bombardier inc. (multinational aerospace and transportation company)] -> (participates in event: Over the
years, Bombardier Inc. has grown into a large manufacturer of regional ) -> "Over the years, Bombardier
Inc. has grown into a large manufacturer of regional airliners."

[bombardier inc. (multinational aerospace and transportation company)] -> (participates in event:
Bombardier Inc. started as a maker of snow machines or snowmobiles.) -> "Bombardier Inc. started as a maker
of snow machines or snowmobiles."

Level 0:

Subquery: What transportation company owns Talbot?
"Talbot is part of Bombardier Inc." -> (has ) -> [bombardier inc. (multinational aerospace and
transportation company)]
```

"Talbot is part of Bombardier Inc." -> PART_OF -> "Over the years, Bombardier Inc. has grown into a large manufacturer of mass transportation equipment."
Logical Conclusions:
Talbot is part of Bombardier Inc., which manufactures mass transportation equipment.

"Talbot is part of Bombardier Inc." -> (has) -> [talbot (builder of NS VIRM trains)]

"Talbot is part of Bombardier Inc." -> HAS_PART -> "The VIRM trains were for the most part built by Talbot."
Logical Conclusions:
Talbot, which built most VIRM trains, is part of Bombardier Inc.

"Talbot is part of Bombardier Inc." -> PART_OF -> "Over the years, Bombardier Inc. has grown into a large manufacturer of business jets."
Logical Conclusions:
Talbot is part of Bombardier Inc., which manufactures business jets.

"Talbot is part of Bombardier Inc." -> PART_OF -> "Bombardier Inc. is based in Montreal, Quebec, Canada."
Logical Conclusions:
Talbot is part of Bombardier Inc., which is based in Montreal, Quebec, Canada.

[talbot (builder of NS VIRM trains)] -> (participates in event: The VIRM trains were for the most part built by Talbot.) -> "The VIRM trains were for the most part built by Talbot."

"The group of companies had interests in transportation services." -> PART_OF -> "Lelio 'Les' Marino grew Modern Continental into a six billion-dollar group of companies."
Logical Conclusions:
The group of companies included interests in transportation services.

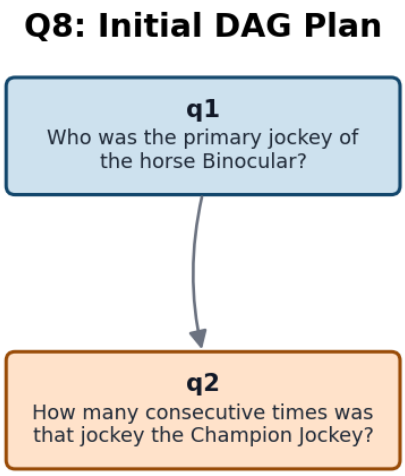
"The group of companies had interests in transportation ser …[truncated for report]

Q8 [bridge / hard] 5a8f51185542992414482a3d

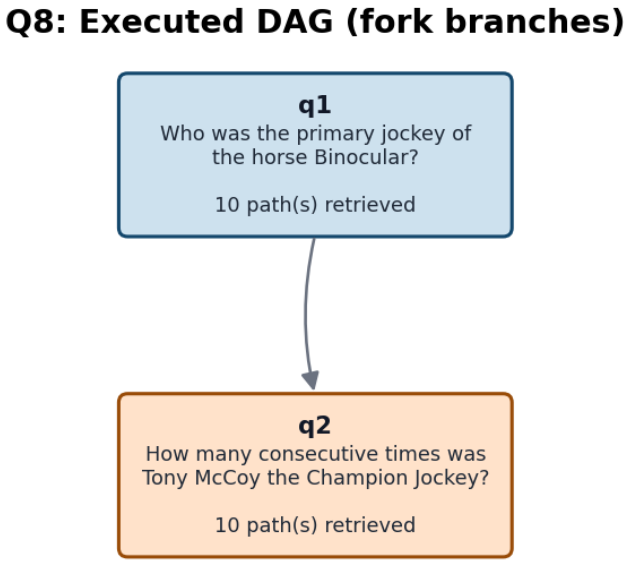
The primary jockey of Binocular was Champion Jockey how many consecutive times?

Gold answer	20
LLM answer	20
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was the primary jockey of the horse Binocular?	10 path(s)
q2	1	q1	How many consecutive times was Tony McCoy the Champion Jockey?	10 path(s)

Final context fed to generation (1025 tokens)

```
Query plan:
- q1: Who was the primary jockey of the horse Binocular?
- q2: How many consecutive times was that jockey the Champion Jockey? (depends on: q1)

Level 1:

Subquery: How many consecutive times was Tony McCoy the Champion Jockey?
"Tony McCoy was Champion Jockey a record 20 consecutive times." -> SUPPORTS -> "Tony McCoy rode a record
4,358 winners."
Logical Conclusions:
Tony McCoy rode a record 4,358 winners and was Champion Jockey a record 20 consecutive times.

"Tony McCoy was Champion Jockey a record 20 consecutive times." -> SUPPORTS -> "Sir Anthony Peter McCoy,
OBE, was born on 4 May 1974."
Logical Conclusions:
Tony McCoy was champion jockey a record 20 consecutive times.

"Tony McCoy was Champion Jockey a record 20 consecutive times." -> (has ) -> [tony mccoy (Northern Irish
former champion jockey)]

"Tony McCoy was Champion Jockey a record 20 consecutive times." -> SUPPORTS -> "Tony McCoy was Champion
Jockey every year he was a professional."
Logical Conclusions:
Tony McCoy was Champion Jockey a record 20 consecutive times and every year he was a professional.

"Tony McCoy was Champion Jockey a record 20 consecutive times." -> SUPPORTS -> "Sir Anthony Peter McCoy,
OBE, is commonly known as A. P. McCoy or Tony McCoy."
Logical Conclusions:
Tony McCoy is known as A. P. McCoy and was Champion Jockey a record 20 consecutive times.

"Tony McCoy was Champion Jockey a record 20 consecutive times." -> CONTRASTS_WITH -> "A prize of £25,000 to
the champion jockey was introduced as part of the 2015 changes."
Logical Conclusions:
The prize for champion jockey was introduced in 2015, while Tony McCoy had already been champion jockey a
record 20 consecutive times before that.

[tony mccoy (Northern Irish former champion jockey)] -> (participates in event: Tony McCoy rode a record
4,358 winners.) -> "Tony McCoy rode a record 4,358 winners."

[tony mccoy (Northern Irish former champion jockey)] -> (participates in event: Tony McCoy was Champion
Jockey every year he was a professional.) -> "Tony McCoy was Champion Jockey every year he was a
professional."

[tony mccoy (Northern Irish former champion jockey)] -> (participates in event: Sir Anthony Peter McCoy,
OBE, is a Northern Irish former horse racing jockey.) -> "Sir Anthony Peter McCoy, OBE, is a Northern Irish
former horse racing jockey."

Level 0:

Subquery: Who was the primary jockey of the horse Binocular?
"Binocular's primary jockey is Tony McCoy." -> SUPPORTS -> "Binocular was foaled on March 17, 2004 in
```

France."
Logical Conclusions:
Binocular was foaled on March 17, 2004 and his primary jockey is Tony McCoy.

"Binocular's primary jockey is Tony McCoy." -> IDENTIFIES_AS -> "Sir Anthony Peter McCoy was born on 4 May 1974."
Logical Conclusions:
Tony McCoy, primary jockey of Binocular, is Sir Anthony Peter McCoy.

"Binocular's primary jockey is Tony McCoy." -> RIDES -> "Binocular is a French thoroughbred retired racehorse."
Logical Conclusions:
Binocular's primary jockey is Tony McCoy.

"Binocular's primary jockey is Tony McCoy." -> ASSOCIATES_WITH -> "Sir Anthony Peter McCoy, OBE, is commonly known as A. P. McCoy or Tony McCoy."
Logical Conclusions:
Binocular's primary jockey is Tony McCoy, who is also known as Sir Anthony Peter McCoy.

"Binocular's primary jockey is Tony McCoy." -> (has) -> [tony mccoy (Northern Irish former champion jockey)]

"Binocular's primary jockey is Tony McCoy." -> SUPPORTS -> "Sir Anthony Peter McCoy is also known as Tony McCoy."
Logical Conclusions:
Binocular's primary jockey is Tony McCoy, who is also known as Sir Anthony Peter McCoy.

[Binocular (French thoroughbred racehorse)] -> (participates in event: Binocular's primary jockey is Tony McCoy.) -> "Binocular's primary jockey is Tony McCoy."

[Binocular (French thoroughbred racehorse)] -> (participates in event: Binocular is trained by Nicky Henderson.) -> "Binocular is trained by Nicky Henderson."

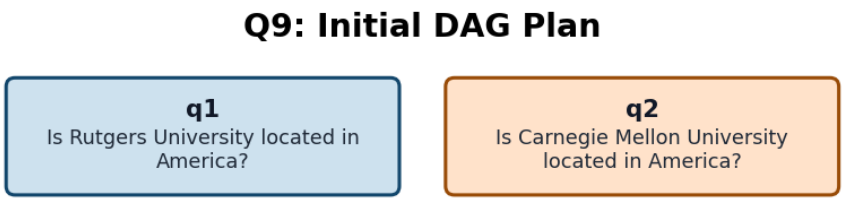
"Binocular is a French thoroughbred retired racehorse." -> TRAINED_BY -> "Binocular is trai …[truncated for report]

Q9 [comparison / hard] 5ab85a1155429934fafe6d7c

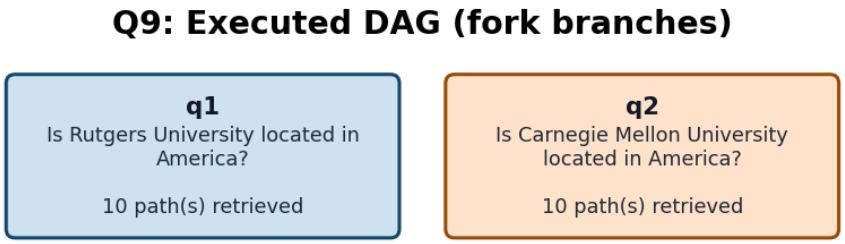
Are both Rutgers University and Carnegie Mellon University located in America?

Gold answer	yes
LLM answer	Yes
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Is Rutgers University located in America?	10 path(s)
q2	0	—	Is Carnegie Mellon University located in America?	10 path(s)

Final context fed to generation (1111 tokens)

```
Query plan:
- q1: Is Rutgers University located in America?
- q2: Is Carnegie Mellon University located in America?

Level 0:

Subquery: Is Rutgers University located in America?
[rutgers university (Public research university in New Jersey)] -> (participates in event: Rutgers
University is an American public research university.) -> "Rutgers University is an American public
research university."

[rutgers university (Public research university in New Jersey)] -> (participates in event: Rutgers
```

University is the largest institution for higher education in New Jersey) -> "Rutgers University is the largest institution for higher education in New Jersey."

[rutgers university (Public research university in New Jersey)] -> (participates in event: Rutgers, The State University of New Jersey is commonly referred to as Rutgers U) -> "Rutgers, The State University of New Jersey is commonly referred to as Rutgers University."

[rutgers university (Public research university in New Jersey)] -> (participates in event: Rutgers, The State University of New Jersey is commonly referred to as Rutgers.) -> "Rutgers, The State University of New Jersey is commonly referred to as Rutgers."

[rutgers university (Public research university in New Jersey)] -> (participates in event: Rutgers, The State University of New Jersey is commonly referred to as RU.) -> "Rutgers, The State University of New Jersey is commonly referred to as RU."

[rutgers university (Public research university in New Jersey)] -> (participates in event: David S. Touretzky received a BA in Computer Science at Rutgers University in 19) -> "David S. Touretzky received a BA in Computer Science at Rutgers University in 1978."

[princeton, new jersey (Location of Japanese weekend school)] -> (participates in event: Urinsuton Nihongo Gakk■ is a Japanese weekend school in the Princeton, New Jerse) -> "Urinsuton Nihongo Gakk■ is a Japanese weekend school in the Princeton, New Jersey area."

[Carnegie Mellon University (University employing Touretzky and others)] -> (participates in event: Carnegie Mellon University in Qatar is a branch campus of Carnegie Mellon Univer) -> "Carnegie Mellon University in Qatar is a branch campus of Carnegie Mellon University."

[Carnegie Mellon University (University employing Touretzky and others)] -> (participates in event: Gloriana St. Clair is currently the official University Liaison to the Pittsburg) -> "Gloriana St. Clair is currently the official University Liaison to the Pittsburgh chapter of the Osher Lifelong Learning Institute at Carnegie Mellon University."

[live acoustic america (acoustic live recording by Howard Jones)] -> (participates in event: 'Live Acoustic America' features Carol Steele on percussion.) -> "'Live Acoustic America' features Carol Steele on percussion."

Subquery: Is Carnegie Mellon University located in America?
"Carnegie Mellon University is located in Pittsburgh, Pennsylvania." -> (has) -> [Carnegie Mellon University (University employing Touretzky and others)]

"Carnegie Mellon University is located in Pittsburgh, Pennsylvania." -> HOSTS -> "Carnegie Mellon University is a private research university."
Logical Conclusions:
Carnegie Mellon University, a private research university, is located in Pittsburgh, Pennsylvania.

[Carnegie Mellon University (University employing Touretzky and others)] -> (participates in event: Carnegie Mellon University in Qatar is Carnegie Mellon's first undergraduate bra) -> "Carnegie Mellon University in Qatar is Carnegie Mellon's first undergraduate branch campus."

[Carnegie Mellon University (University employing Touretzky and others)] -> (participates in event: The projects are related to understanding and improving the usability of privacy) -> "The projects are related to understanding and improving the usability of privacy and security software and systems."

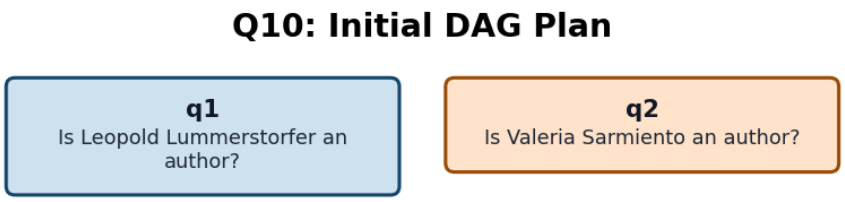
[Carnegie Mellon University (University employing Touretzky and others)] -> (participates in event: CUPS was established to …[truncated for report]

Q10 [comparison / hard] 5ae3790f5542992e3233c415

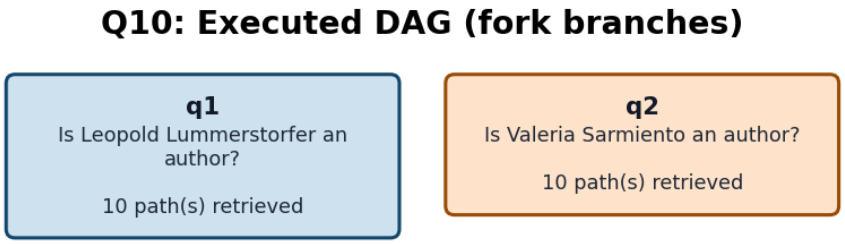
Are Leopold Lummerstorfer and Valeria Sarmiento both authors?

Gold answer	no
LLM answer	Yes
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Is Leopold Lummerstorfer an author?	10 path(s)
q2	0	—	Is Valeria Sarmiento an author?	10 path(s)

Final context fed to generation (1170 tokens)

```
Query plan:
- q1: Is Leopold Lummerstorfer an author?
- q2: Is Valeria Sarmiento an author?

Level 0:

Subquery: Is Leopold Lummerstorfer an author?
"Leopold Lummerstorfer is an Austrian film director, author and producer." -> (has ) -> [Leopold (Various
individuals named Leopold)]

"Leopold Lummerstorfer is an Austrian film director, author and producer." -> BORN_IN -> "Leopold
Lummerstorfer was born in 1968 in Gramastetten, Austria."
```

Logical Conclusions:
Leopold Lummerstorfer is an Austrian film director born in 1968 in Gramastetten, Austria.

"Leopold Lummerstorfer is an Austrian film director, author and producer." -> RESIDES_NEAR -> "Leopold Lummerstorfer resides near Kapuvár."
Logical Conclusions:
Leopold Lummerstorfer resides near Kapuvár.

"Leopold Lummerstorfer is an Austrian film director, author and producer." -> RESIDES_IN -> "Leopold Lummerstorfer resides in Vienna."
Logical Conclusions:
Leopold Lummerstorfer resides in Vienna.

[Leopold (Various individuals named Leopold)] -> (participates in event: Leopold Lummerstorfer resides near Kapuvár.) -> "Leopold Lummerstorfer resides near Kapuvár."

[Leopold (Various individuals named Leopold)] -> (participates in event: Barrie & Rockliff's managing director was Leopold Ullstein.) -> "Barrie & Rockliff's managing director was Leopold Ullstein."

[Leopold (Various individuals named Leopold)] -> (participates in event: Leopold Dippel described it from a cultivated plant as Zelkova japonica var. ver) -> "Leopold Dippel described it from a cultivated plant as Zelkova japonica var. verschaffeltii."

[Leopold (Various individuals named Leopold)] -> (participates in event: The current structure was the site of the Leopold and Loeb murder case trial.) -> "The current structure was the site of the Leopold and Loeb murder case trial."

[Leopold (Various individuals named Leopold)] -> (participates in event: The West Shore was founded by Leopold Samuel.) -> "The West Shore was founded by Leopold Samuel."

Subquery: Is Valeria Sarmiento an author?
"Valeria Sarmiento worked with Raúl Ruiz as a writer." -> COLLABORATED_WITH -> "Valeria Sarmiento is a Chilean screenwriter."
Logical Conclusions:
Valeria Sarmiento worked with Raúl Ruiz as a writer.

"Valeria Sarmiento worked with Raúl Ruiz as a writer." -> RELATED_ROLE -> "Valeria Sarmiento worked with Raúl Ruiz as an editor."
Logical Conclusions:
Valeria Sarmiento worked with Raúl Ruiz both as a writer and as an editor.

"Valeria Sarmiento worked with Raúl Ruiz as a writer." -> SUPPORTS -> "Valeria Sarmiento is a Chilean film director."
Logical Conclusions:
Fact 17 states Valeria Sarmiento is a Chilean film director, and fact 19 states she worked with Raúl Ruiz as a writer, linking her profession to her collaboration.

[valeria sarmiento (Chilean film editor and director)] -> (participates in event: Valeria Sarmiento worked with Raúl Ruiz as a writer.) -> "Valeria Sarmiento worked with Raúl Ruiz as a writer."

"Valeria Sarmiento worked with Raúl Ruiz as a writer." -> SUPPORTS -> "Valeria Sarmiento's film 'Lines of Wellington' competed for the Golden Lion at the 69th Venice International Film Festival."
Logical Conclusions:
Fact 19 states Valeria Sarmiento worked with Raúl Ruiz as a writer, and fact 22 states her film 'Lines of Wellington' competed for the Golden Lion, linking her collaboration to her work.

"Valeria Sarmiento worked with Raúl Ruiz as a writer." -> SUPPORTS -> "Valeria Sarmiento worked with Raúl Ruiz for decades as a regular collaborator."
Logical Conclusions:
Valeria Sarmiento worked with Raúl Ruiz as a writer and was a regular collaborator for decades.

"Valeria Sarmiento worked with Raúl Ruiz as a writer." -> (has) -> [raúl ruiz (Chilean film director and

husband)]

[valeria sarmiento (Chilean film editor and director)] -> (participates in event: Valeria Sarmiento is a Guggenheim Fellow (1988).) -> "Valeria Sarmiento is a Guggenheim Fellow (1988)."

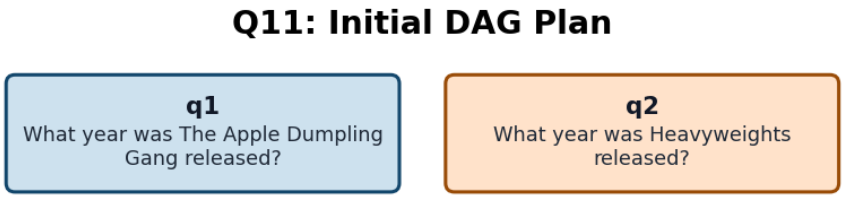
[valeria sarmiento (Chilean film editor and director)] -> (particip …[truncated for report]

Q11 [comparison / hard] 5ae383e15542992f92d822fb

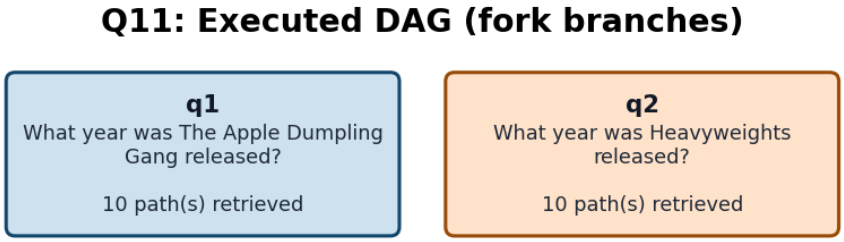
Which film is newer, The Apple Dumpling Gang or Heavyweights?

Gold answer	Heavyweights
LLM answer	Heavyweights
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What year was The Apple Dumpling Gang released?	10 path(s)
q2	0	—	What year was Heavyweights released?	10 path(s)

Final context fed to generation (1132 tokens)

```
Query plan:
- q1: What year was The Apple Dumpling Gang released?
- q2: What year was Heavyweights released?

Level 0:

Subquery: What year was The Apple Dumpling Gang released?
"'The Apple Dumpling Gang' is a 1971 novel by Jack Bickham." -> ADAPTED_INTO -> "The Apple Dumpling Gang is
a 1975 American comedy-western film."
Logical Conclusions:
```

The Apple Dumpling Gang (1975) is based on a 1971 novel by Jack Bickham.

"'The Apple Dumpling Gang' is a 1971 novel by Jack Bickham." -> ADAPTED_INTO -> "Buddy Baker scored 'The Apple Dumpling Gang' in 1975."
Logical Conclusions:
The novel The Apple Dumpling Gang was adapted into a film scored by Buddy Baker.

"'The Apple Dumpling Gang' is a 1971 novel by Jack Bickham." -> (has) -> [The Apple Dumpling Gang (1975 comedy-western film)]

"'The Apple Dumpling Gang' is a 1971 novel by Jack Bickham." -> IDENTIFIED_AS -> "The two novels by Jack Bickham made into movies are 'The Apple Dumpling Gang' and 'Baker's Hawk'."
Logical Conclusions:
The Apple Dumpling Gang is a 1971 novel by Jack Bickham.

"'The Apple Dumpling Gang' is a 1971 novel by Jack Bickham." -> (has) -> [jack bickham (Author of The Apple Dumpling Gang novel)]

"'The Apple Dumpling Gang' is a 1971 novel by Jack Bickham." -> ABOUT -> "'The Apple Dumpling Gang' is about a group of orphaned children during the California gold rush."
Logical Conclusions:
The novel The Apple Dumpling Gang is about a group of orphaned children during the California gold rush.

[The Apple Dumpling Gang (1975 comedy-western film)] -> (participates in event: The Apple Dumpling Gang was produced by Walt Disney Productions.) -> "The Apple Dumpling Gang was produced by Walt Disney Productions."

[The Apple Dumpling Gang (1975 comedy-western film)] -> (participates in event: The Apple Dumpling Gang Rides Again is a sequel to 'The Apple Dumpling Gang' (19) -> "The Apple Dumpling Gang Rides Again is a sequel to 'The Apple Dumpling Gang' (1975)."

[The Apple Dumpling Gang (1975 comedy-western film)] -> (participates in event: The two novels by Jack Bickham made into movies are 'The Apple Dumpling Gang' an) -> "The two novels by Jack Bickham made into movies are 'The Apple Dumpling Gang' and 'Baker's Hawk'."

[The Apple Dumpling Gang (1975 comedy-western film)] -> (participates in event: 'The Apple Dumpling Gang' is about a group of orphaned children during the Calif) -> "'The Apple Dumpling Gang' is about a group of orphaned children during the California gold rush."

Subquery: What year was Heavyweights released?
[heavyweights (1995 comedy-drama film about a fat camp)] -> (participates in event: Heavyweights is a 1995 American comedy-drama film.) -> "Heavyweights is a 1995 American comedy-drama film."

"Heavyweights was directed by Steven Brill." -> DIRECTED -> "Heavyweights is a 1995 American comedy-drama film."
Logical Conclusions:
Heavyweights was directed by Steven Brill.

"Heavyweights was directed by Steven Brill." -> (has) -> [heavyweights (1995 comedy-drama film about a fat camp)]

[heavyweights (1995 comedy-drama film about a fat camp)] -> (participates in event: Heavyweights was written by Steven Brill and Judd Apatow.) -> "Heavyweights was written by Steven Brill and Judd Apatow."

[heavyweights (1995 comedy-drama film about a fat camp)] -> (participates in event: Heavyweights centers around a fat camp for kids.) -> "Heavyweights centers around a fat camp for kids."

"Heavyweights was directed by Steven Brill." -> WRITTEN_BY -> "Heavyweights was written by Steven Brill and Judd Apatow."
Logical Conclusions:
Fact 68 states Heavyweights was written by Steven Brill and Judd Apatow, and fact 31 states it was directed by Steven Brill.

"Heavyweights was directed by Steven Brill." -> (has) -> [steven brill (Director and co-writer of Heavyweights)]

"'Wild at Heart' was released in 1990." -> CONTEMPORARY -> "'Red Rock West' was released in 1993."
Logical Conclusions:
Both facts specify release years for films from the early 1990s.

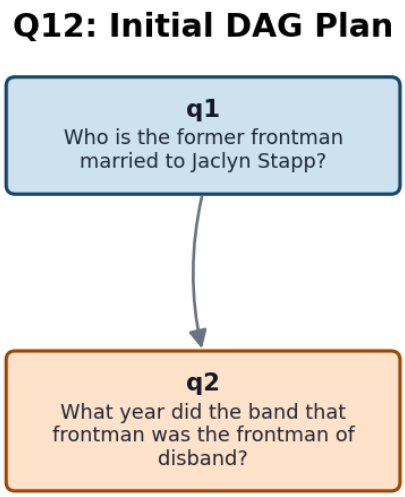
"'Wild at Heart' was r …[truncated for report]

Q12 [bridge / hard] 5a8e27d45542995a26add46a

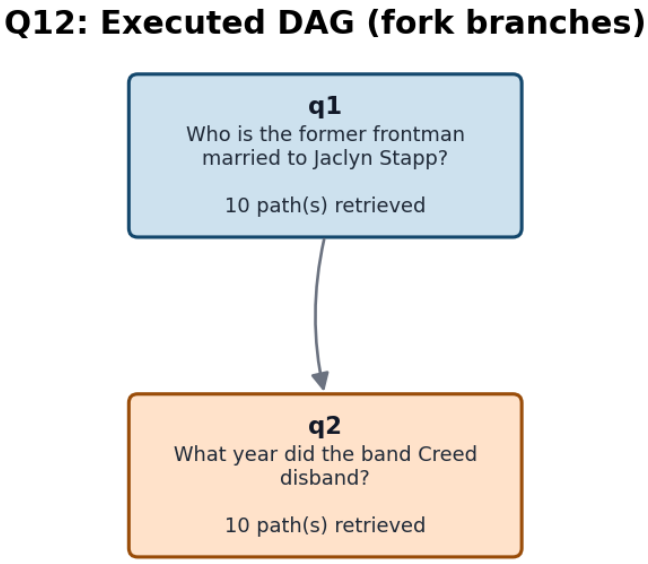
Jaclyn Stapp is married to the former frontman of a band that disbanded in what year?

Gold answer	2004
LLM answer	2004
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is the former frontman married to Jaclyn Stapp?	10 path(s)
q2	1	q1	What year did the band Creed disband?	10 path(s)

Final context fed to generation (1224 tokens)

```
Query plan:
- q1: Who is the former frontman married to Jaclyn Stapp?
- q2: What year did the band that frontman was the frontman of disband? (depends on: q1)

Level 1:

Subquery: What year did the band Creed disband?
"Creed disbanded in 2004." -> DISBANDMENT_OF -> "Creed is an American rock band."
Logical Conclusions:
Creed disbanded in 2004.

"Creed disbanded in 2004." -> (has ) -> [Creed (American rock band)]

"Creed disbanded in 2004." -> FOLLOWED_BY -> "Scott Stapp is the former frontman of the band Creed."
Logical Conclusions:
Creed disbanded in 2004, and Scott Stapp was the former frontman of the band.

"Creed disbanded in 2004." -> FOLLOWS -> "Creed released the studio album 'Human Clay' in 1999."
Logical Conclusions:
Creed released 'Human Clay' in 1999 and later disbanded in 2004.

"Creed disbanded in 2004." -> LED_TO -> "Tremonti, Marshall, and Phillips went on to found Alter Bridge."
Logical Conclusions:
After Creed disbanded, Tremonti, Marshall, and Phillips went on to found Alter Bridge.

"Creed disbanded in 2004." -> LED_TO -> "Scott Stapp followed a solo career."
Logical Conclusions:
After Creed disbanded, Scott Stapp followed a solo career.

[Creed (American rock band)] -> (participates in event: Creed disbanded due to increasing tension between
members.) -> "Creed disbanded due to increasing tension between members."

[Creed (American rock band)] -> (participates in event: Scott Stapp is the former frontman of the band
Creed.) -> "Scott Stapp is the former frontman of the band Creed."

[Creed (American rock band)] -> (participates in event: Tremonti, Marshall, and Phillips went on to found
Alter Bridge.) -> "Tremonti, Marshall, and Phillips went on to found Alter Bridge."

[Creed (American rock band)] -> (participates in event: Lead singer and band nucleus Jonny Radford Mead is
the former frontman of bass-d) -> "Lead singer and band nucleus Jonny Radford Mead is the former frontman
of bass-driven indie rockers Primary."

Level 0:

Subquery: Who is the former frontman married to Jaclyn Stapp?
[Jaclyn Nesheiwat Stapp (beauty queen and wife of Scott Stapp)] -> (participates in event: Jaclyn Nesheiwat
Stapp is married to Scott Stapp.) -> "Jaclyn Nesheiwat Stapp is married to Scott Stapp."

[Jaclyn Nesheiwat Stapp (beauty queen and wife of Scott Stapp)] -> (participates in event: Jaclyn Nesheiwat
Stapp was born on July 29, 1980.) -> "Jaclyn Nesheiwat Stapp was born on July 29, 1980."

[Jaclyn Nesheiwat Stapp (beauty queen and wife of Scott Stapp)] -> (participates in event: Jaclyn Nesheiwat
```

Stapp is executive director of The Scott Stapp With Arms Wide O) -> "Jaclyn Nesheiwat Stapp is executive director of The Scott Stapp With Arms Wide Open Foundation."

[Jaclyn Nesheiwat Stapp (beauty queen and wife of Scott Stapp)] -> (participates in event: Jaclyn Pearson was a member of The Preytells on drums and percussion.) -> "Jaclyn Pearson was a member of The Preytells on drums and percussion."

[Jaclyn Nesheiwat Stapp (beauty queen and wife of Scott Stapp)] -> (participates in event: Jaclyn Nesheiwat Stapp is a beauty queen, philanthropist and fashion model.) -> "Jaclyn Nesheiwat Stapp is a beauty queen, philanthropist and fashion model."

[Jaclyn Nesheiwat Stapp (beauty queen and wife of Scott Stapp)] -> (participates in event: Jaclyn Nesheiwat Stapp has pageant roots in Florida and New York.) -> "Jaclyn Nesheiwat Stapp has pageant roots in Florida and New York."

[Susanna Clark (Songwriter and lyricist)] -> (participates in event: The song 'Come from the Heart' was also released in 1988 by Clark's husband on h) -> "The song 'Come from the Heart' was also released in 1988 by Clark's husband on his album 'Old Friends'."

[Susanna Clark (Songwriter and lyricist)] -> (participates in event: The children of Gertrud Dahlgren and Rolf Dahlgren were Susanna, Helena and Fred) -> "The children of Gertrud Dahlgren and Rolf Dahlgren were Susanna, Helena and Fredrik."

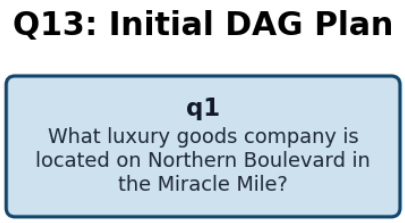
[Jane Rhodes (French opera singer known for Carmen role)] -> (pa …[truncated for report]

Q13 [bridge / hard] 5ab71b5c5542991d3222377e

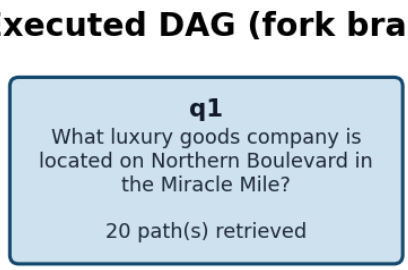
What is a luxury goods company long Northern Boulevard in the Miracle Mile?

Gold answer	Gucci
LLM answer	Hermes, Dior, Cartier, Bottega Veneta, Chanel, Fendi, Gucci, Louis Vuitton, MaxMara, Celine, Tiffany & Co., Prada
EM / F1	**0.00 / 0.12**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What luxury goods company is located on Northern Boulevard in the Miracle Mile?	20 path(s)

Final context fed to generation (1545 tokens)

```
[Miracle Mile (Shopping district near St. Mary's)] -> (participates in event: The area along Northern
Boulevard is well known for its high-end premium open-ai) -> "The area along Northern Boulevard is well
known for its high-end premium open-air shopping center." -> IS -> "The shopping center is the Americana
Manhasset." -> (has ) -> [americana manhasset (Luxury shopping center with designer stores)] ->
(participates in event: The companies at Americana Manhasset include Hermes, Dior, Cartier, Bottega Vene)
-> "The companies at Americana Manhasset include Hermes, Dior, Cartier, Bottega Veneta, Chanel, Fendi,
Gucci, Louis Vuitton, MaxMara, Celine, Tiffany & Co., and Prada."
Logical Conclusions:
The high-end premium open-air shopping center along Northern Boulevard is the Americana Manhasset.
```

"The shopping center is the Americana Manhasset." -> INTRODUCES -> "The companies at Americana Manhasset include Hermes, Dior, Cartier, Bottega Veneta, Chanel, Fendi, Gucci, Louis Vuitton, MaxMara, Celine, Tiffany & Co., and Prada."
Logical Conclusions:
Fact 2 identifies the shopping center as Americana Manhasset, and fact 3 lists the companies located there.

[americana manhasset (Luxury shopping center with designer stores)] -> (participates in event: Americana Manhasset is a site with international fashion designer labels and lux) -> "Americana Manhasset is a site with international fashion designer labels and luxury goods companies."

[americana manhasset (Luxury shopping center with designer stores)] -> (participates in event: Americana Manhasset is on a 220000 sqft site.) -> "Americana Manhasset is on a 220000 sqft site."

[americana manhasset (Luxury shopping center with designer stores)] -> (participates in event: Peter Marino is Americana Manhasset's master architect.) -> "Peter Marino is Americana Manhasset's master architect."

[americana manhasset (Luxury shopping center with designer stores)] -> (participates in event: Americana Manhasset maintains a total of more than 60 retail stores and restaura) -> "Americana Manhasset maintains a total of more than 60 retail stores and restaurants."

[americana manhasset (Luxury shopping center with designer stores)] -> (participates in event: The site of Americana Manhasset is surrounded by gardens designed by Oheme van S) -> "The site of Americana Manhasset is surrounded by gardens designed by Oheme van Sweden."

"The shopping center is the Americana Manhasset." -> IDENTIFIES -> "Americana Manhasset is a site with international fashion designer labels and luxury goods companies."
Logical Conclusions:
Fact 2 identifies the shopping center as Americana Manhasset, and fact 7 describes it as a site with international fashion designer labels and luxury goods companies.

"The shopping center is the Americana Manhasset." -> SPECIFIES -> "Americana Manhasset is on a 220000 sqft site."
Logical Conclusions:
Fact 1 specifies the site size of Americana Manhasset, while fact 2 identifies the shopping center as Americana Manhasset.

"The shopping center is the Americana Manhasset." -> INTRODUCES -> "The site of Americana Manhasset is surrounded by gardens designed by Oheme van Sweden."
Logical Conclusions:
Fact 2 identifies the shopping center as Americana Manhasset, and fact 18 describes its gardens designed by Oheme van Sweden.

"The shopping center is the Americana Manhasset." -> INTRODUCES -> "Americana Manhasset maintains a total of more than 60 retail stores and restaurants."
Logical Conclusions:
Fact 2 identifies the shopping center as Americana Manhasset, and fact 4 states it maintains over 60 retail stores and restaurants.

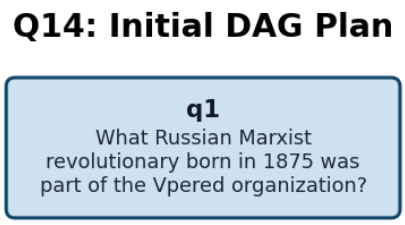
[Miracle Mile (Shopping district near St. Mary's)] -> (participates in event: The Miracle Mile is a prominent shopping district in Manhasset on the North Shor) -> "The Miracle Mile is a prominent shopping district in Manhasset on the North Shore of Long Island in Nassau County, New York." -> KNOWN_FOR -> "The area along Northern Boulevard is well known for its high-end pr …[truncated for report]

Q14 [bridge / hard] 5a7dbc175542995ed0d1666b

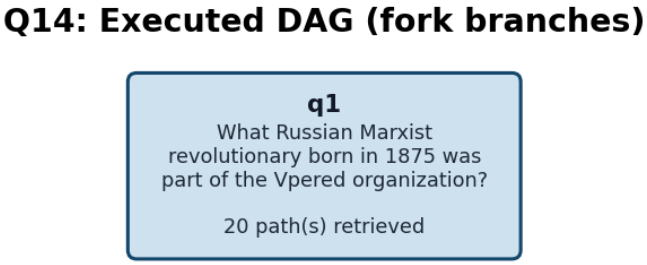
What russian Marxist revolutionary born in 1875 was part of the Vpered organization?

Gold answer	Anatoly Vasilyevich Lunacharsky
LLM answer	Alexander Bogdanov
EM / F1	**0.00 / 0.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What Russian Marxist revolutionary born in 1875 was part of the Vpered organization?	20 path(s)

Final context fed to generation (860 tokens)

```
[Vpered faction members (Russian Marxist revolutionaries)] -> (participates in event: The Vpered group
included Grigory Aleksinsky.) -> "The Vpered group included Grigory Aleksinsky." -> CO_MEMBER -> "The
Vpered group included Alexander Bogdanov." -> MEMBER_OF -> "Vpered was an organization."
Logical Conclusions:
The Vpered group included both Alexander Bogdanov and Grigory Aleksinsky.
The Vpered group included Alexander Bogdanov.

"The Vpered group included Alexander Bogdanov." -> (has ) -> [Vpered (faction within the RSDLP)] ->
(participates in event: Vpered was an organization.) -> "Vpered was an organization."

"The Vpered group included Grigory Aleksinsky." -> MEMBER_OF -> "Vpered was an organization."
```

Logical Conclusions:
The Vpered group included Grigory Aleksinsky.

[Vpered faction members (Russian Marxist revolutionaries)] -> (participates in event: The Vpered group included Stanislav Volski.) -> "The Vpered group included Stanislav Volski." -> CO_MEMBER -> "The Vpered group included Alexander Bogdanov."
Logical Conclusions:
The Vpered group included both Alexander Bogdanov and Stanislav Volski.

[Vpered faction members (Russian Marxist revolutionaries)] -> (participates in event: The Vpered group included Mikhail Pokrovsky.) -> "The Vpered group included Mikhail Pokrovsky." -> CO_MEMBER -> "The Vpered group included Alexander Bogdanov."
Logical Conclusions:
The Vpered group included both Alexander Bogdanov and Mikhail Pokrovsky.

[Vpered faction members (Russian Marxist revolutionaries)] -> (participates in event: The Vpered group included Maxim Gorky.) -> "The Vpered group included Maxim Gorky." -> CO_MEMBER -> "The Vpered group included Alexander Bogdanov."
Logical Conclusions:
The Vpered group included both Alexander Bogdanov and Maxim Gorky.

"The Vpered group included Stanislav Volski." -> MEMBER_OF -> "Vpered was an organization."
Logical Conclusions:
The Vpered group included Stanislav Volski.

"The Vpered group included Grigory Aleksinsky." -> CO_MEMBER -> "The Vpered group included Stanislav Volski."
Logical Conclusions:
The Vpered group included both Stanislav Volski and Grigory Aleksinsky.

"The Vpered group included Maxim Gorky." -> MEMBER_OF -> "Vpered was an organization."
Logical Conclusions:
The Vpered group included Maxim Gorky.

"The Vpered group included Grigory Aleksinsky." -> CO_MEMBER -> "The Vpered group included Anatoly Lunacharsky." -> MEMBER_OF -> "Vpered was an organization."
Logical Conclusions:
The Vpered group included both Anatoly Lunacharsky and Grigory Aleksinsky.
The Vpered group included Anatoly Lunacharsky.

"The Vpered group included Alexander Bogdanov." -> CO_MEMBER -> "The Vpered group included Anatoly Lunacharsky."
Logical Conclusions:
The Vpered group included both Anatoly Lunacharsky and Alexander Bogdanov.

[Vpered faction members (Russian Marxist revolutionaries)] -> (participates in event: The Vpered group included Martyn Liadov.) -> "The Vpered group included Martyn Liadov." -> CO_MEMBER -> "The Vpered group included Alexander Bogdanov."
Logical Conclusions:
The Vpered group included both Alexander Bogdanov and Martyn Liadov.

"The Vpered group included Stanislav Volski." -> CO_MEMBER -> "The Vpered group included Mikhail Pokrovsky." -> MEMBER_OF -> "Vpered was an organization."
Logical Conclusions:
The Vpered group included both Mikhail Pokrovsky and Stanislav Volski.
The Vpered group included Mikhail Pokrovsky.

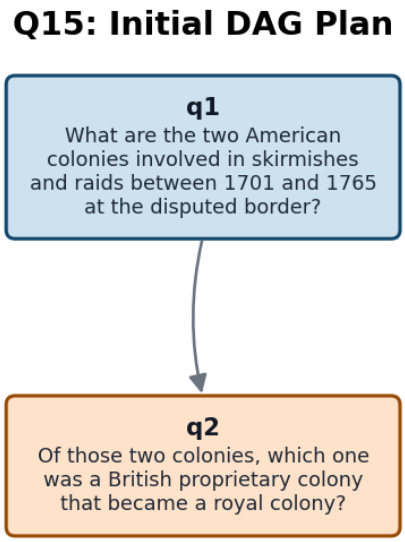
"The Vpered group included Mikhail Pokrovsky." -> CO_MEMBER -> "The Vpered group included Anatoly Lunacharsky."
Logical Conclusions:
The Vpered group included both Anatoly Lunacharsky and Mikhail Pokrovsky.

Q15 [bridge / hard] 5a7a33205542996a35c1712f

Out of two American colonies that had a series of skirmishes and raids between 1701 and 1765 at the disputed border, which British proprietary colony became a royal colony on the northeast coast of North America?

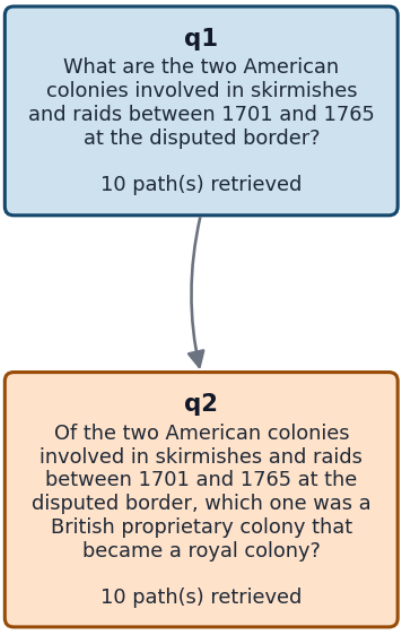
Gold answer	Province of New York
LLM answer	Pennsylvania
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q15: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What are the two American colonies involved in skirmishes and raids between 1701 and 1765 at the disputed border?	10 path(s)
q2	1	q1	Of the two American colonies involved in skirmishes and raids between 1701 and 1765 at the disputed border, which one was a British proprietary colony that became a royal colony?	10 path(s)

Final context fed to generation (1260 tokens)

```
Query plan:
- q1: What are the two American colonies involved in skirmishes and raids between 1701 and 1765 at the
disputed border?
- q2: Of those two colonies, which one was a British proprietary colony that became a royal colony?
(depends on: q1)

Level 1:

Subquery: Of the two American colonies involved in skirmishes and raids between 1701 and 1765 at the
disputed border, which one was a British proprietary colony that became a royal colony?
"A proprietary colony was a type of British colony mostly in North America and the Caribbean in the 17th
century." -> CATEGORIZES -> "All colonial properties were partitioned by royal charter into one of four
types: proprietary, royal, joint stock, or covenant."
Logical Conclusions:
A proprietary colony was one of four types of British colonies partitioned by royal charter.

"A proprietary colony was a type of British colony mostly in North America and the Caribbean in the 17th
century." -> SUPPORTS -> "King Charles II used the proprietary solution to reward allies."
```

Logical Conclusions:
A proprietary colony was a type of British colony in the 17th century, and King Charles II used the proprietary solution to reward allies.

"A proprietary colony was a type of British colony mostly in North America and the Caribbean in the 17th century." -> (has) -> [proprietary colony (Type of British colonial governance)]

"A proprietary colony was a type of British colony mostly in North America and the Caribbean in the 17th century." -> SUPPORTS -> "King Charles II used the proprietary solution to focus his own attention on Britain itself."
Logical Conclusions:
A proprietary colony was a type of British colony in the 17th century, and King Charles II used the proprietary solution to focus his own attention on Britain itself.

[East and West Jersey (Divisions of colonial New Jersey)] -> (participates in event: East Jersey and West Jersey were united as a royal colony in 1702.) -> "East Jersey and West Jersey were united as a royal colony in 1702."

[East and West Jersey (Divisions of colonial New Jersey)] -> (participates in event: In 1683, East Jersey adopted a constitution.) -> "In 1683, East Jersey adopted a constitution."

[Pennsylvania (Colony and later state)] -> (participates in event: The Province of Pennsylvania was one of the two major Restoration colonies.) -> "The Province of Pennsylvania was one of the two major Restoration colonies."

[Pennsylvania (Colony and later state)] -> (participates in event: The Province of Pennsylvania was founded in English North America by William Pen) -> "The Province of Pennsylvania was founded in English North America by William Penn on March 4, 1681."

[Pennsylvania (Colony and later state)] -> (participates in event: During the American Revolution, the Commonwealth of Pennsylvania was created.) -> "During the American Revolution, the Commonwealth of Pennsylvania was created."

[Pennsylvania (Colony and later state)] -> (participates in event: The Commonwealth of Pennsylvania became one of the original thirteen states.) -> "The Commonwealth of Pennsylvania became one of the original thirteen states."

Level 0:

Subquery: What are the two American colonies involved in skirmishes and raids between 1701 and 1765 at the disputed border?
[Pennsylvania (Colony and later state)] -> (participates in event: The Province of Pennsylvania was one of the two major Restoration colonies.) -> "The Province of Pennsylvania was one of the two major Restoration colonies."

[Pennsylvania (Colony and later state)] -> (participates in event: The Province of Pennsylvania was founded in English North America by William Pen) -> "The Province of Pennsylvania was founded in English North America by William Penn on March 4, 1681."

[Pennsylvania (Colony and later state)] -> (participates in event: During the American Revolution, the Commonwealth of Pennsylvania was created.) -> "During the American Revolution, the Commonwealth of Pennsylvania was created."

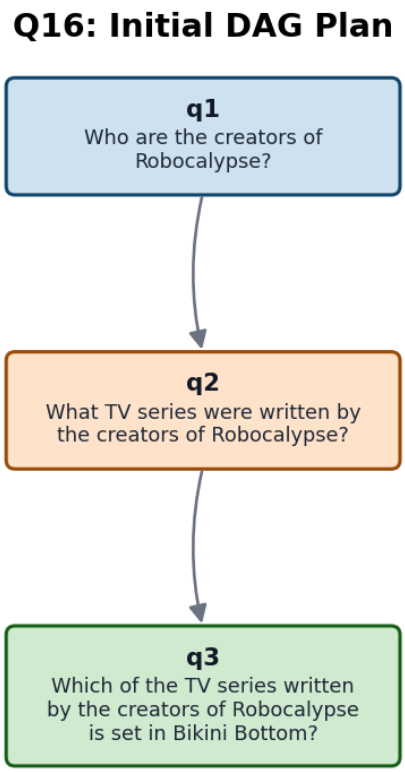
[Pennsylvania (Colony a …[truncated for report]

Q16 [bridge / hard] 5a7730015542994aec3b7210

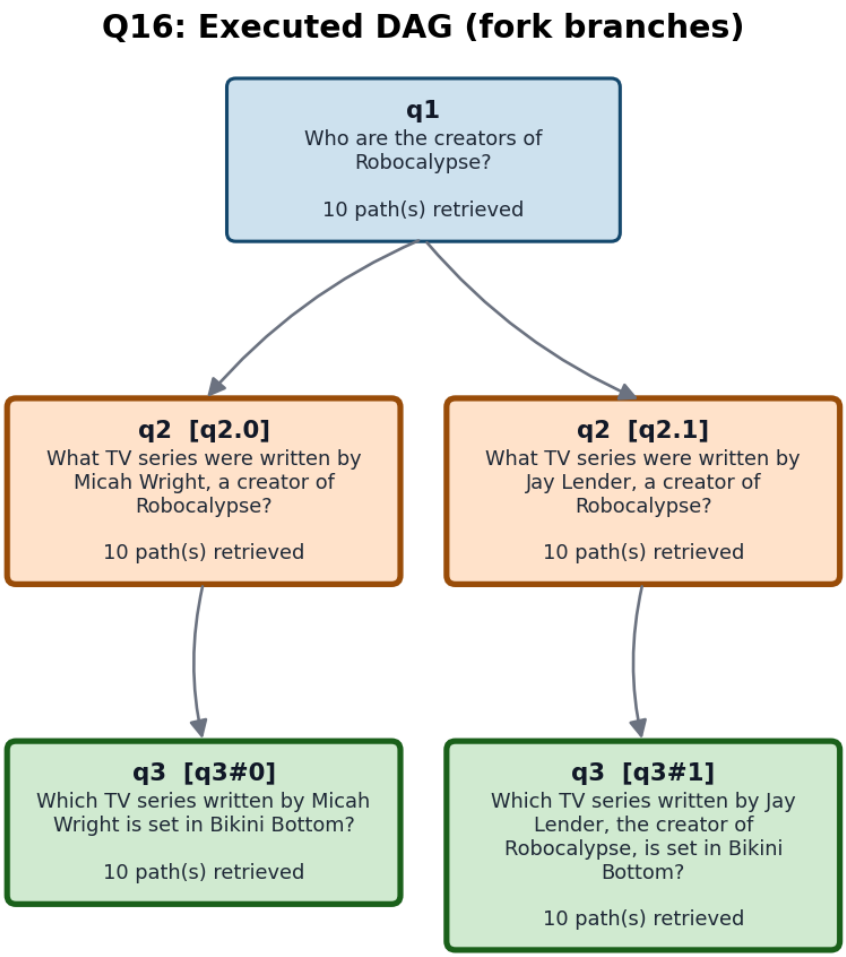
Which TV series, written by the creators of Robocalypse, is set in Bikini Bottom?

Gold answer	SpongeBob SquarePants
LLM answer	SpongeBob SquarePants
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who are the creators of Robocalypse?	10 path(s)
q2.0	1	q1	What TV series were written by Micah Wright, a creator of Robocalypse?	10 path(s)
q2.1	1	q1	What TV series were written by Jay Lender, a creator of Robocalypse?	10 path(s)
q3#0	2	q2.0	Which TV series written by Micah Wright is set in Bikini Bottom?	10 path(s)
q3#1	2	q2.1	Which TV series written by Jay Lender, the creator of Robocalypse, is set in Bikini Bottom?	10 path(s)

Final context fed to generation (2531 tokens)

```
Query plan:
- q1: Who are the creators of Robocalypse?
- q2: What TV series were written by the creators of Robocalypse? (depends on: q1)
- q3: Which of the TV series written by the creators of Robocalypse is set in Bikini Bottom? (depends on:
q2)
```

Level 2:

Subquery: Which TV series written by Micah Wright is set in Bikini Bottom?
"Micah Wright was a former writer for 'SpongeBob SquarePants'." -> (has) -> [SpongeBob SquarePants (Title character of the series)]

[SpongeBob SquarePants (Title character of the series)] -> (participates in event: 'Legends of Bikini Bottom' is an anthology series of six episodes in the America) -> "'Legends of Bikini Bottom' is an anthology series of six episodes in the American animated television series 'SpongeBob SquarePants'."

"Micah Wright was a former writer for 'SpongeBob SquarePants'." -> (has) -> [micah wright (Writer for Robocalypse)]

"Micah Wright was a former writer for 'SpongeBob SquarePants'." -> COMPARES -> "Micah Wright was a former writer for 'The Angry Beavers'."
Logical Conclusions:
Micah Wright wrote for both SpongeBob SquarePants and The Angry Beavers.

"Micah Wright was a former writer for 'SpongeBob SquarePants'." -> EMPLOYED -> "Robocalypse was created by Micah Wright."
Logical Conclusions:
Micah Wright was a former writer for SpongeBob SquarePants.

"Micah Wright was a former writer for 'SpongeBob SquarePants'." -> CONTEXTUALIZES -> "Micah Wright was a former writer for 'Phineas and Ferb'."
Logical Conclusions:
Fact 64 states Micah Wright was a former writer for SpongeBob SquarePants, while fact 73 states he was a former writer for Phineas and Ferb.

"The adventures take place in the fictional underwater city of Bikini Bottom." -> CORROBORATES -> "The series is set in the fictional underwater city of Bikini Bottom."
Logical Conclusions:
Both facts state that the series is set in the fictional underwater city of Bikini Bottom.

"The adventures take place in the fictional underwater city of Bikini Bottom." -> CORROBORATES -> "The series 'SpongeBob SquarePants' chronicles the exploits and adventures of the title character and his various friends in the fictional underwater city of Bikini Bottom."
Logical Conclusions:
Both facts describe the setting of the series as the fictional underwater city of Bikini Bottom.

"The adventures take place in the fictional underwater city of Bikini Bottom." -> IS_SETTING_FOR -> "The series chronicles the adventures and endeavors of the title character and his various friends."
Logical Conclusions:
The adventures take place in Bikini Bottom.

Subquery: Which TV series written by Jay Lender, the creator of Robocalypse, is set in Bikini Bottom?
"Jay Lender was a former writer for 'Phineas and Ferb'." -> EMPLOYED -> "Robocalypse was created by Jay Lender."
Logical Conclusions:
Jay Lender was a former writer for Phineas and Ferb.

"Jay Lender was a former writer for 'Phineas and Ferb'." -> (has) -> [jay lender (Writer for Robocalypse)]

"Jay Lender was a former writer for 'Phineas and Ferb'." -> (has) -> [phineas and ferb (TV series written by Jay Lender and Micah Wright)]

"Jay Lender was a former writer for 'Phineas and Ferb'." -> CONTEXTUALIZES -> "Jay Lender was a former writer for 'SpongeBob SquarePants'."
Logical Conclusions:
Fact 66 states Jay Lender was a former writer for SpongeBob SquarePants, while fact 74 states he was a former writer for Phineas and Ferb.

"Jay Lender was a former writer for 'Phineas and Ferb'." -> SHARES_WORK -> "Micah Wright was a former writer for 'Phineas and Ferb'."
Logical Conclusions:
Both Jay Lender and Micah Wright were former writers for Phineas and Ferb.

[SpongeBob SquarePants (Title character of the series)] -> (participates in event: 'Legends of Bikini Bottom' is an anthology series of six episodes in the America) -> "'Legends of Bikini Bottom' is an anthology series of six episodes in the American animated television series 'SpongeBob SquarePants'."

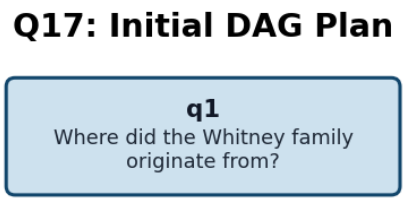
"The adventures take place in the fictional und …[truncated for report]

Q17 [bridge / hard] 5ab90fdd55429916710eb0fc

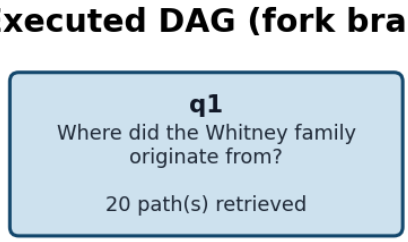
Wheelock Whitney is just one member of the Whitney Family. Where did this American family originate from?

Gold answer	London, England
LLM answer	The Whitney family originated from London, England.
EM / F1	**0.00 / 0.50**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Where did the Whitney family originate from?	20 path(s)

Final context fed to generation (980 tokens)

```
"The Whitney family was founded by John Whitney (1592–1673)." -> PRECEDES -> "John Whitney came from
London, England to Watertown, Massachusetts in 1635." -> (has ) -> [John Hay Whitney (U.S. Ambassador and
publisher)] -> (participates in event: John Hay Whitney was publisher of the 'New York Herald Tribune'.) ->
"John Hay Whitney was publisher of the 'New York Herald Tribune'."
Logical Conclusions:
John Whitney, the founder of the Whitney family, came from London to Massachusetts in 1635.

[John Hay Whitney (U.S. Ambassador and publisher)] -> (participates in event: John Hay Whitney was a member
of the Whitney family.) -> "John Hay Whitney was a member of the Whitney family."

[John Hay Whitney (U.S. Ambassador and publisher)] -> (participates in event: John Hay Whitney was U.S.
Ambassador to the United Kingdom.) -> "John Hay Whitney was U.S. Ambassador to the United Kingdom."

[John Hay Whitney (U.S. Ambassador and publisher)] -> (participates in event: John Hay Whitney was
```

president of the Museum of Modern Art.) -> "John Hay Whitney was president of the Museum of Modern Art."

[John Hay Whitney (U.S. Ambassador and publisher)] -> (participates in event: John Hay Whitney (August 17, 1904 – February 8, 1982) was colloquially known as) -> "John Hay Whitney (August 17, 1904 – February 8, 1982) was colloquially known as Jock Whitney."

"The Whitney family was founded by John Whitney (1592–1673)." -> (has) -> [John Hay Whitney (U.S. Ambassador and publisher)]

"John Whitney came from London, England to Watertown, Massachusetts in 1635." -> ASSOCIATED_WITH -> "The historic Whitney family mansion in Watertown, known as The Elms, was built in 1710."
Logical Conclusions:
The Whitney family mansion in Watertown, built in 1710, relates to John Whitney who arrived in Watertown in 1635.

"The Whitney family was founded by John Whitney (1592–1673)." -> (has) -> [whitney family (Prominent American family in horse racing and business)] -> (participates in event: John Hay Whitney was a member of the Whitney family.) -> "John Hay Whitney was a member of the Whitney family."

[whitney family (Prominent American family in horse racing and business)] -> (participates in event: The Whitney family is an American family notable for their social prominence, we) -> "The Whitney family is an American family notable for their social prominence, wealth, business enterprises and philanthropy." -> FOUNDED_BY -> "The Whitney family was founded by John Whitney (1592–1673)."
Logical Conclusions:
The Whitney family, notable for their wealth and philanthropy, was founded by John Whitney.

[whitney family (Prominent American family in horse racing and business)] -> (participates in event: The historic Whitney family mansion in Watertown, known as The Elms, was built i) -> "The historic Whitney family mansion in Watertown, known as The Elms, was built in 1710." -> FOLLOWS -> "The Whitney family was founded by John Whitney (1592–1673)."
Logical Conclusions:
The Whitney family mansion was built in 1710, after the family was founded by John Whitney.

"The Whitney family is an American family notable for their social prominence, wealth, business enterprises and philanthropy." -> INCLUDES -> "Joan Whitney Payson was a member of the prominent New York City Whitney family."
Logical Conclusions:
The Whitney family, to which Joan Whitney Payson belongs, is known for social prominence and wealth.

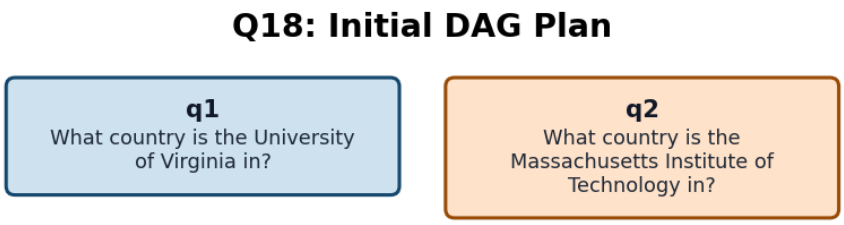
[whitney family (Prominent American family in horse racing and business)] -> (participates in event: Channing managed the mining operations of the Whitney family.) -> "Channing managed the mining operations of the Whitney family." -> (has) -> [Roderick Kennedy, Jr. (Former resident who created and funded the Walk of Fame)] -> (participates in event: Channing and Cornelius Vanderbilt Whitney formed the Hudson Bay Mining and Smelt) -> "Channing and Cornelius Vanderbilt Whitney formed the Hudson Bay Mining and Smelting Company in Flin Flon, Saskatchewan, Canada."

Q18 [comparison / hard] 5a8b82a05542995d1e6f13d0

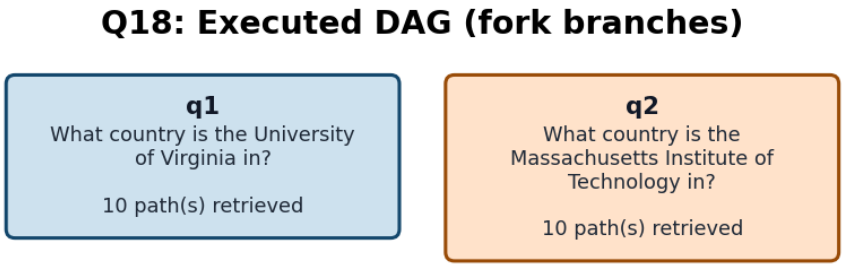
Are University of Virginia and Massachusetts Institute of Technology in the same country?

Gold answer	yes
LLM answer	Yes
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What country is the University of Virginia in?	10 path(s)
q2	0	—	What country is the Massachusetts Institute of Technology in?	10 path(s)

Final context fed to generation (1168 tokens)

```
Query plan:
- q1: What country is the University of Virginia in?
- q2: What country is the Massachusetts Institute of Technology in?

Level 0:

Subquery: What country is the University of Virginia in?
[university of virginia (Public research university founded by Thomas Jefferson)] -> (participates in
event: The University of Virginia is the flagship for the Commonwealth of Virginia.) -> "The University of
```

Virginia is the flagship for the Commonwealth of Virginia."

[university of virginia (Public research university founded by Thomas Jefferson)] -> (participates in event: The University of Virginia is a public research university.) -> "The University of Virginia is a public research university."

[university of virginia (Public research university founded by Thomas Jefferson)] -> (participates in event: The University of Virginia was founded by Declaration of Independence author Tho) -> "The University of Virginia was founded by Declaration of Independence author Thomas Jefferson."

[university of virginia (Public research university founded by Thomas Jefferson)] -> (participates in event: UVA is known for its secret societies.) -> "UVA is known for its secret societies."

[university of virginia (Public research university founded by Thomas Jefferson)] -> (participates in event: UVA is known for its historic foundations.) -> "UVA is known for its historic foundations."

[university of virginia (Public research university founded by Thomas Jefferson)] -> (participates in event: UVA is known for its student-run honor code.) -> "UVA is known for its student-run honor code."

[university of washington (Former faculty employer of Guruswami)] -> (participates in event: Venkatesan Guruswami was a member of the faculty at the University of Washington) -> "Venkatesan Guruswami was a member of the faculty at the University of Washington from 2002 to 2009."

[washington and lee university (Law school attended by Holt)] -> (participates in event: Turning then to law, he entered Washington and Lee University.) -> "Turning then to law, he entered Washington and Lee University."

[duke university (institution where Omid Safi works)] -> (participates in event: Omid Safi is an American Professor of Asian and Middle Eastern Studies at Duke U) -> "Omid Safi is an American Professor of Asian and Middle Eastern Studies at Duke University."

[northwestern university (Institution with business advisory council)] -> (participates in event: Gilbert V. Rohleder was a member of the business advisory council of Northwester) -> "Gilbert V. Rohleder was a member of the business advisory council of Northwestern University."

Subquery: What country is the Massachusetts Institute of Technology in?
"The Massachusetts Institute of Technology (MIT) is located in Cambridge, Massachusetts." -> (has) -> [massachusetts institute of technology (University and employer of the individual)]

"The Massachusetts Institute of Technology (MIT) is located in Cambridge, Massachusetts." -> LOCATION_OF -> "The Massachusetts Institute of Technology (MIT) is a private research university."
Logical Conclusions:
MIT is a private research university located in Cambridge, Massachusetts.

"The Massachusetts Institute of Technology (MIT) is located in Cambridge, Massachusetts." -> LOCATED_IN -> "He is Professor Emeritus at the Department of Aeronautics and Astronautics at the Massachusetts Institute of Technology."
Logical Conclusions:
The individual is a Professor Emeritus at MIT, which is located in Cambridge, Massachusetts.

"The Massachusetts Institute of Technology (MIT) is located in Cambridge, Massachusetts." -> LOCATED_IN -> "He received his S.B. degree from Massachusetts Institute of Technology in 1971."
Logical Conclusions:
The individual earned an S.B. from MIT, located in Cambridge, Massachusetts.

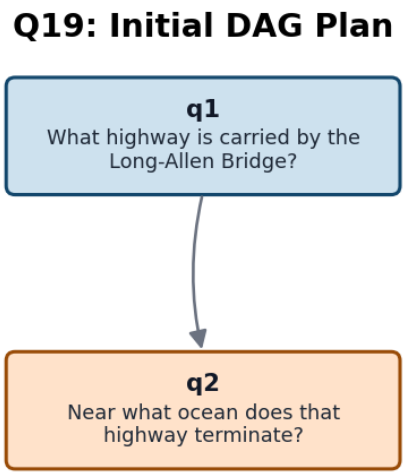
"The Massachusetts Institute of Technology (MIT) is located in Cambridge, Massachusetts." -> LOCATED_IN -> "After Hanover Academy, he attended Massachusetts Institute of Techn …[truncated for report]

Q19 [bridge / hard] 5a76c98855429972597f13d0

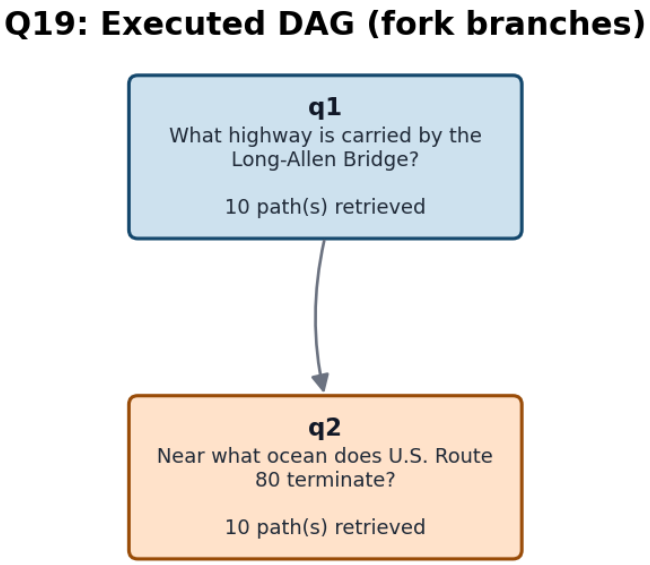
Near what ocean does the highway carried by the Long-Allen bridge terminate near?

Gold answer	the Atlantic Ocean
LLM answer	Atlantic Ocean
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What highway is carried by the Long-Allen Bridge?	10 path(s)
q2	1	q1	Near what ocean does U.S. Route 80 terminate?	10 path(s)

Final context fed to generation (1167 tokens)

```
Query plan:
- q1: What highway is carried by the Long-Allen Bridge?
- q2: Near what ocean does that highway terminate? (depends on: q1)

Level 1:

Subquery: Near what ocean does U.S. Route 80 terminate?
"The eastern terminus of US 80 is near the Atlantic Ocean." -> (has ) -> [U.S. Route 80 (former
transcontinental highway now truncated)]

"The eastern terminus of US 80 is near the Atlantic Ocean." -> SUPPORTED_BY -> "US 80 originally ran from
the Pacific Ocean to the Atlantic Ocean."
Logical Conclusions:
US 80 originally ran from the Pacific Ocean to the Atlantic Ocean, and its eastern terminus is near the
Atlantic Ocean.

"The eastern terminus of US 80 is near the Atlantic Ocean." -> SUPPORTS -> "U.S. Route 80 (US 80) is an
east–west United States Numbered Highway."
Logical Conclusions:
U.S. Route 80 (US 80) is an east–west United States Numbered Highway whose eastern terminus is near the
Atlantic Ocean.

"The eastern terminus of US 80 is near the Atlantic Ocean." -> LOCATED_AT -> "The eastern terminus of US 80
is at the intersection of Butler Avenue, Inlet Avenue, and Tybrisa Street."
Logical Conclusions:
The eastern terminus of US 80 is near the Atlantic Ocean and at the intersection of Butler Avenue, Inlet
Avenue, and Tybrisa Street.

"The eastern terminus of US 80 is near the Atlantic Ocean." -> NEAR -> "The eastern terminus of US 80 is in
Tybee Island, Georgia."
Logical Conclusions:
The eastern terminus in Tybee Island is near the Atlantic Ocean.

[U.S. Route 80 (former transcontinental highway now truncated)] -> (participates in event: US 80 originally
ran from the Pacific Ocean to the Atlantic Ocean.) -> "US 80 originally ran from the Pacific Ocean to the
Atlantic Ocean."

[U.S. Route 80 (former transcontinental highway now truncated)] -> (participates in event: The Ocean To
Ocean Bridge later carried its successor, US 80.) -> "The Ocean To Ocean Bridge later carried its
successor, US 80."

[U.S. Route 80 (former transcontinental highway now truncated)] -> (participates in event: Currently, the
western terminus of US 80 is at an interchange with Interstate 30) -> "Currently, the western terminus of
US 80 is at an interchange with Interstate 30 (I-30) on the Dallas–Mesquite, Texas city line."

[U.S. Route 80 (former transcontinental highway now truncated)] -> (participates in event: The entire
segment of US 80 west of Dallas, Texas, has been decommissioned.) -> "The entire segment of US 80 west of
Dallas, Texas, has been decommissioned."

Level 0:

Subquery: What highway is carried by the Long-Allen Bridge?
```

"The Long–Allen Bridge opened in 1933." -> SUPPORTS -> "The Long–Allen Bridge carries U.S. Route 80 across the Red River."
Logical Conclusions:
The Long–Allen Bridge opened in 1933 and carries U.S. Route 80 across the Red River.

[long–allen bridge (Truss bridge in Shreveport, Louisiana)] -> (participates in event: The Long–Allen Bridge carries U.S. Route 80 across the Red River.) -> "The Long–Allen Bridge carries U.S. Route 80 across the Red River."

"The Long–Allen Bridge opened in 1933." -> SUPPORTS -> "The Long–Allen Bridge is also known as the Texas Street Bridge."
Logical Conclusions:
The Long–Allen Bridge opened in 1933 and is also known as the Texas Street Bridge.

"The Long–Allen Bridge opened in 1933." -> OPENING_YEAR_OF -> "The Long–Allen Bridge is a truss bridge in Shreveport, Louisiana."
Logical Conclusions:
The Long–Allen Bridge opened in 1933.

"The Long–Allen Bridge opened in 1933." -> (has) -> [long–allen bridge (Truss bridge in Shreveport, Louisiana)]

"The Long–Allen Bridge opened in 1933." -> SUPPORTS -> "The Long–Allen Bridge is named for Louisiana governors Huey Long and Oscar K. Allen."
Logical Conclusions:
The Long–Allen Bridge opened in 1933.

[long–allen bridge (Truss bridge in Shreveport, Louisiana)] -> (participates in event: The Long–Allen Bridge is also known as the Texas Street Bridge.) -> "The Long–Allen Bridge is also known as the Texas Street Bridge."

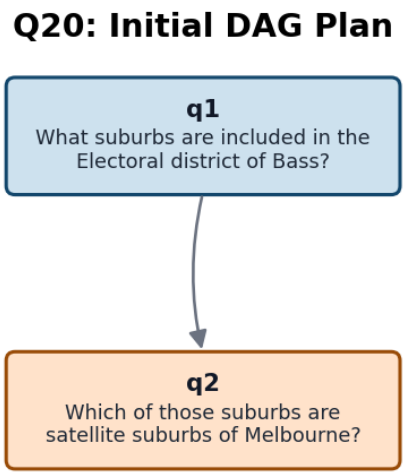
[long–allen bridge (Truss brid …[truncated for report]

Q20 [bridge / hard] 5ae810325542995540e5a572a

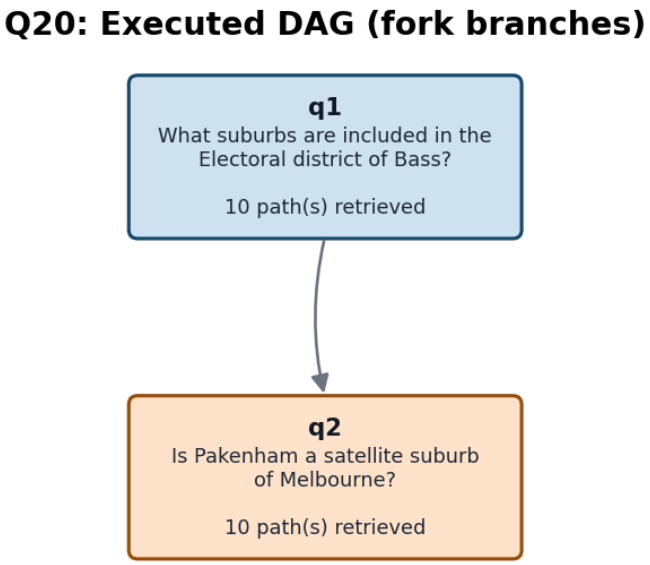
Which satellite suburb of Melbourne is partially included in the Electoral district of Bass?

Gold answer	Pakenham
LLM answer	Pakenham
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What suburbs are included in the Electoral district of Bass?	10 path(s)
q2	1	q1	Is Pakenham a satellite suburb of Melbourne?	10 path(s)

Final context fed to generation (1389 tokens)

```
Query plan:
- q1: What suburbs are included in the Electoral district of Bass?
- q2: Which of those suburbs are satellite suburbs of Melbourne? (depends on: q1)

Level 1:

Subquery: Is Pakenham a satellite suburb of Melbourne?
"Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria,
Australia." -> SUPPORTS -> "The Electoral district of Bass includes part of suburban Pakenham."
Logical Conclusions:
Fact 10 describes Pakenham as a satellite suburb of Melbourne, and fact 13 notes that the Bass district
includes part of suburban Pakenham.

"Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria,
Australia." -> LOCATED_IN -> "Pakenham is 56 km south-east of Melbourne's Central Business District."
Logical Conclusions:
Pakenham is a satellite suburb located 56 km from Melbourne's CBD.

"Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria,
Australia." -> HAS_POPULATION -> "Pakenham recorded a population of 46,421 at the 2016 Census."
Logical Conclusions:
Pakenham had a population of 46,421 at the 2016 Census.

"Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria,
Australia." -> (has ) -> [west gippsland (Region of Victoria)]

"Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria,
Australia." -> SIMILAR_TO -> "Officer is a satellite suburb of Melbourne, Victoria, Australia."
Logical Conclusions:
Both Pakenham and Officer are satellite suburbs of Melbourne within the Shire of Cardinia.

"Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria,
Australia." -> PART_OF -> "Pakenham is located within the Shire of Cardinia local government area."
Logical Conclusions:
Pakenham is a suburb within the Shire of Cardinia local government area.

[pakenham (Satellite suburb of Melbourne)] -> (participates in event: Pakenham is a satellite suburb of
Melbourne on the edge of the West Gippsland re) -> "Pakenham is a satellite suburb of Melbourne on the edge
of the West Gippsland region of Victoria, Australia."

[pakenham (Satellite suburb of Melbourne)] -> (participates in event: Pakenham is 56 km south-east of
Melbourne's Central Business District.) -> "Pakenham is 56 km south-east of Melbourne's Central Business
District."

[pakenham (Satellite suburb of Melbourne)] -> (participates in event: Pakenham is located within the Shire
of Cardinia local government area.) -> "Pakenham is located within the Shire of Cardinia local government
area."

Level 0:

Subquery: What suburbs are included in the Electoral district of Bass?
"The Electoral district of Bass includes part of suburban Pakenham." -> (has ) -> [pakenham (Satellite
```

suburb of Melbourne)]

"The Electoral district of Bass includes part of suburban Pakenham." -> SUPPORTS -> "Pakenham is a satellite suburb of Melbourne on the edge of the West Gippsland region of Victoria, Australia."
Logical Conclusions:
Fact 10 describes Pakenham as a satellite suburb of Melbourne, and fact 13 notes that the Bass district includes part of suburban Pakenham.

"The Electoral district of Bass includes part of suburban Pakenham." -> CONTEXTUALIZES -> "The Electoral district of Bass is located south east of Melbourne."
Logical Conclusions:
Fact 1 states the Bass district is southeast of Melbourne, and fact 13 specifies that it includes part of suburban Pakenham.

"The Electoral district of Bass includes part of suburban Pakenham." -> SUPPORTS -> "The Electoral district of Bass includes the towns of Bass, Cape Patterson, Corinella, Cowes, Grantville, Inverloch, Kilcunda, Koo Wee Rup, Lang Lang, Rhyll, San Remo, Ventnor, Wonthaggi."
Logical Conclusions:
Fact 94 states Bass includes part of suburban Pakenham, and fact 121 lists the towns it includes.

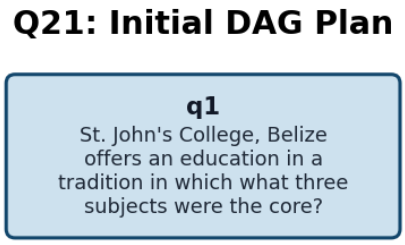
"The Electoral district of Bass includes part of suburban Pakenham." -> PART_OF -> "The Electoral district of Bass is …[truncated for report]

Q21 [bridge / hard] 5a7f244255429934daa2fcec

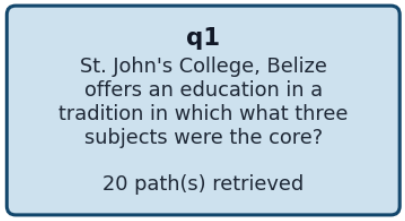
St. John's College, Belize offers an education in a tradition in which what three subjects were the core?

Gold answer	Grammar, logic, and rhetoric
LLM answer	Latin, Greek, and mathematics
EM / F1	**0.00 / 0.25**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	St. John's College, Belize offers an education in a tradition in which what three subjects were the core?	20 path(s)

Final context fed to generation (1133 tokens)

```
"Through its three divisions, St. John's College offers a wide variety of liberal arts and science
courses." -> OFFERED_BY -> "St. John's College is a Roman Catholic institution in the Jesuit tradition." ->
(has ) -> [St. John's College, Belize (Roman Catholic educational institution in Belize)] -> (participates
in event: The third form classes at St. John's College High School follow the Caribbean Se) -> "The third
form classes at St. John's College High School follow the Caribbean Secondary Education Certificate (CSEC)
curricula."
Logical Conclusions:
St. John's College, a Roman Catholic institution in the Jesuit tradition, offers a wide variety of liberal
```

arts and science courses through its three divisions.

[St. John's College, Belize (Roman Catholic educational institution in Belize)] -> (participates in event: The third and fourth form classes sit the regionally administered examinations a) -> "The third and fourth form classes sit the regionally administered examinations at the end of their fourth year."

[St. John's College, Belize (Roman Catholic educational institution in Belize)] -> (participates in event: Through its three divisions, St. John's College offers a wide variety of liberal) -> "Through its three divisions, St. John's College offers a wide variety of liberal arts and science courses." -> SUPPORTS -> "St. John's College has a number of central academic centres and activities." -> PART_OF -> "St. John's College is a Roman Catholic institution in the Jesuit tradition."
Logical Conclusions:
St. John's College offers a wide variety of liberal arts and science courses through its three divisions and has central academic centres and activities.
St. John's College, a Roman Catholic institution in the Jesuit tradition, has a number of central academic centres and activities.

[St. John's College, Belize (Roman Catholic educational institution in Belize)] -> (participates in event: St. John's College is one of the most diverse educational institutions in Belize) -> "St. John's College is one of the most diverse educational institutions in Belize."

[St. John's College, Belize (Roman Catholic educational institution in Belize)] -> (participates in event: The school's curriculum at St. John's College High School is complemented by spo) -> "The school's curriculum at St. John's College High School is complemented by sports."

"Through its three divisions, St. John's College offers a wide variety of liberal arts and science courses." -> SUPPORTS -> "St. John's College is one of the largest educational institutions in Belize." -> CHARACTERISTIC_OF -> "St. John's College is a Roman Catholic institution in the Jesuit tradition."
Logical Conclusions:
St. John's College offers a wide variety of liberal arts and science courses and is one of the largest educational institutions in Belize.
St. John's College is a Roman Catholic institution in the Jesuit tradition and is one of the largest educational institutions in Belize.

"St. John's College is one of the largest educational institutions in Belize." -> (has) -> [St. John's College, Belize (Roman Catholic educational institution in Belize)]

"St. John's College is a Roman Catholic institution in the Jesuit tradition." -> IS -> "St. John's College is one of the oldest educational institutions in Belize."
Logical Conclusions:
St. John's College is a Roman Catholic institution in the Jesuit tradition and is one of the oldest educational institutions in Belize.

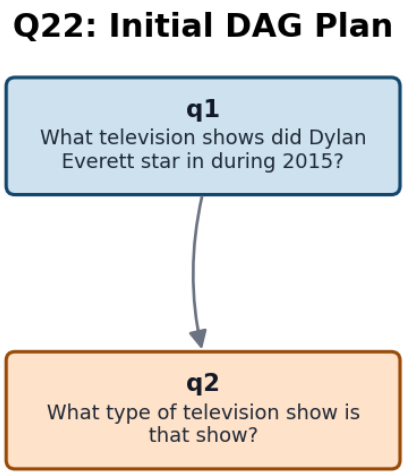
"Through its three divisions, St. John's College offers a wide variety of liberal arts and science courses." -> INCLUDES -> "St. John's College offers courses at the United States junior college levels." -> OFFERED_BY -> "St. John's College is a Roman Catholic institution in the Jesuit tradition."
Logical Conclusions:
St. John's College offers a wide variety of liberal arts and science courses through its three divisions, including courses at the US junior college levels.
St. John's College, a Roman Catholic institut …[truncated for report]

Q22 [bridge / hard] 5a8fae205542997ba9cb32ec

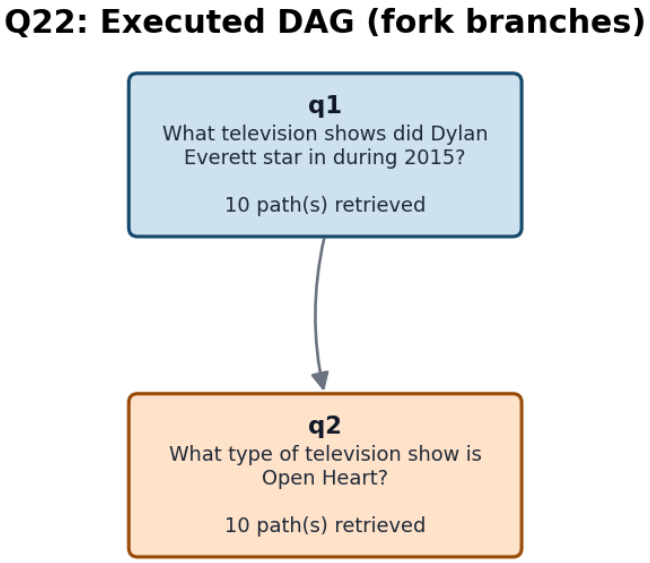
What type of television show did Dylan Everett star in, in 2015?

Gold answer	American-Canadian mystery-drama
LLM answer	mystery-drama
EM / F1	**0.00 / 0.67**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What television shows did Dylan Everett star in during 2015?	10 path(s)
q2	1	q1	What type of television show is Open Heart?	10 path(s)

Final context fed to generation (1140 tokens)

```
Query plan:
- q1: What television shows did Dylan Everett star in during 2015?
- q2: What type of television show is that show? (depends on: q1)

Level 1:

Subquery: What type of television show is Open Heart?
"Open Heart is a 2015 American-Canadian mystery-drama television series." -> PRODUCED_BY -> "Open Heart is
produced by the Epitome Pictures unit of DHX Media in association with marblemedia."
Logical Conclusions:
Open Heart is a 2015 American-Canadian mystery-drama television series produced by Epitome Pictures and
marblemedia.

"Open Heart is a 2015 American-Canadian mystery-drama television series." -> (has ) -> [Dylan Everett TV
Shows (TV series featuring Dylan Everett)]

"Open Heart is a 2015 American-Canadian mystery-drama television series." -> AIRED_ON -> "Open Heart
previously aired on YTV in Canada."
Logical Conclusions:
Open Heart previously aired on YTV in Canada.

"Open Heart is a 2015 American-Canadian mystery-drama television series." -> AIRED_ON -> "Open Heart first
aired on January 20, 2015, on TeenNick in the United States."
Logical Conclusions:
Open Heart first aired on January 20, 2015, on TeenNick in the United States.

"Open Heart is a 2015 American-Canadian mystery-drama television series." -> CO_OCCURS -> "Karaoke Star Jr.
is a reality television show for children."
Logical Conclusions:
Both Open Heart and Karaoke Star Jr. are television series mentioned in the same context.

"Open Heart is a 2015 American-Canadian mystery-drama television series." -> AIRS_ON -> "Open Heart
currently airs on ABC Spark in Canada."
Logical Conclusions:
Open Heart currently airs on ABC Spark in Canada.

"Open Heart previously aired on YTV in Canada." -> (has ) -> [Dylan Everett TV Shows (TV series featuring
Dylan Everett)]

"Open Heart previously aired on YTV in Canada." -> PRECEDES -> "Open Heart currently airs on ABC Spark in
Canada."
Logical Conclusions:
Open Heart previously aired on YTV and currently airs on ABC Spark in Canada.

"Open Heart previously aired on YTV in Canada." -> PREMIERED_ON -> "Open Heart first aired on January 20,
2015, on TeenNick in the United States."
Logical Conclusions:
Fact 20 gives the premiere date and network for Open Heart, and fact 38 states where it previously aired.

Level 0:

Subquery: What television shows did Dylan Everett star in during 2015?
```

[Dylan Everett TV Shows (TV series featuring Dylan Everett)] -> (participates in event: Dylan Everett is best known for his role in 'Open Heart' (2015).) -> "Dylan Everett is best known for his role in 'Open Heart' (2015)."

[Dylan Everett TV Shows (TV series featuring Dylan Everett)] -> (participates in event: Open Heart first aired on January 20, 2015, on TeenNick in the United States.) -> "Open Heart first aired on January 20, 2015, on TeenNick in the United States."

[Dylan Everett TV Shows (TV series featuring Dylan Everett)] -> (participates in event: Open Heart is a 2015 American-Canadian mystery-drama television series.) -> "Open Heart is a 2015 American-Canadian mystery-drama television series."

"Dylan Everett is best known for his role in 'Open Heart' (2015)." -> (has) -> [dylan everett (Canadian actor with multiple TV roles)]

"Dylan Everett is best known for his role in 'Open Heart' (2015)." -> CONTRASTS -> "Dylan Everett is best known for his role in '' (2012–2013)."
Logical Conclusions:
Dylan Everett is best known for his role in 'Open Heart' and also for a role in an unnamed show from 2012–2013.

"Dylan Everett is best known for his role in 'Open Heart' (2015)." -> RELATES_TO -> "Dylan Everett is best known for his role in 'How To Be Indie' (2009–2011)."
Logical Conclusions:
Dylan Everett is best known for his roles in 'How To Be Indie' and 'Open Heart'.

"Dylan Everett is best known for his role in 'Open Heart' (2015)." -> RELATES_TO -> "Dylan Everett is best known for his role in 'Wingin' It' (2010–2013)."
Logical Conclusions:
Dylan Everett is best known for his roles in 'Wingin' It' and 'Open Heart'.

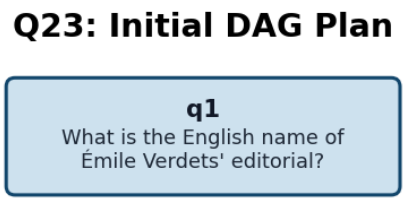
"Dylan Everett is …[truncated for report]

Q23 [bridge / hard] 5ade7fa3554299728e26c727

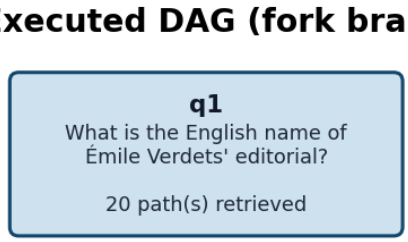
what is the english name of Émile Verdets editorial?

Gold answer	Annals of Chemistry and of Physics
LLM answer	Annales of Chemistry and Physics
EM / F1	**0.00 / 0.73**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the English name of Émile Verdets' editorial?	20 path(s)

Final context fed to generation (1095 tokens)

```
[Émile Verdet (French physicist)] -> (participates in event: Verdet championed the theory through his
editorial supervision of the 'Annales d) -> "Verdet championed the theory through his editorial supervision
of the 'Annales de chimie et de physique'." -> EDITOR -> "In 1815, it became the 'Annales de chimie et de
physique'." -> PUBLISHED_AS -> "Annales de chimie et de physique was published under that name for the next
100 years."
Logical Conclusions:
Verdet championed the theory through his editorial supervision of the 'Annales de chimie et de physique',
which became that name in 1815.
Annales de chimie et de physique was published under that name for the next 100 years.

"In 1815, it became the 'Annales de chimie et de physique'." -> (has ) -> [annales de chimie et de physique
(Scientific journal founded in Paris)] -> (participates in event: Annales de chimie et de physique was
```

published under that name for the next 100) -> "Annales de chimie et de physique was published under that name for the next 100 years."

"Verdet championed the theory through his editorial supervision of the 'Annales de chimie et de physique'." -> (has) -> [annales de chimie et de physique (Scientific journal founded in Paris)]

"In 1815, it became the 'Annales de chimie et de physique'." -> IS_A -> "Annales de chimie et de physique is a scientific journal."
Logical Conclusions:
Annales de chimie et de physique, a scientific journal, became known by that name in 1815.

"In 1815, it became the 'Annales de chimie et de physique'." -> RENAMED_FROM -> "Annales de chimie et de physique was founded under the title 'Annales de chimie'."
Logical Conclusions:
In 1815, 'Annales de chimie' became the 'Annales de chimie et de physique'.

[annales de chimie et de physique (Scientific journal founded in Paris)] -> (participates in event: One of the early editors of Annales de chimie et de physique was the French chem) -> "One of the early editors of Annales de chimie et de physique was the French chemist Antoine Lavoisier."

"In 1815, it became the 'Annales de chimie et de physique'." -> FOLLOWS_FOUNDING -> "Annales de chimie et de physique was founded in 1789."
Logical Conclusions:
Annales de chimie et de physique, which became that name in 1815, was founded in 1789.

[annales de chimie et de physique (Scientific journal founded in Paris)] -> (participates in event: Annales de chimie et de physique is a scientific journal.) -> "Annales de chimie et de physique is a scientific journal."

"Marcel Émile Verdet edited the works of Augustin-Jean Fresnel." -> (has) -> [Émile Verdet (French physicist)]

[annales de chimie et de physique (Scientific journal founded in Paris)] -> (participates in event: Annales de chimie et de physique was founded under the title 'Annales de chimie') -> "Annales de chimie et de physique was founded under the title 'Annales de chimie'."

[annales de chimie et de physique (Scientific journal founded in Paris)] -> (participates in event: Annales de chimie et de physique was founded in 1789.) -> "Annales de chimie et de physique was founded in 1789."

[annales de chimie et de physique (Scientific journal founded in Paris)] -> (participates in event: Annales de chimie et de physique was founded in Paris, France.) -> "Annales de chimie et de physique was founded in Paris, France."

"Verdet championed the theory through his editorial supervision of the 'Annales de chimie et de physique'." -> FOLLOWS -> "One of the early editors of Annales de chimie et de physique was the French chemist Antoine Lavoisier."
Logical Conclusions:
Antoine Lavoisier was an early editor of Annales de chimie et de physique, and Verdet later championed theory through his editorial supervision of the same journal.

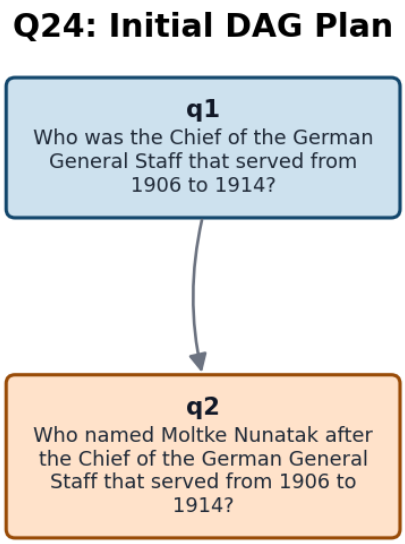
"Annales de chimie et de physique was published under that name for the next 100 years." -> IS_A -> "Annales de chimie et de physique is a scientific journal."
Logical Conclusions:
Annales de chimie et de physique, a scientific journal, was published under t …[truncated for report]

Q24 [bridge / hard] 5ac514d05542994611c8b372

What is the name of the person that named Moltke Nunatak after the Chief of the German General staff that served from 1906 to 1914?

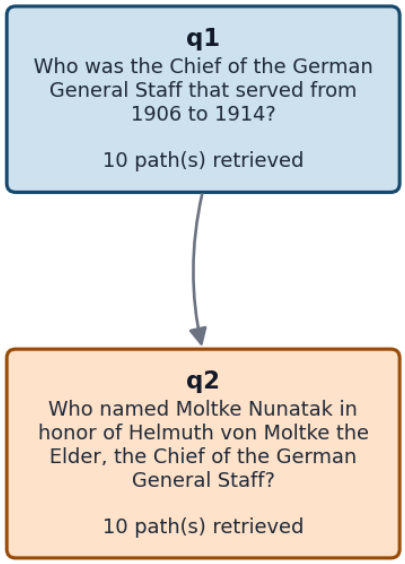
Gold answer	Second German Antarctic Expedition of 1911–12 under Wilhelm Filchner.
LLM answer	Wilhelm Filchner
EM / F1	**0.00 / 0.36**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q24: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was the Chief of the German General Staff that served from 1906 to 1914?	10 path(s)
q2	1	q1	Who named Moltke Nunatak in honor of Helmuth von Moltke the Elder, the Chief of the German General Staff?	10 path(s)

Final context fed to generation (1499 tokens)

```
Query plan:
- q1: Who was the Chief of the German General Staff that served from 1906 to 1914?
- q2: Who named Moltke Nunatak after the Chief of the German General Staff that served from 1906 to 1914?
(depends on: q1)

Level 1:

Subquery: Who named Moltke Nunatak in honor of Helmuth von Moltke the Elder, the Chief of the German
General Staff?
"Wilhelm Filchner named the nunatak for General Helmuth von Moltke." -> (has ) -> [Helmuth von Moltke the
Elder (Chief of the German General Staff)]

"Wilhelm Filchner named the nunatak for General Helmuth von Moltke." -> IDENTIFIES_AS -> "General Helmuth
von Moltke was Chief of the German General Staff."
Logical Conclusions:
General Helmuth von Moltke was the Chief of the German General Staff, for whom the nunatak was named.

"Wilhelm Filchner named the nunatak for General Helmuth von Moltke." -> HONORED_BY -> "One nunatak was
first roughly mapped and named 'Moltke Nunatak' by the Second German Antarctic Expedition of 1911–12."
Logical Conclusions:
The nunatak was named for General Helmuth von Moltke by Wilhelm Filchner during the expedition.
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"Wilhelm Filchner named the nunatak for General Helmuth von Moltke." -> (has) -> [German explorers (German explorers and scientists)]

"Wilhelm Filchner named the nunatak for General Helmuth von Moltke." -> SUPPORTS -> "The Moltke Nunataks are a chain of north–south trending nunataks close to the northeastern end of the Filchner Ice Shelf, Antarctica."
Logical Conclusions:
The Moltke Nunataks are a chain of north–south trending nunataks and Wilhelm Filchner named the nunatak for General Helmuth von Moltke.

[Otto Struve (Astronomer of the Struve family)] -> (participates in event: Otto Struve was the great-grandson of Friedrich Georg Wilhelm von Struve.) -> "Otto Struve was the great-grandson of Friedrich Georg Wilhelm von Struve."

[Randy W. Schekman (Co-recipient of 2013 prize)] -> (participates in event: In 2013, the prize was awarded to Thomas C. Südhof.) -> "In 2013, the prize was awarded to Thomas C. Südhof."

[Upper Saxon German (East Central German dialect)] -> (participates in event: Upper Saxon German is colloquially called 'Saxon'.) -> "Upper Saxon German is colloquially called 'Saxon'."

[Alexander (Various historical figures named Alexander)] -> (participates in event: Alexander Bogdanov lived from 1873 to 1928.) -> "Alexander Bogdanov lived from 1873 to 1928."

[Alexander (Various historical figures named Alexander)] -> (participates in event: The Vpered group included Alexander Bogdanov.) -> "The Vpered group included Alexander Bogdanov."

Level 0:

Subquery: Who was the Chief of the German General Staff that served from 1906 to 1914?
"Helmuth Johann Ludwig von Moltke served as the Chief of the German General Staff from 1906 to 1914." -> PRECEDES -> "Helmuth von Moltke the Younger was dismissed after the First Battle of the Marne (5–12 September 1914)."
Logical Conclusions:
Helmuth von Moltke the Younger served as Chief of the General Staff from 1906 to 1914 and was dismissed after the First Battle of the Marne.

"Helmuth Johann Ludwig von Moltke served as the Chief of the German General Staff from 1906 to 1914." -> (has) -> [Helmuth von Moltke the Elder (Chief of the German General Staff)]

"Helmuth Johann Ludwig von Moltke served as the Chief of the German General Staff from 1906 to 1914." -> SERVED_AS -> "Helmuth Johann Ludwig von Moltke is also known as Moltke the Younger."
Logical Conclusions:
Helmuth Johann Ludwig von Moltke served as Chief of the German General Staff from 1906 to 1914 and is known as Moltke the Younger.

"Helmuth Johann Ludwig von Moltke served as the Chief of the German General Staff from 1906 to 1914." -> SPECIFIES -> "Helmuth von Moltke the Younger was the Commander-in-Chief of the German army after Schlieffen retired in 1906."
Logical Conclusions:
Helmuth von Moltke the Younger was Commander-in-Chief after Schlieffen retired in 1906 and served as Chief of the General Staff from 1906 to 1914.

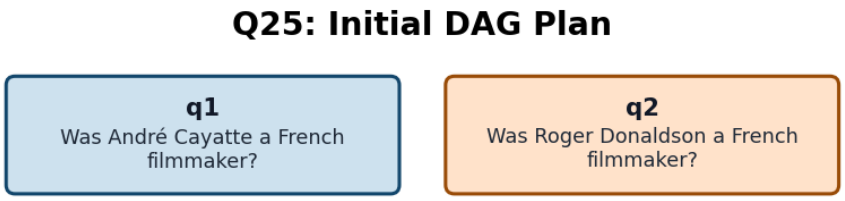
"Helmuth Jo …[truncated for report]

Q25 [comparison / hard] 5abb8afe5542993f40c73b1e

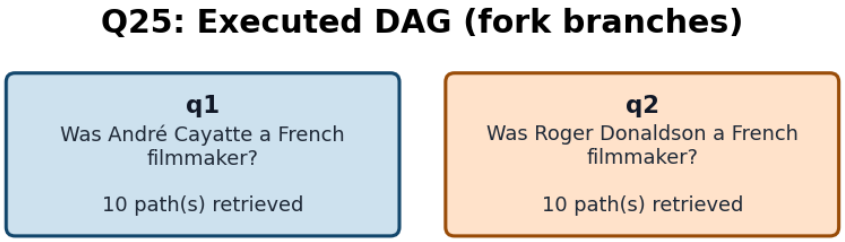
Were both Roger Donaldson and André Cayatte french filmmakers?

Gold answer	no
LLM answer	No, only André Cayatte was a French filmmaker. Roger Donaldson was a New Zealand film director.
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Was André Cayatte a French filmmaker?	10 path(s)
q2	0	—	Was Roger Donaldson a French filmmaker?	10 path(s)

Final context fed to generation (1044 tokens)

```
Query plan:
- q1: Was André Cayatte a French filmmaker?
- q2: Was Roger Donaldson a French filmmaker?

Level 0:

Subquery: Was André Cayatte a French filmmaker?
"André Cayatte was a French filmmaker." -> KNOWN_FOR -> "André Cayatte became known for his films centering
on themes of crime, justice, and moral responsibility."
Logical Conclusions:
```

André Cayatte was a French filmmaker known for films centering on themes of crime, justice, and moral responsibility.

"André Cayatte was a French filmmaker." -> CONTEXT_FOR -> "André Cayatte was born on 3 February 1909 in Carcassonne."
Logical Conclusions:
André Cayatte was born in Carcassonne and was a French filmmaker.

"André Cayatte was a French filmmaker." -> CO_DIRECTED -> "Return to Life was directed by Georges Lampin, André Cayatte, Henri-Georges Clouzot and Jean Dréville."
Logical Conclusions:
Return to Life was co-directed by André Cayatte, who was a French filmmaker.

"André Cayatte was a French filmmaker." -> DIRECTED -> "'Black Dossier' was directed by André Cayatte."
Logical Conclusions:
André Cayatte directed Black Dossier and was a French filmmaker.

"André Cayatte was a French filmmaker." -> (has) -> [andré cayatte (French filmmaker and former lawyer)]

"André Cayatte was a French filmmaker." -> DIRECTED -> "'Before the Deluge' was directed by André Cayatte."
Logical Conclusions:
Before the Deluge was directed by André Cayatte, who was a French filmmaker.

[andré cayatte (French filmmaker and former lawyer)] -> (participates in event: André Cayatte was a former attorney when he made 'We Are All Murderers'.) -> "André Cayatte was a former attorney when he made 'We Are All Murderers'."

[andré cayatte (French filmmaker and former lawyer)] -> (participates in event: 'Before the Deluge' was directed by André Cayatte.) -> "'Before the Deluge' was directed by André Cayatte."

Subquery: Was Roger Donaldson a French filmmaker?
"Roger Donaldson was born on 15 November 1945." -> SPECIFIES -> "Roger Donaldson is a New Zealand film director."
Logical Conclusions:
Roger Donaldson was born on 15 November 1945 and is a New Zealand film director.

"Roger Donaldson was born on 15 November 1945." -> (has) -> [roger donaldson (Australian-born New Zealand film director)]

"Roger Donaldson was born on 15 November 1945." -> SPECIFIES -> "Roger Donaldson is Australian-born."
Logical Conclusions:
Roger Donaldson was born on 15 November 1945 and is Australian-born.

"Roger Donaldson's films include a run of titles shot in the United States." -> SUPPORTS -> "Roger Donaldson is a New Zealand writer."
Logical Conclusions:
Roger Donaldson is a New Zealand writer whose films include a run of titles shot in the United States.

"Roger Donaldson's films include a run of titles shot in the United States." -> (has) -> [roger donaldson (Australian-born New Zealand film director)]

"Roger Donaldson's films include a run of titles shot in the United States." -> SUPPORTS -> "Roger Donaldson is Australian-born."
Logical Conclusions:
Roger Donaldson, who is Australian-born, has films including a run of titles shot in the United States.

"Roger Donaldson's films include a run of titles shot in the United States." -> INCLUDES -> "The run of titles shot in the United States includes the Kevin Costner film 'Thirteen Days' (2000)."
Logical Conclusions:
The run of titles shot in the United States includes the Kevin Costner film 'Thirteen Days' (2000).

"Roger Donaldson's films include a run of titles shot in the United States." -> INCLUDES -> "The run of

titles shot in the United States includes the 1997 disaster film 'Dante's Peak'."
Logical Conclusions:
The run of titles shot in the United States includes the 1997 disaster film 'Dante's Peak'.

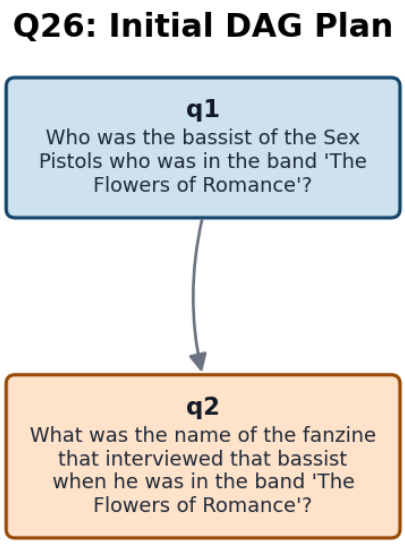
"Roger Donaldson's films include a run of titles shot in the United States." -> INCLUDES -> "The run of titles shot in the United States includes the Kevin Costner film 'No Way Out' (1987)."
Logical Conclusions:
The run of titles shot in the United States includes the Ke …[truncated for report]

Q26 [bridge / hard] 5adea9ea55429939a52fe934

What was the name of the fanzine that interviewed the Sex Pistols bassist when he was in the band "The Flowers of Romance?"

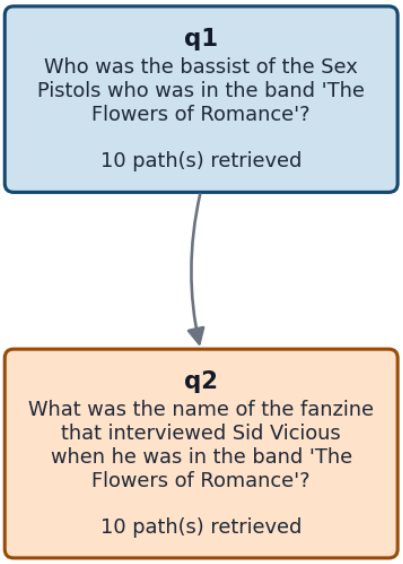
Gold answer	SKUM
LLM answer	SKUM
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q26: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was the bassist of the Sex Pistols who was in the band 'The Flowers of Romance'?	10 path(s)
q2	1	q1	What was the name of the fanzine that interviewed Sid Vicious when he was in the band 'The Flowers of Romance'?	10 path(s)

Final context fed to generation (1299 tokens)

```
Query plan:
- q1: Who was the bassist of the Sex Pistols who was in the band 'The Flowers of Romance'?
- q2: What was the name of the fanzine that interviewed that bassist when he was in the band 'The Flowers of Romance'? (depends on: q1)

Level 1:

Subquery: What was the name of the fanzine that interviewed Sid Vicious when he was in the band 'The Flowers of Romance'?
"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> SUPPORTS -> "Despite never playing live, The Flowers of Romance were interviewed by a fanzine named 'SKUM'."
Logical Conclusions:
The Flowers of Romance band members included Sid Vicious, and despite never playing live, the band was interviewed by a fanzine named 'SKUM'.

"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> ELABORATES -> "Sid Vicious proclaimed 'I'll just be the yob that I am now' in the fanzine 'SKUM'."
Logical Conclusions:
Sid Vicious was a member of The Flowers of Romance and proclaimed 'I'll just be the yob that I am now' in the fanzine 'SKUM'.

"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> SUPPORTS -> "The Flowers
```

of Romance were formed in mid-1976 by Jo Faull and Sarah Hall."
Logical Conclusions:
The Flowers of Romance band members included Sid Vicious, and the band was formed in mid-1976 by Jo Faull and Sarah Hall.

"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> MEMBER_OF -> "The Flowers of Romance are more famed for the number of band members that later became well known."
Logical Conclusions:
Sid Vicious of the Sex Pistols was a member of The Flowers of Romance.

[the flowers of romance (Early punk band formed in 1976)] -> (participates in event: Despite never playing live, The Flowers of Romance were interviewed by a fanzine) -> "Despite never playing live, The Flowers of Romance were interviewed by a fanzine named 'SKUM'."

[the flowers of romance (Early punk band formed in 1976)] -> (participates in event: The Flowers of Romance band members included Sid Vicious of the Sex Pistols.) -> "The Flowers of Romance band members included Sid Vicious of the Sex Pistols."

"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> SUPPORTS -> "The Flowers of Romance band members included Viv Albertine, who went on to play in The Slits."
Logical Conclusions:
The Flowers of Romance band members included Sid Vicious and Viv Albertine, who went on to play in The Slits.

"Viv Albertine was interviewed by a fanzine named 'SKUM'." -> (has) -> [skum (fanzine that interviewed Sid Vicious)]

"Viv Albertine was interviewed by a fanzine named 'SKUM'." -> SUPPORTS -> "The Flowers of Romance band members included Viv Albertine, who went on to play in The Slits."
Logical Conclusions:
Viv Albertine was interviewed by 'SKUM' and was a member of The Flowers of Romance.

Level 0:

Subquery: Who was the bassist of the Sex Pistols who was in the band 'The Flowers of Romance'?
"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> HAS_MEMBER -> "Sid Vicious was born John Simon Ritchie on 10 May 1957."
Logical Conclusions:
Sid Vicious, born John Simon Ritchie, was a member of The Flowers of Romance.

"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> SUPPORTS -> "The Flowers of Romance were formed in mid-1976 by Jo Faull and Sarah Hall."
Logical Conclusions:
The Flowers of Romance band members included Sid Vicious, and the band was formed in mid-1976 by Jo Faull and Sarah Hall.

"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> SHARES_MEMBER -> "Sid Vicious is most famous as a member of the influential punk rock band the Sex Pistols."
Logical Conclusions:
Sid Vicious, most famous as a Sex Pistol, was also a member of The Flowers of Romance.

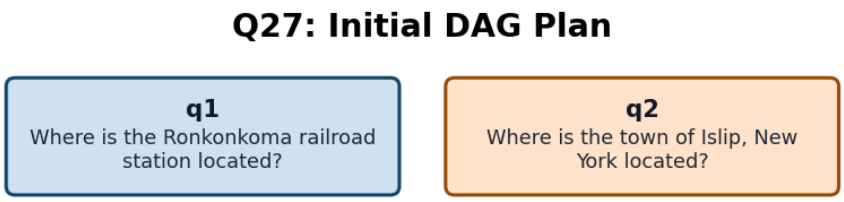
"The Flowers of Romance band members included Sid Vicious of the Sex Pistols." -> (has) -> [the flowers of romance (Early punk band fo …[truncated for report]

Q27 [bridge / hard] 5a7f38e255429930675136b6

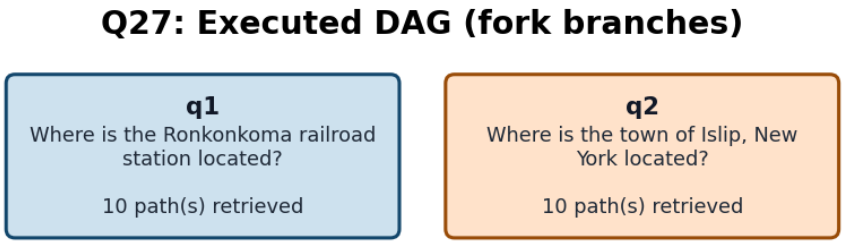
The Ronkonkama railroad station and the town of Islip, New York are near?

Gold answer	Long Island
LLM answer	Ronkonkoma, New York
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Where is the Ronkonkoma railroad station located?	10 path(s)
q2	0	—	Where is the town of Islip, New York located?	10 path(s)

Final context fed to generation (1410 tokens)

```
Query plan:
- q1: Where is the Ronkonkoma railroad station located?
- q2: Where is the town of Islip, New York located?

Level 0:

Subquery: Where is the Ronkonkoma railroad station located?
"Ronkonkoma is a major railroad station and transportation hub along the Main Line of the Long Island Rail
Road." -> LOCATED_IN -> "Ronkonkoma is located in Ronkonkoma, New York."
Logical Conclusions:
```

Ronkonkoma station is a major railroad station located in Ronkonkoma, New York.

[Ronkonkoma (Major railroad station in New York)] -> (participates in event: Ronkonkoma is located in Ronkonkoma, New York.) -> "Ronkonkoma is located in Ronkonkoma, New York."

"The station in Ronkonkoma is the eastern terminus of the Ronkonkoma Branch." -> LOCATED_IN -> "Ronkonkoma is located in Ronkonkoma, New York."
Logical Conclusions:
Ronkonkoma station is located in Ronkonkoma, New York.

"Ronkonkoma is a major railroad station and transportation hub along the Main Line of the Long Island Rail Road." -> (has) -> [Ronkonkoma (Major railroad station in New York)]

"Ronkonkoma is a major railroad station and transportation hub along the Main Line of the Long Island Rail Road." -> LOCATES -> "Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York, United States."
Logical Conclusions:
Fact 1 identifies Ronkonkoma as a railroad station, while fact 10 describes it as a hamlet and CDP on Long Island.

"Ronkonkoma is a major railroad station and transportation hub along the Main Line of the Long Island Rail Road." -> SUPPORTS -> "Main Line of the Long Island Rail Road is in Ronkonkoma, New York."
Logical Conclusions:
Both facts describe Ronkonkoma station, with one noting it is a major hub on the Main Line and the other stating the Main Line is in Ronkonkoma.

[Ronkonkoma (Major railroad station in New York)] -> (participates in event: The station in Ronkonkoma is the western terminus of the out Branch.) -> "The station in Ronkonkoma is the western terminus of the out Branch."

[Ronkonkoma (Major railroad station in New York)] -> (participates in event: Ronkonkoma station is the eastern terminus of the Ronkonkoma Branch.) -> "Ronkonkoma station is the eastern terminus of the Ronkonkoma Branch."

"The station in Ronkonkoma is the eastern terminus of the Ronkonkoma Branch." -> (has) -> [Ronkonkoma (Major railroad station in New York)]

[Ronkonkoma (Major railroad station in New York)] -> (participates in event: Main Line of the Long Island Rail Road is in Ronkonkoma, New York.) -> "Main Line of the Long Island Rail Road is in Ronkonkoma, New York."

Subquery: Where is the town of Islip, New York located?
"Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York, United States." -> (has) -> [Ronkonkoma (Major railroad station in New York)]

"Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York, United States." -> DESCRIBES -> "Ronkonkoma is located in Ronkonkoma, New York."
Logical Conclusions:
Both facts describe Ronkonkoma, with one stating its location and the other providing its classification as a hamlet and CDP.

"Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York, United States." -> LOCATION_OF -> "The station in Ronkonkoma is the eastern terminus of the Ronkonkoma Branch."
Logical Conclusions:
Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York.

"Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York, United States." -> LOCATES -> "The station in Ronkonkoma is the western terminus of the out Branch."
Logical Conclusions:
Fact 9 describes the station as the western terminus of the out Branch, while fact 10 describes Ronkonkoma as a hamlet and CDP.

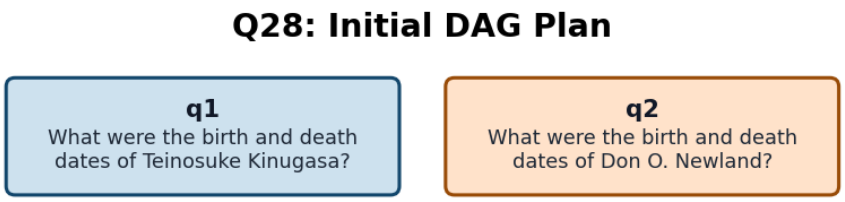
"Ronkonkoma is a hamlet and census-designated place (CDP) on Long Island in the Town of Islip, New York, United States." …[truncated for report]

Q28 [comparison / hard] 5a7ae38c5542992d025e6721

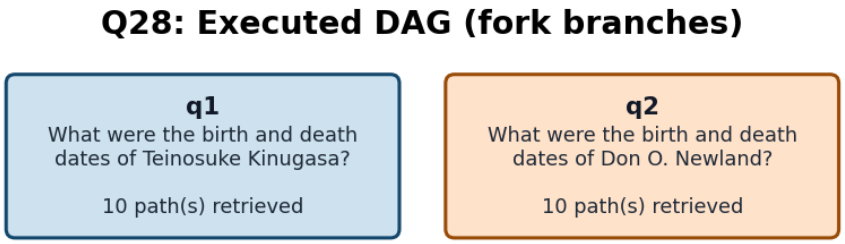
Who was older when they died, Teinosuke Kinugasa or Don O. Newland

Gold answer	Teinosuke Kinugasa
LLM answer	Teinosuke Kinugasa was older when he died.
EM / F1	**0.00 / 0.44**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What were the birth and death dates of Teinosuke Kinugasa?	10 path(s)
q2	0	—	What were the birth and death dates of Don O. Newland?	10 path(s)

Final context fed to generation (1248 tokens)

```
Query plan:
- q1: What were the birth and death dates of Teinosuke Kinugasa?
- q2: What were the birth and death dates of Don O. Newland?

Level 0:

Subquery: What were the birth and death dates of Teinosuke Kinugasa?
"Teinosuke Kinugasa was born on 1 January 1896." -> LIFESPAN -> "Teinosuke Kinugasa died on 26 February
1982."
```

Logical Conclusions:
Teinosuke Kinugasa was born on 1 January 1896 and died on 26 February 1982.

"Teinosuke Kinugasa was born on 1 January 1896." -> PRECEDES -> "Teinosuke Kinugasa died in Kyoto."
Logical Conclusions:
Fact 41 states Teinosuke Kinugasa was born on 1 January 1896, and fact 54 states he died in Kyoto, providing his birthplace and death location.

"Teinosuke Kinugasa died in Kyoto." -> SUPPORTS -> "Teinosuke Kinugasa died on 26 February 1982."
Logical Conclusions:
Fact 54 states Teinosuke Kinugasa died in Kyoto, and fact 58 states he died on 26 February 1982, providing his death location and date.

"Teinosuke Kinugasa was born on 1 January 1896." -> PRECEDES -> "Teinosuke Kinugasa won the 1954 Palme d'or at Cannes for 'Jigokumon' ('The Gate of Hell')."
Logical Conclusions:
Fact 41 states Teinosuke Kinugasa was born on 1 January 1896, and fact 66 states he won the 1954 Palme d'or, linking his birth to his achievement.

"Teinosuke Kinugasa was born on 1 January 1896." -> BIRTH_DATE_OF -> "Teinosuke Kinugasa was a Japanese actor and film director."
Logical Conclusions:
Teinosuke Kinugasa, a Japanese actor and film director, was born on 1 January 1896.

"Teinosuke Kinugasa was born on 1 January 1896." -> (has) -> [teinosuke kinugasa (Japanese film director and actor)]

"Teinosuke Kinugasa was born on 1 January 1896." -> BORN_IN -> "Teinosuke Kinugasa was born in Kameyama, Mie Prefecture."
Logical Conclusions:
Teinosuke Kinugasa, born on 1 January 1896, was born in Kameyama, Mie Prefecture.

"Teinosuke Kinugasa died in Kyoto." -> SUPPORTS -> "Teinosuke Kinugasa was born in Kameyama, Mie Prefecture."
Logical Conclusions:
Fact 54 states Teinosuke Kinugasa died in Kyoto, and fact 69 states he was born in Kameyama, Mie Prefecture, providing his birthplace and death location.

"Teinosuke Kinugasa died in Kyoto." -> DEATHPLACE_OF -> "Teinosuke Kinugasa was a Japanese actor and film director."
Logical Conclusions:
Teinosuke Kinugasa died in Kyoto.

Subquery: What were the birth and death dates of Don O. Newland?
"Don Oliver Newland lived from 1896 to 1951." -> PRECEDES -> "Roger Joseph Ebert was born on June 18, 1942."
Logical Conclusions:
Both facts provide lifespan or birth year information, allowing for a chronological comparison of their lifetimes.

"Don Oliver Newland lived from 1896 to 1951." -> PRECEDES -> "George Lynn was born in 1915."
Logical Conclusions:
Both facts provide lifespan or birth year information, allowing for a chronological comparison of their lifetimes.

[don o. newland (American film director and producer)] -> (participates in event: Don Oliver Newland lived from 1896 to 1951.) -> "Don Oliver Newland lived from 1896 to 1951."

"Don Oliver Newland lived from 1896 to 1951." -> LIFESPAN_OF -> "Don Oliver Newland was an American film director and producer."
Logical Conclusions:
Don Oliver Newland lived from 1896 to 1951, establishing his lifespan.

[don o. newland (American film director and producer)] -> (participates in event: From the 1920s until his death, Don Oliver Newland traveled to cities throughout) -> "From the 1920s until his death, Don Oliver Newland traveled to cities throughout the United States making films."

[don o. newland (American film director and producer)] -> (participates in event: Don Oliver Newland was an American film director and producer.) -> "Don Oliver Newland was an American film director and producer."

[don o. newland (American film director and producer)] -> (participates in event: Don Oliver Newland used a standard script.) -> "Don Oliver Newland used a standard script."

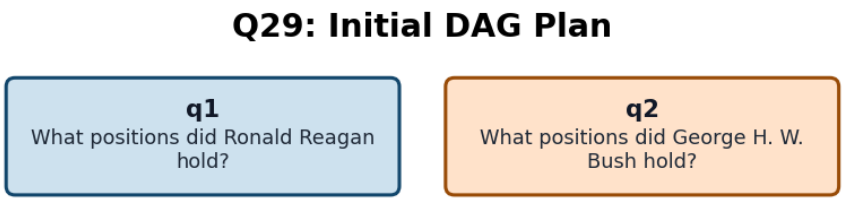
[don o. newland (American film director and producer)] -> (participates …[truncated for report]

Q29 [bridge / hard] 5abb30755542996cc5e49fd8

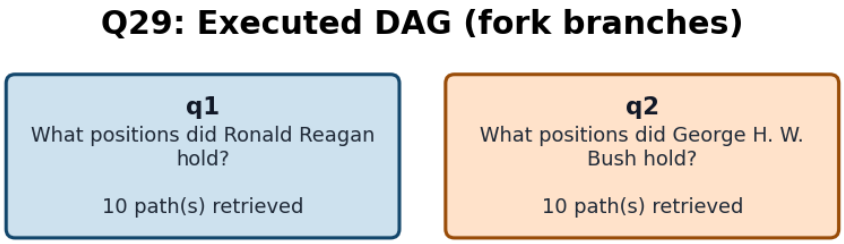
Ronald Reagan and George H. W. Bush both held which position in office?

Gold answer	President of the United States
LLM answer	President of the United States
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What positions did Ronald Reagan hold?	10 path(s)
q2	0	—	What positions did George H. W. Bush hold?	10 path(s)

Final context fed to generation (1171 tokens)

```
Query plan:
- q1: What positions did Ronald Reagan hold?
- q2: What positions did George H. W. Bush hold?

Level 0:

Subquery: What positions did Ronald Reagan hold?
[ronald reagan (40th President of the United States)] -> (participates in event: The Presidents under whom
Martin held senior appointments and advisory positions) -> "The Presidents under whom Martin held senior
appointments and advisory positions include Ronald Reagan."

"Abrams served in foreign policy positions for President Ronald Reagan." -> (has ) -> [ronald reagan (40th
```

President of the United States)]

[ronald reagan (40th President of the United States)] -> (participates in event: Lawrence Eagleburger had served in lesser capacities under President Ronald Reag) -> "Lawrence Eagleburger had served in lesser capacities under President Ronald Reagan."

[ronald reagan (40th President of the United States)] -> (participates in event: Ronald Reagan was President of the United States.) -> "Ronald Reagan was President of the United States."

[ronald reagan (40th President of the United States)] -> (participates in event: The inauguration marked the commencement of the second four-year term of Ronald) -> "The inauguration marked the commencement of the second four-year term of Ronald Reagan as President."

"Abrams served in foreign policy positions for President Ronald Reagan." -> PRECEDES -> "Abrams served in foreign policy positions for President George W. Bush."
Logical Conclusions:
Abrams served in foreign policy positions for President Ronald Reagan and also for President George W. Bush.

"Abrams served in foreign policy positions for President Ronald Reagan." -> LEADS_TO -> "Abrams was convicted of withholding information from Congress about the Iran–Contra affair while serving under Reagan."
Logical Conclusions:
While serving in foreign policy positions under Reagan, Abrams was convicted of withholding information about the Iran–Contra affair.

"Abrams served in foreign policy positions for President Ronald Reagan." -> IDENTIFIES -> "Elliott Abrams is a former American diplomat, lawyer and political scientist."
Logical Conclusions:
Elliott Abrams is a former American diplomat, and he served in foreign policy positions for President Ronald Reagan.

"Abrams served in foreign policy positions for President Ronald Reagan." -> (has) -> [elliott abrams (Former diplomat convicted in Iran-Contra affair)]

[White House (Location of private inauguration ceremony)] -> (participates in event: The second inauguration of Ronald Reagan as President of the United States was h) -> "The second inauguration of Ronald Reagan as President of the United States was held privately on Sunday, January 20, 1985 in the Grand Foyer of the White House."

Subquery: What positions did George H. W. Bush hold?
[george h. w. bush (Vice President and later President)] -> (participates in event: The Presidents under whom Martin held senior appointments and advisory positions) -> "The Presidents under whom Martin held senior appointments and advisory positions include George H.W. Bush."

"George H. W. Bush was previously a congressman." -> (has) -> [george h. w. bush (Vice President and later President)]

[george h. w. bush (Vice President and later President)] -> (participates in event: Lawrence Eagleburger served briefly as the Secretary of State under President Ge) -> "Lawrence Eagleburger served briefly as the Secretary of State under President George H. W. Bush."

[george h. w. bush (Vice President and later President)] -> (participates in event: Eight of the twelve seats eventually were filled by appointees of President Geor) -> "Eight of the twelve seats eventually were filled by appointees of President George H. W. Bush."

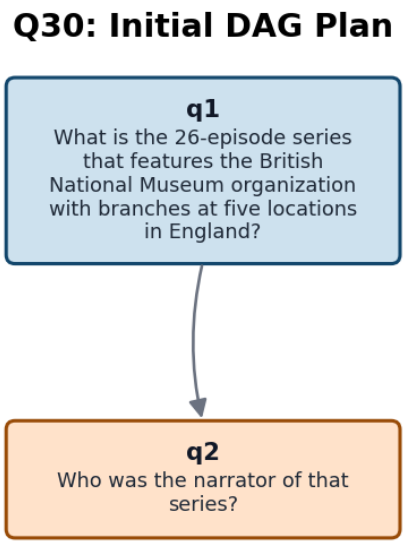
[george h. w. bush (Vice President and later President)] -> (participates in event: The inauguration marked the commencement of the second four-year term of George) -> "The inauguration marked the commencement of the second four-year term of George H. W. Bush as V …[truncated for report]

Q30 [bridge / hard] 5a8efba955429918e830d173

Who was the narrator of the 26-episode series which featured the British national museum organization with branches at five locations in England?

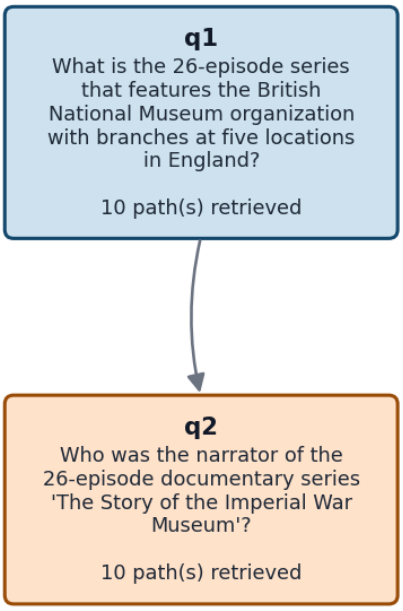
Gold answer	Michael Redgrave
LLM answer	Sebastian Shaw
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q30: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the 26-episode series that features the British National Museum organization with branches at five locations in England?	10 path(s)
q2	1	q1	Who was the narrator of the 26-episode documentary series 'The Story of the Imperial War Museum'?	10 path(s)

Final context fed to generation (1227 tokens)

```
Query plan:
- q1: What is the 26-episode series that features the British National Museum organization with branches at five locations in England?
- q2: Who was the narrator of that series? (depends on: q1)

Level 1:

Subquery: Who was the narrator of the 26-episode documentary series 'The Story of the Imperial War Museum'?
[Sebastian Shaw (Reader in The Great War documentary)] -> (participates in event: Readings in 'The Great War' were by Marius Goring, Ralph Richardson, Cyril Luckh) -> "Readings in 'The Great War' were by Marius Goring, Ralph Richardson, Cyril Luckham, Sebastian Shaw, and Emlyn Williams."

[Edmund Robert Harris (Principal benefactor of Harris Museum and schools)] -> (participates in event: Sir Edmund Robert Bartley Bartley-Denniss KC was a barrister.) -> "Sir Edmund Robert Bartley Bartley-Denniss KC was a barrister."

[Edmund Robert Harris (Principal benefactor of Harris Museum and schools)] -> (participates in event: Edmund Robert Anthony de Unger was an art collector.) -> "Edmund Robert Anthony de Unger was an art collector."
```

[Edmund Robert Harris (Principal benefactor of Harris Museum and schools)] -> (participates in event: Edmund Robert Anthony de Unger was a Hungarian-born property developer.) -> "Edmund Robert Anthony de Unger was a Hungarian-born property developer."

[Edmund Robert Harris (Principal benefactor of Harris Museum and schools)] -> (participates in event: Edmund Robert Harris was an English lawyer from Preston, Lancashire, UK.) -> "Edmund Robert Harris was an English lawyer from Preston, Lancashire, UK."

[Edmund Robert Harris (Principal benefactor of Harris Museum and schools)] -> (participates in event: In London, Edmund Robert Anthony de Unger built up the Keir Collection.) -> "In London, Edmund Robert Anthony de Unger built up the Keir Collection."

[Edward Thiel (Examiner and sketcher of nunatak)] -> (participates in event: The nunatak was examined and sketched by Edward Thiel.) -> "The nunatak was examined and sketched by Edward Thiel."

[Marisha Chamberlain (American writer born January 6, 1952)] -> (participates in event: Marisha Anne Chamberlain is an American writer.) -> "Marisha Anne Chamberlain is an American writer."

[Robert Totten (Director of Gunsmoke installments with a small part)] -> (participates in event: 'Independence Day' was directed by Robert Mandel.) -> "'Independence Day' was directed by Robert Mandel."

[Nick Robinson (Actor in Frenemies)] -> (participates in event: Charles Whitney, Levi L. Whitney, and Orville E. Thompson were attributed with s) -> "Charles Whitney, Levi L. Whitney, and Orville E. Thompson were attributed with successfully running the first factory of its kind in Chicago."

Level 0:

Subquery: What is the 26-episode series that features the British National Museum organization with branches at five locations in England?
[Imperial War Museum (British national museum organization)] -> (participates in event: The Imperial War Museum has branches at five locations in England.) -> "The Imperial War Museum has branches at five locations in England."

[Imperial War Museum (British national museum organization)] -> (participates in event: The project was part of The Space project from the BBC and the Arts Council Engl) -> "The project was part of The Space project from the BBC and the Arts Council England."

[Imperial War Museum (British national museum organization)] -> (participates in event: The Imperial War Museum is a British national museum organisation.) -> "The Imperial War Museum is a British national museum organisation."

[Imperial War Museum (British national museum organization)] -> (participates in event: The museum's remit has expanded to include all conflicts in which British or Com) -> "The museum's remit has expanded to include all conflicts in which British or Commonwealth forces have been involved since 1914."

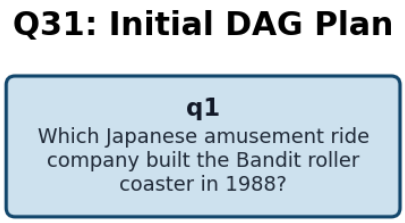
[Imperial War Museum (British national museum organization)] -> (participates in event: As of 2012, the museum aims 'to …[truncated for report]

Q31 [bridge / hard] 5a8b83bc55429949d91db5c0

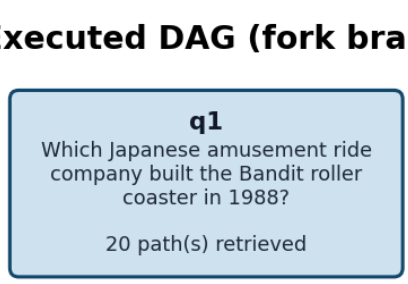
Bandit was built in 1988 by which Japanese amusement ride company that built roller coasters, giant wheels, carousels, flumes, dark rides, sky cycles and other amusement rides?

Gold answer	TOGO company
LLM answer	TOGO
EM / F1	**0.00 / 0.67**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which Japanese amusement ride company built the Bandit roller coaster in 1988?	20 path(s)

Final context fed to generation (415 tokens)

```
[bandit (Steel roller coaster at Yomiuriland)] -> (participates in event: Bandit was built in 1988.) ->
"Bandit was built in 1988." -> BUILT_BY -> "Bandit was built by the TOGO company." -> CONTEXTUALIZES ->
"TOGO was a Japanese amusement ride company."
Logical Conclusions:
Bandit was built in 1988 by the TOGO company.
Bandit was built by the TOGO company, and TOGO was a Japanese amusement ride company.

"Bandit was built by the TOGO company." -> (has ) -> [togo (Japanese amusement ride company)] ->
(participates in event: TOGO was a Japanese amusement ride company.) -> "TOGO was a Japanese amusement ride
company."
```

[bandit (Steel roller coaster at Yomiuriland)] -> (participates in event: Bandit was built by the TOGO company.) -> "Bandit was built by the TOGO company."

[togo (Japanese amusement ride company)] -> (participates in event: TOGO built other amusement rides.) -> "TOGO built other amusement rides."

[togo (Japanese amusement ride company)] -> (participates in event: TOGO built dark rides.) -> "TOGO built dark rides."

[togo (Japanese amusement ride company)] -> (participates in event: TOGO built roller coasters.) -> "TOGO built roller coasters."

[togo (Japanese amusement ride company)] -> (participates in event: TOGO built giant wheels.) -> "TOGO built giant wheels."

[togo (Japanese amusement ride company)] -> (participates in event: TOGO built sky cycles.) -> "TOGO built sky cycles."

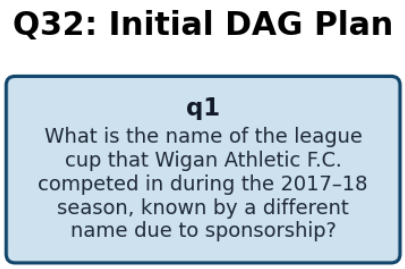
"Bandit was built by the TOGO company." -> SUPPORTS -> "TOGO built other amusement rides."
Logical Conclusions:
Both facts involve TOGO building rides, with one specifying a roller coaster and the other mentioning other amusement rides.

Q32 [bridge / hard] 5aba7cfe554299232ef4a2fd

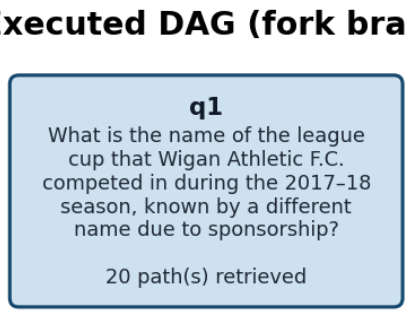
The 2017–18 Wigan Athletic F.C. season will be a year in which the team competes in the league cup known as what for sponsorship reasons?

Gold answer	Carabao Cup
LLM answer	Carabao Cup
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the name of the league cup that Wigan Athletic F.C. competed in during the 2017–18 season, known by a different name due to sponsorship?	20 path(s)

Final context fed to generation (963 tokens)

```
"The 2017–18 season is Wigan Athletic's 86th year in existence." -> INCLUDES -> "Along with competing in the league, the club will also participate in the EFL Cup." -> (has ) -> [EFL Cup (annual knockout football competition)] -> (participates in event: The EFL Cup is currently known as the Carabao Cup for sponsorship reasons.) -> "The EFL Cup is currently known as the Carabao Cup for sponsorship reasons."
```

Logical Conclusions:
During the 2017–18 season, Wigan Athletic competed in the league and the EFL Cup.

[EFL Cup (annual knockout football competition)] -> (participates in event: The EFL Cup is referred to historically and colloquially as simply the League Cu) -> "The EFL Cup is referred to historically and colloquially as simply the League Cup."

"The 2017–18 season is Wigan Athletic's 86th year in existence." -> (has) -> [Wigan Athletic F.C. (football club in League One)] -> (participates in event: Towards the end of the 1978–79 season, a second consecutive promotion seemed pos) -> "Towards the end of the 1978–79 season, a second consecutive promotion seemed possible for Wigan Athletic F.C."

[Wigan Athletic F.C. (football club in League One)] -> (participates in event: Wigan Borough F.C. folded due to financial problems.) -> "Wigan Borough F.C. folded due to financial problems."

[Wigan Athletic F.C. (football club in League One)] -> (participates in event: The departure of much of the previous coaching staff occurred the day before Pau) -> "The departure of much of the previous coaching staff occurred the day before Paul Cook's appointment."

[Wigan Athletic F.C. (football club in League One)] -> (participates in event: Paul Cook signed a 3 year deal.) -> "Paul Cook signed a 3 year deal."

"Along with competing in the league, the club will also participate in the EFL Cup." -> (has) -> [Wigan Athletic F.C. (football club in League One)]

[Wigan Athletic F.C. (football club in League One)] -> (participates in event: The club was nominated to apply for Football League status after finishing 2nd i) -> "The club was nominated to apply for Football League status after finishing 2nd in the Northern Premier League."

[Wigan Athletic F.C. (football club in League One)] -> (participates in event: Along with competing in the league, the club will also participate in the FA Cup) -> "Along with competing in the league, the club will also participate in the FA Cup." -> PART_OF -> "The 2017–18 season is Wigan Athletic's 86th year in existence."
Logical Conclusions:
During the 2017–18 season, Wigan Athletic competed in the league and the FA Cup.

[Wigan Athletic F.C. (football club in League One)] -> (participates in event: Springfield Park was the home ground of Wigan Athletic F.C. until the club moved) -> "Springfield Park was the home ground of Wigan Athletic F.C. until the club moved to the new JJB Stadium."

"The 2017–18 season is Wigan Athletic's 86th year in existence." -> SPECIFIES -> "The 2017–18 season is Wigan Athletic's first back in League One."
Logical Conclusions:
The 2017–18 season was Wigan Athletic's 86th year and their first back in League One.

"The 2017–18 season is Wigan Athletic's 86th year in existence." -> INCLUDES -> "Along with competing in the league, the club will also participate in the EFL Trophy."
Logical Conclusions:
During the 2017–18 season, Wigan Athletic competed in the league and the EFL Trophy.

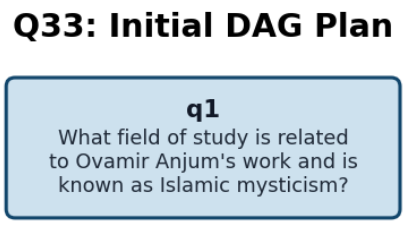
"The 2017–18 season is Wigan Athletic's 86th year in existence." -> FOLLOWS -> "Wigan Athletic moved to the new JJB Stadium after the 1998–99 season."
Logical Conclusions:
Wigan Athletic moved to the JJB Stadium after the 1998–99 season, which is long before the 2017–18 season.

Q33 [bridge / hard] 5a8ad6db55429950cd6afba7

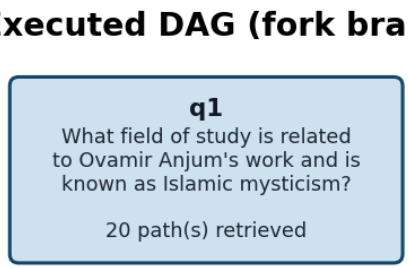
What field of study is related to Ovamir Anjum's work and is known as Islamic mysticism?

Gold answer	Sufism
LLM answer	Sufism
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What field of study is related to Ovamir Anjum's work and is known as Islamic mysticism?	20 path(s)

Final context fed to generation (1075 tokens)

```
[ovamir anjum (Imam Khattab Chair of Islamic Studies)] -> (participates in event: Ovamir Anjum's related
fields of study include Islamic philosophy and Sufism.) -> "Ovamir Anjum's related fields of study include
Islamic philosophy and Sufism." -> (has ) -> [islamic mysticism (subject of discussion on cognition and
love)] -> (participates in event: Islamic mysticism is distinct from conventional mysticism.) -> "Islamic
mysticism is distinct from conventional mysticism." -> DISTINCT_FROM -> "Islamic mysticism has been
intertwined with Sufism."
Logical Conclusions:
Islamic mysticism is distinct from conventional mysticism but has been intertwined with Sufism.

[islamic mysticism (subject of discussion on cognition and love)] -> (participates in event: Islamic
mysticism has been intertwined with Sufism.) -> "Islamic mysticism has been intertwined with Sufism."
```

"Islamic mysticism is distinct from conventional mysticism." -> IDENTIFIES -> "Islamic mysticism is cognition and knowledge."
Logical Conclusions:
Islamic mysticism is cognition and knowledge, and is distinct from conventional mysticism.

"Islamic mysticism is distinct from conventional mysticism." -> CONTRADICTS -> "Islamic mysticism can be considered as a vast range that engulfs theoretical and practical and conventional mysticism."
Logical Conclusions:
Fact 11 describes Islamic mysticism as a vast range that engulfs theoretical, practical, and conventional mysticism, while fact 12 states it is distinct from conventional mysticism.

[islamic mysticism (subject of discussion on cognition and love)] -> (participates in event: Islamic mysticism is cognition and knowledge.) -> "Islamic mysticism is cognition and knowledge."

"Islamic mysticism has been intertwined with Sufism." -> RESTATES -> "Islamic mysticism has been intertwined with sufism."
Logical Conclusions:
Both facts state that Islamic mysticism has been intertwined with Sufism.

"Ovamir Anjum's related fields of study include Islamic philosophy and Sufism." -> (has) -> [sufism (Mystical trend in Islam)] -> (participates in event: Dr. Safi specializes in Islamic mysticism (Sufism).) -> "Dr. Safi specializes in Islamic mysticism (Sufism)." -> (has) -> [islamic mysticism (subject of discussion on cognition and love)]

[islamic mysticism (subject of discussion on cognition and love)] -> (participates in event: Islamic mysticism is assumed as one of the Islamic sciences.) -> "Islamic mysticism is assumed as one of the Islamic sciences." -> CONTEXTUALIZES -> "Islamic mysticism has been intertwined with Sufism."
Logical Conclusions:
Fact 4 states Islamic mysticism is an Islamic science, while fact 10 notes its intertwining with Sufism.

"Islamic mysticism has been intertwined with Sufism." -> (has) -> [sufism (Mystical trend in Islam)] -> (participates in event: Islamic mysticism has been intertwined with sufism.) -> "Islamic mysticism has been intertwined with sufism."

"Islamic mysticism is cognition and knowledge." -> INCLUDES -> "Love has been intertwined through Islamic mysticism."
Logical Conclusions:
Islamic mysticism involves both cognition and knowledge, and love is intertwined through it.

"Islamic mysticism is cognition and knowledge." -> EMBODIES -> "Islamic mysticism has a structure of revelation in Islam."
Logical Conclusions:
Islamic mysticism has a structure of revelation and is also a form of cognition and knowledge.

[islamic mysticism (subject of discussion on cognition and love)] -> (participates in event: Islamic mysticism is assumed alongside theology and philosophy.) -> "Islamic mysticism is assumed alongside theology and philosophy." -> CONTEXTUALIZES -> "Islamic mysticism can be considered as a vast range that engulfs theoretical and practical and conventional mysticism."
Logical Conclusions:
Fact 5 positions Islamic mysticism alongside theology and philosophy, while fact 11 describes it as a vast range encompassing theoretical and practical mysticism.

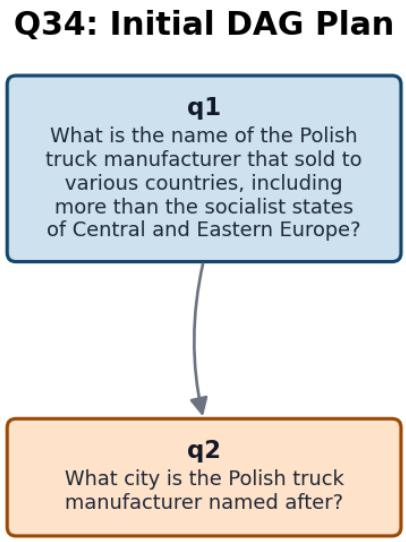
"Islamic mysticism has been intertwined w …[truncated for report]

Q34 [bridge / hard] 5a7faed65542994857a767b6

What city did the name come from for the Polish truck manufacturer who sold to various countries, including more than the socialist states of Central and Eastern Europe?

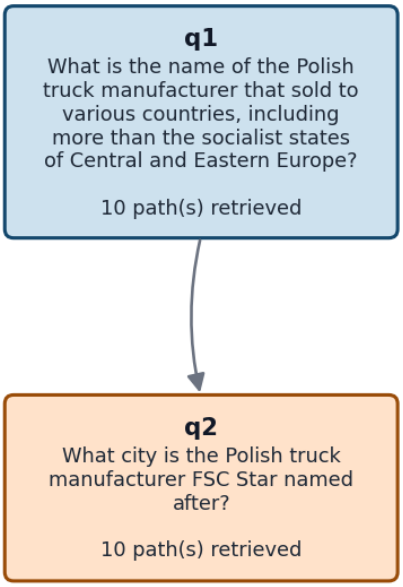
Gold answer	City of Starachowice
LLM answer	Starachowice
EM / F1	**0.00 / 0.50**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q34: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the name of the Polish truck manufacturer that sold to various countries, including more than the socialist states of Central and Eastern Europe?	10 path(s)
q2	1	q1	What city is the Polish truck manufacturer FSC Star named after?	10 path(s)

Final context fed to generation (1054 tokens)

```
Query plan:
- q1: What is the name of the Polish truck manufacturer that sold to various countries, including more than the socialist states of Central and Eastern Europe?
- q2: What city is the Polish truck manufacturer named after? (depends on: q1)

Level 1:

Subquery: What city is the Polish truck manufacturer FSC Star named after?
"Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer." -> NAMED_AFTER -> "FSC Star's name comes from the City of Starachowice."
Logical Conclusions:
FSC Star's name comes from the City of Starachowice.

"FSC Star's name comes from the City of Starachowice." -> (has ) -> [FSC Star (Polish truck manufacturer)]

"Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer." -> FIRST_VEHICLE -> "FSC Star's first vehicle was the 'Star 20' in 1948."
Logical Conclusions:
FSC Star's first vehicle was the 'Star 20' in 1948.
```

"Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer." -> WAS_STATE_OWNED -> "For many years, FSC Star was a state-owned company."
Logical Conclusions:
For many years, FSC Star was a state-owned company.

"Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer." -> (has) -> [FSC Star (Polish truck manufacturer)]

"Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer." -> MOST_POPULAR_PRODUCT -> "FSC Star's most popular product was the 'Star 266'."
Logical Conclusions:
FSC Star's most popular product was the 'Star 266'.

"FSC Star's name comes from the City of Starachowice." -> NAME_FROM -> "For many years, FSC Star was a state-owned company."
Logical Conclusions:
For many years, FSC Star was a state-owned company, and its name comes from the City of Starachowice.

"FSC Star's name comes from the City of Starachowice." -> CORROBORATES -> "The FSC Star factory is located in Starachowice."
Logical Conclusions:
FSC Star's name and factory location both relate to Starachowice.

"Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer." -> NOW_OWNED_BY -> "Star is now owned by MAN AG."
Logical Conclusions:
Star is now owned by MAN AG.

Level 0:

Subquery: What is the name of the Polish truck manufacturer that sold to various countries, including more than the socialist states of Central and Eastern Europe?
[FSC Star (Polish truck manufacturer)] -> (participates in event: The Star 266 was sold in various countries for many years.) -> "The Star 266 was sold in various countries for many years."

"The Star 266 was sold not only in the Eastern Bloc." -> (has) -> [FSC Star (Polish truck manufacturer)]

[FSC Star (Polish truck manufacturer)] -> (participates in event: Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer.) -> "Fabryka Samochodów Ci■■arowych 'Star' was a Polish truck manufacturer."

[FSC Star (Polish truck manufacturer)] -> (participates in event: For many years, FSC Star was a state-owned company.) -> "For many years, FSC Star was a state-owned company."

[FSC Star (Polish truck manufacturer)] -> (participates in event: The Star 266 model offered very good quality.) -> "The Star 266 model offered very good quality."

[FSC Star (Polish truck manufacturer)] -> (participates in event: MAN AG eliminated the Star brand in January 2009.) -> "MAN AG eliminated the Star brand in January 2009."

[FSC Star (Polish truck manufacturer)] -> (participates in event: The FSC Star factory is located in Starachowice.) -> "The FSC Star factory is located in Starachowice."

"CoMEcon was the regional economic organization for the socialist states of Central and Eastern Europe." -> COMPLEMENT_TO -> "The Warsaw Pact was the military complement to the Council for Mutual Economic Assistance (CoMEcon)."
Logical Conclusions:
CoMEcon was the regional economic organization for the socialist states of Central and Eastern Europe.

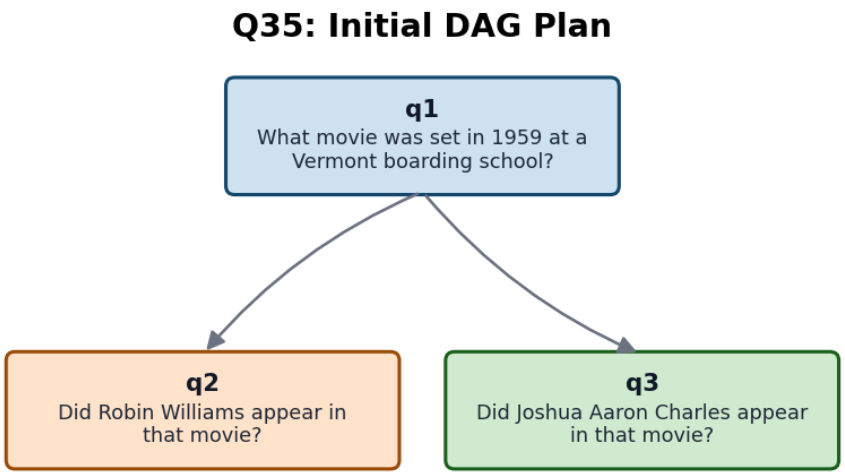
"CoMEcon was the regional economic organization for the socialist states of Central and Eastern Europe." -> (has) -> [council for mutual economic assistance (Regional econom …[truncated for report]

Q35 [bridge / hard] 5a72af155542992359bc316c

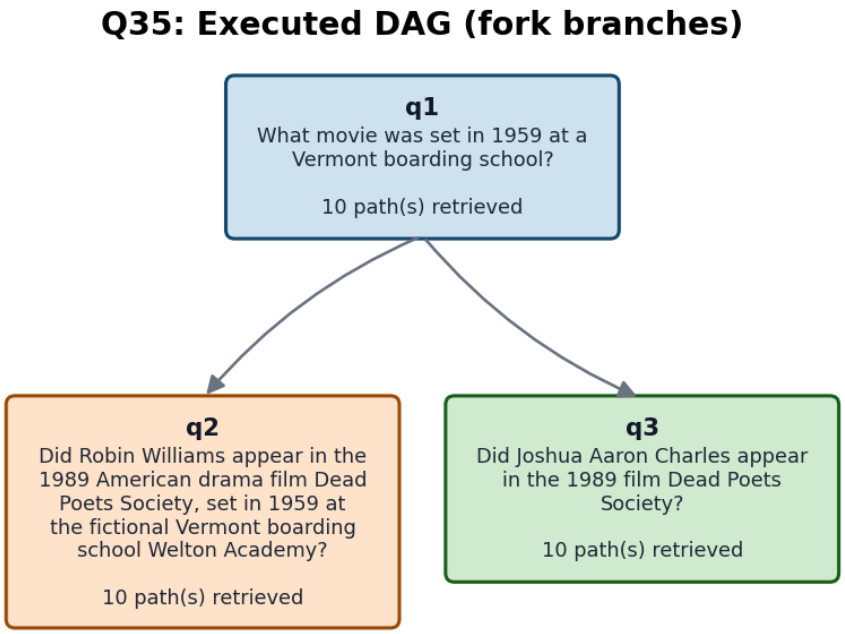
What movie included Robin Williams along with Joshua Aaron Charles, set in 1959 at a Vermont boarding school?

Gold answer	Dead Poets Society
LLM answer	Dead Poets Society
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What movie was set in 1959 at a Vermont boarding school?	10 path(s)
q2	1	q1	Did Robin Williams appear in the 1989 American drama film Dead Poets Society, set in 1959 at the fictional Vermont boarding school Welton Academy?	10 path(s)
q3	1	q1	Did Joshua Aaron Charles appear in the 1989 film Dead Poets Society?	10 path(s)

Final context fed to generation (1829 tokens)

```
Query plan:
- q1: What movie was set in 1959 at a Vermont boarding school?
- q2: Did Robin Williams appear in that movie? (depends on: q1)
- q3: Did Joshua Aaron Charles appear in that movie? (depends on: q1)

Level 1:

Subquery: Did Robin Williams appear in the 1989 American drama film Dead Poets Society, set in 1959 at the
fictional Vermont boarding school Welton Academy?
[dead poets society (1989 American drama film about a teacher)] -> (participates in event: Dead Poets
Society is set at the fictional elite conservative Vermont boarding s) -> "Dead Poets Society is set at the
fictional elite conservative Vermont boarding school Welton Academy."

[dead poets society (1989 American drama film about a teacher)] -> (participates in event: Dead Poets
Society stars Robin Williams.) -> "Dead Poets Society stars Robin Williams."

[dead poets society (1989 American drama film about a teacher)] -> (participates in event: Dead Poets
Society is a 1989 American drama film.) -> "Dead Poets Society is a 1989 American drama film."

[dead poets society (1989 American drama film about a teacher)] -> (participates in event: Dead Poets
Society was directed by Peter Weir.) -> "Dead Poets Society was directed by Peter Weir."

[dead poets society (1989 American drama film about a teacher)] -> (participates in event: Dead Poets
Society tells the story of an English teacher who inspires his studen) -> "Dead Poets Society tells the
story of an English teacher who inspires his students through his teaching of poetry."

[dead poets society (1989 American drama film about a teacher)] -> (participates in event: Dead Poets
Society was written by Tom Schulman.) -> "Dead Poets Society was written by Tom Schulman."

[Robin Williams (Actor starring in Dead Poets Society)] -> (participates in event: Dead Poets Society stars
Robin Williams.) -> "Dead Poets Society stars Robin Williams."

[Robin Williams (Actor starring in Dead Poets Society)] -> (participates in event: 'Unbreakable' stars
Robin Wright.) -> "'Unbreakable' stars Robin Wright."

[Robin Williams (Actor starring in Dead Poets Society)] -> (participates in event: 'Wonder Woman' stars
Robin Wright.) -> "'Wonder Woman' stars Robin Wright."

[Robin Williams (Actor starring in Dead Poets Society)] -> (participates in event: A Christmas Carol (2009
film) features supporting roles done by Gary Oldman, Col) -> "A Christmas Carol (2009 film) features
supporting roles done by Gary Oldman, Colin Firth, Bob Hoskins, Robin Wright, and Cary Elwes."

Subquery: Did Joshua Aaron Charles appear in the 1989 film Dead Poets Society?
```

"Josh Charles is known for his early work as Knox Overstreet in 'Dead Poets Society'." -> CONTRASTS_WITH -> "Josh Charles is best known for the role of Will Gardner on 'The Good Wife'."
Logical Conclusions:
Josh Charles is known for his early role in 'Dead Poets Society' and his later role on 'The Good Wife'.

"Josh Charles is known for his early work as Knox Overstreet in 'Dead Poets Society'." -> ROLE_OF -> "Josh Charles is an American stage, film and television actor."
Logical Conclusions:
Josh Charles is known for his early work as Knox Overstreet in 'Dead Poets Society'.

"Josh Charles is known for his early work as Knox Overstreet in 'Dead Poets Society'." -> CONTRASTS_WITH -> "Josh Charles is best known for the role of Dan Rydell on 'Sports Night'."
Logical Conclusions:
Both facts describe different notable roles played by Josh Charles, one from his early career and one from a later television series.

"Josh Charles is known for his early work as Knox Overstreet in 'Dead Poets Society'." -> SUPPORTS -> "The role of Will Gardner on 'The Good Wife' earned Josh Charles two Primetime Emmy Award nominations."
Logical Conclusions:
Josh Charles's role on 'The Good Wife' earned him Emmy nominations, while he is also known for 'Dead Poets Society'.

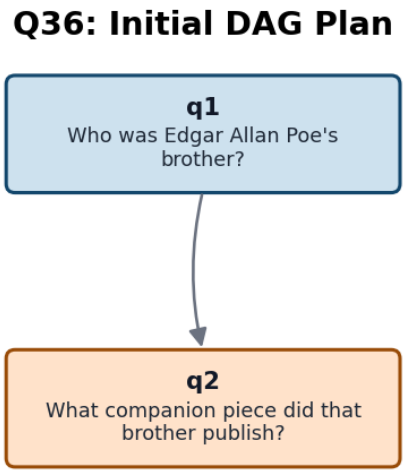
"Josh Charles is known for his early work as Knox Overstreet in 'Dead Poets Society'." -> (has) -> [josh charles (American actor known for TV …[truncated for report]

Q36 [bridge / hard] 5abd05e655429965836004f1

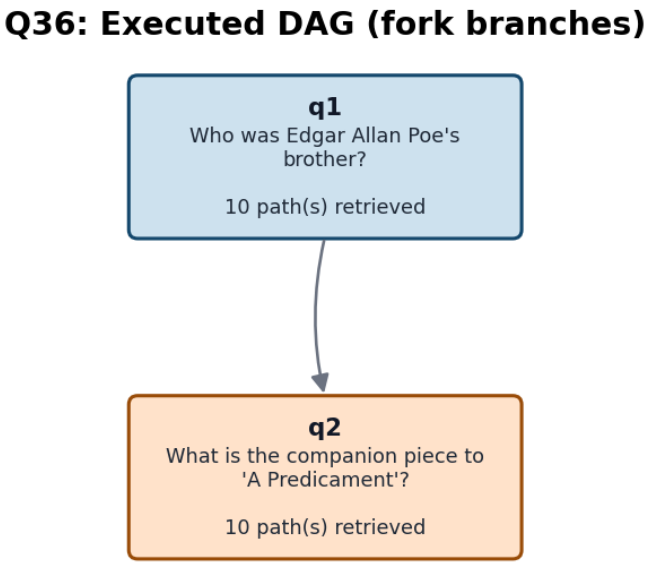
What was the companion piece that Poe's brother published?

Gold answer	"The Whalestoe Letters"
LLM answer	Edgar Allan Poe's brother was William Henry Poe, but he did not publish a companion piece to any of Edgar Allan Poe's works. The companion piece "How to Write a Blackwood Article" was written by Edgar Allan Poe himself, not his brother. Therefore, there is no companion piece published by Poe's brother.
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was Edgar Allan Poe's brother?	10 path(s)
q2	1	q1	What is the companion piece to 'A Predicament'?	10 path(s)

Final context fed to generation (1358 tokens)

```
Query plan:
- q1: Who was Edgar Allan Poe's brother?
- q2: What companion piece did that brother publish? (depends on: q1)

Level 1:

Subquery: What is the companion piece to 'A Predicament'?
"'A Predicament' is usually combined with its companion piece 'How to Write a Blackwood Article.'" ->
PAIRED_WITH -> "'A Predicament' is a humorous short story by Edgar Allan Poe."
Logical Conclusions:
A Predicament is usually combined with its companion piece How to Write a Blackwood Article.

"'A Predicament' is usually combined with its companion piece 'How to Write a Blackwood Article.'" -> (has
) -> [edgar allan poe (Author of A Predicament)]

"'A Predicament' is usually combined with its companion piece 'How to Write a Blackwood Article.'" ->
PARODIES -> "The paired stories 'A Predicament' and 'How to Write a Blackwood Article' parody the Gothic
sensation tale."
Logical Conclusions:
The paired stories parody the Gothic sensation tale.

"'A Predicament' was originally titled 'The Scythe of Time'." -> WAS_ORIGINAL_TITLE -> "'A Predicament' is
a humorous short story by Edgar Allan Poe."
Logical Conclusions:
A Predicament was originally titled The Scythe of Time.

"'A Predicament' was originally titled 'The Scythe of Time'." -> PART_OF -> "The paired stories 'A
Predicament' and 'How to Write a Blackwood Article' parody the Gothic sensation tale."
Logical Conclusions:
'A Predicament' was originally titled 'The Scythe of Time' and is paired with 'How to Write a Blackwood
Article' to parody the Gothic sensation tale.

"'A Predicament' was originally titled 'The Scythe of Time'." -> (has ) -> [edgar allan poe (Author of A
Predicament)]

"Atkinson calls A God in Ruins the 'companion piece' rather than a sequel to Life After Life." ->
PUBLISHED_IN -> "A God in Ruins was published in 2015."
Logical Conclusions:
Fact 1 identifies 'A God in Ruins' as a companion piece to 'Life After Life', and fact 15 states it was
published in 2015.

"Atkinson calls A God in Ruins the 'companion piece' rather than a sequel to Life After Life." ->
SHARES_THEME -> "Both A God in Ruins and Life After Life are about World War II."
Logical Conclusions:
Fact 1 describes 'A God in Ruins' as a companion piece to 'Life After Life', and fact 20 states both novels
are about World War II.

"Atkinson calls A God in Ruins the 'companion piece' rather than a sequel to Life After Life." ->
IS_COMPANION_TO -> "A God in Ruins is the ninth novel by Kate Atkinson."
Logical Conclusions:
Atkinson calls A God in Ruins a 'companion piece' to Life After Life, not a sequel.
```

"Atkinson calls A God in Ruins the 'companion piece' rather than a sequel to Life After Life." -> AWARDED -> "A God in Ruins won the Costa Book Award for Novel in 2015."
Logical Conclusions:
Fact 1 describes 'A God in Ruins' as a companion piece, and fact 21 states it won the Costa Book Award for Novel in 2015.

Level 0:

Subquery: Who was Edgar Allan Poe's brother?
[edgar allan poe (Author of A Predicament)] -> (participates in event: 'A Predicament' is a humorous short story by Edgar Allan Poe.) -> "'A Predicament' is a humorous short story by Edgar Allan Poe."

[edgar allan poe (Author of A Predicament)] -> (participates in event: 'A Predicament' was originally titled 'The Scythe of Time'.) -> "'A Predicament' was originally titled 'The Scythe of Time'."

[edgar allan poe (Author of A Predicament)] -> (participates in event: The paired stories 'A Predicament' and 'How to Write a Blackwood Article' parody) -> "The paired stories 'A Predicament' and 'How to Write a Blackwood Article' parody the Gothic sensation tale."

[edgar allan poe (Author of A Predicament)] -> (participates in event: 'A Predicament' is usually combined with its companion piece 'How to Write a Bla) -> "'A Predicament' is usually combined with its companion piece 'How to Write a Blackwood Article.'"

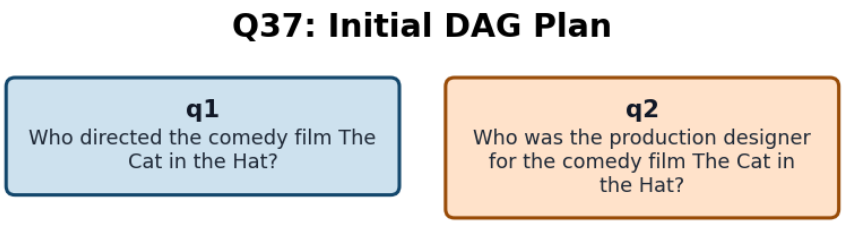
[henry melville whitney (Son of James Scollay Whitney)] -> (participates in event: James Scollay Whitney was the father of Henry Melville Whitney.) …[truncated for report]

Q37 [bridge / hard] 5ae6152d5542996de7b71b38

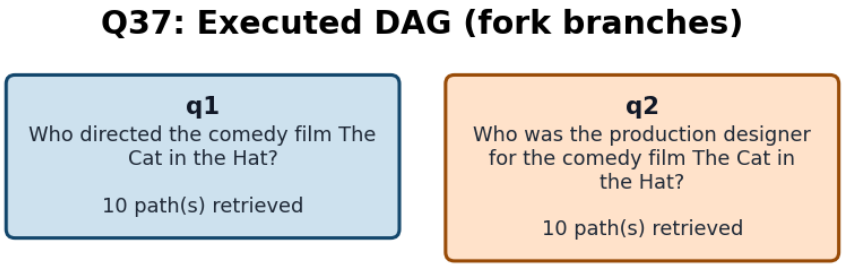
The Cat in the Hat was a comedy film directed by which production designer and director?

Gold answer	Robert W. "Bo" Welch III
LLM answer	Bo Welch
EM / F1	0.00 / 0.57
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who directed the comedy film The Cat in the Hat?	10 path(s)
q2	0	—	Who was the production designer for the comedy film The Cat in the Hat?	10 path(s)

Final context fed to generation (1373 tokens)

```
Query plan:
- q1: Who directed the comedy film The Cat in the Hat?
- q2: Who was the production designer for the comedy film The Cat in the Hat?

Level 0:

Subquery: Who directed the comedy film The Cat in the Hat?
[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event:
Dr. Seuss' The Cat in the Hat was directed by Bo Welch.) -> "Dr. Seuss' The Cat in the Hat was directed by
```

Bo Welch."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: The film Dr. Seuss' The Cat in the Hat stars Mike Myers in the title role of the) -> "The film Dr. Seuss' The Cat in the Hat stars Mike Myers in the title role of the Cat in the Hat."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: The film Dr. Seuss' The Cat in the Hat stars Dakota Fanning as Sally.) -> "The film Dr. Seuss' The Cat in the Hat stars Dakota Fanning as Sally."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Conrad is portrayed by Spencer Breslin.) -> "Conrad is portrayed by Spencer Breslin."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Dr. Seuss' The Cat in the Hat is the second feature-length Dr. Seuss adaptation) -> "Dr. Seuss' The Cat in the Hat is the second feature-length Dr. Seuss adaptation after the 2000 holiday film How the Grinch Stole Christmas."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Sally's brother is unnamed in the book and the 1971 TV special.) -> "Sally's brother is unnamed in the book and the 1971 TV special."

"Roger la Honte was directed by André Cayatte." -> DIRECTED -> "'Before the Deluge' was directed by André Cayatte."
Logical Conclusions:
Roger la Honte was directed by André Cayatte.

"Roger la Honte was directed by André Cayatte." -> (has) -> [andré cayatte (French filmmaker and former lawyer)]

"Roger la Honte was directed by André Cayatte." -> CORROBORATES -> "'Roger la Honte' was also directed by André Cayatte."
Logical Conclusions:
Both facts confirm that André Cayatte directed Roger la Honte.

"Roger la Honte was directed by André Cayatte." -> CORROBORATES -> "Roger la Honte stars Lucien Coëdel, María Casares and Paul Bernard."
Logical Conclusions:
Both facts confirm that André Cayatte directed Roger la Honte.

Subquery: Who was the production designer for the comedy film The Cat in the Hat?
[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Dr. Seuss' The Cat in the Hat was directed by Bo Welch.) -> "Dr. Seuss' The Cat in the Hat was directed by Bo Welch."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: The film Dr. Seuss' The Cat in the Hat stars Mike Myers in the title role of the) -> "The film Dr. Seuss' The Cat in the Hat stars Mike Myers in the title role of the Cat in the Hat."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Conrad is portrayed by Spencer Breslin.) -> "Conrad is portrayed by Spencer Breslin."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Dr. Seuss' The Cat in the Hat is the second feature-length Dr. Seuss adaptation) -> "Dr. Seuss' The Cat in the Hat is the second feature-length Dr. Seuss adaptation after the 2000 holiday film How the Grinch Stole Christmas."

[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: The film Dr. Seuss' The Cat in the Hat stars Dakota Fanning as Sally.) -> "The film Dr. Seuss' The Cat in the Hat stars Dakota Fanning as Sally."

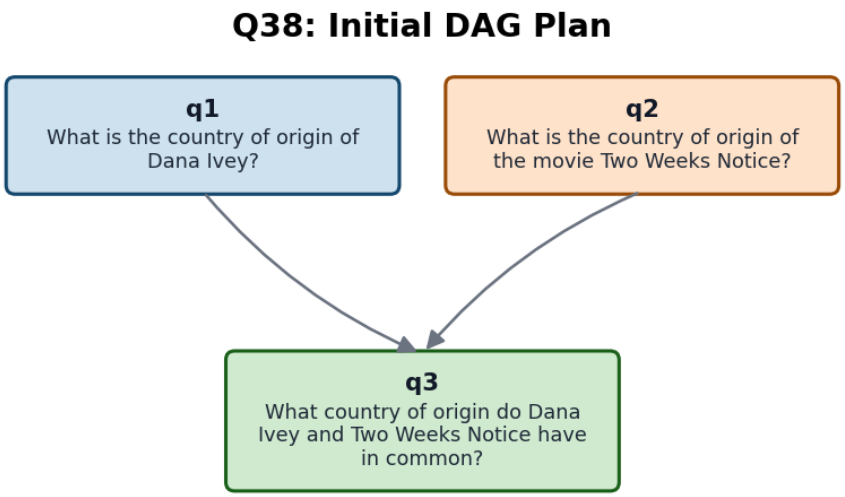
[the cat in the hat (2003 American family comedy film based on Dr. Seuss book)] -> (participates in event: Sally's brother is unnam …[truncated for report]

Q38 [bridge / hard] 5ab9d6ef5542994dbf01985b

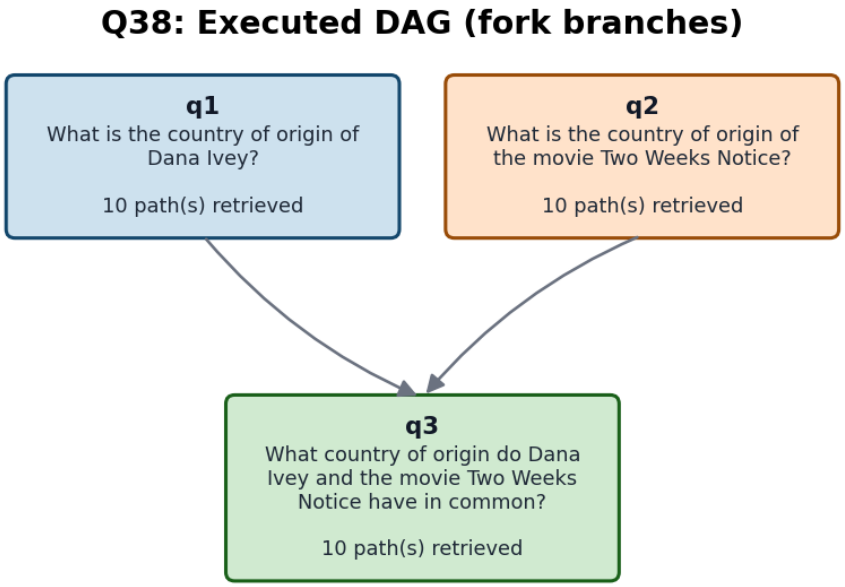
What country of origin does Dana Ivey and Two Weeks Notice have in common?

Gold answer	American
LLM answer	United States
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the country of origin of Dana Ivey?	10 path(s)
q2	0	—	What is the country of origin of the movie Two Weeks Notice?	10 path(s)
q3	1	q1, q2	What country of origin do Dana Ivey and the movie Two Weeks Notice have in common?	10 path(s)

Final context fed to generation (1639 tokens)

```
Query plan:
- q1: What is the country of origin of Dana Ivey?
- q2: What is the country of origin of the movie Two Weeks Notice?
- q3: What country of origin do Dana Ivey and Two Weeks Notice have in common? (depends on: q1, q2)

Level 1:

Subquery: What country of origin do Dana Ivey and the movie Two Weeks Notice have in common?
"Dana Ivey's film appearances include 'Two Weeks Notice' (2002)." -> EXEMPLIFIES -> "Dana Ivey is an
American actress."
Logical Conclusions:
Dana Ivey is an American actress who appeared in 'Two Weeks Notice'.

"Dana Ivey's film appearances include 'Two Weeks Notice' (2002)." -> LISTS -> "Dana Ivey's film appearances
include 'The Addams Family' (1991)."
Logical Conclusions:
Dana Ivey's film appearances include 'The Addams Family' (1991) and 'Two Weeks Notice' (2002).

"Dana Ivey's film appearances include 'Two Weeks Notice' (2002)." -> (has ) -> [dana ivey (American actress
and Tony nominee)]

"Dana Ivey's film appearances include 'Two Weeks Notice' (2002)." -> LISTS -> "Dana Ivey's film appearances
include 'Rush Hour 3' (2007)."
Logical Conclusions:
Dana Ivey's film appearances include 'Two Weeks Notice' (2002) and 'Rush Hour 3' (2007).

"Dana Ivey's film appearances include 'Two Weeks Notice' (2002)." -> LISTS -> "Dana Ivey's film appearances
include 'The Color Purple' (1985)."
Logical Conclusions:
Dana Ivey's film appearances include 'The Color Purple' (1985) and 'Two Weeks Notice' (2002).

"Dana Ivey's film appearances include 'Two Weeks Notice' (2002)." -> SUPPORTS -> "Dana Ivey won the 1997
Drama Desk Award for Outstanding Featured Actress in a Play."
Logical Conclusions:
Dana Ivey won a Drama Desk Award and appeared in 'Two Weeks Notice'.

[dana (Surname with multiple origins)] -> (participates in event: Dana is a common forename in Ireland.) ->
"Dana is a common forename in Ireland."

[dana ivey (American actress and Tony nominee)] -> (participates in event: The film features an ensemble
cast including Sally Field, Regina King, Jennifer ) -> "The film features an ensemble cast including Sally
Field, Regina King, Jennifer Coolidge, Bruce McGill, Dana Ivey, Mary Lynn Rajskub, Bob Newhart, and Luke
Wilson."

[dana ivey (American actress and Tony nominee)] -> (participates in event: Dana Ivey's film appearances
include 'Rush Hour 3' (2007).) -> "Dana Ivey's film appearances include 'Rush Hour 3' (2007)."

[dana ivey (American actress and Tony nominee)] -> (participates in event: Dana Robins Ivey was born on
August 12, 1941.) -> "Dana Robins Ivey was born on August 12, 1941."
```

Level 0:

Subquery: What is the country of origin of Dana Ivey?
"Dana Ivey is an American actress." -> SPECIFIES -> "Dana Robins Ivey was born on August 12, 1941."
Logical Conclusions:
Dana Robins Ivey, born on August 12, 1941, is an American actress.

"Dana Ivey is an American actress." -> EXEMPLIFIES -> "Dana Ivey's film appearances include 'The Addams Family' (1991)."
Logical Conclusions:
Dana Ivey is an American actress who appeared in 'The Addams Family'.

"Dana Ivey is an American actress." -> EXEMPLIFIES -> "Dana Ivey's film appearances include 'The Color Purple' (1985)."
Logical Conclusions:
Dana Ivey is an American actress who appeared in 'The Color Purple'.

"Dana Ivey is an American actress." -> (has) -> [dana ivey (American actress and Tony nominee)]

"Dana Ivey is an American actress." -> HIGHLIGHTS -> "Dana Ivey is a five-time Tony Award nominee for her work on Broadway."
Logical Conclusions:
Dana Ivey is an American actress and a five-time Tony Award nominee.

"The surname Dana can be found mostly in the Piedmont region of Italy." -> GENERALIZES -> "Dana as a surname may have several origins."
Logical Conclusions:
Dana as a surname may have several origins, and the surname Dana can be found mostly in the Piedmont region of Italy.

"The surname Dana can be found mostly in the Piedmont region of Italy." -> (has) -> [dana (Surname with multiple origins)]

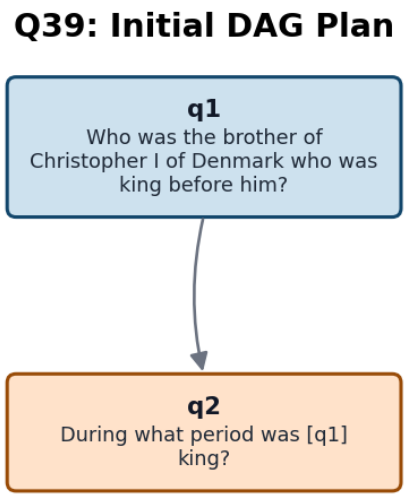
"The surname Dana can be found mostly in the Piedmont re …[truncated for report]

Q39 [bridge / hard] 5ac308135542990b17b154f4

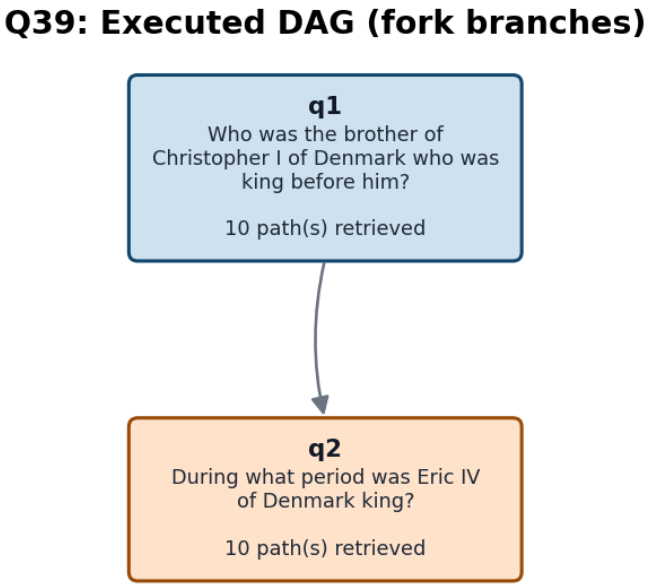
Christopher I of Denmark succeeded his brother, who was king during what period?

Gold answer	1241 until his death in 1250
LLM answer	1241–1250
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was the brother of Christopher I of Denmark who was king before him?	10 path(s)
q2	1	q1	During what period was Eric IV of Denmark king?	10 path(s)

Final context fed to generation (1242 tokens)

```
Query plan:
- q1: Who was the brother of Christopher I of Denmark who was king before him?
- q2: During what period was [q1] king? (depends on: q1)

Level 1:

Subquery: During what period was Eric IV of Denmark king?
[Eric IV of Denmark (King of Denmark known as Ploughpenny)] -> (participates in event: Eric IV was king of
Denmark from 1241 until his death in 1250.) -> "Eric IV was king of Denmark from 1241 until his death in
1250."

"Eric IV of Denmark was born in 1216." -> FOLLOWS -> "Eric IV was king of Denmark from 1241 until his death
in 1250."
Logical Conclusions:
Eric IV of Denmark was born in 1216 and reigned from 1241 until his death in 1250.

"Eric IV of Denmark died on 10 August 1250." -> SUPPORTS -> "Eric IV was king of Denmark from 1241 until his
death in 1250."
Logical Conclusions:
Eric IV of Denmark died on 10 August 1250, having reigned from 1241 until his death.

[Eric III of Denmark (King of Denmark from 1137 to 1146)] -> (participates in event: Eric III Lamb's
succession led to a period of civil war between Sweyn III, Canut) -> "Eric III Lamb's succession led to a
period of civil war between Sweyn III, Canute V, and Valdemar I."

[Eric III of Denmark (King of Denmark from 1137 to 1146)] -> (participates in event: Eric III Lamb was the
King of Denmark from 1137 until 1146.) -> "Eric III Lamb was the King of Denmark from 1137 until 1146."

[Eric III of Denmark (King of Denmark from 1137 to 1146)] -> (participates in event: Eric the Memorable was
promptly succeeded by his nephew Eric III of Denmark.) -> "Eric the Memorable was promptly succeeded by his
nephew Eric III of Denmark."

[Eric III of Denmark (King of Denmark from 1137 to 1146)] -> (participates in event: Eric III Lamb
succeeded Eric II of Denmark on the throne.) -> "Eric III Lamb succeeded Eric II of Denmark on the throne."

[Eric III of Denmark (King of Denmark from 1137 to 1146)] -> (participates in event: Eric III Lamb was the
grandson of Eric I of Denmark.) -> "Eric III Lamb was the grandson of Eric I of Denmark."

"Eric III Lamb succeeded Eric II of Denmark on the throne." -> PRECEDES -> "Eric III Lamb was the King of
Denmark from 1137 until 1146."
Logical Conclusions:
Eric III Lamb succeeded Eric II of Denmark on the throne and reigned from 1137 until 1146.

Level 0:

Subquery: Who was the brother of Christopher I of Denmark who was king before him?
[Christopher I of Denmark (King of Denmark 1252-1259)] -> (participates in event: Eric IV was the brother
of King Christopher I.) -> "Eric IV was the brother of King Christopher I."

"Christopher I of Denmark succeeded his brothers Eric IV Plovpenning and Abel of Denmark on the throne." ->
(has ) -> [Christopher I of Denmark (King of Denmark 1252-1259)]
```

[Christopher I of Denmark (King of Denmark 1252-1259)] -> (participates in event: Christopher I of Denmark was elected King upon the death of his older brother Ab) -> "Christopher I of Denmark was elected King upon the death of his older brother Abel in the summer of 1252."

[Christopher I of Denmark (King of Denmark 1252-1259)] -> (participates in event: Christopher I of Denmark was also known as 'Christoffer I'.) -> "Christopher I of Denmark was also known as 'Christoffer I'."

"Christopher I of Denmark succeeded his brothers Eric IV Plovpenning and Abel of Denmark on the throne." -> EXPLAINS -> "Christopher I of Denmark was elected King upon the death of his older brother Abel in the summer of 1252."
Logical Conclusions:
Christopher I of Denmark succeeded his brothers and was elected king upon Abel's death in summer 1252.

"Christopher I of Denmark succeeded his brothers Eric IV Plovpenning and Abel of Denmark on the throne." -> SUPPORTS -> "Christopher I of Denmark was King of Denmark between 1252 and 1259."
Logical Conclusions:
Christopher I of Denmark succeeded his brothers Eric IV and Abel, reigning from 1252 to 1259.

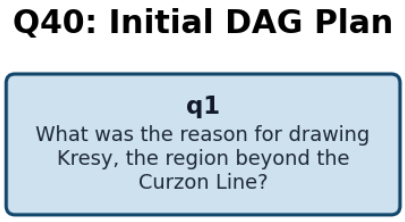
"Christopher I of Denmark succeeded his brothers Eric IV Plovpenning and Abel of Denmark on the throne." -> (has) -> [Eric IV …[truncated for report]

Q40 [bridge / hard] 5ade636255429939a52fe896

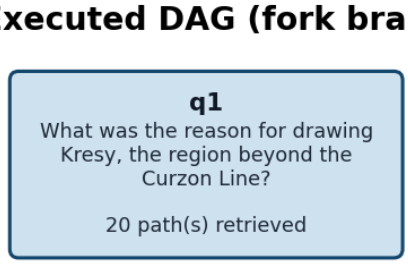
Kresy, which roughly was a part of the land beyond the so-called Curson Line, was drawn for what reason?

Gold answer	demarcation line between the newly emerging states, the Second Polish Republic, and the Soviet Union
LLM answer	Kresy, which roughly corresponded to the lands beyond the so-called Curzon Line, was drawn as a proposed eastern border of the re-emerging sovereign Republic of Poland following the century of partitions, to define the boundary between Poland and the Soviet Union after World War I.
EM / F1	**0.00 / 0.24**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What was the reason for drawing Kresy, the region beyond the Curzon Line?	20 path(s)

Final context fed to generation (1141 tokens)

```
"In the Second Polish Republic, the term 'Kresy' roughly equated with the lands beyond the so-called Curzon
Line." -> RELATES_TO -> "The Curzon Line in its proposed form never materialized because the war went on."
-> PRECEDES -> "The Curzon Line was drawn for the first time by the Supreme War Council."
Logical Conclusions:
The Curzon Line, which Kresy roughly equated with, never materialized because the war went on.
The Curzon Line was drawn for the first time by the Supreme War Council but never materialized because the
```

war went on.

[Kresy (historical eastern borderlands of Poland)] -> (participates in event: In the Second Polish Republic, the term 'Kresy' roughly equated with the lands b) -> "In the Second Polish Republic, the term 'Kresy' roughly equated with the lands beyond the so-called Curzon Line."

"In the Second Polish Republic, the term 'Kresy' roughly equated with the lands beyond the so-called Curzon Line." -> (has) -> [curzon line (Proposed border between Poland and Soviet Union)] -> (participates in event: The Curzon Line in its proposed form never materialized because the war went on.) -> "The Curzon Line in its proposed form never materialized because the war went on."

[curzon line (Proposed border between Poland and Soviet Union)] -> (participates in event: The Curzon Line was drawn for the first time by the Supreme War Council.) -> "The Curzon Line was drawn for the first time by the Supreme War Council."

"In the Second Polish Republic, the term 'Kresy' roughly equated with the lands beyond the so-called Curzon Line." -> (has) -> [second polish republic (state containing the Kresy region)] -> (participates in event: Kresy Wschodnie or Kresy was a region of the Second Polish Republic during the i) -> "Kresy Wschodnie or Kresy was a region of the Second Polish Republic during the interwar period."

[second polish republic (state containing the Kresy region)] -> (participates in event: Kresy Wschodnie or Kresy was a region of the Second Polish Republic during the i) -> "Kresy Wschodnie or Kresy was a region of the Second Polish Republic during the interwar period."

"In the Second Polish Republic, the term 'Kresy' roughly equated with the lands beyond the so-called Curzon Line." -> RELATES_TO -> "The Curzon Line was the demarcation line between the Second Polish Republic and the Soviet Union." -> (has) -> [curzon line (Proposed border between Poland and Soviet Union)]
Logical Conclusions:
The Curzon Line, which Kresy roughly equated with, was the demarcation line between the Second Polish Republic and the Soviet Union.

[second polish republic (state containing the Kresy region)] -> (participates in event: Kresy constituted nearly half of the territory of the Second Polish Republic.) -> "Kresy constituted nearly half of the territory of the Second Polish Republic."

"In the Second Polish Republic, the term 'Kresy' roughly equated with the lands beyond the so-called Curzon Line." -> CONTEXTUALIZES -> "The Curzon Line was suggested as the eastern border of the re-emerging sovereign Republic following the century of partitions." -> (has) -> [curzon line (Proposed border between Poland and Soviet Union)]
Logical Conclusions:
Both facts discuss the Curzon Line in relation to Poland, with one stating it was suggested as the eastern border and the other stating Kresy equated with lands beyond it.

[second polish republic (state containing the Kresy region)] -> (participates in event: The Curzon Line was the demarcation line between the Second Polish Republic and) -> "The Curzon Line was the demarcation line between the Second Polish Republic and the Soviet Union."

[second polish republic (state containing the Kresy region)] -> (participates in event: The Curzon Line was suggested as the eastern border of the re-emerging sovereign) -> "The Curzon Line was suggested as the eastern border of the re-emerging sovereign Republic following the century of partitions."

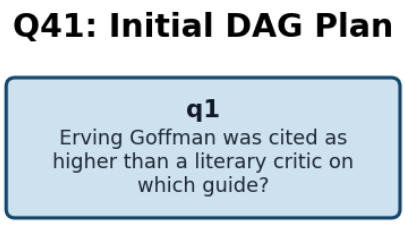
"In the Second Polish Republic, the term 'Kresy …[truncated for report]

Q41 [bridge / hard] 5ac09d265542992a796ded39

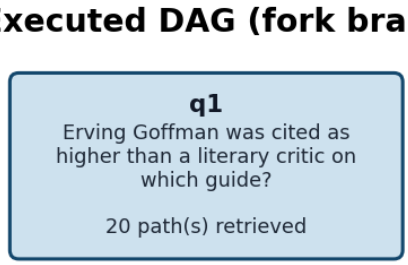
Erving Goffman was cited as higher than a literary critic on which guide?

Gold answer	The Times Higher Education Guide
LLM answer	The Times Higher Education Guide
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Erving Goffman was cited as higher than a literary critic on which guide?	20 path(s)

Final context fed to generation (1147 tokens)

```
"In 2007, Erving Goffman was listed by 'The Times Higher Education Guide' as the sixth most-cited author in the humanities and social sciences." -> SPECIFIES -> "Erving Goffman was listed behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault." -> (has ) -> [michel foucault (French philosopher and social theorist)] -> (participates in event: Michel Foucault was a literary critic.) -> "Michel Foucault was a literary critic."
Logical Conclusions:
Goffman was listed behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault.

[michel foucault (French philosopher and social theorist)] -> (participates in event: Michel Foucault was a French philosopher.) -> "Michel Foucault was a French philosopher."
```

"In 2007, Erving Goffman was listed by 'The Times Higher Education Guide' as the sixth most-cited author in the humanities and social sciences." -> SPECIFIES -> "Erving Goffman was listed ahead of Jürgen Habermas." -> CONTRASTS -> "Erving Goffman was listed behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault."
Logical Conclusions:
Goffman was listed ahead of Jürgen Habermas.
Goffman was listed ahead of Jürgen Habermas, and behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault.

"Erving Goffman was listed behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault." -> (has) -> [anthony giddens (Cited author in humanities and social sciences)]

"Erving Goffman was listed behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault." -> (has) -> [pierre bourdieu (Cited author in humanities and social sciences)]

"Erving Goffman was listed behind Anthony Giddens, Pierre Bourdieu, and Michel Foucault." -> (has) -> [erving goffman (Canadian-American sociologist and writer)] -> (participates in event: 'Kinesics and Context' 'would not have appeared if it had not been envisaged by) -> "'Kinesics and Context' 'would not have appeared if it had not been envisaged by Erving Goffman'."

[erving goffman (Canadian-American sociologist and writer)] -> (participates in event: Erving Goffman was listed ahead of Jürgen Habermas.) -> "Erving Goffman was listed ahead of Jürgen Habermas."

[erving goffman (Canadian-American sociologist and writer)] -> (participates in event: Erving Goffman would later acknowledge Kenneth Burke as an influence.) -> "Erving Goffman would later acknowledge Kenneth Burke as an influence."

"In 2007, Erving Goffman was listed by 'The Times Higher Education Guide' as the sixth most-cited author in the humanities and social sciences." -> (has) -> [erving goffman (Canadian-American sociologist and writer)]

"In 2007, Erving Goffman was listed by 'The Times Higher Education Guide' as the sixth most-cited author in the humanities and social sciences." -> (has) -> [the times higher education guide (Publication listing most-cited authors)]

"Erving Goffman was listed ahead of Jürgen Habermas." -> (has) -> [German intellectuals (German philosophers and educators)] -> (participates in event: Oswalt Kolle was a German sex educator.) -> "Oswalt Kolle was a German sex educator."

[German intellectuals (German philosophers and educators)] -> (participates in event: Oswalt Kolle's work was translated into all major languages.) -> "Oswalt Kolle's work was translated into all major languages."

[erving goffman (Canadian-American sociologist and writer)] -> (participates in event: IMT is based on Erving Goffman's 'Interaction ritual: Essays on face-to-face beh) -> "IMT is based on Erving Goffman's 'Interaction ritual: Essays on face-to-face behavior' (1967)."

[German intellectuals (German philosophers and educators)] -> (participates in event: In his 1997 book 'Open to Both Sides' Oswalt Kolle came out as bisexual.) -> "In his 1997 book 'Open to Both Sides' Oswalt Kolle came out as bisexual."

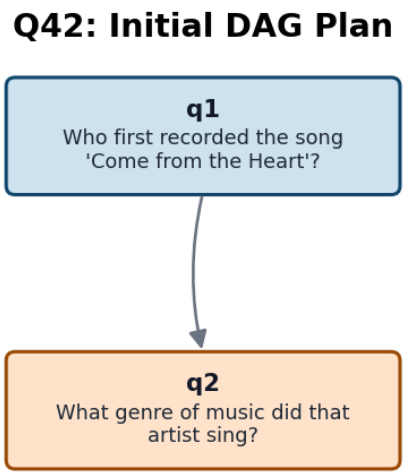
[German intellectuals (German philosophers and educators)] -> (participates in event: Oswalt Kolle became famous for his numerous pioneering books and films on human) -> "Oswalt Kolle became famous for his numerous pion …[truncated for report]

Q42 [bridge / hard] 5ae705725542995703ce8bbc

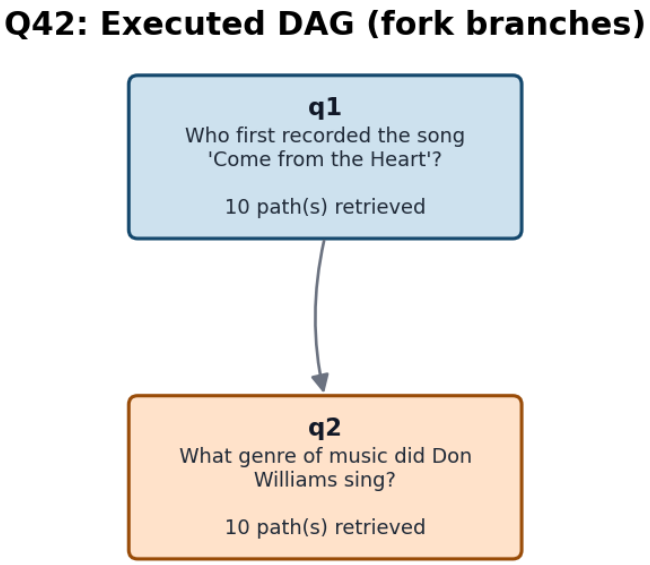
Come from the Heart:was first recorded by an artist that sang what genre of music?

Gold answer	country
LLM answer	country
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who first recorded the song 'Come from the Heart'?	10 path(s)
q2	1	q1	What genre of music did Don Williams sing?	10 path(s)

Final context fed to generation (1504 tokens)

```
Query plan:
- q1: Who first recorded the song 'Come from the Heart'?
- q2: What genre of music did that artist sing? (depends on: q1)

Level 1:

Subquery: What genre of music did Don Williams sing?
"Don Williams discography is a detailed discography for American country music artist Don Williams." ->
(has ) -> [don williams (American country music artist with discography)]

[Robbie Williams (Co-writer and performer of Shame)] -> (participates in event: 'Shame' is a pop song with
country and electro music influences.) -> "'Shame' is a pop song with country and electro music
influences."

[Robbie Williams (Co-writer and performer of Shame)] -> (participates in event: 'Shame' is a song written
and recorded by English singers Robbie Williams and Ga) -> "'Shame' is a song written and recorded by
English singers Robbie Williams and Gary Barlow."

[Robbie Williams (Co-writer and performer of Shame)] -> (participates in event: 'Shame' marks the first
time Williams and Barlow collaborated on a song together) -> "'Shame' marks the first time Williams and
Barlow collaborated on a song together solely."

[Robbie Williams (Co-writer and performer of Shame)] -> (participates in event: Two reviewers noted that
'Shame' contains an acoustic guitar part similar to the) -> "Two reviewers noted that 'Shame' contains an
acoustic guitar part similar to the one of the Beatles' 1968 song, 'Blackbird'."

[Robbie Williams (Co-writer and performer of Shame)] -> (participates in event: The lyrical content of
'Shame' revolves around singers' broken relationship and ) -> "The lyrical content of 'Shame' revolves
around singers' broken relationship and fixing things up."

"The music was fairly taken over by the Nashville Sound in the 1960s." -> FOLLOWS -> "The lyrics of 'Leave
Them Boys Alone' echo the sentiment that the outlaw singers and their current escapades were predated by
the hard living honky tonkers of the 1950s."
Logical Conclusions:
The music was fairly taken over by the Nashville Sound in the 1960s, contrasting with the earlier honky
tonk style.

"The music was fairly taken over by the Nashville Sound in the 1960s." -> CONTRASTS_WITH -> "The Motown
Sound is a style of soul music with a distinct pop influence."
Logical Conclusions:
The Motown Sound is a style of soul music with pop influence, while the Nashville Sound took over country
music in the 1960s.

"The music was fairly taken over by the Nashville Sound in the 1960s." -> CONTRASTS_WITH -> "In the late
1960s and early 1970s, mainstream country music was dominated by the rock-influenced Bakersfield sound of
artists like Merle Haggard."
Logical Conclusions:
Fact 1 states the Nashville Sound dominated in the 1960s, while fact 12 notes it was challenged by the
Bakersfield sound in the late 1960s and early 1970s.

"The music was fairly taken over by the Nashville Sound in the 1960s." -> CORROBORATES -> "In the late
1960s and early 1970s, mainstream country music was dominated by the slick Nashville sound."
```

Logical Conclusions:
Both facts describe the dominance of the Nashville Sound in country music during overlapping time periods.

Level 0:

Subquery: Who first recorded the song 'Come from the Heart'?
"The song 'Come from the Heart' was first recorded and released on the 1987 Don Williams album 'Traces'." -> SUPPORTS -> "The song 'Come from the Heart' was also released in 1988 by Clark's husband on his album 'Old Friends'."
Logical Conclusions:
The song 'Come from the Heart' was first recorded on Don Williams' album 'Traces' in 1987 and later released by Clark's husband on his album 'Old Friends' in 1988.

"The song 'Come from the Heart' was first recorded and released on the 1987 Don Williams album 'Traces'." -> SUPPORTS -> "'Come from the Heart' is most known through the 1989 single by Kathy Mattea."
Logical Conclusions:
The song 'Come from the Heart' was first recorded and released on the 1987 Don Williams album 'Traces'.

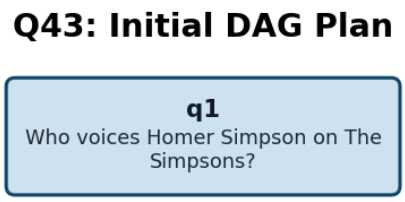
"The song 'Come from the Heart' was first recorded and released on t …[truncated for report]

Q43 [bridge / hard] 5a83a7e15542992ef85e2347

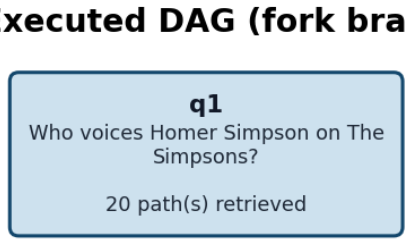
Who voices Homer Simpson's character on the animated television series?

Gold answer	Daniel Louis Castellaneta
LLM answer	Dan Castellaneta
EM / F1	**0.00 / 0.40**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who voices Homer Simpson on The Simpsons?	20 path(s)

Final context fed to generation (1158 tokens)

```
"Dan Castellaneta voices Homer Simpson in 'The Simpsons'." -> VOICES -> "Dan Castellaneta voices Mayor
Quimby on 'The Simpsons'." -> VOICES -> "Homer Jay Simpson is voiced by Dan Castellaneta."
Logical Conclusions:
Dan Castellaneta voices Homer Simpson and Mayor Quimby on 'The Simpsons'.
Homer Jay Simpson is voiced by Dan Castellaneta, who also voices Mayor Quimby.

"Dan Castellaneta voices Mayor Quimby on 'The Simpsons'." -> (has ) -> [dan castellaneta (Voice actor for
Homer Simpson)] -> (participates in event: Homer Jay Simpson is voiced by Dan Castellaneta.) -> "Homer Jay
Simpson is voiced by Dan Castellaneta."

"Dan Castellaneta voices Homer Simpson in 'The Simpsons'." -> VOICED_BY -> "Daniel Louis Castellaneta was
born on October 29, 1957." -> VOICES -> "Homer Jay Simpson is voiced by Dan Castellaneta."
Logical Conclusions:
Dan Castellaneta was born on October 29, 1957, and voices Homer Simpson in 'The Simpsons'.
```

Dan Castellaneta was born on October 29, 1957, and voices Homer Jay Simpson.

"Dan Castellaneta voices Homer Simpson in 'The Simpsons'." -> VOICES -> "Dan Castellaneta also voices many other characters for 'The Simpsons'." -> VOICES -> "Homer Jay Simpson is voiced by Dan Castellaneta."
Logical Conclusions:
Dan Castellaneta voices Homer Simpson and also voices many other characters for 'The Simpsons'.
Homer Jay Simpson is voiced by Dan Castellaneta, who also voices many other characters.

"Dan Castellaneta voices Homer Simpson in 'The Simpsons'." -> (has) -> [homer simpson (Main protagonist of The Simpsons)] -> (participates in event: Homer Jay Simpson is voiced by Dan Castellaneta.) -> "Homer Jay Simpson is voiced by Dan Castellaneta." -> VOICED_BY -> "Dan Castellaneta is an American actor, voice actor, comedian and screenwriter."
Logical Conclusions:
Dan Castellaneta is an American actor who voices Homer Jay Simpson.

"Dan Castellaneta voices Homer Simpson in 'The Simpsons'." -> VOICES -> "Homer Jay Simpson is the main protagonist of the American animated television series 'The Simpsons'." -> VOICES -> "Homer Jay Simpson is voiced by Dan Castellaneta."
Logical Conclusions:
Homer Jay Simpson is the main protagonist and is voiced by Dan Castellaneta.
Homer Simpson is voiced by Dan Castellaneta.

"Homer Jay Simpson is voiced by Dan Castellaneta." -> VOICES -> "Homer Jay Simpson is the patriarch of the eponymous family in 'The Simpsons'."
Logical Conclusions:
Homer Jay Simpson is the patriarch of the Simpson family and is voiced by Dan Castellaneta.

[dan castellaneta (Voice actor for Homer Simpson)] -> (participates in event: Dan Castellaneta voices Hans Moleman on 'The Simpsons'.) -> "Dan Castellaneta voices Hans Moleman on 'The Simpsons'."

"Dan Castellaneta voices Mayor Quimby on 'The Simpsons'." -> EXEMPLIFIES -> "Dan Castellaneta also voices many other characters for 'The Simpsons'."
Logical Conclusions:
Another character Castellaneta voices is Mayor Quimby.

"Daniel Louis Castellaneta was born on October 29, 1957." -> (has) -> [dan castellaneta (Voice actor for Homer Simpson)]

"Dan Castellaneta voices Mayor Quimby on 'The Simpsons'." -> (has) -> [mayor quimby (Character voiced by Castellaneta)]

"Dan Castellaneta also voices many other characters for 'The Simpsons'." -> (has) -> [dan castellaneta (Voice actor for Homer Simpson)]

"Homer Jay Simpson is the main protagonist of the American animated television series 'The Simpsons'." -> CREATED -> "Homer was created and designed by cartoonist Matt Groening."
Logical Conclusions:
Homer was created and designed by Matt Groening.

"Daniel Louis Castellaneta was born on October 29, 1957." -> VOICES -> "Dan Castellaneta voices Hans Moleman on 'The Simpsons'."
Logical Conclusions:
Dan Castellaneta was born on October 29, 1957, and voices Hans Moleman on 'The Simpsons'.

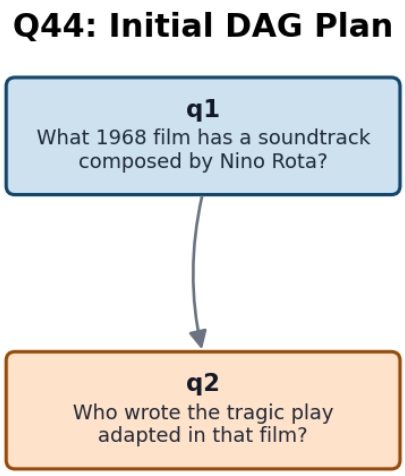
"Dan Castellaneta also voices many other characters for 'The Simpsons'." -> IS_EXAMPLE_OF -> "Dan Castellaneta voices Groundskeeper Willie on 'The Simpsons'."
Logical Conclusions:
Another characte …[truncated for report]

Q44 [bridge / hard] 5ae2eda355429928c4239570

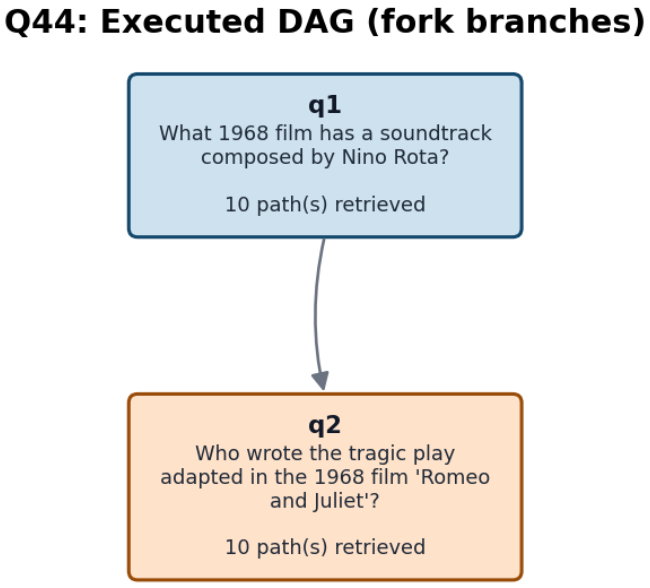
Who wrote the tragic play which also has a 1968 film with a soundtrack by Nino Rota?

Gold answer	William Shakespeare
LLM answer	William Shakespeare
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What 1968 film has a soundtrack composed by Nino Rota?	10 path(s)
q2	1	q1	Who wrote the tragic play adapted in the 1968 film 'Romeo and Juliet'?	10 path(s)

Final context fed to generation (1434 tokens)

```
Query plan:
- q1: What 1968 film has a soundtrack composed by Nino Rota?
- q2: Who wrote the tragic play adapted in that film? (depends on: q1)

Level 1:

Subquery: Who wrote the tragic play adapted in the 1968 film 'Romeo and Juliet'?
"Romeo and Juliet is a 1968 British-Italian romance film." -> BASED_ON -> "Romeo and Juliet is based on the
tragic play of the same name (1591–1595) by William Shakespeare."
Logical Conclusions:
Romeo and Juliet is based on the tragic play by William Shakespeare.

"Romeo and Juliet is a 1968 British-Italian romance film." -> SUPPORTS -> "The soundtrack for the 1968 film
'Romeo and Juliet' was composed and conducted by Nino Rota."
Logical Conclusions:
Both facts identify the 1968 film 'Romeo and Juliet'.

"Romeo and Juliet is a 1968 British-Italian romance film." -> (has ) -> [romeo and juliet (1968 film) (Film
based on Shakespeare play)]

"Romeo and Juliet is a 1968 British-Italian romance film." -> SUPPORTS -> "The title of the third work
described is 'Romeo and Juliet (1968 film soundtrack)'."
Logical Conclusions:
Both facts identify the 1968 film 'Romeo and Juliet'.

"Romeo and Juliet is a 1968 British-Italian romance film." -> HAS_SOUNDTRACK -> "The soundtrack for the
1968 film 'Romeo and Juliet' was originally released as a vinyl record."
Logical Conclusions:
The 1968 film 'Romeo and Juliet' had its soundtrack originally released as a vinyl record.

[Love Theme from Romeo and Juliet (musical piece by Nino Rota)] -> (participates in event: Many of the
stories on 'Our Tune' are read out over Nino Rota's Love Theme from ) -> "Many of the stories on 'Our Tune'
are read out over Nino Rota's Love Theme from 'Romeo and Juliet'."

[Love Theme from Romeo and Juliet (musical piece by Nino Rota)] -> (participates in event: 'The Love Theme'
from the first film makes a brief appearance during a flashback) -> "'The Love Theme' from the first film
makes a brief appearance during a flashback sequence in The Godfather Part II."

"Kenneth Burke's notions of dramatism derive from Shakespeare." -> (has ) -> [kenneth burke (Influential
dramatism theorist)]

"Kenneth Burke's notions of dramatism derive from Shakespeare." -> INSPIRES -> "Kenneth Burke presented his
notions of dramatism in 1945."
Logical Conclusions:
Burke's dramatism notions derive from Shakespeare.

[The Rose Variations (debut novel by Marisha Chamberlain)] -> (participates in event: Brian Ralston's
latest film is the 2017 drama Rose.) -> "Brian Ralston's latest film is the 2017 drama Rose."

Level 0:
```

Subquery: What 1968 film has a soundtrack composed by Nino Rota?
"The title of the third work described is 'Romeo and Juliet (1968 film soundtrack)'." -> COMPOSED_BY -> "The soundtrack for the 1968 film 'Romeo and Juliet' was composed and conducted by Nino Rota."
Logical Conclusions:
Nino Rota composed and conducted the soundtrack for the 1968 film 'Romeo and Juliet'.

"'8½' features a soundtrack by Nino Rota." -> SHARES_COMPOSER -> "The soundtrack for the 1968 film 'Romeo and Juliet' was composed and conducted by Nino Rota."
Logical Conclusions:
Both facts refer to film scores composed by Nino Rota, for '8½' and 'Romeo and Juliet' respectively.

"The title of the third work described is 'Romeo and Juliet (1968 film soundtrack)'." -> SUPPORTS -> "The most notable entry on the soundtrack for the 1968 film 'Romeo and Juliet' is the song 'What Is a Youth'."
Logical Conclusions:
Both facts describe the soundtrack for the 1968 film 'Romeo and Juliet'.

"The title of the third work described is 'Romeo and Juliet (1968 film soundtrack)'." -> SUPPORTS -> "The soundtrack for the 1968 film 'Romeo and Juliet' contains nine entries."
Logical Conclusions:
Both facts describe the soundtrack for the 1968 film 'Romeo and Juliet'.

"The title of the third work described is 'Romeo and Juliet (1968 film soundtrack)'." -> (has) -> [romeo and juliet (1968 film soundtrack) (Soundtrack for 1968 film)]

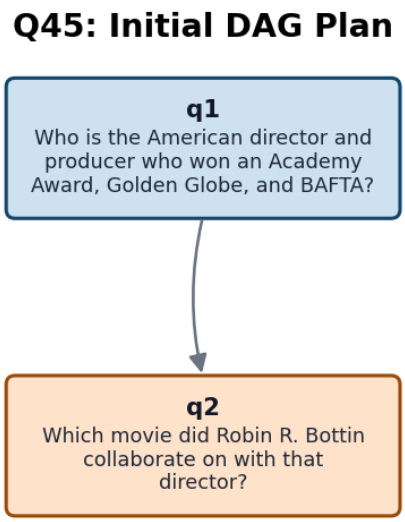
"'8½' features a soundtrack by Nino Rota." -> SOUNDTRACK_OF -> "'8½ …[truncated for report]

Q45 [bridge / hard] 5ab5574b554299488d4d994e

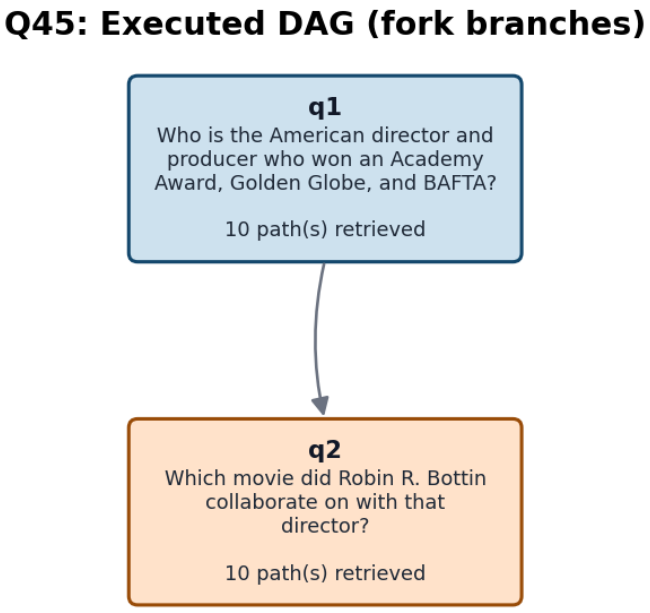
Robin R. Bottin is known for his collaboration with an American director and producer who won an Academy Award, Golden Globe and BAFTA award for what movie?

Gold answer	The Social Network
LLM answer	RoboCop
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is the American director and producer who won an Academy Award, Golden Globe, and BAFTA?	10 path(s)
q2	1	q1	Which movie did Robin R. Bottin collaborate on with that director?	10 path(s)

Final context fed to generation (1243 tokens)

```
Query plan:
- q1: Who is the American director and producer who won an Academy Award, Golden Globe, and BAFTA?
- q2: Which movie did Robin R. Bottin collaborate on with that director? (depends on: q1)

Level 1:

Subquery: Which movie did Robin R. Bottin collaborate on with that director?
"Robin R. Bottin is known for his collaborations with directors John Carpenter, Paul Verhoeven and David
Fincher." -> EXEMPLIFIES -> "Robin R. Bottin worked with Paul Verhoeven on 'RoboCop', 'Total Recall' and
'Basic Instinct'."
Logical Conclusions:
Robin Bottin's collaboration with Paul Verhoeven includes work on 'RoboCop', 'Total Recall', and 'Basic
Instinct'.

"Robin R. Bottin is known for his collaborations with directors John Carpenter, Paul Verhoeven and David
Fincher." -> EXEMPLIFIES -> "Robin R. Bottin worked with John Carpenter on both 'The Fog' and 'The Thing'."
Logical Conclusions:
Robin Bottin's collaboration with John Carpenter includes work on 'The Fog' and 'The Thing'.

"Robin R. Bottin is known for his collaborations with directors John Carpenter, Paul Verhoeven and David
Fincher." -> EXEMPLIFIES -> "Robin R. Bottin worked with David Fincher on 'Se7en' and 'Fight Club'."
Logical Conclusions:
Robin Bottin's collaboration with David Fincher includes work on 'Se7en' and 'Fight Club'.

"Robin R. Bottin is known for his collaborations with directors John Carpenter, Paul Verhoeven and David
Fincher." -> (has ) -> [john carpenter (Director collaborated with Rob Bottin)]

"Robin R. Bottin is known for his collaborations with directors John Carpenter, Paul Verhoeven and David
Fincher." -> (has ) -> [paul verhoeven (Director collaborated with Rob Bottin)]

"Robin R. Bottin is known for his collaborations with directors John Carpenter, Paul Verhoeven and David
Fincher." -> SUPPORTS -> "Robin R. Bottin is an American special make-up effects creator."
Logical Conclusions:
Robin R. Bottin, an American special make-up effects creator, is known for collaborations with directors
like John Carpenter and David Fincher.

[Robert Totten (Director of Gunsmoke installments with a small part)] -> (participates in event:
'Independence Day' was directed by Robert Mandel.) -> "'Independence Day' was directed by Robert Mandel."

[Robert Totten (Director of Gunsmoke installments with a small part)] -> (participates in event: Robert
Totten directed installments of 'Gunsmoke'.) -> "Robert Totten directed installments of 'Gunsmoke'."

[Robert Totten (Director of Gunsmoke installments with a small part)] -> (participates in event: Robert
Totten had a small part in The Apple Dumpling Gang Rides Again.) -> "Robert Totten had a small part in The
Apple Dumpling Gang Rides Again."

[Robert T. Schimke (Mentor of Daniel Haber at Stanford)] -> (participates in event: Valeria Sarmiento has
edited films for Robert Kramer.) -> "Valeria Sarmiento has edited films for Robert Kramer."
```

Level 0:

Subquery: Who is the American director and producer who won an Academy Award, Golden Globe, and BAFTA?
[academy honorary award (special award for career achievements)] -> (participates in event: Kirk Douglas is a Classic Hollywood star who received the Academy Honorary Award) -> "Kirk Douglas is a Classic Hollywood star who received the Academy Honorary Award."

[academy honorary award (special award for career achievements)] -> (participates in event: West Side Story won a special award for Jerome Robbins.) -> "West Side Story won a special award for Jerome Robbins."

[academy honorary award (special award for career achievements)] -> (participates in event: The company or its employees received many Academy Awards for scientific or tech) -> "The company or its employees received many Academy Awards for scientific or technical achievements."

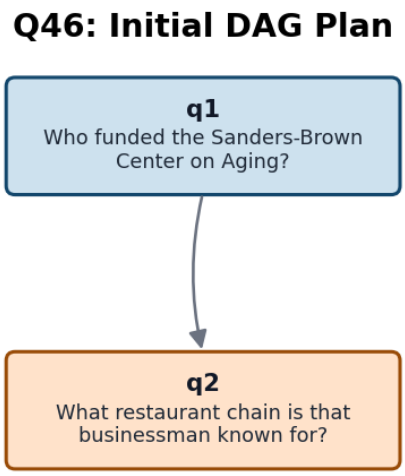
[academy honorary award (special award for career achievements)] -> (participates in event: Prior winners of competitive Academy Awards are not excluded from receiving the) -> "Prior winners of competitive Academy Awards are not excluded from …[truncated for report]

Q46 [bridge / hard] 5ab913205542991b5579f0e0

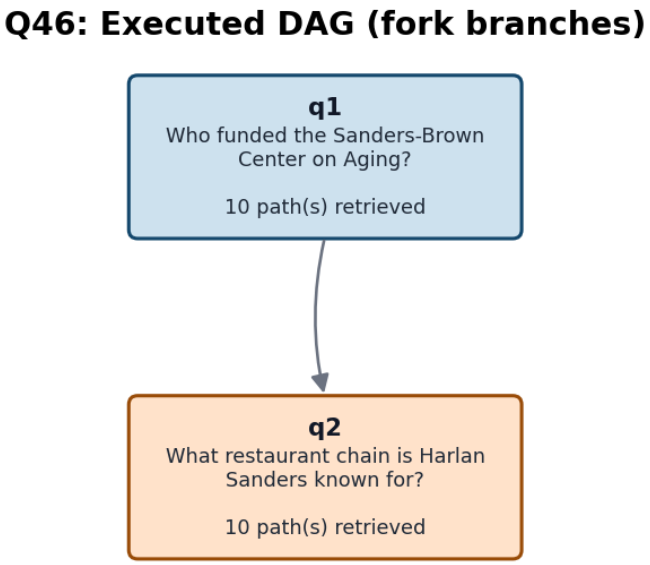
The Sanders-Brown Center on Aging was funded by a grant from a businessman known for what restaurant chain?

Gold answer	Kentucky Fried Chicken (KFC)
LLM answer	KFC
EM / F1	**0.00 / 0.40**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who funded the Sanders-Brown Center on Aging?	10 path(s)
q2	1	q1	What restaurant chain is Harlan Sanders known for?	10 path(s)

Final context fed to generation (1111 tokens)

```
Query plan:
- q1: Who funded the Sanders-Brown Center on Aging?
- q2: What restaurant chain is that businessman known for? (depends on: q1)

Level 1:

Subquery: What restaurant chain is Harlan Sanders known for?
[harlan sanders (business partner and center namesake)] -> (participates in event: The center was named for the Browns and his then-business partner, (Col.) Harlan) -> "The center was named for the Browns and his then-business partner, (Col.) Harlan Sanders."

[KFC (American fast food chain specializing in fried chicken)] -> (participates in event: John Young Brown Jr. may be best known for building Kentucky Fried Chicken (KFC)) -> "John Young Brown Jr. may be best known for building Kentucky Fried Chicken (KFC) into a multimillion-dollar restaurant chain."

[KFC (American fast food chain specializing in fried chicken)] -> (participates in event: KFC is the world's second-largest restaurant chain as measured by sales.) -> "KFC is the world's second-largest restaurant chain as measured by sales."

[KFC (American fast food chain specializing in fried chicken)] -> (participates in event: KFC is an American fast food restaurant chain.) -> "KFC is an American fast food restaurant chain."

[KFC (American fast food chain specializing in fried chicken)] -> (participates in event: KFC is headquartered in Louisville, Kentucky.) -> "KFC is headquartered in Louisville, Kentucky."

[KFC (American fast food chain specializing in fried chicken)] -> (participates in event: KFC specializes in fried chicken.) -> "KFC specializes in fried chicken."

[KFC (American fast food chain specializing in fried chicken)] -> (participates in event: KFC is a subsidiary of Yum! Brands.) -> "KFC is a subsidiary of Yum! Brands."

[john altman (Actor in EastEnders and band frontman)] -> (participates in event: John Altman is perhaps best known for playing "Nasty" Nick Cotton in the popular) -> "John Altman is perhaps best known for playing "Nasty" Nick Cotton in the popular BBC soap opera "EastEnders"."

"The group of companies had interests in restaurants." -> PARALLEL_INTEREST -> "The group of companies had interests in marinas."
Logical Conclusions:
Both facts describe the diverse business interests of the same group of companies, with one mentioning restaurants and the other mentioning marinas.

"The group of companies had interests in restaurants." -> PART_OF -> "Lelio 'Les' Marino grew Modern Continental into a six billion-dollar group of companies."
Logical Conclusions:
The group of companies included interests in restaurants.

Level 0:

Subquery: Who funded the Sanders-Brown Center on Aging?
"The Sanders–Brown Center on Aging is responsible for research, education and service programs in aging." -> NAMESAKE -> "The center was named for the Browns and his then-business partner, (Col.) Harlan Sanders."
Logical Conclusions:
```

The center was named for the Browns and Harlan Sanders, and it is the Sanders–Brown Center on Aging.

"The Sanders–Brown Center on Aging is responsible for research, education and service programs in aging." -> FOUNDED_IN -> "The program was founded in 1963 with the creation of the Council on Aging."
Logical Conclusions:
The program was founded in 1963 with the creation of the Council on Aging.

[sanders–brown center on aging (aging research center at University of Kentucky)] -> (participates in event: The center was named for the Browns and his then-business partner, (Col.) Harlan) -> "The center was named for the Browns and his then-business partner, (Col.) Harlan Sanders."

"The Sanders–Brown Center on Aging is responsible for research, education and service programs in aging." -> MISSION_OF -> "The Sanders–Brown Center on Aging is at the University of Kentucky in Lexington, Kentucky."
Logical Conclusions:
The Sanders–Brown Center on Aging at the University of Kentucky is responsible for research, education and service programs in aging.

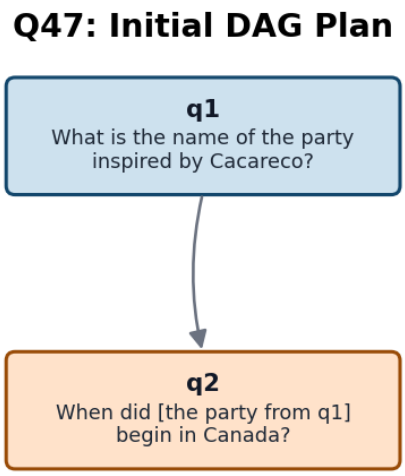
"The Sanders–Brown Center on Aging is responsible for research, education and service pr …[truncated for report]

Q47 [bridge / hard] 5a886b7555429938390d3f4d

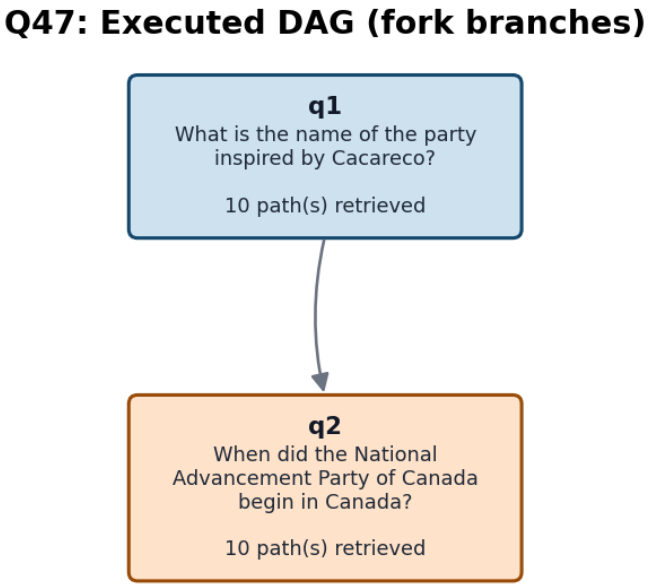
When did the party inspired by Cacareco begin in Canada?

Gold answer	1960s
LLM answer	The party inspired by Cacareco is the Rhinoceros Party, which began in Canada in 1963.
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the name of the party inspired by Cacareco?	10 path(s)
q2	1	q1	When did the National Advancement Party of Canada begin in Canada?	10 path(s)

Final context fed to generation (1188 tokens)

```
Query plan:
- q1: What is the name of the party inspired by Cacareco?
- q2: When did [the party from q1] begin in Canada? (depends on: q1)

Level 1:

Subquery: When did the National Advancement Party of Canada begin in Canada?
[National Advancement Party of Canada (Minor federal political party)] -> (participates in event: The party
was registered with Elections Canada in 2015.) -> "The party was registered with Elections Canada in 2015."

[National Advancement Party of Canada (Minor federal political party)] -> (participates in event: The party
was initially registered under the name Democratic Advancement Party o) -> "The party was initially
registered under the name Democratic Advancement Party of Canada."

"The National Advancement Party of Canada is a minor federal political party in Canada." -> (has ) ->
[National Advancement Party of Canada (Minor federal political party)]

[National Advancement Party of Canada (Minor federal political party)] -> (participates in event: The party
is headquartered in Calgary, Alberta.) -> "The party is headquartered in Calgary, Alberta."

"The National Advancement Party of Canada is a minor federal political party in Canada." -> FORMERLY_NAMED
-> "The party was initially registered under the name Democratic Advancement Party of Canada."
Logical Conclusions:
The party was initially registered under the name Democratic Advancement Party of Canada.

"The National Advancement Party of Canada is a minor federal political party in Canada." ->
HEADQUARTERED_IN -> "The party is headquartered in Calgary, Alberta."
Logical Conclusions:
The National Advancement Party of Canada is headquartered in Calgary, Alberta.

"The National Advancement Party of Canada is a minor federal political party in Canada." -> DESCRIBES_AS ->
"The party describes itself as for 'advancement of the Canadian people and Canada'."
Logical Conclusions:
The National Advancement Party of Canada describes itself as for the 'advancement of the Canadian people
and Canada'.

"The National Advancement Party of Canada is a minor federal political party in Canada." -> LED_BY -> "The
party is led by writer Stephen Garvey."
Logical Conclusions:
The National Advancement Party of Canada is led by writer Stephen Garvey.

[first peoples national party of canada (Eligible political party in Canada)] -> (participates in event:
The First Peoples National Party of Canada became an eligible political party on) -> "The First Peoples
National Party of Canada became an eligible political party on December 6, 2005."

[first peoples national party of canada (Eligible political party in Canada)] -> (participates in event: In
late October 2005, the APP sought to unite with the First Peoples National Pa) -> "In late October 2005,
the APP sought to unite with the First Peoples National Party of Canada."
```

Level 0:

Subquery: What is the name of the party inspired by Cacareco?
"Cacareco eventually won 100,000 votes." -> (has) -> [Rhinoceros Party (protest party with animal candidates)]

"Cacareco eventually won 100,000 votes." -> SPECIFIES -> "Cacareco won more votes than any other party in the 1958 São Paulo city council election."
Logical Conclusions:
Cacareco won 100,000 votes in the 1958 São Paulo city council election, more than any other party.

"Cacareco eventually won 100,000 votes." -> VOTES_FOR -> "Cacareco became famous as a candidate for the 1958 São Paulo city council elections."
Logical Conclusions:
Cacareco eventually won 100,000 votes.

"Cacareco eventually won 100,000 votes." -> CONTRASTS -> "Electoral officials did not accept Cacareco's candidacy."
Logical Conclusions:
Electoral officials did not accept Cacareco's candidacy, yet she eventually won 100,000 votes.

[National Advancement Party of Canada (Minor federal political party)] -> (participates in event: The party was initially registered under the name Democratic Advancement Party o) -> "The party was initially registered under the name Democratic Advancement Party of Canada."

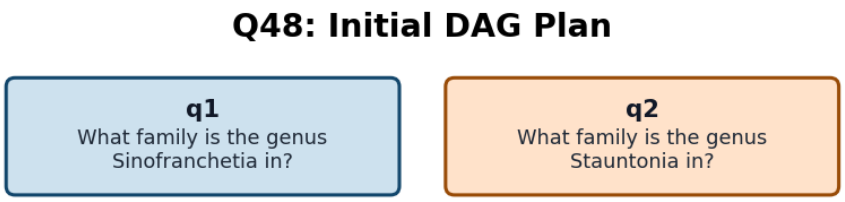
[Na …[truncated for report]

Q48 [comparison / hard] 5ab6d04c554299710c8d1f70

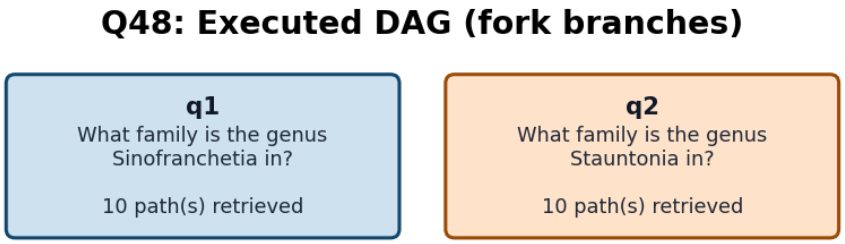
What family are the genus' Sinofranchetia and Stauntonia from?

Gold answer	a genus of flowering plant in the Lardizabalaceae family
LLM answer	Lardizabalaceae
EM / F1	**0.00 / 0.25**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What family is the genus Sinofranchetia in?	10 path(s)
q2	0	—	What family is the genus Stauntonia in?	10 path(s)

Final context fed to generation (999 tokens)

```
Query plan:
- q1: What family is the genus Sinofranchetia in?
- q2: What family is the genus Stauntonia in?

Level 0:

Subquery: What family is the genus Sinofranchetia in?
"Sinofranchetia is in the Lardizabalaceae family." -> CONTAINS -> "Sinofranchetia is a genus of flowering
plant."
Logical Conclusions:
Sinofranchetia is a genus of flowering plant in the Lardizabalaceae family.
```

"Sinofranchetia is in the Lardizabalaceae family." -> (has) -> [Sinofranchetia (Genus of flowering plant)]

"Sinofranchetia is in the Lardizabalaceae family." -> CONTEXTUALIZES -> "Sinofranchetia contains a single species."
Logical Conclusions:
Sinofranchetia contains a single species, and Sinofranchetia is in the Lardizabalaceae family.

"Sinofranchetia is in the Lardizabalaceae family." -> CONTEXTUALIZES -> "The single species is Sinofranchetia chinensis."
Logical Conclusions:
The single species is Sinofranchetia chinensis, and Sinofranchetia is in the Lardizabalaceae family.

"Sinofranchetia is in the Lardizabalaceae family." -> SAME_FAMILY -> "Stauntonia is in the Lardizabalaceae family."
Logical Conclusions:
Both Sinofranchetia and Stauntonia are classified in the Lardizabalaceae family.

[Sinofranchetia (Genus of flowering plant)] -> (participates in event: Sinofranchetia is a genus of flowering plant.) -> "Sinofranchetia is a genus of flowering plant."

[Sinofranchetia (Genus of flowering plant)] -> (participates in event: The single species is Sinofranchetia chinensis.) -> "The single species is Sinofranchetia chinensis."

"Sinofranchetia contains a single species." -> IDENTIFIES -> "The single species is Sinofranchetia chinensis."
Logical Conclusions:
The single species contained in Sinofranchetia is Sinofranchetia chinensis.

Subquery: What family is the genus Stauntonia in?
"Stauntonia is in the Lardizabalaceae family." -> CONTAINS -> "Stauntonia is a genus of flowering plant."
Logical Conclusions:
Stauntonia is a genus of flowering plant in the Lardizabalaceae family.

"Stauntonia is in the Lardizabalaceae family." -> (has) -> [stauntonia (genus of flowering plant)]

"Stauntonia is in the Lardizabalaceae family." -> SAME_FAMILY -> "Sinofranchetia is in the Lardizabalaceae family."
Logical Conclusions:
Both Sinofranchetia and Stauntonia are classified in the Lardizabalaceae family.

"Stauntonia is in the Lardizabalaceae family." -> CONTEXTUALIZES -> "George Staunton brought Stauntonia to Britain from China."
Logical Conclusions:
George Staunton brought Stauntonia to Britain from China, and Stauntonia is in the Lardizabalaceae family.

"Stauntonia is in the Lardizabalaceae family." -> CONTEXTUALIZES -> "George Staunton brought Stauntonia to Britain in the 19th Century."
Logical Conclusions:
George Staunton brought Stauntonia to Britain in the 19th Century, and Stauntonia is in the Lardizabalaceae family.

[stauntonia (genus of flowering plant)] -> (participates in event: Stauntonia is a genus of flowering plant.) -> "Stauntonia is a genus of flowering plant."

"George Staunton brought Stauntonia to Britain from China." -> CONTEXTUALIZES -> "Stauntonia is a genus of flowering plant."
Logical Conclusions:
Stauntonia is a genus of flowering plant, and George Staunton brought Stauntonia to Britain from China.

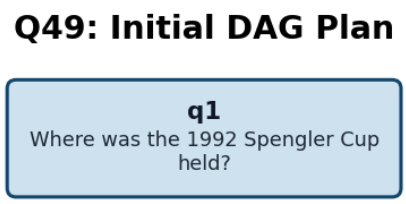
[stauntonia (genus of flowering plant)] -> (participates in event: Stauntonia is named after George Staunton.) -> "Stauntonia is named after George Staunton."

Q49 [bridge / hard] 5ae234385542992decbdcc59

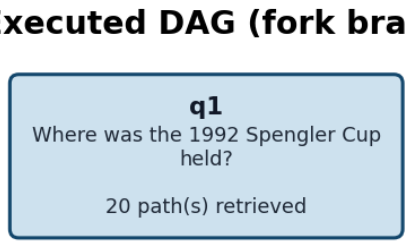
in 1992 Spengler Cup all matches were played at the home arena of what professional Swiss ice hockey club

Gold answer	HC Davos
LLM answer	HC Davos
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Where was the 1992 Spengler Cup held?	20 path(s)

Final context fed to generation (1332 tokens)

```
"The 1992 Spengler Cup was held between December 26, 1992, and December 31, 1992." -> HOSTS -> "The 1992
Spengler Cup was held in Davos, Switzerland." -> PLAYED_AT -> "All matches of the 1992 Spengler Cup were
played at HC Davos's home arena, Eisstadion Davos."
Logical Conclusions:
The 1992 Spengler Cup was held in Davos, Switzerland, between December 26 and December 31, 1992.
The 1992 Spengler Cup was held in Davos, Switzerland, with all matches at Eisstadion Davos.

"All matches of the 1992 Spengler Cup were played at HC Davos's home arena, Eisstadion Davos." -> SUPPORTS
-> "The 1992 Spengler Cup was held between December 26, 1992, and December 31, 1992."
Logical Conclusions:
Fact 18 provides the dates of the 1992 Spengler Cup (Dec 26-31, 1992), and fact 21 states all matches were
played at HC Davos's home arena.

"The 1992 Spengler Cup was held between December 26, 1992, and December 31, 1992." -> (has ) -> [Spengler
```

Cup (Annual invitational ice hockey tournament)] -> (participates in event: The Spengler Cup is held in Davos, Switzerland.) -> "The Spengler Cup is held in Davos, Switzerland." -> GENERALIZES -> "The 1992 Spengler Cup was held in Davos, Switzerland."
Logical Conclusions:
Fact 23 states the 1992 Spengler Cup was held in Davos, and fact 24 states the Spengler Cup is held in Davos generally.

"The 1992 Spengler Cup was held in Davos, Switzerland." -> (has) -> [Spengler Cup (Annual invitational ice hockey tournament)] -> (participates in event: Every year the Spengler Cup is played in Vaillant Arena.) -> "Every year the Spengler Cup is played in Vaillant Arena."

"The 1992 Spengler Cup was held in Davos, Switzerland." -> HAS_FINAL -> "The final of the 1992 Spengler Cup was won 6-5 by Team Canada."
Logical Conclusions:
The 1992 Spengler Cup final was won 6-5 by Team Canada.

[Spengler Cup (Annual invitational ice hockey tournament)] -> (participates in event: The groups are named.) -> "The groups are named."

"The 1992 Spengler Cup was held between December 26, 1992, and December 31, 1992." -> SUPPORTS -> "The final of the 1992 Spengler Cup was won by Team Canada over Färjestad BK."
Logical Conclusions:
Fact 18 provides the dates of the 1992 Spengler Cup, and fact 27 states the final was won by Team Canada over Färjestad BK.

"The 1992 Spengler Cup was held between December 26, 1992, and December 31, 1992." -> SUPPORTS -> "The final of the 1992 Spengler Cup was won 6-5 by Team Canada."
Logical Conclusions:
Fact 18 provides the dates of the 1992 Spengler Cup, and fact 25 states the final was won 6-5 by Team Canada.

[Spengler Cup (Annual invitational ice hockey tournament)] -> (participates in event: The final of the 1992 Spengler Cup was won by Team Canada over Färjestad BK.) -> "The final of the 1992 Spengler Cup was won by Team Canada over Färjestad BK."

"The 1992 Spengler Cup was held in Davos, Switzerland." -> (has) -> [HC Davos (Professional Swiss ice hockey club)] -> (participates in event: Vaillant Arena is also known as Eisstadion Davos.) -> "Vaillant Arena is also known as Eisstadion Davos."

[Spengler Cup (Annual invitational ice hockey tournament)] -> (participates in event: The Spengler Cup is often cited as the oldest invitational ice hockey tournament) -> "The Spengler Cup is often cited as the oldest invitational ice hockey tournament in the world."

"All matches of the 1992 Spengler Cup were played at HC Davos's home arena, Eisstadion Davos." -> (has) -> [Spengler Cup (Annual invitational ice hockey tournament)]

[Spengler Cup (Annual invitational ice hockey tournament)] -> (participates in event: The Spengler Cup is an annual invitational ice hockey tournament.) -> "The Spengler Cup is an annual invitational ice hockey tournament."

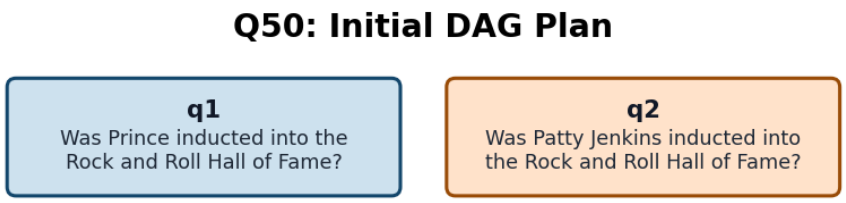
"The Spengler Cup is held in Davos, Switzerland." -> LOCATION_OF -> "Currently, all Spengler Cup games are held at Vaillant Arena."
Logical Conclusions:
Currently, all Spengler Cup games are held at Vaillant Arena, and the Spengler Cup is held in Davos, Switzerland. …[truncated for report]

Q50 [comparison / hard] 5a80caac554299260e20a161

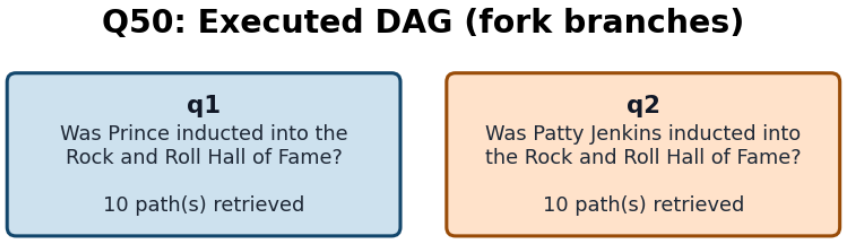
Were both Prince and Patty Jenkins have been inducted into the Rock and Roll Hall of Fame?

Gold answer	no
LLM answer	No, only Prince was inducted into the Rock and Roll Hall of Fame. Patty Jenkins was not.
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Was Prince inducted into the Rock and Roll Hall of Fame?	10 path(s)
q2	0	—	Was Patty Jenkins inducted into the Rock and Roll Hall of Fame?	10 path(s)

Final context fed to generation (1129 tokens)

```
Query plan:
- q1: Was Prince inducted into the Rock and Roll Hall of Fame?
- q2: Was Patty Jenkins inducted into the Rock and Roll Hall of Fame?

Level 0:

Subquery: Was Prince inducted into the Rock and Roll Hall of Fame?
"Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in his first year of eligibility."
```

-> (has) -> [Prince (American singer-songwriter and musician)]

"Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in his first year of eligibility." -> (has) -> [Rock and Roll Hall of Fame (Hall of Fame for rock music)]

"Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in his first year of eligibility." -> YEAR_OF -> "Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in 2004."
Logical Conclusions:
Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in 2004 in his first year of eligibility.

[Prince (American singer-songwriter and musician)] -> (participates in event: Prince Rogers Nelson was an American multi-instrumentalist.) -> "Prince Rogers Nelson was an American multi-instrumentalist."

[Prince (American singer-songwriter and musician)] -> (participates in event: Prince Rogers Nelson won seven Grammy Awards.) -> "Prince Rogers Nelson won seven Grammy Awards."

[Prince (American singer-songwriter and musician)] -> (participates in event: Prince Rogers Nelson was known for his flamboyant stage presence.) -> "Prince Rogers Nelson was known for his flamboyant stage presence."

[Prince (American singer-songwriter and musician)] -> (participates in event: Prince Rogers Nelson has sold over 100 million records worldwide.) -> "Prince Rogers Nelson has sold over 100 million records worldwide."

[Prince (American singer-songwriter and musician)] -> (participates in event: Romeo Nelson's recordings from 1929 are regarded as the fastest boogie woogie sh) -> "Romeo Nelson's recordings from 1929 are regarded as the fastest boogie woogie showpieces on record."

[Roy Stevens (Teacher of embouchure method to Roman)] -> (participates in event: After his injury, Roy Roman was introduced to Roy Stevens.) -> "After his injury, Roy Roman was introduced to Roy Stevens."

Subquery: Was Patty Jenkins inducted into the Rock and Roll Hall of Fame?
[Rock and Roll Hall of Fame (Hall of Fame for rock music)] -> (participates in event: Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in his fir) -> "Prince Rogers Nelson was inducted into the Rock and Roll Hall of Fame in his first year of eligibility."

[Rock and Roll Hall of Fame (Hall of Fame for rock music)] -> (participates in event: The notable figures recognized by the Rock and Roll Hall of Fame have had some m) -> "The notable figures recognized by the Rock and Roll Hall of Fame have had some major influence on the development of rock and roll."

[Rock and Roll Hall of Fame (Hall of Fame for rock music)] -> (participates in event: In 1986, Cleveland was chosen as the Hall of Fame.) -> "In 1986, Cleveland was chosen as the Hall of Fame."

[Rock and Roll Hall of Fame (Hall of Fame for rock music)] -> (participates in event: The Rock and Roll Hall of Fame Foundation was established by Atlantic Records fo) -> "The Rock and Roll Hall of Fame Foundation was established by Atlantic Records founder and chairman Ahmet Ertegun."

[Rock and Roll Hall of Fame (Hall of Fame for rock music)] -> (participates in event: The Rock and Roll Hall of Fame recognizes and archives the history of the best-k) -> "The Rock and Roll Hall of Fame recognizes and archives the history of the best-known and most influential artists, producers, engineers, and other notable figures."

[Rock and Roll Hall of Fame (Hall of Fame for rock music)] -> (participates in event: The Rock and Roll Hall of Fame has had a cumulative economic impact estimated at) -> "The Rock and Roll Hall of Fame has had a cumulative economic impact estimated at more than $1.8 billion."

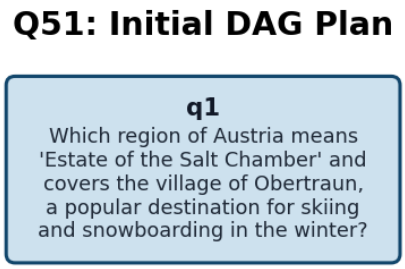
[patty jenkins (American film director and scr …[truncated for report]

Q51 [bridge / hard] 5abc23c45542993a06baf896

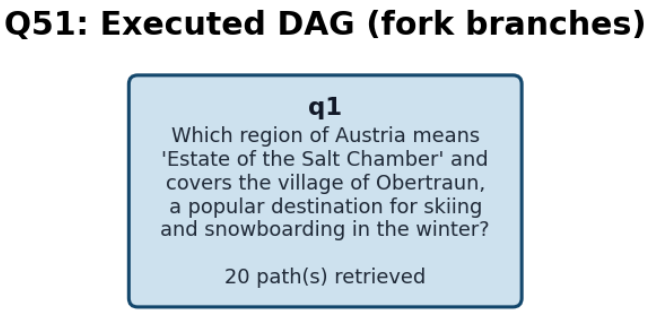
Which region of Austria means 'Estate of the Salt Chamber' and covers the village of Obertraun, a popular destination for skiing and snowboarding in the winter?

Gold answer	Salzkammergut
LLM answer	Salzkammergut
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which region of Austria means 'Estate of the Salt Chamber' and covers the village of Obertraun, a popular destination for skiing and snowboarding in the winter?	20 path(s)

Final context fed to generation (1352 tokens)

```
[Obertraun (Village in Upper Austria)] -> (participates in event: Obertraun offers activities such as skiing and snowboarding in the winter.) -> "Obertraun offers activities such as skiing and snowboarding in the winter." -> OFFERED_BY -> "Obertraun, Upper Austria is a village in the Salzkammergut." -> (has ) -> [Salzkammergut (Resort area in Austria)] -> (participates in event: The name 'Salzkammergut' literally
```

means 'Estate of the Salt Chamber'.) -> "The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'."
Logical Conclusions:
Obertraun, Upper Austria is a village in the Salzkammergut and offers activities such as skiing and snowboarding in the winter.

[Obertraun (Village in Upper Austria)] -> (participates in event: Obertraun, Upper Austria is a village in the Salzkammergut.) -> "Obertraun, Upper Austria is a village in the Salzkammergut."

"The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'." -> DERIVES_FROM -> "The Salzkammergut stretches from the City of Salzburg eastwards along the Austrian Alpine Foreland and the Northern Limestone Alps to the peaks of the Dachstein Mountains."
Logical Conclusions:
The name 'Salzkammergut' literally means 'Estate of the Salt Chamber', and the region stretches from the City of Salzburg eastwards to the peaks of the Dachstein Mountains.

[Obertraun (Village in Upper Austria)] -> (participates in event: Obertraun is a popular holiday destination.) -> "Obertraun is a popular holiday destination." -> LOCATED_IN -> "Obertraun, Upper Austria is a village in the Salzkammergut."
Logical Conclusions:
Obertraun is a popular holiday destination located in the Salzkammergut region.

"The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'." -> DERIVES_FROM -> "The main river of the Salzkammergut region is the Traun."
Logical Conclusions:
The name 'Salzkammergut' literally means 'Estate of the Salt Chamber', and the main river of the region is the Traun.

"The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'." -> DERIVES_FROM -> "The Salzkammergut is a UNESCO World Heritage Site."
Logical Conclusions:
The name 'Salzkammergut' literally means 'Estate of the Salt Chamber', and the Salzkammergut is a UNESCO World Heritage Site.

"The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'." -> DERIVES_FROM -> "The Salzkammergut spans the federal states of Upper Austria, Salzburg, and Styria."
Logical Conclusions:
The name 'Salzkammergut' literally means 'Estate of the Salt Chamber', and it spans the federal states of Upper Austria, Salzburg, and Styria.

"The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'." -> DERIVES_FROM -> "The name 'Salzkammergut' derives from the Imperial Salt Chamber."
Logical Conclusions:
The name 'Salzkammergut' literally means 'Estate of the Salt Chamber' and derives from the Imperial Salt Chamber.

[Salzkammergut (Resort area in Austria)] -> (participates in event: The main river of the Salzkammergut region is the Traun.) -> "The main river of the Salzkammergut region is the Traun."

"The name 'Salzkammergut' literally means 'Estate of the Salt Chamber'." -> DERIVES_FROM -> "The Salzkammergut is a resort area located in Austria."
Logical Conclusions:
The name 'Salzkammergut' literally means 'Estate of the Salt Chamber', and it is a resort area located in Austria.

[Salzkammergut (Resort area in Austria)] -> (participates in event: The Salzkammergut stretches from the City of Salzburg eastwards along the Austri) -> "The Salzkammergut stretches from the City of Salzburg eastwards along the Austrian Alpine Foreland and the Northern Limestone Alps to the peaks of the Dachstein Mountains."

"Obertraun, Upper Austria is a village in the Salzkammergut." -> LOCATED_IN -> "The Salzkammergut is a resort area located in Austria." -> (has) -> [Salzkammergut (Resort area in Austria)]
Logical Conclusions:

Obertraun is a village in the Salzkammergut, which is a resort area in Austria.

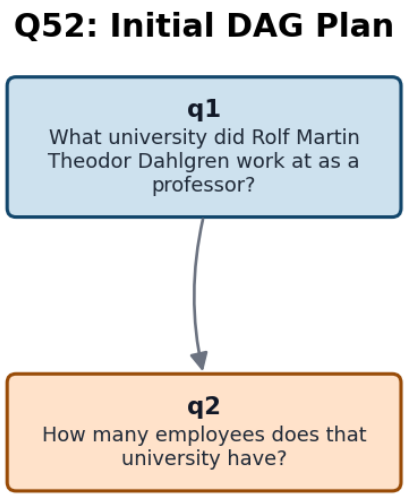
"Obertrau …[truncated for report]

Q52 [bridge / hard] 5a89e47f5542992e4fca8457

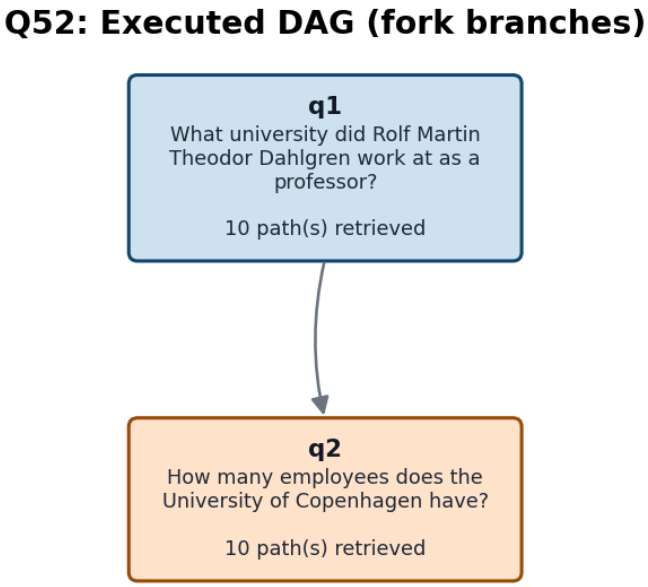
The university where Rolf Martin Theodor Dahlgren worked as a professor has how many employees?

Gold answer	9,000
LLM answer	The University of Copenhagen has over 9,000 employees.
EM / F1	**0.00 / 0.25**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What university did Rolf Martin Theodor Dahlgren work at as a professor?	10 path(s)
q2	1	q1	How many employees does the University of Copenhagen have?	10 path(s)

Final context fed to generation (1186 tokens)

```
Query plan:
- q1: What university did Rolf Martin Theodor Dahlgren work at as a professor?
- q2: How many employees does that university have? (depends on: q1)

Level 1:

Subquery: How many employees does the University of Copenhagen have?
"The University of Copenhagen (UCPH) has over 9,000 employees." -> EMPLOYEE_COUNT -> "The University of
Copenhagen (UCPH) is the oldest university in Denmark."
Logical Conclusions:
The University of Copenhagen has over 9,000 employees.

"The University of Copenhagen (UCPH) has over 9,000 employees." -> (has ) -> [university of copenhagen
(oldest university and research institution in Denmark)]

[university of copenhagen (oldest university and research institution in Denmark)] -> (participates in
event: The University of Copenhagen (UCPH) has several thousands of foreign students.) -> "The University
of Copenhagen (UCPH) has several thousands of foreign students."

[university of copenhagen (oldest university and research institution in Denmark)] -> (participates in
event: The headquarters of the University of Copenhagen (UCPH) is located in central Co) -> "The
headquarters of the University of Copenhagen (UCPH) is located in central Copenhagen."

[university of copenhagen (oldest university and research institution in Denmark)] -> (participates in
event: Many courses at the University of Copenhagen (UCPH) are also offered in English.) -> "Many courses
at the University of Copenhagen (UCPH) are also offered in English."

[university of copenhagen (oldest university and research institution in Denmark)] -> (participates in
event: The University of Copenhagen (UCPH) is the second oldest institution for higher ) -> "The University
of Copenhagen (UCPH) is the second oldest institution for higher education in Scandinavia."

[university of copenhagen (oldest university and research institution in Denmark)] -> (participates in
event: Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen fro) -> "Rolf Martin
Theodor Dahlgren was a professor at the University of Copenhagen from 1973 to his death."

[kwame nkrumah university of science and technology (university where Prempeh College topped
matriculation)] -> (participates in event: Prempeh College topped matriculation at the Kwame Nkrumah
University of Science ) -> "Prempeh College topped matriculation at the Kwame Nkrumah University of Science
and Technology in 2004 with 441 students admitted."

[university of washington (Former faculty employer of Guruswami)] -> (participates in event: Venkatesan
Guruswami was a member of the faculty at the University of Washington) -> "Venkatesan Guruswami was a
member of the faculty at the University of Washington from 2002 to 2009."

Level 0:

Subquery: What university did Rolf Martin Theodor Dahlgren work at as a professor?
"Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen." -> (has ) -> [university of
copenhagen (oldest university and research institution in Denmark)]

"Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen." -> WORKED_DURING -> "Rolf
```

Martin Theodor Dahlgren was a professor at the University of Copenhagen from 1973 to his death."
Logical Conclusions:
Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen from 1973 to his death.

"Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen." -> EMPLOYED -> "Rolf Martin Theodor Dahlgren was born on 7 July 1932."
Logical Conclusions:
Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen.

"Rolf Martin Theodor Dahlgren was a professor at the University of Copenhagen." -> (has) -> [Rolf (Swedish or Swiss scientist)]

"Gertrud Dahlgren graduated from university at Helsingborg." -> (has) -> [Various cities (Cities in Europe)]

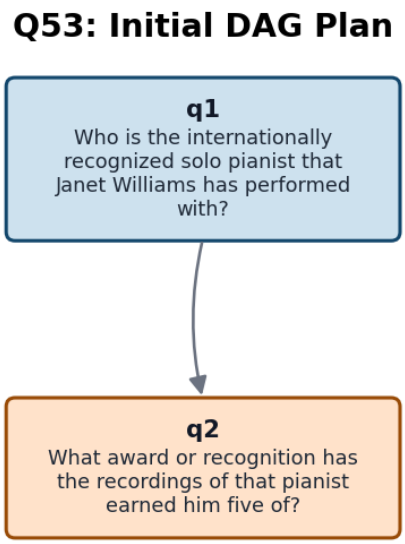
"Gertrud Dahlgren graduated from university at Helsingborg." -> ALUMNA_OF -> "Gertrud Dahlgren was a Swedish botanist."
Logical Conclusions:
Gertrud Dahlgren graduated from university at Helsin …[truncated for report]

Q53 [bridge / hard] 5ab7de455542992aa3b8c864

Janet Williams has performed with an internationally recognized solo pianist whose recordings have earned him five what?

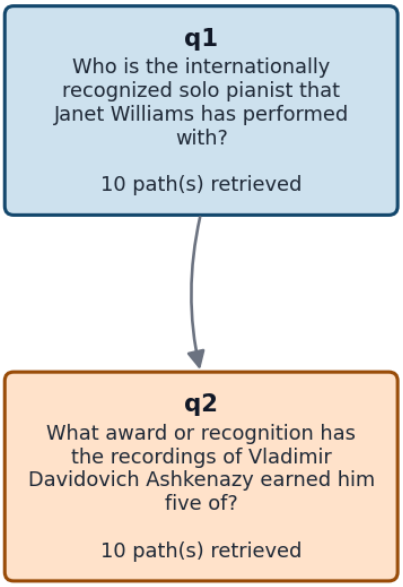
Gold answer	Grammy awards
LLM answer	Grammy awards
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q53: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is the internationally recognized solo pianist that Janet Williams has performed with?	10 path(s)
q2	1	q1	What award or recognition has the recordings of Vladimir Davidovich Ashkenazy earned him five of?	10 path(s)

Final context fed to generation (1193 tokens)

```
Query plan:
- q1: Who is the internationally recognized solo pianist that Janet Williams has performed with?
- q2: What award or recognition has the recordings of that pianist earned him five of? (depends on: q1)

Level 1:

Subquery: What award or recognition has the recordings of Vladimir Davidovich Ashkenazy earned him five of?
"Vladimir Davidovich Ashkenazy's recordings have earned him five Grammy awards." -> EARNED_BY -> "Vladimir
Davidovich Ashkenazy is an internationally recognized solo pianist, chamber music performer, and
conductor."
Logical Conclusions:
Vladimir Ashkenazy's recordings have earned him five Grammy awards.

"Vladimir Davidovich Ashkenazy's recordings have earned him five Grammy awards." -> CORROBORATES ->
"Vladimir Davidovich Ashkenazy's recordings have earned him Iceland's Order of the Falcon."
Logical Conclusions:
Vladimir Ashkenazy's recordings earned him both Grammy awards and Iceland's Order of the Falcon.

"Vladimir Davidovich Ashkenazy's recordings have earned him five Grammy awards." -> CAUSED_BY -> "Vladimir
Davidovich Ashkenazy has recorded a large storehouse of classical and romantic works."
Logical Conclusions:
Vladimir Ashkenazy's recordings of classical and romantic works earned him Grammy awards.
```

"Vladimir Davidovich Ashkenazy's recordings have earned him five Grammy awards." -> (has) -> [Musicians (Pianists and composers)]

"Vladimir Davidovich Ashkenazy's recordings have earned him five Grammy awards." -> FOLLOWS -> "Vladimir Davidovich Ashkenazy was born on 6 July 1937."
Logical Conclusions:
Vladimir Davidovich Ashkenazy was born on 6 July 1937 and his recordings have earned him five Grammy awards.

"Vladimir Davidovich Ashkenazy's recordings have earned him five Grammy awards." -> EXEMPLIFIES -> "Vladimir Davidovich Ashkenazy has held Icelandic citizenship since 1972."
Logical Conclusions:
Vladimir Davidovich Ashkenazy has held Icelandic citizenship since 1972 and his recordings have earned him five Grammy awards.

"Vladimir Davidovich Ashkenazy's recordings have earned him Iceland's Order of the Falcon." -> EARNED_BY -> "Vladimir Davidovich Ashkenazy is an internationally recognized solo pianist, chamber music performer, and conductor."
Logical Conclusions:
Vladimir Ashkenazy's recordings have earned him Iceland's Order of the Falcon.

"Vladimir Davidovich Ashkenazy's recordings have earned him Iceland's Order of the Falcon." -> CAUSED_BY -> "Vladimir Davidovich Ashkenazy has recorded a large storehouse of classical and romantic works."
Logical Conclusions:
Vladimir Ashkenazy's recordings earned him Iceland's Order of the Falcon.

Level 0:

Subquery: Who is the internationally recognized solo pianist that Janet Williams has performed with?
"Janet Williams has won international critical acclaim for performances at Cologne." -> (has) -> [janet williams (American soprano with international acclaim)]

"Janet Williams has won international critical acclaim for performances at Cologne." -> HOSTED -> "Janet Williams is an American soprano."
Logical Conclusions:
Janet Williams, an American soprano, has won acclaim for performances at Cologne.

"Janet Williams has won international critical acclaim for performances at Cologne." -> CORROBORATES -> "Janet Williams has won international critical acclaim for performances at Théâtre des Champs-Élysées."
Logical Conclusions:
Janet Williams has won acclaim at both Théâtre des Champs-Élysées and Cologne.

"Janet Williams has won international critical acclaim for performances at Cologne." -> CORROBORATES -> "Janet Williams has won international critical acclaim for performances at Opera de Lyon."
Logical Conclusions:
Janet Williams has won acclaim at both Opera de Lyon and Cologne.

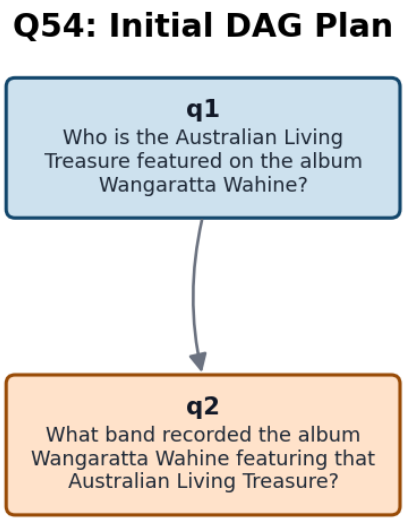
"Janet Williams has won international critical acclaim for performances at Cologne." -> CORROBORATES -> "Janet Williams has won international critical acclaim for performances at Frankfurt Opera."
Logical Conclusions:
Janet Williams has won acclaim at both Frankfurt Ope …[truncated for report]

Q54 [bridge / hard] 5ae39e9c5542990afbd1e195

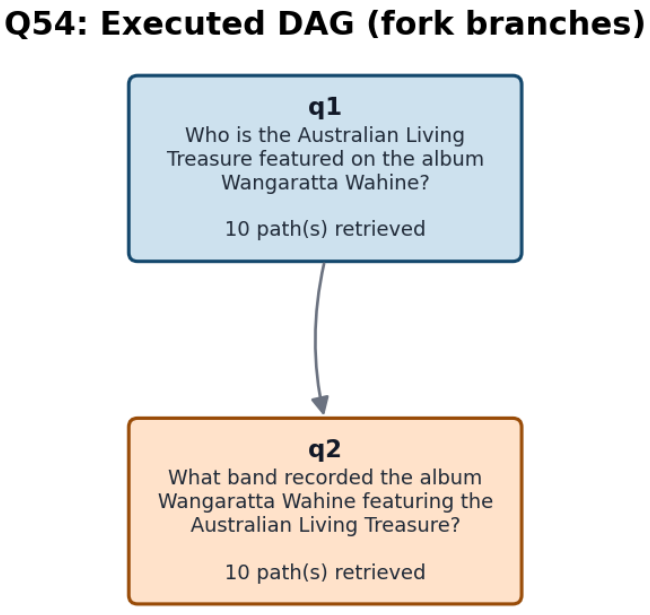
Wangaratta Wahine, an album featuring the work of an Australian Living Treasure, was recorded by what band?

Gold answer	The Captain Matchbox Whoopee Band
LLM answer	The Captain Matchbox Whoopee Band
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is the Australian Living Treasure featured on the album Wangaratta Wahine?	10 path(s)
q2	1	q1	What band recorded the album Wangaratta Wahine featuring the Australian Living Treasure?	10 path(s)

Final context fed to generation (1175 tokens)

```
Query plan:
- q1: Who is the Australian Living Treasure featured on the album Wangaratta Wahine?
- q2: What band recorded the album Wangaratta Wahine featuring that Australian Living Treasure? (depends
on: q1)

Level 1:

Subquery: What band recorded the album Wangaratta Wahine featuring the Australian Living Treasure?
"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." ->
RECORDING_LOCATION -> "Wangaratta Wahine is an album by Australian band The Captain Matchbox Whoopee Band."
Logical Conclusions:
Wangaratta Wahine was recorded at Armstrong Studios in Melbourne.

"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." -> IDENTIFIES ->
"Wangaratta Wahine was the second album by The Captain Matchbox Whoopee Band."
Logical Conclusions:
Wangaratta Wahine, the second album by The Captain Matchbox Whoopee Band, was recorded at Armstrong Studios
in Melbourne.

[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)] -> (participates in event: Wangaratta
Wahine is an album by Australian band The Captain Matchbox Whoopee Ba) -> "Wangaratta Wahine is an album by
Australian band The Captain Matchbox Whoopee Band."

"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." -> (has ) ->
[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)]

"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." -> SUPPORTS ->
"Wangaratta Wahine was released in 1974."
Logical Conclusions:
'Wangaratta Wahine' was recorded at top recording studio Armstrong Studios in Melbourne.

[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)] -> (participates in event: The cover art
for Wangaratta Wahine was by famed Australian artist Michael Leuni) -> "The cover art for Wangaratta Wahine
was by famed Australian artist Michael Leunig."

[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)] -> (participates in event: Wangaratta
Wahine was released in 1974.) -> "Wangaratta Wahine was released in 1974."

[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)] -> (participates in event: Wangaratta
Wahine was the second album by The Captain Matchbox Whoopee Band.) -> "Wangaratta Wahine was the second
album by The Captain Matchbox Whoopee Band."

"An unnamed person was named an Australian Living Treasure for their humanitarian work for the people of
East Timor in their later years." -> IDENTIFIES -> "Mavis Taylor was named an Australian Living Treasure."
Logical Conclusions:
Mavis Taylor was named an Australian Living Treasure for her humanitarian work for the people of East
Timor.

Level 0:
```

Subquery: Who is the Australian Living Treasure featured on the album Wangaratta Wahine?
[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)] -> (participates in event: The cover art for Wangaratta Wahine was by famed Australian artist Michael Leuni) -> "The cover art for Wangaratta Wahine was by famed Australian artist Michael Leunig."

[wangaratta wahine (Album by The Captain Matchbox Whoopee Band)] -> (participates in event: Wangaratta Wahine is an album by Australian band The Captain Matchbox Whoopee Ba) -> "Wangaratta Wahine is an album by Australian band The Captain Matchbox Whoopee Band."

"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." -> (has) -> [wangaratta wahine (Album by The Captain Matchbox Whoopee Band)]

"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." -> RECORDING_LOCATION -> "Wangaratta Wahine is an album by Australian band The Captain Matchbox Whoopee Band."
Logical Conclusions:
Wangaratta Wahine was recorded at Armstrong Studios in Melbourne.

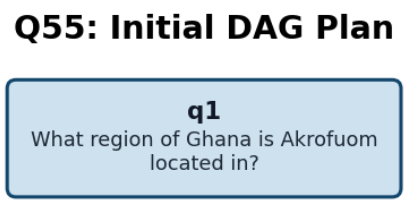
"Wangaratta Wahine was recorded at top recording studio Armstrong Studios in Melbourne." -> IDENTIFIES -> "Wangaratta Wahine was the second album by The Captain Matchbox Whoopee Band."
Logical Conclusions:
Wangaratta Wahine, the second album by The Capta …[truncated for report]

Q55 [bridge / hard] 5ac325495542995ef918c12e

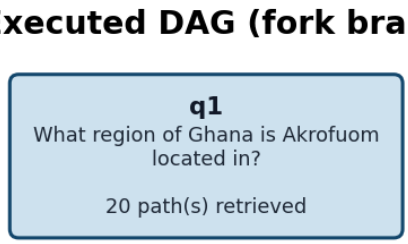
Which region of Ghana was the city where Akrofuom is located?

Gold answer	Ashanti Region of Ghana
LLM answer	Ashanti Region
EM / F1	0.00 / 0.67
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What region of Ghana is Akrofuom located in?	20 path(s)

Final context fed to generation (1232 tokens)

```
"Akrofuom is located in the Obuasi Municipal district of the Ashanti Region of Ghana." -> (has ) ->
[ashanti region (administrative region in south Ghana)] -> (participates in event: The Ashanti Region is
located in south Ghana.) -> "The Ashanti Region is located in south Ghana." -> PARALLELS -> "The Western
Region is located in south Ghana."
Logical Conclusions:
Both the Ashanti Region and the Western Region are located in south Ghana.

"The Ashanti Region is located in south Ghana." -> (has ) -> [ghana (country in West Africa)] ->
(participates in event: The Western Region is located in south Ghana.) -> "The Western Region is located in
south Ghana."

[ghana (country in West Africa)] -> (participates in event: The Western Region includes Ghana's
southernmost location, Cape Three Points.) -> "The Western Region includes Ghana's southernmost location,
Cape Three Points."
```

"The Ashanti Region is located in south Ghana." -> SUPPORTS -> "The Ashanti Region occupies a total land surface of 24389 km2."
Logical Conclusions:
Fact 21 states the Ashanti Region is located in south Ghana, and fact 172 gives its total land surface area of 24,389 km2.

[ghana (country in West Africa)] -> (participates in event: Taifa is located in southern Ghana near the capital Accra.) -> "Taifa is located in southern Ghana near the capital Accra."

[ghana (country in West Africa)] -> (participates in event: The Ashanti Region occupies 10.2 per cent of the total land area of Ghana.) -> "The Ashanti Region occupies 10.2 per cent of the total land area of Ghana."

"The Ashanti Region is located in south Ghana." -> SUPPORTS -> "The Ashanti Region occupies 10.2 per cent of the total land area of Ghana."
Logical Conclusions:
Fact 21 locates the Ashanti Region in south Ghana, and fact 234 states it occupies 10.2% of the total land area of Ghana.

[ghana (country in West Africa)] -> (participates in event: The Western Region's coastline stretches from South Ghana's border with Ivory Co) -> "The Western Region's coastline stretches from South Ghana's border with Ivory Coast to the Western region's boundary with the Central Region on the east."

"Akrofuom is located in the Obuasi Municipal district of the Ashanti Region of Ghana." -> LOCATED_IN -> "The Obuasi Municipality District is one of the twenty-seven districts of the Ashanti Region of Ghana." -> (has) -> [ashanti region (administrative region in south Ghana)]
Logical Conclusions:
Akrofuom is located in the Obuasi Municipal district of the Ashanti Region.

"The Ashanti Region is located in south Ghana." -> SUPPORTS -> "The Ashanti Region is the third largest of 10 administrative regions in Ghana."
Logical Conclusions:
Fact 21 places the Ashanti Region in south Ghana, and fact 233 notes it is the third largest of 10 administrative regions in Ghana.

"The Ashanti Region is located in south Ghana." -> SUPPORTS -> "In terms of population, the Ashanti Region is the most populated region in Ghana."
Logical Conclusions:
Fact 21 places the Ashanti Region in south Ghana, and fact 221 notes it is the most populated region in Ghana.

[ashanti region (administrative region in south Ghana)] -> (participates in event: The regional capital of the Ashanti Region is Kumasi.) -> "The regional capital of the Ashanti Region is Kumasi."

"The regional capital of the Ashanti Region is Kumasi." -> (has) -> [kumasi (regional capital of Ashanti Region)] -> (participates in event: Tafo is near the regional capital Kumasi.) -> "Tafo is near the regional capital Kumasi."

[ashanti region (administrative region in south Ghana)] -> (participates in event: The Ashanti Region is known for its major gold bar production.) -> "The Ashanti Region is known for its major gold bar production." -> CO_OCCURS_WITH -> "The Ashanti Region is known for its major cocoa production."
Logical Conclusions:
The Ashanti Region is known for both major gold bar production and major cocoa production.

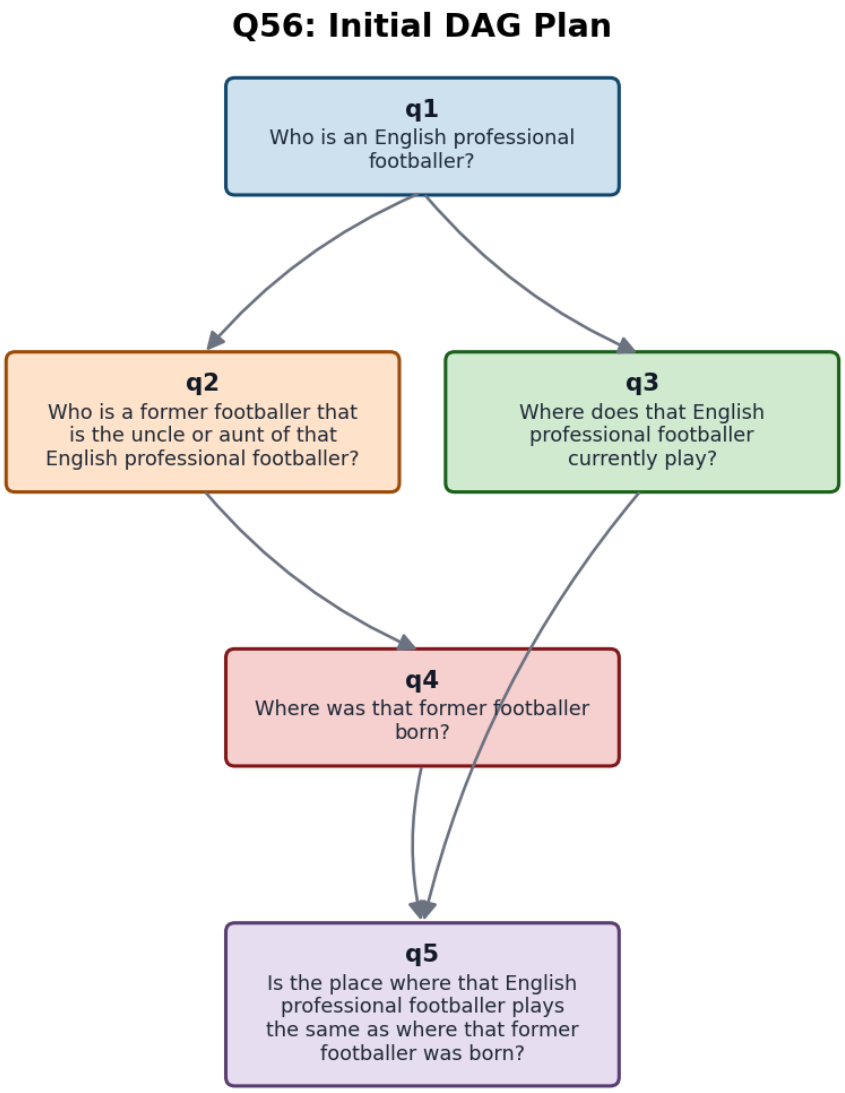
"The regional capital of the Ashanti Region is Kumasi." -> …[truncated for report]

Q56 [bridge / hard] 5a8af1e255429950cd6afc28

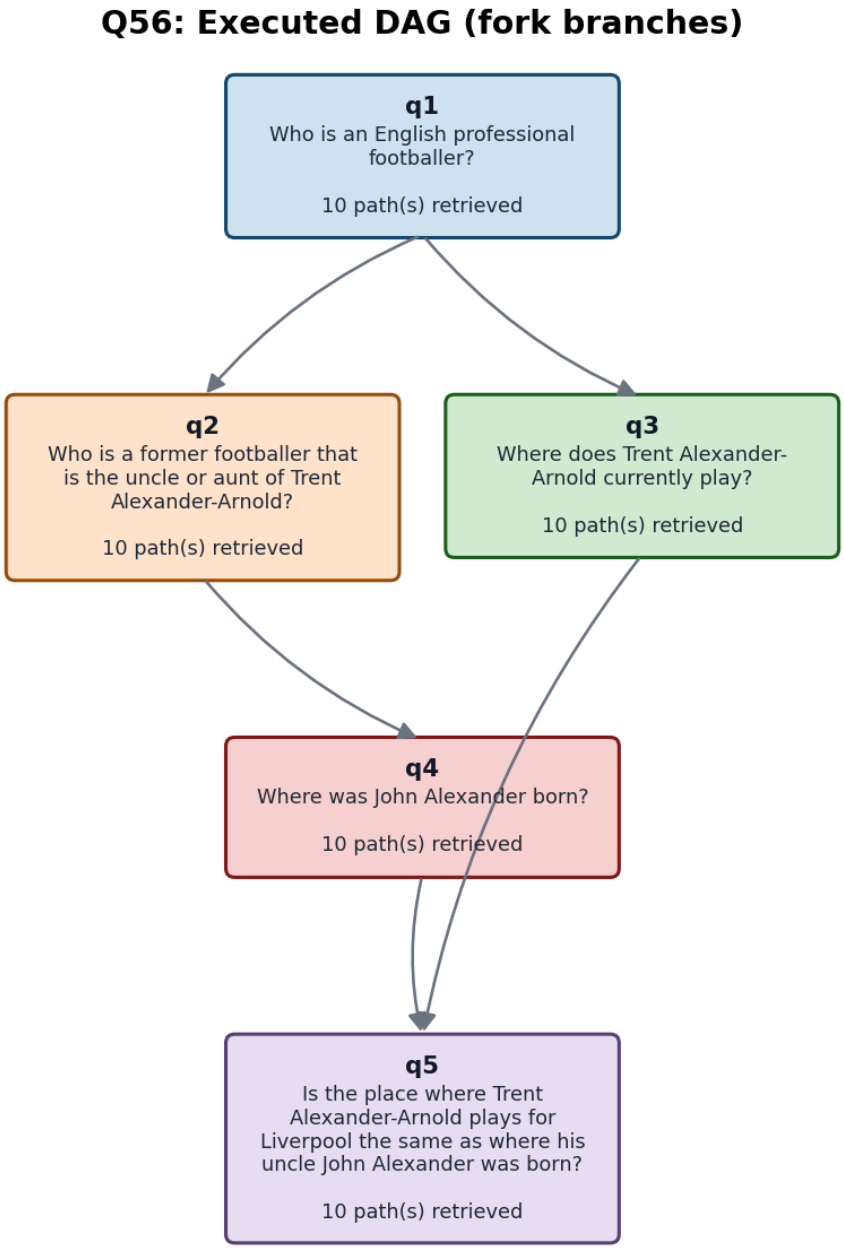
Who is the English Professional Footballer that is the niece of a former footballer born in the same place as where he plays?

Gold answer	Trent Alexander-Arnold
LLM answer	Trent Alexander-Arnold
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is an English professional footballer?	10 path(s)
q2	1	q1	Who is a former footballer that is the uncle or aunt of Trent Alexander-Arnold?	10 path(s)
q3	1	q1	Where does Trent Alexander-Arnold currently play?	10 path(s)
q4	2	q2	Where was John Alexander born?	10 path(s)

Branch	Level	Parents	Refined query	Result
q5	3	q3, q4	Is the place where Trent Alexander-Arnold plays for Liverpool the same as where his uncle John Alexander was born?	10 path(s)

Final context fed to generation (2845 tokens)

```
Query plan:
- q1: Who is an English professional footballer?
- q2: Who is a former footballer that is the uncle or aunt of that English professional footballer?
(depends on: q1)
- q3: Where does that English professional footballer currently play? (depends on: q1)
- q4: Where was that former footballer born? (depends on: q2)
- q5: Is the place where that English professional footballer plays the same as where that former
footballer was born? (depends on: q3, q4)

Level 3:

Subquery: Is the place where Trent Alexander-Arnold plays for Liverpool the same as where his uncle John
Alexander was born?
"John Alexander is the uncle of Liverpool player Trent Alexander-Arnold." -> RELATES_TO -> "Trent
Alexander-Arnold plays as a right-back for Premier League club Liverpool."
Logical Conclusions:
John Alexander is the uncle of Trent Alexander-Arnold, who plays as a right-back for Liverpool.

"John Alexander is the uncle of Liverpool player Trent Alexander-Arnold." -> SUPPORTS -> "Due to the length
of his name, Trent Alexander-Arnold's surname is shortened to Alexander for Premier League fixtures."
Logical Conclusions:
John Alexander being the uncle of Trent Alexander-Arnold and Trent's surname being shortened for Premier
League fixtures are related family and naming facts.

"John Alexander is the uncle of Liverpool player Trent Alexander-Arnold." -> (has ) -> [trent
alexander-arnold (English professional right-back for Liverpool)]

"John Alexander is the uncle of Liverpool player Trent Alexander-Arnold." -> SUPPORTS -> "Trent
Alexander-Arnold is an English professional footballer."
Logical Conclusions:
Trent Alexander-Arnold being an English professional footballer and being the nephew of John Alexander are
related biographical facts.

"John Alexander is the uncle of Liverpool player Trent Alexander-Arnold." -> SUPPORTS -> "John Alexander is
an English former footballer."
Logical Conclusions:
John Alexander being an English former footballer and being the uncle of Trent Alexander-Arnold are related
biographical facts.

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander
is the uncle of Liverpool player Trent Alexander-Arnold.) -> "John Alexander is the uncle of Liverpool
player Trent Alexander-Arnold."

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander
was born on 5 October 1955.) -> "John Alexander was born on 5 October 1955."

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander
moved to Reading.) -> "John Alexander moved to Reading."

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander
holds the post of club secretary at Manchester United.) -> "John Alexander holds the post of club secretary
at Manchester United."
```

Level 2:

Subquery: Where was John Alexander born?
[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander was born in Liverpool.) -> "John Alexander was born in Liverpool."

"John Alexander was born on 5 October 1955." -> (has) -> [john alexander (English former footballer and club secretary)]

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander is the uncle of Liverpool player Trent Alexander-Arnold.) -> "John Alexander is the uncle of Liverpool player Trent Alexander-Arnold."

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander began his football career with a club called Ulysses.) -> "John Alexander began his football career with a club called Ulysses."

[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander is an English former footballer.) -> "John Alexander is an English former footballer."

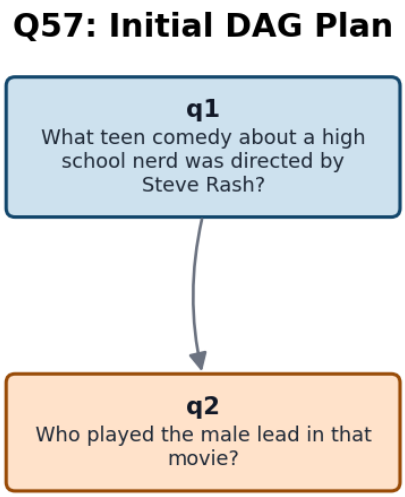
[john alexander (English former footballer and club secretary)] -> (participates in event: John Alexander re …[truncated for report]

Q57 [bridge / hard] 5ac18b585542994ab5c67d76

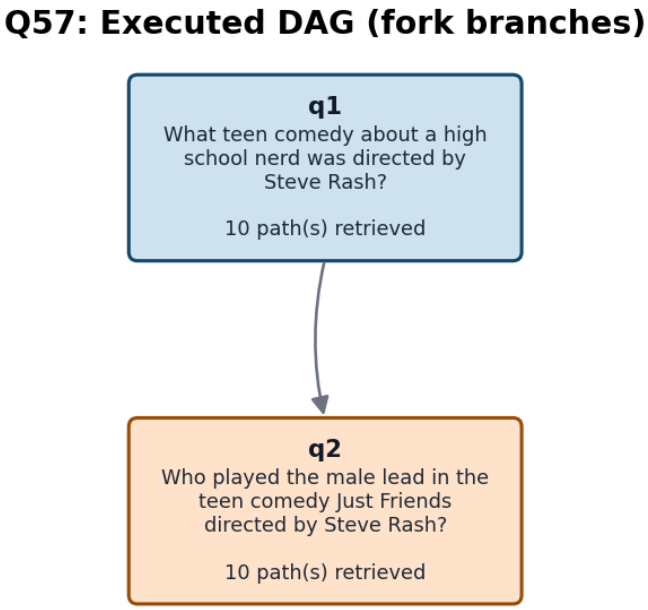
What actor played the male lead in a teen comedy about a high school nerd directed by Steve Rash?

Gold answer	Patrick Dempsey
LLM answer	Ryan Reynolds
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What teen comedy about a high school nerd was directed by Steve Rash?	10 path(s)
q2	1	q1	Who played the male lead in the teen comedy Just Friends directed by Steve Rash?	10 path(s)

Final context fed to generation (1090 tokens)

```
Query plan:
- q1: What teen comedy about a high school nerd was directed by Steve Rash?
- q2: Who played the male lead in that movie? (depends on: q1)

Level 1:

Subquery: Who played the male lead in the teen comedy Just Friends directed by Steve Rash?
"Just Friends is a 2005 American romantic comedy film." -> (has ) -> [Steve Rash film (Film directed by
Steve Rash)]

"Just Friends is a 2005 American romantic comedy film." -> SAME_DIRECTOR -> "Bring It On: All or Nothing is
a 2006 cheerleading comedy film."
Logical Conclusions:
Both films are comedies released in 2005 and 2006 respectively, and are associated with director Steve
Rash.

"Just Friends is a 2005 American romantic comedy film." -> STARRED -> "Just Friends stars Ryan Reynolds."
Logical Conclusions:
Just Friends is a 2005 romantic comedy starring Ryan Reynolds.

"Just Friends is a 2005 American romantic comedy film." -> WRITTEN_BY -> "Just Friends was written by Adam
'Tex' Davis."
Logical Conclusions:
Just Friends is a 2005 romantic comedy written by Adam 'Tex' Davis.

[Ryan Reynolds (Actor in Just Friends)] -> (participates in event: The plot of Just Friends focuses on a
formerly overweight high school nerd (Reyn) -> "The plot of Just Friends focuses on a formerly overweight
high school nerd (Reynolds)."

[Ryan Reynolds (Actor in Just Friends)] -> (participates in event: Just Friends stars Ryan Reynolds.) ->
"Just Friends stars Ryan Reynolds."

[Ryan Reynolds (Actor in Just Friends)] -> (participates in event: Don't Fade Away stars Ryan Kwanten.) ->
"Don't Fade Away stars Ryan Kwanten."

[Ryan Reynolds (Actor in Just Friends)] -> (participates in event: James Garner starred in 'The Notebook'
(2004) with Ryan Gosling and Rachel McAda) -> "James Garner starred in 'The Notebook' (2004) with Ryan
Gosling and Rachel McAdams."

[Ryan Reynolds (Actor in Just Friends)] -> (participates in event: The Notebook stars Ryan Gosling and
Rachel McAdams as a young couple.) -> "The Notebook stars Ryan Gosling and Rachel McAdams as a young
couple."

"American Pie Presents: Band Camp was directed by Steve Rash." -> CONTEXTUALIZES -> "The film American Pie
Presents: Band Camp centers on Matt Stifler."
Logical Conclusions:
Both facts relate to the film 'American Pie Presents: Band Camp', with one stating its director and the
other specifying the central character.
```

Level 0:

Subquery: What teen comedy about a high school nerd was directed by Steve Rash?
[Steve Rash film (Film directed by Steve Rash)] -> (participates in event: The plot of Just Friends focuses on a formerly overweight high school nerd (Reyn) -> "The plot of Just Friends focuses on a formerly overweight high school nerd (Reynolds)."

[Steve Rash film (Film directed by Steve Rash)] -> (participates in event: The nerd attempts to free himself from the friend zone after reconnecting with h) -> "The nerd attempts to free himself from the friend zone after reconnecting with his lifelong crush and best friend (Smart)."

[Steve Rash film (Film directed by Steve Rash)] -> (participates in event: Steve Rash is best known for directing 'The Buddy Holly Story'.) -> "Steve Rash is best known for directing 'The Buddy Holly Story'."

[Steve Rash film (Film directed by Steve Rash)] -> (participates in event: Bob Degus directed 'Pleasantville'.) -> "Bob Degus directed 'Pleasantville'."

[Steve Rash film (Film directed by Steve Rash)] -> (participates in event: Just Friends stars Anna Faris.) -> "Just Friends stars Anna Faris."

[Steve Rash film (Film directed by Steve Rash)] -> (participates in event: The reconnection happens while visiting his hometown for Christmas.) -> "The reconnection happens while visiting his hometown for Christmas."

[Ryan Reynolds (Actor in Just Friends)] -> (participates in event: The plot of Just Friends focuses on a formerly overweight high school nerd (Reyn) -> "The plot of Just Friends focuses on a formerly overweight high school nerd (Reynolds)."

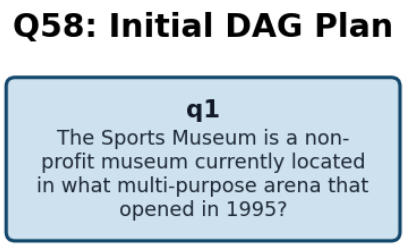
[Ryan Reynolds (Actor in Just Friends)] -> (participates i …[truncated for report]

Q58 [bridge / hard] 5a8241a855429940e5e1a819

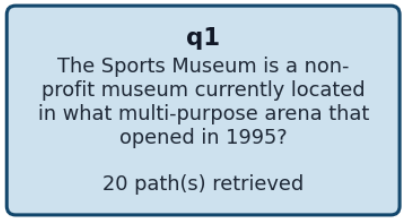
The Sports Museum is a non-profit museum currently located in what multi-purpose arena that opened in 1995?

Gold answer	TD Garden
LLM answer	TD Garden
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	The Sports Museum is a non-profit museum currently located in what multi-purpose arena that opened in 1995?	20 path(s)

Final context fed to generation (867 tokens)

```
[The Sports Museum (non-profit museum in TD Garden)] -> (participates in event: The Sports Museum is currently located in the TD Garden in Boston, Massachusetts) -> "The Sports Museum is currently located in the TD Garden in Boston, Massachusetts." -> (has ) -> [td garden (Multi-purpose arena in Boston)] -> (participates in event: TD Garden opened in 1995.) -> "TD Garden opened in 1995." -> OPENING_YEAR -> "TD Garden is a multi-purpose arena in Boston, Massachusetts."
Logical Conclusions:
TD Garden opened in 1995.
```

"TD Garden opened in 1995." -> REPLACED -> "TD Garden opened as a replacement for the original Boston Garden."
Logical Conclusions:
TD Garden opened as a replacement for the original Boston Garden.

"The Sports Museum is currently located in the TD Garden in Boston, Massachusetts." -> HOUSES -> "The Garden has been home to Celtics and Bruins for twenty years." -> (has) -> [td garden (Multi-purpose arena in Boston)]
Logical Conclusions:
The Garden has been home to Celtics and Bruins for twenty years.

[td garden (Multi-purpose arena in Boston)] -> (participates in event: TD Garden opened as a replacement for the original Boston Garden.) -> "TD Garden opened as a replacement for the original Boston Garden."

"The Sports Museum is currently located in the TD Garden in Boston, Massachusetts." -> (has) -> [Boston (City in Massachusetts)] -> (participates in event: The Garden has been home to Celtics and Bruins for twenty years.) -> "The Garden has been home to Celtics and Bruins for twenty years."

[The Sports Museum (non-profit museum in TD Garden)] -> (participates in event: The Sports Museum may close when an event or game is scheduled in the Garden.) -> "The Sports Museum may close when an event or game is scheduled in the Garden." -> (has) -> [td garden (Multi-purpose arena in Boston)]

[td garden (Multi-purpose arena in Boston)] -> (participates in event: TD Garden is a multi-purpose arena in Boston, Massachusetts.) -> "TD Garden is a multi-purpose arena in Boston, Massachusetts."

[td garden (Multi-purpose arena in Boston)] -> (participates in event: The arena is located directly above the MBTA's North Station.) -> "The arena is located directly above the MBTA's North Station." -> LOCATED_BELOW -> "TD Garden is a multi-purpose arena in Boston, Massachusetts."
Logical Conclusions:
TD Garden is located directly above the MBTA's North Station.

[td garden (Multi-purpose arena in Boston)] -> (participates in event: TD Garden has been known as FleetCenter.) -> "TD Garden has been known as FleetCenter." -> LATER_KNOWN_AS -> "TD Garden is a multi-purpose arena in Boston, Massachusetts."
Logical Conclusions:
TD Garden has been known as FleetCenter.

"TD Garden opened as a replacement for the original Boston Garden." -> SUCCEEDED -> "The Garden has been home to Celtics and Bruins for twenty years."
Logical Conclusions:
TD Garden opened as a replacement for the original Boston Garden and has been home to the Celtics and Bruins for twenty years.

"The arena is located directly above the MBTA's North Station." -> (has) -> [mbta's north station (Transit station located below the arena)]

"The Sports Museum may close when an event or game is scheduled in the Garden." -> LOCATED_AT -> "The Sports Museum eventually moved to Boston to the Garden."
Logical Conclusions:
The Sports Museum may close when an event or game is scheduled in the Garden, and it eventually moved to Boston to the Garden.

[Boston (City in Massachusetts)] -> (participates in event: The Sports Museum eventually moved to Boston to the Garden.) -> "The Sports Museum eventually moved to Boston to the Garden."

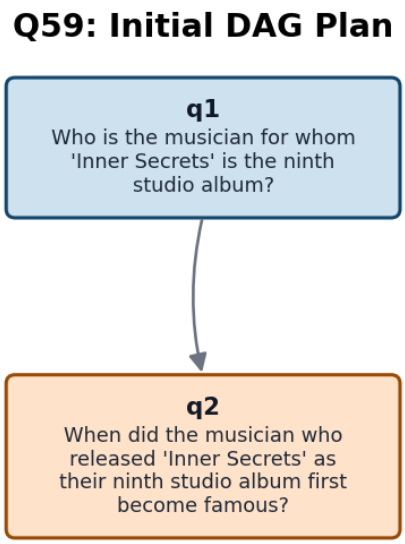
"The Sports Museum may close when an event or game is scheduled in the Garden." -> MAY_CLOSE_DURING -> "The Sports Museum is located on a subway stop."
Logical Conclusions:
The Sports Museum is located on a subway stop and may close when an event or game is scheduled in the Garden.

Q59 [bridge / hard] 5ae612f3554299546bf83065

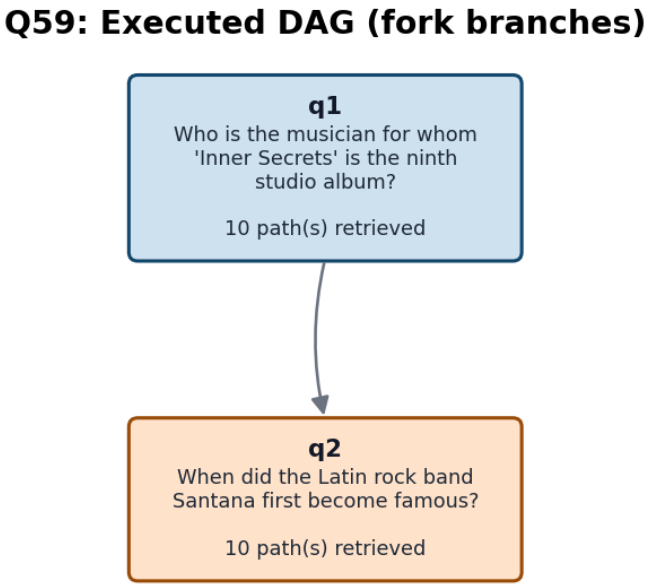
Which musician who first became famous in the late 1960s and early 1970s is Inner Secrets the ninth studio album by

Gold answer	Carlos Santana
LLM answer	Carlos Santana
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is the musician for whom 'Inner Secrets' is the ninth studio album?	10 path(s)
q2	1	q1	When did the Latin rock band Santana first become famous?	10 path(s)

Final context fed to generation (1175 tokens)

```
Query plan:
- q1: Who is the musician for whom 'Inner Secrets' is the ninth studio album?
- q2: When did the musician who released 'Inner Secrets' as their ninth studio album first become famous?
(depends on: q1)

Level 1:

Subquery: When did the Latin rock band Santana first become famous?
[Santana (Latin rock band)] -> (participates in event: Carlos Santana first became famous in the late 1960s
and early 1970s with his ba) -> "Carlos Santana first became famous in the late 1960s and early 1970s with
his band, Santana."

"Carlos Santana first became famous in the late 1960s and early 1970s with his band, Santana." -> PIONEERED
-> "Santana pioneered a fusion of rock and Latin American music."
Logical Conclusions:
Santana pioneered a fusion of rock and Latin American music.

"Carlos Santana first became famous in the late 1960s and early 1970s with his band, Santana." -> PRECEDES
-> "Inner Secrets marks the start of the phase of Santana's career where he moved away from the fusion of
Latin, jazz, rock and blues that marked his previous records."
Logical Conclusions:
Fact 18 states Santana first became famous in the late 1960s/early 1970s, and fact 2 describes a shift in
his career marked by Inner Secrets.

"Carlos Santana first became famous in the late 1960s and early 1970s with his band, Santana." ->
FOLLOWED_BY -> "Santana experienced a resurgence of popularity and critical acclaim in the late 1990s."
Logical Conclusions:
Santana experienced a resurgence of popularity in the late 1990s after their initial fame.

"Carlos Santana first became famous in the late 1960s and early 1970s with his band, Santana." -> FEATURED
-> "The band's sound featured percussion instruments such as timbales and congas."
Logical Conclusions:
The band's sound featured percussion instruments such as timbales and congas.

"Carlos Santana first became famous in the late 1960s and early 1970s with his band, Santana." -> (has ) ->
[carlos santana (Mexican-American musician and bandleader)]

"Carlos Santana first became famous in the late 1960s and early 1970s with his band, Santana." ->
FAME_PERIOD_OF -> "Carlos Santana is a Mexican and American musician."
Logical Conclusions:
Carlos Santana, a Mexican and American musician, first became famous with his band Santana.

"Santana pioneered a fusion of rock and Latin American music." -> PRECEDES -> "Inner Secrets marks the
start of the phase of Santana's career where he moved away from the fusion of Latin, jazz, rock and blues
that marked his previous records."
Logical Conclusions:
Fact 17 describes Santana's pioneering fusion style, while fact 2 describes his later shift away from that
fusion.
```

Level 0:

Subquery: Who is the musician for whom 'Inner Secrets' is the ninth studio album?
"Inner Secrets is the ninth studio album by Santana." -> (has) -> [Santana (Latin rock band)]

"Inner Secrets is the ninth studio album by Santana." -> HAS_QUALITY -> "The album's quality is widely disputed among fans."
Logical Conclusions:
The quality of Inner Secrets is widely disputed among fans.

"Inner Secrets is the ninth studio album by Santana." -> RELEASED_IN -> "Inner Secrets was released in 1978."
Logical Conclusions:
Inner Secrets, the ninth studio album by Santana, was released in 1978.

"Inner Secrets is the ninth studio album by Santana." -> HAS_SINGLE -> "In The Netherlands, 'Well All Right' was released as a single from Inner Secrets."
Logical Conclusions:
'Well All Right' was released as a single from Inner Secrets in The Netherlands.

"Inner Secrets is the ninth studio album by Santana." -> HAS_SINGLE -> "'Stormy' was a hit single from Inner Secrets."
Logical Conclusions:
'Stormy' was a hit single from Inner Secrets.

"Inner Secrets is the ninth studio album by Santana." -> HAS_SINGLE -> "'One Chain (Don't Make No Prison)' was a hit single from Inner Secrets."
Logical Conclusions:
'One Chain (Don't Make No Prison)' was a hit single from Inner Secrets.

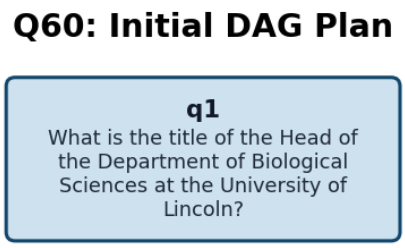
[Jimmy Bryant (Singer and musician)] -> (participates …[truncated for report]

Q60 [bridge / hard] 5a753215554299l6b0164290

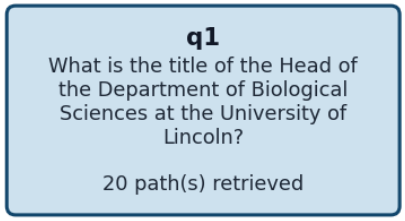
What is the title of the person who is Head of the Department of Biological Sciences at the University of Lincoln

Gold answer	Doctor
LLM answer	Head of the Department of Biological Sciences
EM / F1	**0.00 / 0.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the title of the Head of the Department of Biological Sciences at the University of Lincoln?	20 path(s)

Final context fed to generation (1195 tokens)

```
"Daniel Simon Mills has been Head of the Department of Biological Sciences at the University of Lincoln."
-> (has ) -> [department of biological sciences (Academic department at UCT)] -> (participates in event:
The Department of Biological Sciences was formed from the merger of the Departme) -> "The Department of
Biological Sciences was formed from the merger of the Department of Botany and the Department of Zoology."
-> (has ) -> [department of zoology (Former academic department at UCT)] -> (participates in event: The
merger of the Department of Botany and the Department of Zoology occurred at) -> "The merger of the
Department of Botany and the Department of Zoology occurred at the start of the 2013 academic year."

[department of biological sciences (Academic department at UCT)] -> (participates in event: The merger of
```

the Department of Botany and the Department of Zoology occurred at) -> "The merger of the Department of Botany and the Department of Zoology occurred at the start of the 2013 academic year."

"The Department of Biological Sciences was formed from the merger of the Department of Botany and the Department of Zoology." -> (has) -> [department of botany (Former academic department at UCT)] -> (participates in event: The merger of the Department of Botany and the Department of Zoology occurred at) -> "The merger of the Department of Botany and the Department of Zoology occurred at the start of the 2013 academic year."

"The Department of Biological Sciences was formed from the merger of the Department of Botany and the Department of Zoology." -> OCCURRED_AT -> "The merger of the Department of Botany and the Department of Zoology occurred at the start of the 2013 academic year."
Logical Conclusions:
The merger of the Department of Botany and the Department of Zoology occurred at the start of the 2013 academic year.

"The Department of Biological Sciences was formed from the merger of the Department of Botany and the Department of Zoology." -> MERGED_INTO -> "The Animal Demography Unit is located within the Department of Biological Sciences of UCT."
Logical Conclusions:
The Department of Biological Sciences was formed from the merger of the Department of Botany and the Department of Zoology.

[department of biological sciences (Academic department at UCT)] -> (participates in event: The Animal Demography Unit is located within the Department of Biological Scienc) -> "The Animal Demography Unit is located within the Department of Biological Sciences of UCT."

"The merger of the Department of Botany and the Department of Zoology occurred at the start of the 2013 academic year." -> HOUSES -> "The Animal Demography Unit is located within the Department of Biological Sciences of UCT."
Logical Conclusions:
The Department of Biological Sciences, which houses the Animal Demography Unit, was formed from the merger of Botany and Zoology departments at the start of the 2013 academic year.

"Daniel Simon Mills has been Head of the Department of Biological Sciences at the University of Lincoln." -> FOLLOWS -> "Daniel Simon Mills subsequently became a Principal Lecturer in animal behaviour at the University of Lincoln." -> (has) -> [university of lincoln (Public university in England)] -> (participates in event: Daniel Simon Mills is based at the University of Lincoln, United Kingdom.) -> "Daniel Simon Mills is based at the University of Lincoln, United Kingdom."
Logical Conclusions:
Daniel Simon Mills became Head of the Department of Biological Sciences at the University of Lincoln after being Principal Lecturer.

"Daniel Simon Mills subsequently became a Principal Lecturer in animal behaviour at the University of Lincoln." -> PROMOTED_FROM -> "Daniel Simon Mills subsequently became a Senior Lecturer in animal behaviour at the University of Lincoln."
Logical Conclusions:
Daniel Simon Mills progressed from Senior Lecturer to Principal Lecturer in animal behaviour at the University of Lincoln.

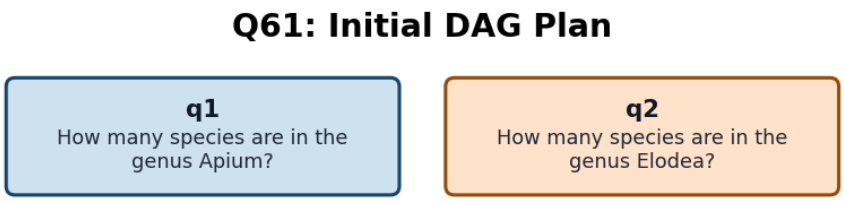
"Daniel Simon Mills subsequently b …[truncated for report]

Q61 [comparison / hard] 5adf39295542992d7e9f92df

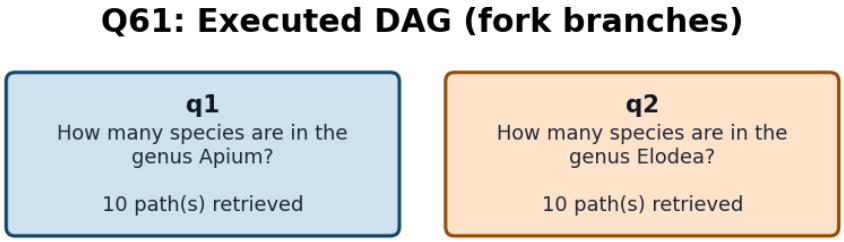
Which genus has more species, Apium or Elodea?

Gold answer	Apium
LLM answer	Apium has more species.
EM / F1	**0.00 / 0.40**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	How many species are in the genus Apium?	10 path(s)
q2	0	—	How many species are in the genus Elodea?	10 path(s)

Final context fed to generation (1094 tokens)

```
Query plan:
- q1: How many species are in the genus Apium?
- q2: How many species are in the genus Elodea?

Level 0:

Subquery: How many species are in the genus Apium?
"Apium is a genus of about 20 species of flowering plants." -> BELONGS_TO -> "Apium is in the family
Apiaceae."
Logical Conclusions:
The genus Apium is in the family Apiaceae.
```

"Apium is a genus of about 20 species of flowering plants." -> (has) -> [apium (genus of flowering plants including celery)]

"Apium is a genus of about 20 species of flowering plants." -> HAS_PROPERTY -> "Apium species grow up to 1 m high."
Logical Conclusions:
Apium is a genus of about 20 species of flowering plants, and Apium species grow up to 1 m high.

"Apium is a genus of about 20 species of flowering plants." -> INCLUDES -> "Apium includes celery and the marshworts."
Logical Conclusions:
Apium includes celery and the marshworts.

"Apium is a genus of about 20 species of flowering plants." -> HAS_SPECIES -> "Apium annuum is one of the 20 species of the genus Apium."
Logical Conclusions:
Apium annuum is one of the 20 species in the genus Apium.

"Apium is a genus of about 20 species of flowering plants." -> HAS_DISTRIBUTION -> "Apium has a subcosmopolitan distribution in Europe, Asia, Africa, South America and Australia."
Logical Conclusions:
Apium has a subcosmopolitan distribution in Europe, Asia, Africa, South America and Australia.

[Apium graveolens (Species including celery and celeriac)] -> (participates in event: Apium graveolens is an edible species.) -> "Apium graveolens is an edible species."

[Apium graveolens (Species including celery and celeriac)] -> (participates in event: Apium graveolens includes the commercially important vegetables celery, celeriac) -> "Apium graveolens includes the commercially important vegetables celery, celeriac and Chinese celery."

[Apium graveolens (Species including celery and celeriac)] -> (participates in event: Apium bermejoi is not closely related to the wild form of celery, A. graveolens.) -> "Apium bermejoi is not closely related to the wild form of celery, A. graveolens."

[Apium bermejoi (Critically endangered flowering plant)] -> (participates in event: Apium bermejoi has fewer than 100 individuals left.) -> "Apium bermejoi has fewer than 100 individuals left."

Subquery: How many species are in the genus Elodea?
[elodea (Genus of aquatic plants)] -> (participates in event: Elodea is a genus of 6 species of aquatic plants.) -> "Elodea is a genus of 6 species of aquatic plants."

"Elodea was described as a genus in 1803." -> DESCRIBES -> "Elodea is a genus of 6 species of aquatic plants."
Logical Conclusions:
Elodea was described as a genus in 1803.

"Elodea was described as a genus in 1803." -> (has) -> [elodea (Genus of aquatic plants)]

[elodea (Genus of aquatic plants)] -> (participates in event: 'Elodea' is native to North and South America.) -> "'Elodea' is native to North and South America."

[elodea (Genus of aquatic plants)] -> (participates in event: Elodea is often called the waterweeds.) -> "Elodea is often called the waterweeds."

"Elodea was described as a genus in 1803." -> CONTEXTUALIZES -> "Elodea is often called the waterweeds."
Logical Conclusions:
Elodea was described as a genus in 1803, and Elodea is often called the waterweeds.

"Elodea was described as a genus in 1803." -> CONTEXTUALIZES -> "'Elodea' is native to North and South America."
Logical Conclusions:
Elodea was described as a genus in 1803, and Elodea is native to North and South America.

"Elodea was described as a genus in 1803." -> CONTEXTUALIZES -> "'Elodea' is also widely used as aquarium vegetation."
Logical Conclusions:
Elodea was described as a genus in 1803, and Elodea is also widely used as aquarium vegetation.

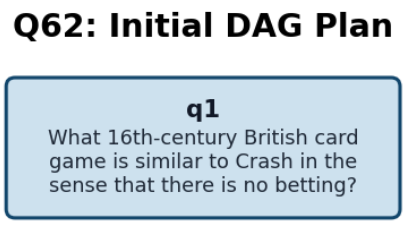
"Elodea was described as a genus in 1803." -> CONTEXTUALIZES -> "Elodea lives in fresh water."
Logical Conclusions:
Elodea was described as a genus in 1803, and Elodea lives in fresh water.

Q62 [bridge / hard] 5ab4774d5542991751b4d781

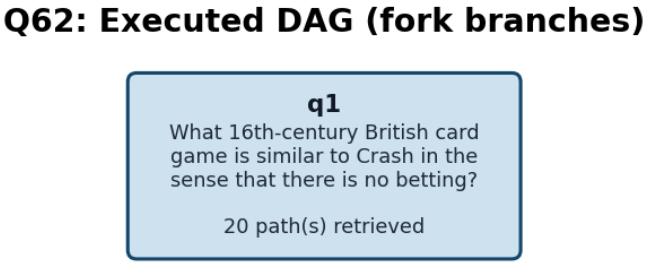
Crash is similar to what 16th century British card game in the sense that there is no betting?

Gold answer	In Crash, there is no betting, as in Brag
LLM answer	Nine-card Brag
EM / F1	**0.00 / 0.18**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What 16th-century British card game is similar to Crash in the sense that there is no betting?	20 path(s)

Final context fed to generation (510 tokens)

```
"Crash is a British card game extension of Nine-card Brag." -> DERIVES_FROM -> "In Crash, there is no
betting." -> SUPPORTS -> "In Crash, points are gained over successive deals."
Logical Conclusions:
Crash is an extension of Nine-card Brag with no betting.
Both describe Crash, with one stating points are gained over successive deals and the other stating there
is no betting.

"Crash is a British card game extension of Nine-card Brag." -> (has ) -> [crash (British card game
extension of Nine-card Brag)] -> (participates in event: In Crash, there is no betting.) -> "In Crash,
there is no betting."
```

"Crash is a British card game extension of Nine-card Brag." -> SUPPORTS -> "In Crash, points are gained over successive deals."
Logical Conclusions:
Both describe Crash, with one stating points are gained over successive deals and the other stating it is an extension of Nine-card Brag.

[crash (British card game extension of Nine-card Brag)] -> (participates in event: In Crash, points are gained over successive deals.) -> "In Crash, points are gained over successive deals."

"In Crash, there is no betting." -> HAS_RULES -> "In Crash, players aim to reach a total of 11 points."
Logical Conclusions:
In Crash, there is no betting, and players aim to reach a total of 11 points.

"Crash is a British card game extension of Nine-card Brag." -> EXTENDS -> "In Crash, players aim to reach a total of 11 points."
Logical Conclusions:
Crash is a British card game extension of Nine-card Brag, and in Crash players aim to reach 11 points.

[crash (British card game extension of Nine-card Brag)] -> (participates in event: In Crash, players aim to reach a total of 11 points.) -> "In Crash, players aim to reach a total of 11 points."

[crash (British card game extension of Nine-card Brag)] -> (participates in event: 'Crash' won the 2005 Academy Award for Best Picture.) -> "'Crash' won the 2005 Academy Award for Best Picture."

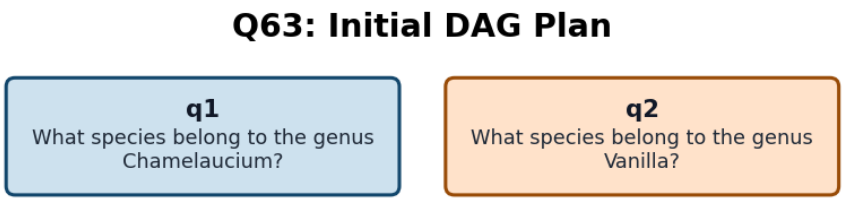
"Crash is a British card game extension of Nine-card Brag." -> (has) -> [nine-card brag (Base game for Crash)]

Q63 [comparison / hard] 5ac41b9e5542995c82c4acc4

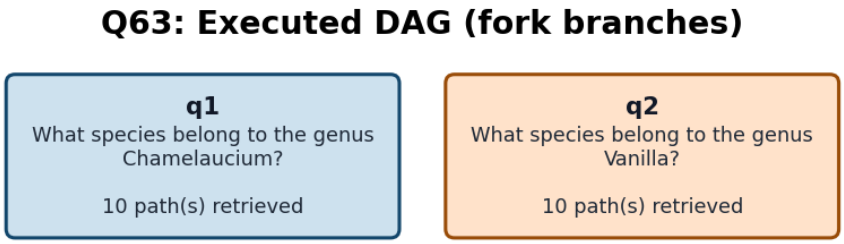
what species of plants are Chamelaucium and Vanilla derived from?

Gold answer	flowers
LLM answer	Chamelaucium uncinatum, Vanilla planifolia
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What species belong to the genus Chamelaucium?	10 path(s)
q2	0	—	What species belong to the genus Vanilla?	10 path(s)

Final context fed to generation (1143 tokens)

```
Query plan:
- q1: What species belong to the genus Chamelaucium?
- q2: What species belong to the genus Vanilla?

Level 0:

Subquery: What species belong to the genus Chamelaucium?
"The most well-known species of Chamelaucium is the Geraldton Wax, Chamelaucium uncinatum." -> EXEMPLIFIES
-> "Chamelaucium belongs to the myrtle family Myrtaceae."
Logical Conclusions:
```

Fact 11 gives the family of Chamelaucium, while fact 15 names the most well-known species.

[chamelaucium (Genus of shrubs endemic to Western Australia)] -> (participates in event: The most well-known species of Chamelaucium is the Geraldton Wax, Chamelaucium u) -> "The most well-known species of Chamelaucium is the Geraldton Wax, Chamelaucium uncinatum."

"The most well-known species of Chamelaucium is the Geraldton Wax, Chamelaucium uncinatum." -> EXEMPLIFIES -> "Chamelaucium is a genus of shrubs."
Logical Conclusions:
The most well-known species of Chamelaucium, a genus of shrubs, is the Geraldton Wax.

"Chamelaucium belongs to the myrtle family Myrtaceae." -> CLASSIFIES -> "Chamelaucium is a genus of shrubs."
Logical Conclusions:
Fact 11 gives the family of Chamelaucium, while fact 12 states it is a genus of shrubs.

"Chamelaucium belongs to the myrtle family Myrtaceae." -> (has) -> [chamelaucium (Genus of shrubs endemic to Western Australia)]

"Chamelaucium belongs to the myrtle family Myrtaceae." -> CLASSIFIES -> "Chamelaucium is a genus of shrubs endemic to south western Western Australia."
Logical Conclusions:
Fact 9 describes the endemic region of Chamelaucium, while fact 11 gives its family.

"The most well-known species of Chamelaucium is the Geraldton Wax, Chamelaucium uncinatum." -> EXEMPLIFIES -> "Chamelaucium is a genus of shrubs endemic to south western Western Australia."
Logical Conclusions:
Fact 9 describes the endemic region of Chamelaucium, while fact 15 names the most well-known species.

"The most well-known species of Chamelaucium is the Geraldton Wax, Chamelaucium uncinatum." -> IDENTIFIES -> "Chamelaucium uncinatum is cultivated widely for its large attractive flowers."
Logical Conclusions:
Chamelaucium uncinatum, the Geraldton Wax, is cultivated widely for its large attractive flowers.

Subquery: What species belong to the genus Vanilla?
"Vanilla forms a genus of about 110 species." -> SPECIFIES -> "The most widely known member of Vanilla is the flat-leaved vanilla (V. planifolia)."
Logical Conclusions:
Fact 21 states Vanilla forms a genus of about 110 species, and fact 24 identifies V. planifolia as the most widely known member.

"Vanilla forms a genus of about 110 species." -> HAS_ALIAS -> "Vanilla is known as the vanilla orchids."
Logical Conclusions:
Vanilla forms a genus of about 110 species known as the vanilla orchids.

"Vanilla forms a genus of about 110 species." -> EXEMPLIFIES -> "V. planifolia is native to Mexico."
Logical Conclusions:
Vanilla forms a genus of about 110 species, and V. planifolia is native to Mexico.

"Vanilla forms a genus of about 110 species." -> (has) -> [Vanilla (Orchid genus cultivated for pods)]

"Vanilla forms a genus of about 110 species." -> IS_SPECIES_OF -> "Vanilla is a flowering plant genus."
Logical Conclusions:
Vanilla forms a genus of about 110 species.

"Vanilla forms a genus of about 110 species." -> HAS_USE -> "Vanilla is the only orchid widely used for industrial purposes in flavoring products."
Logical Conclusions:
Vanilla forms a genus of about 110 species, and Vanilla is the only orchid widely used for industrial purposes in flavoring products.

[Vanilla (Orchid genus cultivated for pods)] -> (participates in event: The most widely known member of

Vanilla is the flat-leaved vanilla (V. planifoli) -> "The most widely known member of Vanilla is the flat-leaved vanilla (V. planifolia)."

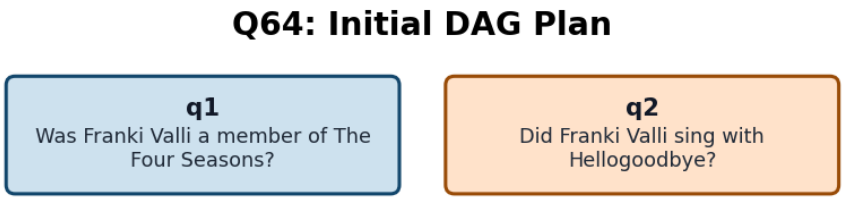
"Vanilla is a flavoring derived from orchids of the genus Vanilla." -> CONTEXTUALIZES -> "Vanilla forms a genus of about 110 species."
Logical Conclusions:
Vanilla is a flavoring derived from orchids of the …[truncated for report]

Q64 [comparison / hard] 5ab5b6c1554299637185c5e5

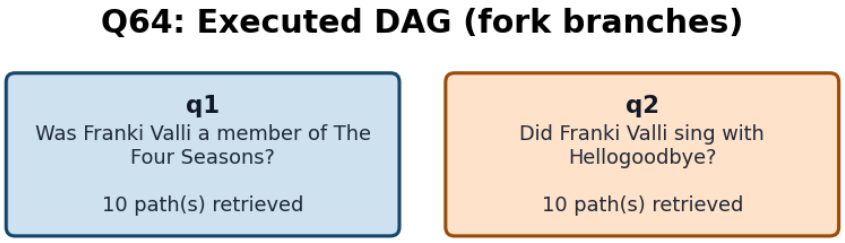
Did Franki Valli sing with Hellogoodbye or The Four Seasons?

Gold answer	The Four Seasons
LLM answer	Yes, Franki Valli sang with The Four Seasons. No, Franki Valli did not sing with Hellogoodbye.
EM / F1	**0.00 / 0.24**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Was Franki Valli a member of The Four Seasons?	10 path(s)
q2	0	—	Did Franki Valli sing with Hellogoodbye?	10 path(s)

Final context fed to generation (1249 tokens)

```
Query plan:
- q1: Was Franki Valli a member of The Four Seasons?
- q2: Did Franki Valli sing with Hellogoodbye?

Level 0:

Subquery: Was Franki Valli a member of The Four Seasons?
[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: Bobby Fox originated the
role of Franki Valli in the Australian production of "J) -> "Bobby Fox originated the role of Franki Valli
in the Australian production of "Jersey Boys"."
```

[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: Roy Roman performed lead with Franki Valli.) -> "Roy Roman performed lead with Franki Valli."

[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: Bob Gaudio was a bandmate of Frankie Valli in The Four Seasons.) -> "Bob Gaudio was a bandmate of Frankie Valli in The Four Seasons."

[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: The owner/founder of Private Stock Records wanted only Frankie Valli's name on t) -> "The owner/founder of Private Stock Records wanted only Frankie Valli's name on the label for 'My Eyes Adored You'."

[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: Wonder Who? recordings generally feature falsetto singing by Frankie Valli.) -> "Wonder Who? recordings generally feature falsetto singing by Frankie Valli."

[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: The song "Can't Take My Eyes Off You" was among Frankie Valli's biggest hits.) -> "The song "Can't Take My Eyes Off You" was among Frankie Valli's biggest hits."

[The Four Seasons (American rock and pop band)] -> (participates in event: Other names used by The Four Seasons included Frankie Valli (as a solo artist) a) -> "Other names used by The Four Seasons included Frankie Valli (as a solo artist) and The Valli Boys."

[The Four Seasons (American rock and pop band)] -> (participates in event: Nick Massi was on electric bass and bass vocals in the Four Seasons.) -> "Nick Massi was on electric bass and bass vocals in the Four Seasons."

"Valli was selected in the 1st round, 6th overall, by the Lions in the 2006 CFL draft." -> IDENTIFIES -> "He played four full seasons with the Clan from 2002 to 2005."
Logical Conclusions:
He played four full seasons with the Clan from 2002 to 2005 and was selected in the 1st round, 6th overall, by the Lions in the 2006 CFL draft.

[Frankie Valli Works (Album and songs by Frankie Valli)] -> (participates in event: Valli was the sixth solo LP album by Frankie Valli of The Four Seasons.) -> "Valli was the sixth solo LP album by Frankie Valli of The Four Seasons."

Subquery: Did Franki Valli sing with Hellogoodbye?
[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: Roy Roman performed lead with Franki Valli.) -> "Roy Roman performed lead with Franki Valli."

[Frankie Valli (Singer and lead of The Four Seasons)] -> (participates in event: Bobby Fox originated the role of Franki Valli in the Australian production of "J) -> "Bobby Fox originated the role of Franki Valli in the Australian production of "Jersey Boys"."

[hellogoodbye (Pop rock band formed in 2001)] -> (participates in event: Hellogoodbye toured as the opening act for Paramore's The Self-Titled Tour.) -> "Hellogoodbye toured as the opening act for Paramore's The Self-Titled Tour."

[hellogoodbye (Pop rock band formed in 2001)] -> (participates in event: The band was formed by singer Forrest Kline.) -> "The band was formed by singer Forrest Kline."

[Hellogoodbye (band that opened for Paramore)] -> (participates in event: Hellogoodbye toured as the opening act for Paramore's The Self-Titled Tour.) -> "Hellogoodbye toured as the opening act for Paramore's The Self-Titled Tour."

[hellogoodbye (Pop rock band formed in 2001)] -> (participates in event: They were signed to Drive-Thru Records.) -> "They were signed to Drive-Thru Records."

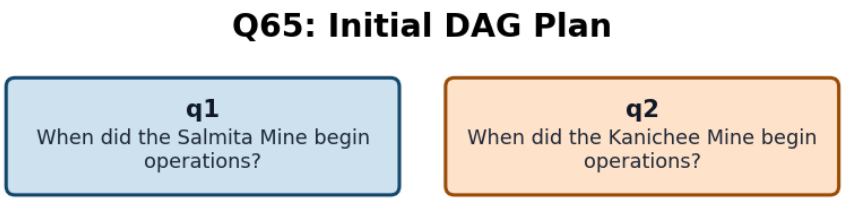
[hellogoodbye (Pop rock band formed in 2001)] -> (participates …[truncated for report]

Q65 [comparison / hard] 5a739323554299623ed4abff

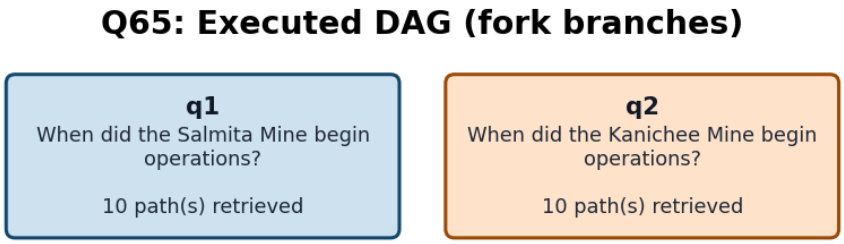
Which famous mine was in service first, the Salmita Mine or the Kanichee Mine?

Gold answer	Kanichee Mine
LLM answer	The Kanichee Mine was in service first.
EM / F1	0.00 / 0.50
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	When did the Salmita Mine begin operations?	10 path(s)
q2	0	—	When did the Kanichee Mine begin operations?	10 path(s)

Final context fed to generation (1397 tokens)

```
Query plan:
- q1: When did the Salmita Mine begin operations?
- q2: When did the Kanichee Mine begin operations?

Level 0:

Subquery: When did the Salmita Mine begin operations?
"The Salmita Mine operated during 1983 to 1987." -> FOLLOWS -> "Underground exploration at the Salmita Mine
was carried out in 1951-1952."
Logical Conclusions:
Underground exploration at the Salmita Mine in 1951-1952 preceded its operation from 1983 to 1987.
```

"The Salmita Mine operated during 1983 to 1987." -> (has) -> [Salmita Mine (Gold producer in Northwest Territories)]

"The Salmita Mine operated during 1983 to 1987." -> FOLLOWS -> "The Salmita Mine was reactivated for exploration by Giant Yellowknife Mines Limited in 1975."
Logical Conclusions:
The Salmita Mine was reactivated for exploration in 1975, and later operated from 1983 to 1987.

"The Salmita Mine operated during 1983 to 1987." -> FOLLOWS -> "The Salmita Mine used the old camp and milling plant of the abandoned Tundra Mine."
Logical Conclusions:
The Salmita Mine used the old camp and milling plant of the Tundra Mine during its operation from 1983 to 1987.

"The Salmita Mine operated during 1983 to 1987." -> OPERATION_OF -> "The Salmita Mine was a gold producer in the Northwest Territories, Canada."
Logical Conclusions:
The Salmita Mine operated during 1983 to 1987.

[Salmita Mine (Gold producer in Northwest Territories)] -> (participates in event: Underground exploration at the Salmita Mine was carried out in 1951-1952.) -> "Underground exploration at the Salmita Mine was carried out in 1951-1952."

"The Salmita Mine operated during 1983 to 1987." -> FOLLOWS -> "The Tundra Mine was located a few kilometres to the south of the Salmita Mine."
Logical Conclusions:
The Tundra Mine, located near the Salmita Mine, was abandoned before the Salmita Mine operated from 1983 to 1987.

[Salmita Mine (Gold producer in Northwest Territories)] -> (participates in event: The Salmita Mine entered production in 1983.) -> "The Salmita Mine entered production in 1983."

[Salmita Mine (Gold producer in Northwest Territories)] -> (participates in event: The Salmita Mine deposit was first discovered in 1945.) -> "The Salmita Mine deposit was first discovered in 1945."

[Salmita Mine (Gold producer in Northwest Territories)] -> (participates in event: The Salmita Mine was reactivated for exploration by Giant Yellowknife Mines Limi) -> "The Salmita Mine was reactivated for exploration by Giant Yellowknife Mines Limited in 1975."

Subquery: When did the Kanichee Mine begin operations?
"The most recent operation of the Kanichee Mine was from 1973 to 1976." -> IS_PART_OF -> "During the 20th century, the Kanichee Mine operated and closed down at least three times."
Logical Conclusions:
The most recent operation of the Kanichee Mine was from 1973 to 1976, part of its multiple closures.

"The most recent operation of the Kanichee Mine was from 1973 to 1976." -> (has) -> [Kanichee Mine (Abandoned base and precious metal mine in Ontario)]

"The most recent operation of the Kanichee Mine was from 1973 to 1976." -> FOLLOWS -> "The Kanichee Mine zone has been explored and mined discontinuously from as early as 1910."
Logical Conclusions:
The Kanichee Mine zone has been explored and mined discontinuously from as early as 1910, with the most recent operation from 1973 to 1976.

"The most recent operation of the Kanichee Mine was from 1973 to 1976." -> ALSO_KNOWN_AS -> "The Kanichee Mine is also less commonly known as the Ajax Mine."
Logical Conclusions:
The most recent operation of the Kanichee Mine, also known as the Ajax Mine, was from 1973 to 1976.

"The most recent operation of the Kanichee Mine was from 1973 to 1976." -> LOCATED_IN -> "The Kanichee Mine is located in the Temagami region of northeastern Ontario, Canada."
Logical Conclusions:

The most recent operation of the Kanichee Mine in the Temagami region was from 1973 to 1976.

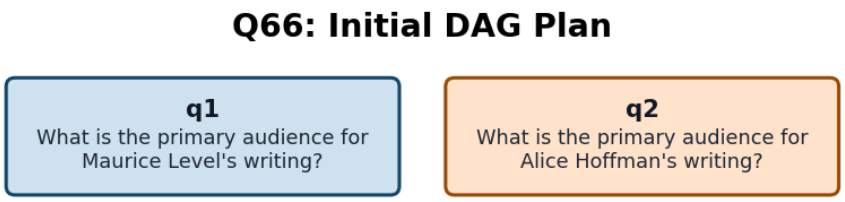
"The most recent operation of the Kanichee Mine was from 1973 to 1976." -> CONTRIBUTE …[truncated for report]

Q66 [comparison / hard] 5abaa954554299232ef4a362

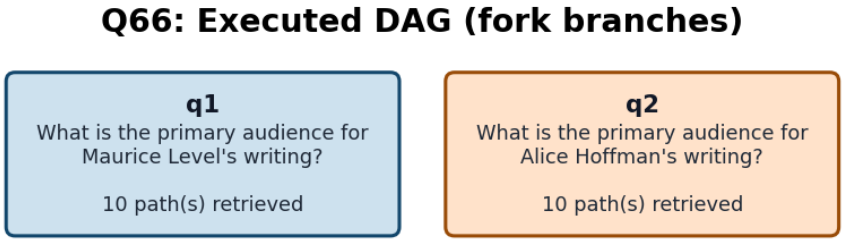
Are the writing from Maurice Level and Alice Hoffman focused on different audieces?

Gold answer	yes
LLM answer	Yes
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the primary audience for Maurice Level's writing?	10 path(s)
q2	0	—	What is the primary audience for Alice Hoffman's writing?	10 path(s)

Final context fed to generation (1106 tokens)

```
Query plan:
- q1: What is the primary audience for Maurice Level's writing?
- q2: What is the primary audience for Alice Hoffman's writing?

Level 0:

Subquery: What is the primary audience for Maurice Level's writing?
"Maurice Level's short stories were printed regularly in the columns of Paris newspapers." -> SUPPORTS ->
"Maurice Level was a French writer of fiction and drama."
```

Logical Conclusions:
Maurice Level's short stories being printed in newspapers is consistent with his career as a French writer.

"Maurice Level's short stories were printed regularly in the columns of Paris newspapers." -> (has) -> [maurice level (French writer of macabre fiction)]

[maurice level (French writer of macabre fiction)] -> (participates in event: Maurice Level was a French writer of fiction and drama.) -> "Maurice Level was a French writer of fiction and drama."

"Maurice Level's short stories were printed regularly in the columns of Paris newspapers." -> PUBLISHED -> "Maurice Level specialized in short stories of the macabre."
Logical Conclusions:
Level's macabre short stories were printed regularly in Paris newspapers.

"Maurice Level was born on August 29, 1875." -> BIRTH_DATE -> "Maurice Level was a French writer of fiction and drama."
Logical Conclusions:
Maurice Level was born on August 29, 1875.

[maurice level (French writer of macabre fiction)] -> (participates in event: Maurice Level's short stories were sometimes staged by 'le Théâtre du Grand-Guig) -> "Maurice Level's short stories were sometimes staged by 'le Théâtre du Grand-Guignol'."

[maurice level (French writer of macabre fiction)] -> (participates in event: Lady Harrington is based on a novel by Maurice Level.) -> "Lady Harrington is based on a novel by Maurice Level."

[maurice level (French writer of macabre fiction)] -> (participates in event: Maurice Level specialized in short stories of the macabre.) -> "Maurice Level specialized in short stories of the macabre."

[maurice level (French writer of macabre fiction)] -> (participates in event: Maurice Level was born on August 29, 1875.) -> "Maurice Level was born on August 29, 1875."

Subquery: What is the primary audience for Alice Hoffman's writing?
"Alice Hoffman is a young-adult writer." -> SUPPORTS -> "Alice Hoffman is a children's writer."
Logical Conclusions:
Both facts categorize Alice Hoffman's writing, with one as children's writer and the other as young-adult writer.

"Alice Hoffman is a young-adult writer." -> SUPPORTS -> "Alice Hoffman referred to 'Angels' as 'a mixture'."
Logical Conclusions:
Alice Hoffman referred to 'Angels' as 'a mixture', and she is a young-adult writer.

"Alice Hoffman is a young-adult writer." -> (has) -> [alice hoffman (American novelist and children's writer)]

"Alice Hoffman is a young-adult writer." -> SUPPORTS -> "'Aquamarine' is a novel by Alice Hoffman."
Logical Conclusions:
'Aquamarine' is a novel by Alice Hoffman, and she is a young-adult writer.

"Alice Hoffman is a young-adult writer." -> SUPPORTS -> "Many of Alice Hoffman's works contain non-standard romances and relationships."
Logical Conclusions:
Alice Hoffman is a young-adult writer, and many of her works contain non-standard romances and relationships.

"Alice Hoffman is a young-adult writer." -> SUPPORTS -> "Many of Alice Hoffman's works contain elements of irony."
Logical Conclusions:
Alice Hoffman is a young-adult writer, and many of her works contain elements of irony.

[Sebastian Shaw (Reader in The Great War documentary)] -> (participates in event: Readings in 'The Great

War' were by Marius Goring, Ralph Richardson, Cyril Luckh) -> "Readings in 'The Great War' were by Marius Goring, Ralph Richardson, Cyril Luckham, Sebastian Shaw, and Emlyn Williams."

[John Pattison (Editorial staff at Barrie & Rockliff)] -> (participates in event: Barrie & Rockliff's editorial staff included John Pattison.) -> "Barrie & Rockliff's editorial staff included John Pattison."

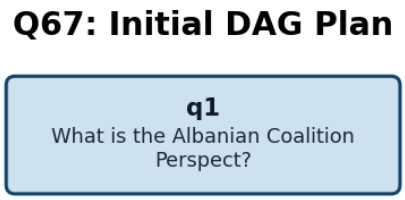
[hollywood, california (City where In Living Col …[truncated for report]

Q67 [bridge / hard] 5a711c535542994082a3e59b

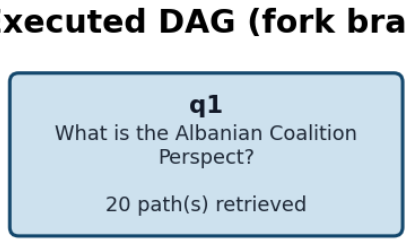
Albanians in Montenegro which make up 4.91 of the county's total population could be members of the Albanian Coalation Perspect which is a what?

Gold answer	Albanian political party in Montenegro
LLM answer	Albanian political party in Montenegro
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the Albanian Coalition Perspect?	20 path(s)

Final context fed to generation (1321 tokens)

```
"The Albanian Coalition 'Perspective' is an Albanian political party in Montenegro." -> (has ) -> [albanian
coalition "perspective" (Albanian political party in Montenegro)]

"The Albanian Coalition 'Perspective' is an Albanian political party in Montenegro." -> CONTEXTUALIZES ->
"Relations between the state and the Albanian minority strained after the Orlov let event." -> (has ) ->
[Sports figures (Athletes and sport shooters)] -> (participates in event: The group's main operative was
the destroying of cultural and religious building) -> "The group's main operative was the destroying of
cultural and religious buildings (Orthodox) in Albanian-inhabited territory."
Logical Conclusions:
Fact 24 mentions an Albanian political party in Montenegro, and fact 12 discusses strained relations
between the state and the Albanian minority in Montenegro.
```

[Sports figures (Athletes and sport shooters)] -> (participates in event: The silver medal in the 800 metres at the 1963 Pan American Games was behind fel) -> "The silver medal in the 800 metres at the 1963 Pan American Games was behind fellow Canadian Don Bertoia."

[Sports figures (Athletes and sport shooters)] -> (participates in event: The arrest of the group occurred on the night of 10/11 September 2006.) -> "The arrest of the group occurred on the night of 10/11 September 2006."

"Relations between the state and the Albanian minority strained after the Orlov let event." -> FOLLOWS_FROM -> "Orlov let was the name of an anti-terrorist operation conducted by the Montenegrin police."
Logical Conclusions:
The Orlov let operation strained relations between the state and the Albanian minority.

[Sports figures (Athletes and sport shooters)] -> (participates in event: Rolf Martin Anker Hagen is a native of Oslo.) -> "Rolf Martin Anker Hagen is a native of Oslo."

"Relations between the state and the Albanian minority strained after the Orlov let event." -> (has) -> [Albanians (Ethnic group native to Albania)] -> (participates in event: The largest Albanian communities in Macedonia are in the municipalities of Tetov) -> "The largest Albanian communities in Macedonia are in the municipalities of Tetovo."

[Sports figures (Athletes and sport shooters)] -> (participates in event: Siegmar Ohlemann represented Canada.) -> "Siegmar Ohlemann represented Canada."

"Relations between the state and the Albanian minority strained after the Orlov let event." -> GENERALIZES -> "Relations between the state and the Albanian minority strained after the event."
Logical Conclusions:
Both facts discuss strained relations after an event, with one mentioning the state and Albanian minority and the other specifying the Orlov let event.

[Albanians (Ethnic group native to Albania)] -> (participates in event: Relations between the state and the Albanian minority strained after the event.) -> "Relations between the state and the Albanian minority strained after the event."

[Albanians (Ethnic group native to Albania)] -> (participates in event: According to Noel Malcolm, the Albanians did not welcome Serbian rule.) -> "According to Noel Malcolm, the Albanians did not welcome Serbian rule."

[Albanians (Ethnic group native to Albania)] -> (participates in event: Albanians share a common culture, ancestry, and speak the Albanian language as a) -> "Albanians share a common culture, ancestry, and speak the Albanian language as a native tongue."

[Albanians (Ethnic group native to Albania)] -> (participates in event: Over the centuries, populations of ethnic Serbs and Albanians tended to shift fo) -> "Over the centuries, populations of ethnic Serbs and Albanians tended to shift following territorial handovers."

[Sports figures (Athletes and sport shooters)] -> (participates in event: Orlov let was the name of an anti-terrorist operation conducted by the Montenegr) -> "Orlov let was the name of an anti-terrorist operation conducted by the Montenegrin police."

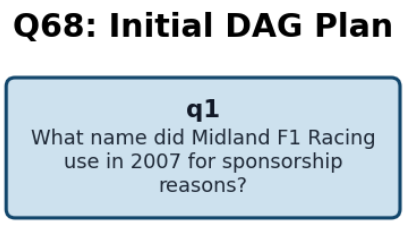
[Albanians (Ethnic group native to Albania)] -> (participate …[truncated for report]

Q68 [bridge / hard] 5ab3c270554299233954ff9e

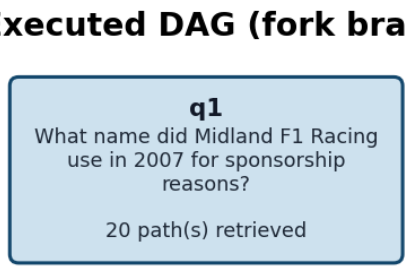
Midland F1 Racing was a racing team that, in 2007, competed under what name for sponsorship reasons?

Gold answer	Spyker F1
LLM answer	Etihad Aldar Spyker F1 Team
EM / F1	**0.00 / 0.57**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What name did Midland F1 Racing use in 2007 for sponsorship reasons?	20 path(s)

Final context fed to generation (1382 tokens)

```
"In 2007, the team competed as Spyker F1." -> EXEMPLIFIES -> "The Spyker F1 Team was known as the Etihad
Aldar Spyker F1 Team for sponsorship reasons." -> (has ) -> [Spyker F1 (Formula One racing team)] ->
(participates in event: The take-over from Midland F1 happened part way through 2006.) -> "The take-over
from Midland F1 happened part way through 2006."
Logical Conclusions:
The Spyker F1 Team was also known as the Etihad Aldar Spyker F1 Team for sponsorship reasons.

[Spyker F1 (Formula One racing team)] -> (participates in event: Spyker F1 took over from the struggling
Midland F1 team.) -> "Spyker F1 took over from the struggling Midland F1 team."

[Midland F1 Racing (Formula One constructor and team)] -> (participates in event: The Spyker F1 Team was
```

created by Spyker Cars after their buyout of the short-li) -> "The Spyker F1 Team was created by Spyker Cars after their buyout of the short-lived Midland F1 team." -> RESULT_OF -> "In 2007, the team competed as Spyker F1." -> (has) -> [Spyker F1 (Formula One racing team)]
Logical Conclusions:
The Spyker F1 Team was created by Spyker Cars after their buyout of Midland F1.

"The take-over from Midland F1 happened part way through 2006." -> (has) -> [Midland F1 Racing (Formula One constructor and team)] -> (participates in event: Midland F1 Racing was also known as MF1 Racing.) -> "Midland F1 Racing was also known as MF1 Racing."

[Spyker F1 (Formula One racing team)] -> (participates in event: In 2007, the team competed as Spyker.) -> "In 2007, the team competed as Spyker."

[Spyker F1 (Formula One racing team)] -> (participates in event: The F8-VII was the only car constructed by Spyker F1 in their own right.) -> "The F8-VII was the only car constructed by Spyker F1 in their own right."

"The Spyker F1 Team was created by Spyker Cars after their buyout of the short-lived Midland F1 team." -> SUPPORTS -> "In 2007, the team competed as Spyker."
Logical Conclusions:
In 2007, the team competed as Spyker and the Spyker F1 Team was created by Spyker Cars after their buyout of Midland F1.

[Midland F1 Racing (Formula One constructor and team)] -> (participates in event: Towards the end of the 2006 season, the team was sold to Spyker Cars N.V.) -> "Towards the end of the 2006 season, the team was sold to Spyker Cars N.V." -> SUPPORTS -> "Spyker F1 took over from the struggling Midland F1 team."
Logical Conclusions:
Spyker F1 took over from the struggling Midland F1 team and towards the end of the 2006 season, the team was sold to Spyker Cars N.V.

"Spyker F1 took over from the struggling Midland F1 team." -> (has) -> [Midland F1 Racing (Formula One constructor and team)]

"In 2007, the team competed as Spyker." -> SUPPORTS -> "The Spyker F1 Team was known as the Etihad Aldar Spyker F1 Team for sponsorship reasons."
Logical Conclusions:
In 2007, the team competed as Spyker and was known as the Etihad Aldar Spyker F1 Team for sponsorship reasons.

"In 2007, the team competed as Spyker F1." -> MODIFIES -> "The change to the Spyker name was accompanied by a switch in racing livery from the red and white previously used by Midland to an orange and silver scheme." -> SUPPORTS -> "Orange is the national colour and the auto racing colour of the Netherlands."
Logical Conclusions:
The change to the Spyker name was accompanied by a switch to an orange and silver racing livery.
Orange is the national colour of the Netherlands, which relates to the livery choice.

"The change to the Spyker name was accompanied by a switch in racing livery from the red and white previously used by Midland to an orange and silver scheme." -> CONTRASTS -> "The record company subsequently replaced the lost drawings with less than stellar car photos."
Logical Conclusions:
The Spyker F1 team changed its livery, while a record company replaced lost drawings with car photos.

"The Jordan team continued into 2006 under the name Midland F1 Racing." -> (has) -> [Midland F1 Racing (Formula One constructor and team)]

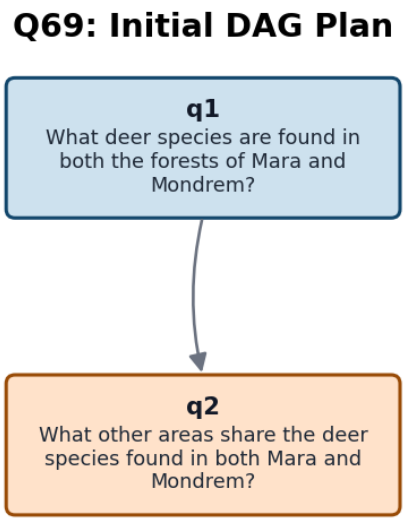
"T …[truncated for report]

Q69 [bridge / hard] 5ae54c085542992663a4f1c4

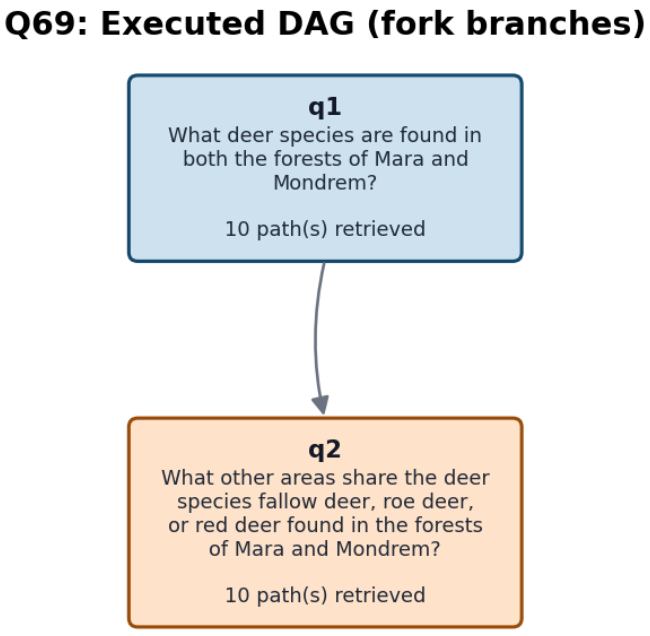
What are some other areas that share a common deer species with the forests of Mara and Mondrem?

Gold answer	New Zealand, Canada, the United States, the Falkland Islands, and Peru.
LLM answer	The Forests of Mara and Mondrem, the Gowy, the Weaver, the Mersey, and Nantwich.
EM / F1	**0.00 / 0.11**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What deer species are found in both the forests of Mara and Mondrem?	10 path(s)
q2	1	q1	What other areas share the deer species fallow deer, roe deer, or red deer found in the forests of Mara and Mondrem?	10 path(s)

Final context fed to generation (1410 tokens)

```
Query plan:
- q1: What deer species are found in both the forests of Mara and Mondrem?
- q2: What other areas share the deer species found in both Mara and Mondrem? (depends on: q1)

Level 1:

Subquery: What other areas share the deer species fallow deer, roe deer, or red deer found in the forests of Mara and Mondrem?
"Game in the Forests of Mara and Mondrem included fallow deer." -> CO_OCCURS_WITH -> "Game in the Forests of Mara and Mondrem included roe deer."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer and roe deer.

"Game in the Forests of Mara and Mondrem included fallow deer." -> CO_OCCURS_WITH -> "Game in the Forests of Mara and Mondrem included red deer."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included red deer and fallow deer.

"Game in the Forests of Mara and Mondrem included fallow deer." -> ASSOCIATES -> "The Forests of Mara and Mondrem stretched from the Mersey in the north almost to Nantwich in the south."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer.

"Game in the Forests of Mara and Mondrem included fallow deer." -> ASSOCIATES -> "The Forests of Mara and Mondrem stretched from the Gowy in the west to the Weaver in the east."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer.

"Game in the Forests of Mara and Mondrem included fallow deer." -> (has ) -> [forests of mara and mondrem (Adjacent medieval hunting forests in Cheshire)]

"Game in the Forests of Mara and Mondrem included fallow deer." -> SIMILAR_TO -> "Game in the Forests of Mara and Mondrem included wild boar."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer and wild boar.

[red deer (One of the largest deer species)] -> (participates in event: Game in the Forests of Mara and Mondrem included red deer.) -> "Game in the Forests of Mara and Mondrem included red deer."

"In the 11th century, the Forests of Mara and Mondrem extended to over 60 sqmi." -> (has ) -> [forests of mara and mondrem (Adjacent medieval hunting forests in Cheshire)]

"In the 11th century, the Forests of Mara and Mondrem extended to over 60 sqmi." -> DESCRIBES -> "The Forests of Mara and Mondrem were adjacent medieval forests in Cheshire, England."
Logical Conclusions:
The Forests of Mara and Mondrem, which were adjacent medieval forests in Cheshire, England, extended to over 60 square miles in the 11th century.
```

"In the 11th century, the Forests of Mara and Mondrem extended to over 60 sqmi." -> HISTORY -> "Mara and Mondrem might earlier have been an Anglo-Saxon hunting forest."
Logical Conclusions:
Mara and Mondrem might earlier have been an Anglo-Saxon hunting forest and extended to over 60 sqmi in the 11th century.

Level 0:

Subquery: What deer species are found in both the forests of Mara and Mondrem?
"Game in the Forests of Mara and Mondrem included fallow deer." -> CO_OCCURS_WITH -> "Game in the Forests of Mara and Mondrem included roe deer."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer and roe deer.

"Game in the Forests of Mara and Mondrem included fallow deer." -> CO_OCCURS_WITH -> "Game in the Forests of Mara and Mondrem included red deer."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included red deer and fallow deer.

"Game in the Forests of Mara and Mondrem included fallow deer." -> SIMILAR_TO -> "Game in the Forests of Mara and Mondrem included wild boar."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer and wild boar.

"Game in the Forests of Mara and Mondrem included fallow deer." -> ASSOCIATES -> "The Forests of Mara and Mondrem stretched from the Mersey in the north almost to Nantwich in the south."
Logical Conclusions:
Game in the Forests of Mara and Mondrem included fallow deer.

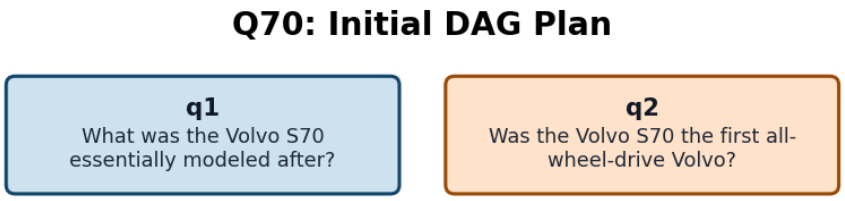
"Game in the Forests of Mara and Mondrem included fallow deer." -> ASSOCIATES -> "The Forests of Mara and Mondrem stretched from the Gowy in th …[truncated for report]

Q70 [bridge / hard] 5a7220f055429971e9dc92ac

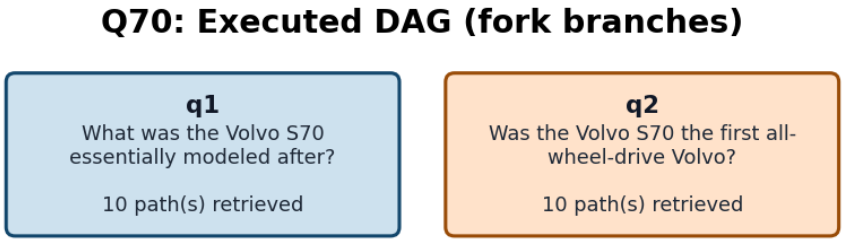
What was the Volvo S70 essentially modeled after and was the first all wheel drive Volvo?

Gold answer	Volvo 850
LLM answer	The Volvo S70 was essentially modeled after the Volvo 850 saloon, and it was not the first all-wheel-drive Volvo.
EM / F1	**0.00 / 0.22**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What was the Volvo S70 essentially modeled after?	10 path(s)
q2	0	—	Was the Volvo S70 the first all-wheel-drive Volvo?	10 path(s)

Final context fed to generation (1337 tokens)

```
Query plan:
- q1: What was the Volvo S70 essentially modeled after?
- q2: Was the Volvo S70 the first all-wheel-drive Volvo?

Level 0:

Subquery: What was the Volvo S70 essentially modeled after?
[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo S70 was essentially a
facelifted 850 saloon.) -> "The Volvo S70 was essentially a facelifted 850 saloon."
```

"The Volvo S70 was produced by Volvo Cars from 1996 to 2000." -> DERIVED_FROM -> "The Volvo S70 was essentially a facelifted 850 saloon."
Logical Conclusions:
The Volvo S70 was produced from 1996 to 2000 and was essentially a facelifted 850 saloon.

[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo S70 was a compact executive car.) -> "The Volvo S70 was a compact executive car."

"The Volvo S70 was produced by Volvo Cars from 1996 to 2000." -> (has) -> [volvo v70 (Replacement for Volvo 850 estate)]

"The Volvo S70 was produced by Volvo Cars from 1996 to 2000." -> PRODUCTION_PERIOD -> "The Volvo S70 was a compact executive car."
Logical Conclusions:
The Volvo S70 was a compact executive car produced by Volvo Cars from 1996 to 2000.

"The Volvo S70 was produced by Volvo Cars from 1996 to 2000." -> SUCCEEDS -> "The Volvo S70 was replaced with the Volvo S60."
Logical Conclusions:
The Volvo S70 was produced from 1996 to 2000 and was then replaced by the Volvo S60.

"The Volvo S70 was produced by Volvo Cars from 1996 to 2000." -> SUPPORTS -> "The Volvo 850 range was replaced for 1997 by the Volvo V70."
Logical Conclusions:
The Volvo 850 range was replaced for 1997 by the Volvo V70, and the S70 was produced from 1996 to 2000.

"The Volvo S70 was produced by Volvo Cars from 1996 to 2000." -> SUPPORTS -> "The Volvo 850 range was replaced for 1997 by the Volvo S70."
Logical Conclusions:
The Volvo 850 range was replaced for 1997 by the Volvo S70, which was produced from 1996 to 2000.

"The same basic chassis of the Volvo P80 platform was used as the underpinnings for the C70." -> MODIFIED_BY -> "The C70's chassis was heavily modified by TWR."
Logical Conclusions:
The C70's chassis, based on the P80 platform, was heavily modified by TWR.

Subquery: Was the Volvo S70 the first all-wheel-drive Volvo?
[volvo (Swedish automaker producing cars and buses)] -> (participates in event: The Volvo 480 was the first front-wheel drive car made by Volvo.) -> "The Volvo 480 was the first front-wheel drive car made by Volvo."

[volvo (Swedish automaker producing cars and buses)] -> (participates in event: It was the first front-wheel drive car made by the automaker.) -> "It was the first front-wheel drive car made by the automaker."

[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo S70 was produced by Volvo Cars from 1996 to 2000.) -> "The Volvo S70 was produced by Volvo Cars from 1996 to 2000."

[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo S70 was a compact executive car.) -> "The Volvo S70 was a compact executive car."

[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo S70 was essentially a facelifted 850 saloon.) -> "The Volvo S70 was essentially a facelifted 850 saloon."

[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo S70 was replaced with the Volvo S60.) -> "The Volvo S70 was replaced with the Volvo S60."

[volvo v70 (Replacement for Volvo 850 estate)] -> (participates in event: The Volvo 850 range was replaced for 1997 by the Volvo S70.) -> "The Volvo 850 range was replaced for 1997 by the Volvo S70."

[volvo s80 (First car to debut the P2 platform)] -> (participates in event: The Volvo P2 platform debuted with the 1998 Volvo S80.) -> "The Volvo P2 platform debuted with the 1998 Volvo S80."

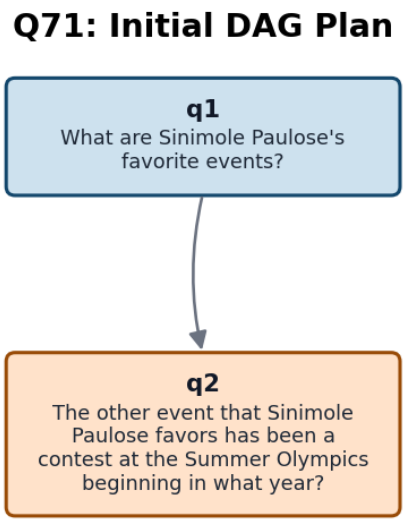
[volvo s90 (executive sedan manufactured since 2016)] -> (participates in event: The Volvo C30 is a variant of the Volvo S40/V50/C70 range.) -> "The Volvo C30 is a variant of the Volvo S40/V50/C70 range."
…[truncated for report]

Q71 [bridge / hard] 5a80b35c5542992bc0c4a7b8

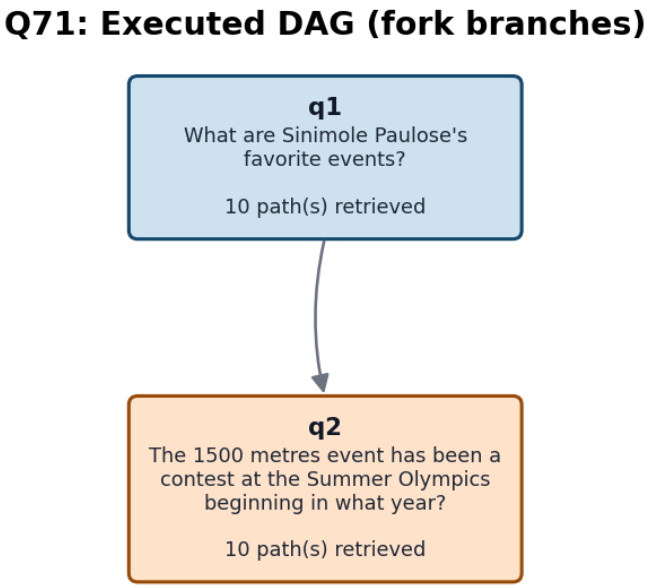
Sinimole Paulose's favorite events are the 800 metres and another event that has been a contest at the Summer Olympics begining in what year?

Gold answer	1896
LLM answer	1896
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What are Sinimole Paulose's favorite events?	10 path(s)
q2	1	q1	The 1500 metres event has been a contest at the Summer Olympics beginning in what year?	10 path(s)

Final context fed to generation (1516 tokens)

```
Query plan:
- q1: What are Sinimole Paulose's favorite events?
- q2: The other event that Sinimole Paulose favors has been a contest at the Summer Olympics beginning in
what year? (depends on: q1)

Level 1:

Subquery: The 1500 metres event has been a contest at the Summer Olympics beginning in what year?
"The 1500 metres has been contested at the Summer Olympics since 1896." -> PRECEDES -> "The 1500 metres has
been contested at the World Championships in Athletics since 1983."
Logical Conclusions:
Both facts provide historical context for the 1500 metres event, with one noting its Olympic debut in 1896
and the other its World Championships debut in 1983.

"The 1500 metres has been contested at the Summer Olympics since 1896." -> FOLLOWS -> "The 2016 Summer
Olympics were held from 5 to 21 August 2016."
Logical Conclusions:
The 2016 Summer Olympics were held from 5 to 21 August 2016 and the 1500 metres has been contested at the
Summer Olympics since 1896.

"The 1500 metres has been contested at the Summer Olympics since 1896." -> INCLUDES -> "The 1500 metres is
the foremost middle distance track event in athletics."
Logical Conclusions:
The 1500 metres, the foremost middle distance track event, has been contested at the Summer Olympics since
1896.

"The 1500 metres has been contested at the Summer Olympics since 1896." -> SUPPORTS -> "The 1500 metres is
equivalent to 1.5 kilometers."
Logical Conclusions:
Fact 5 states the 1500 metres has been contested at the Summer Olympics since 1896, while fact 9 provides a
metric conversion for the distance.

[Sports figures (Athletes and sport shooters)] -> (participates in event: The three events Siegmar Ohlemann
competed in at the 1960 Summer Olympics were t) -> "The three events Siegmar Ohlemann competed in at the
1960 Summer Olympics were the 800 metres, 4 × 100 metres, and 4 × 400 metres."

[Sports figures (Athletes and sport shooters)] -> (participates in event: The silver medal in the 800
metres at the 1963 Pan American Games was behind fel) -> "The silver medal in the 800 metres at the 1963 Pan
American Games was behind fellow Canadian Don Bertoia."

[Caster Semenya (South African middle-distance runner)] -> (participates in event: Semenya won the gold
medal in the 800 metre event at the 2016 Summer Olympics.) -> "Semenya won the gold medal in the 800 metre
event at the 2016 Summer Olympics."

[Caster Semenya (South African middle-distance runner)] -> (participates in event: Caster Semenya won
silver medals at the 2012 Summer Olympics in the 800 metres.) -> "Caster Semenya won silver medals at the
2012 Summer Olympics in the 800 metres."

[Winter Olympic Games (Event hosting snowboarding competitions)] -> (participates in event: Snowboarding
has been contested at the Winter Olympic Games since the 1998 Winte) -> "Snowboarding has been contested at
```

the Winter Olympic Games since the 1998 Winter Olympics."

[Winter Olympic Games (Event hosting snowboarding competitions)] -> (participates in event: Snowboarding was the only new sport or discipline added between 1992 and 2002 no) -> "Snowboarding was the only new sport or discipline added between 1992 and 2002 not to have been a previous medal or demonstration event."

Level 0:

Subquery: What are Sinimole Paulose's favorite events?
"Sinimol Paulose's favorite events are 800 metres and 1500 metres." -> SUPPORTS -> "Sinimol Paulose represented India at the Beijing Olympics."
Logical Conclusions:
Sinimol Paulose represented India at the Beijing Olympics and her favorite events are 800 metres and 1500 metres.

"Sinimol Paulose's favorite events are 800 metres and 1500 metres." -> SUPPORTS -> "In 2009, Sinimol Paulose received the Arjuna Award for her contribution to Indian athletics."
Logical Conclusions:
Sinimol Paulose received the Arjuna Award in 2009 and her favorite events are 800 metres and 1500 metres.

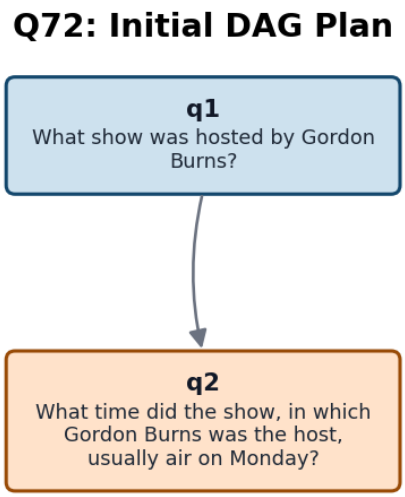
"Sinimol Paulose's favorite events are 800 metres and 1500 metres." -> IS_FAVORED_BY -> "Sinimol Paulose is an Indian middle distan …[truncated for report]

Q72 [bridge / hard] 5abced075542993a06baf9ac

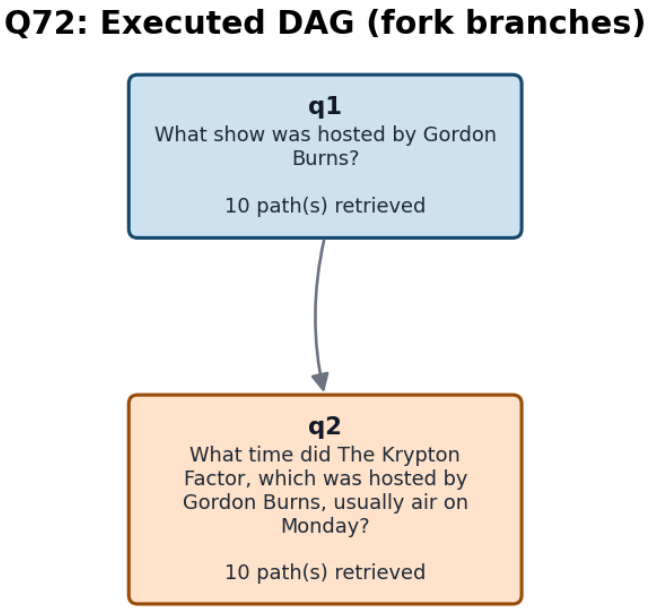
What time did the show, in which Gordon Burns was the host, usually air on Monday?

Gold answer	7pm
LLM answer	7pm
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What show was hosted by Gordon Burns?	10 path(s)
q2	1	q1	What time did The Krypton Factor, which was hosted by Gordon Burns, usually air on Monday?	10 path(s)

Final context fed to generation (1146 tokens)

```
Query plan:
- q1: What show was hosted by Gordon Burns?
- q2: What time did the show, in which Gordon Burns was the host, usually air on Monday? (depends on: q1)

Level 1:

Subquery: What time did The Krypton Factor, which was hosted by Gordon Burns, usually air on Monday?
"The Krypton Factor was usually broadcast on the ITV network on Mondays at 7pm." -> IDENTIFIES -> "The
Krypton Factor was hosted by Gordon Burns."
Logical Conclusions:
Both facts describe the same British game show, with one providing its broadcast schedule and the other
naming its host.

"The Krypton Factor was usually broadcast on the ITV network on Mondays at 7pm." -> BROADCAST_ON -> "The
Krypton Factor was broadcast on ITV."
Logical Conclusions:
The Krypton Factor was usually broadcast on ITV on Mondays at 7pm.

"The Krypton Factor was usually broadcast on the ITV network on Mondays at 7pm." -> IDENTIFIES -> "The
Krypton Factor is a British game show."
Logical Conclusions:
Both facts describe the same British game show, with one stating it is a game show and the other providing
its broadcast schedule.

"The Krypton Factor was usually broadcast on the ITV network on Mondays at 7pm." -> IDENTIFIES -> "The
Krypton Factor was produced by Granada."
Logical Conclusions:
Both facts describe the same British game show, with one providing its broadcast schedule and the other
naming its producer.

"The Krypton Factor was hosted by Gordon Burns." -> IDENTIFIES -> "The Krypton Factor was broadcast on
ITV."
Logical Conclusions:
Both facts describe the same British game show, with one stating its broadcast network and the other naming
its host.

"The Krypton Factor was hosted by Gordon Burns." -> HOSTED_BY -> "The Krypton Factor is a British game
show."
Logical Conclusions:
The Krypton Factor was hosted by Gordon Burns.

"The Krypton Factor was hosted by Gordon Burns." -> IDENTIFIES -> "The Krypton Factor was produced by
Granada."
Logical Conclusions:
Both facts describe the same British game show, with one naming its host and the other naming its producer.

[the krypton factor (Game show hosted by Gordon Burns)] -> (participates in event: Gordon Burns was the
host of 'The Krypton Factor' for its original 18-year run () -> "Gordon Burns was the host of 'The Krypton
Factor' for its original 18-year run (1977–1995)."
```

[Java (island with the cape Tanjung Layar)] -> (participates in event: Java Head has deep water close to the shore.) -> "Java Head has deep water close to the shore."

Level 0:

Subquery: What show was hosted by Gordon Burns?
"The Krypton Factor was hosted by Gordon Burns." -> HOSTED_BY -> "The Krypton Factor is a British game show."
Logical Conclusions:
The Krypton Factor was hosted by Gordon Burns.

"The Krypton Factor was hosted by Gordon Burns." -> IDENTIFIES -> "The Krypton Factor was broadcast on ITV."
Logical Conclusions:
Both facts describe the same British game show, with one stating its broadcast network and the other naming its host.

"The Krypton Factor was hosted by Gordon Burns." -> IDENTIFIES -> "The Krypton Factor was usually broadcast on the ITV network on Mondays at 7pm."
Logical Conclusions:
Both facts describe the same British game show, with one providing its broadcast schedule and the other naming its host.

"The Krypton Factor was hosted by Gordon Burns." -> IDENTIFIES -> "The Krypton Factor was produced by Granada."
Logical Conclusions:
Both facts describe the same British game show, with one naming its host and the other naming its producer.

[gordon burns (Host of Password on ITV and other shows)] -> (participates in event: Gordon Burns was the host of 'The Krypton Factor' for its original 18-year run () -> "Gordon Burns was the host of 'The Krypton Factor' for its original 18-year run (1977–1995)."

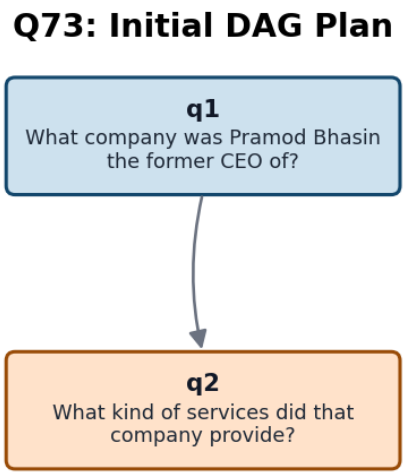
"Gordon Burns was the chief anchorman for the BBC regional news programme 'North West Tonight' from January 1997 to October 2011." -> FOLLOWS -> "Gordon Burns was the host of 'The Krypton Factor' for its original 18-year run (1977–199 …[truncated for report]

Q73 [bridge / hard] 5a86fde6554299211dda2b7f

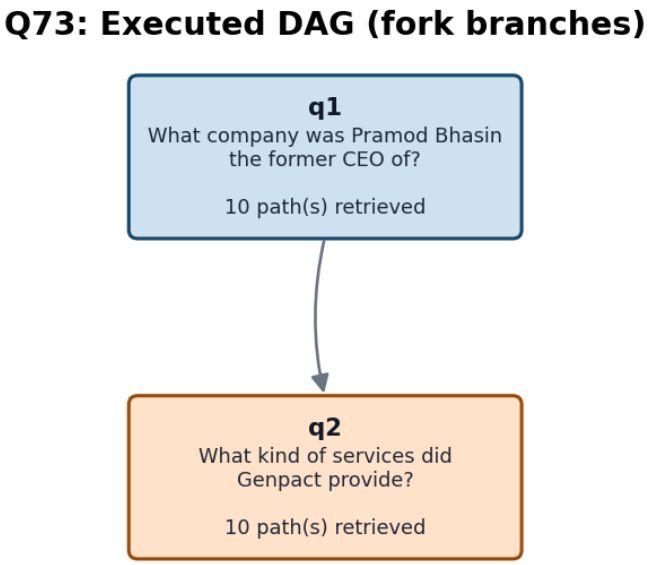
Pramod Bhasin was the former CEO of a company that provided what kind of services?

Gold answer	IT
LLM answer	consulting services
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What company was Pramod Bhasin the former CEO of?	10 path(s)
q2	1	q1	What kind of services did Genpact provide?	10 path(s)

Final context fed to generation (1243 tokens)

```
Query plan:
- q1: What company was Pramod Bhasin the former CEO of?
- q2: What kind of services did that company provide? (depends on: q1)

Level 1:

Subquery: What kind of services did Genpact provide?
"Genpact provides consulting services to hundreds of clients." -> SERVICES_FROM -> "Genpact is a global
business process management and services and IT corporation."
Logical Conclusions:
Genpact provides consulting services to hundreds of clients.

"Genpact provides consulting services to hundreds of clients." -> EXEMPLIFIES -> "Genpact is a global
business process management and services and IT corporation."
Logical Conclusions:
Genpact's identity as a global business process management corporation supports its ability to provide
consulting services to hundreds of clients.

"Genpact provides consulting services to hundreds of clients." -> PROVIDES -> "Genpact's clients include
approximately one-fifth of the Fortune Global 500."
Logical Conclusions:
Genpact's clients include approximately one-fifth of the Fortune Global 500 and it provides consulting
services to hundreds of clients.

"Genpact provides consulting services to hundreds of clients." -> SUPPORTS -> "In 2016, Genpact had more
than 77,000 employees."
Logical Conclusions:
Genpact's consulting services to hundreds of clients are supported by its workforce of more than 77,000
employees in 2016.

"Genpact provides consulting services to hundreds of clients." -> (has ) -> [genpact (Global business
process management and IT corporation)]

"Genpact provides consulting services to hundreds of clients." -> CONTRIBUTES_TO -> "In 2016, Genpact
reported net revenues of US$2.57 billion."
Logical Conclusions:
Genpact's consulting services to hundreds of clients contribute to its 2016 net revenues of US$2.57
billion.

[genpact (Global business process management and IT corporation)] -> (participates in event: Genpact is a
global business process management and services and IT corporation.) -> "Genpact is a global business
process management and services and IT corporation."

[genpact (Global business process management and IT corporation)] -> (participates in event: Genpact is
India's largest business process outsourcing (BPO) company.) -> "Genpact is India's largest business
process outsourcing (BPO) company."

[genpact (Global business process management and IT corporation)] -> (participates in event: Genpact
operates in 20 countries.) -> "Genpact operates in 20 countries."

Level 0:
```

Subquery: What company was Pramod Bhasin the former CEO of?
[pramod bhasin (Former President and CEO of Genpact)] -> (participates in event: Pramod Bhasin stepped down as President and CEO of Genpact.) -> "Pramod Bhasin stepped down as President and CEO of Genpact."

[pramod bhasin (Former President and CEO of Genpact)] -> (participates in event: Pramod Bhasin became non-executive Vice Chairman of Genpact in 2011.) -> "Pramod Bhasin became non-executive Vice Chairman of Genpact in 2011."

[Intamin (company Giovanola subcontracted for)] -> (participates in event: During its last years in business, Giovanola fashioned steel behind the scenes f) -> "During its last years in business, Giovanola fashioned steel behind the scenes for companies such as Intamin and Bolliger & Mabillard."

[Intamin (company Giovanola subcontracted for)] -> (participates in event: During its last years in business, Giovanola fashioned steel behind the scenes f) -> "During its last years in business, Giovanola fashioned steel behind the scenes for companies such as Bolliger & Mabillard."

[Intamin (company Giovanola subcontracted for)] -> (participates in event: During its last years in business, Giovanola fashioned steel behind the scenes f) -> "During its last years in business, Giovanola fashioned steel behind the scenes for companies such as Intamin."

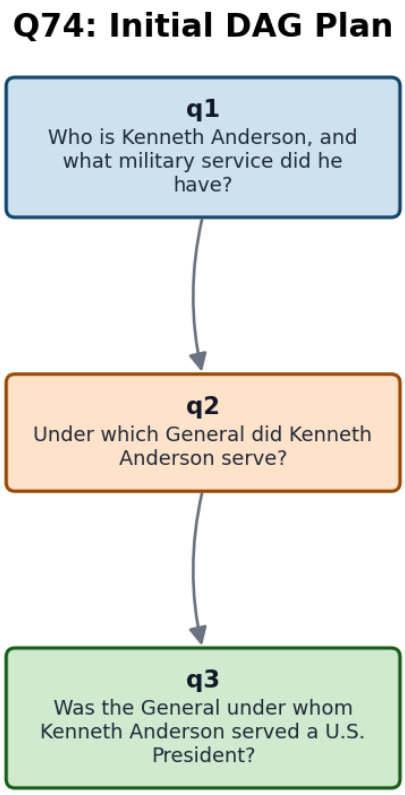
[Intamin (company Giovanola subcontracted for)] -> (participates in event: Giovanola served as a subcontractor to Intamin.) -> "Giovanola served as a subcontractor to Intamin …[truncated for report]

Q74 [bridge / hard] 5a8459aa5542996488c2e51e

Kenneth Anderson served under the General who was which President of the United States?

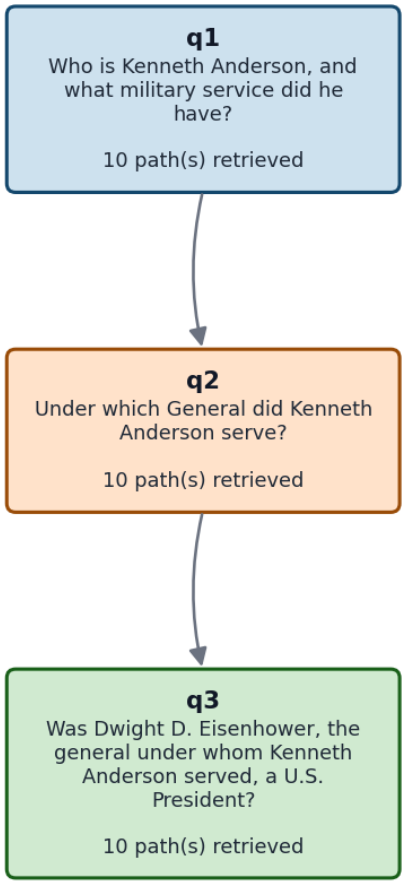
Gold answer	34th
LLM answer	Dwight D. Eisenhower, the 34th President of the United States.
EM / F1	**0.00 / 0.22**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)

Q74: Executed DAG (fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is Kenneth Anderson, and what military service did he have?	10 path(s)
q2	1	q1	Under which General did Kenneth Anderson serve?	10 path(s)
q3	2	q2	Was Dwight D. Eisenhower, the general under whom Kenneth Anderson served, a U.S. President?	10 path(s)

Final context fed to generation (1548 tokens)

```
Query plan:
- q1: Who is Kenneth Anderson, and what military service did he have?
- q2: Under which General did Kenneth Anderson serve? (depends on: q1)
- q3: Was the General under whom Kenneth Anderson served a U.S. President? (depends on: q2)

Level 2:

Subquery: Was Dwight D. Eisenhower, the general under whom Kenneth Anderson served, a U.S. President?
```

[dwight d. eisenhower (34th President and WWII general)] -> (participates in event: General Dwight D. Eisenhower was Kenneth Anderson's American superior.) -> "General Dwight D. Eisenhower was Kenneth Anderson's American superior."

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> (has) -> [kenneth anderson (Senior British Army officer)]

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> RELATES_TO -> "Kenneth Anderson is mainly remembered as the commander of the British First Army during Operation Torch."
Logical Conclusions:
Eisenhower, as Anderson's superior, was involved in Operation Torch, for which Anderson is remembered as commander of the British First Army.

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> SUBORDINATE_OF -> "General Sir Kenneth Arthur Noel Anderson was a senior British Army officer."
Logical Conclusions:
General Dwight D. Eisenhower was Kenneth Anderson's American superior.

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> RELATES_TO -> "Richard Mead wrote that Kenneth Anderson handled a difficult campaign more competently than his critics suggest."
Logical Conclusions:
Richard Mead's defense of Anderson's campaign competence relates to his relationship with his superior Eisenhower.

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> DESCRIBED -> "Dwight D. Eisenhower wrote that Kenneth Anderson was 'blunt, at times to the point of rudeness'."
Logical Conclusions:
Dwight D. Eisenhower wrote that Kenneth Anderson was 'blunt, at times to the point of rudeness'.

[dwight d. eisenhower (34th President and WWII general)] -> (participates in event: Dwight D. Eisenhower served as the 34th President of the United States from 1953) -> "Dwight D. Eisenhower served as the 34th President of the United States from 1953 until 1961."

[dwight d. eisenhower (34th President and WWII general)] -> (participates in event: Dwight D. Eisenhower was an American politician and Army general.) -> "Dwight D. Eisenhower was an American politician and Army general."

[dwight d. eisenhower (34th President and WWII general)] -> (participates in event: During World War II, Dwight D. Eisenhower served as Supreme Commander of the All) -> "During World War II, Dwight D. Eisenhower served as Supreme Commander of the Allied Expeditionary Forces in Europe."

Level 1:

Subquery: Under which General did Kenneth Anderson serve?
"Kenneth Anderson saw service in both World wars." -> SERVICE_OF -> "General Sir Kenneth Arthur Noel Anderson was a senior British Army officer."
Logical Conclusions:
General Sir Kenneth Arthur Noel Anderson saw service in both World Wars.

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> SUBORDINATE_OF -> "General Sir Kenneth Arthur Noel Anderson was a senior British Army officer."
Logical Conclusions:
General Dwight D. Eisenhower was Kenneth Anderson's American superior.

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> (has) -> [kenneth anderson (Senior British Army officer)]

"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> RELATES_TO -> "Kenneth Anderson is mainly remembered as the commander of the British First Army during Operation Torch."
Logical Conclusions:
Eisenhower, as Anderson's superior, was involved in Operation Torch, for which Anderson is remembered as commander of the British First Army.

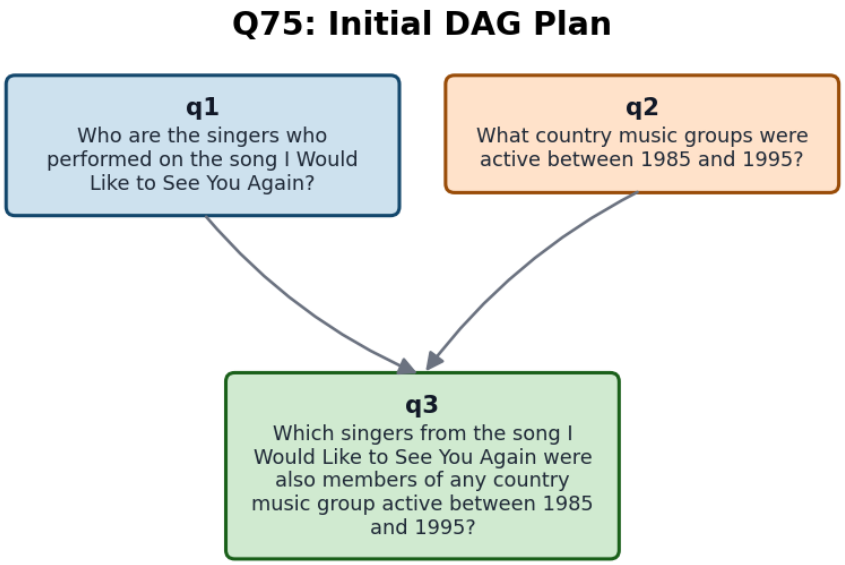
"General Dwight D. Eisenhower was Kenneth Anderson's American superior." -> RELATES_TO -> "Richard Mead wrote that Kenneth Anderson handled a difficult …[truncated for report]

Q75 [bridge / hard] 5ade6eb65542992fa25da782

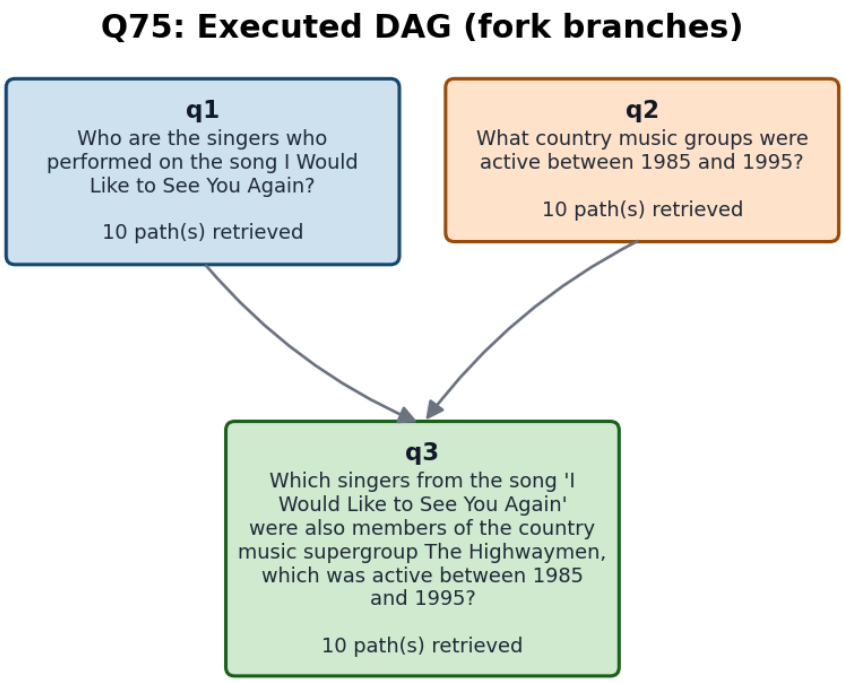
What two singers who worked together on the song I Would Like to See You Again also performed as part of a country music group which was active between 1985 and 1995?

Gold answer	Johnny Cash, Waylon Jennings
LLM answer	Johnny Cash, Waylon Jennings
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who are the singers who performed on the song I Would Like to See You Again?	10 path(s)
q2	0	—	What country music groups were active between 1985 and 1995?	10 path(s)
q3	1	q1, q2	Which singers from the song 'I Would Like to See You Again' were also members of the country music supergroup The Highwaymen, which was active between 1985 and 1995?	10 path(s)

Final context fed to generation (2064 tokens)

```
Query plan:
- q1: Who are the singers who performed on the song I Would Like to See You Again?
- q2: What country music groups were active between 1985 and 1995?
- q3: Which singers from the song I Would Like to See You Again were also members of any country music group
active between 1985 and 1995? (depends on: q1, q2)

Level 1:

Subquery: Which singers from the song 'I Would Like to See You Again' were also members of the country
music supergroup The Highwaymen, which was active between 1985 and 1995?
[Johnny Cash (American country singer and songwriter)] -> (participates in event: 'I Would Like to See You
Again' is the 57th album by American country singer Joh) -> "'I Would Like to See You Again' is the 57th
album by American country singer Johnny Cash."

[Johnny Cash (American country singer and songwriter)] -> (participates in event: The album 'I Would Like
to See You Again' features a pair.) -> "The album 'I Would Like to See You Again' features a pair."
```

"'I Would Like to See You Again' was released on Columbia Records in 1978." -> (has) -> [Johnny Cash (American country singer and songwriter)]

"'I Would Like to See You Again' was released on Columbia Records in 1978." -> RELEASED -> "'I Would Like to See You Again' is the 57th album by American country singer Johnny Cash."
Logical Conclusions:
The album 'I Would Like to See You Again' was released on Columbia Records in 1978.

[Johnny Cash (American country singer and songwriter)] -> (participates in event: The members of The Highwaymen were Johnny Cash, Waylon Jennings, and Willie Nels) -> "The members of The Highwaymen were Johnny Cash, Waylon Jennings, and Willie Nelson."

"The Highwaymen was active as a group between 1985 and 1995." -> ACTIVE_PERIOD -> "The four artists were Johnny Cash, Waylon Jennings, Willie Nelson, and Kris Kristofferson."
Logical Conclusions:
Fact 18 states The Highwaymen were active between 1985 and 1995, and fact 45 lists the four artists as Johnny Cash, Waylon Jennings, Willie Nelson, and Kris Kristofferson.

"The Highwaymen was active as a group between 1985 and 1995." -> (has) -> [The Highwaymen (Country music supergroup)]

[Johnny Cash and Waylon Jennings Song (Song and its chart performance)] -> (participates in event: "There Ain't No Good Chain Gang" was released as the second single from the albu) -> ""There Ain't No Good Chain Gang" was released as the second single from the album "I Would Like to See You Again"."

"'I Would Like to See You Again' was released on Columbia Records in 1978." -> SUPPORTS -> "Columbia Records released an album in 1978."
Logical Conclusions:
'I Would Like to See You Again' was released on Columbia Records in 1978.

"'I Would Like to See You Again' was released on Columbia Records in 1978." -> (has) -> [columbia records (Record label for Cash's album)]

Level 0:

Subquery: Who are the singers who performed on the song I Would Like to See You Again?
"'I Would Like to See You Again' is the 57th album by American country singer Johnny Cash." -> HAS_SINGLE -> "The title track of 'I Would Like to See You Again' peaked at #12 on the singles chart."
Logical Conclusions:
The title track of the album peaked at #12 on the singles chart.

"'I Would Like to See You Again' is the 57th album by American country singer Johnny Cash." -> IDENTIFIES_ALBUM -> "One of the duets with Waylon Jennings was the single 'There Ain't No Good Chain Gang'."
Logical Conclusions:
Fact 29 identifies the album, and fact 47 states that one of the duets with Waylon Jennings was the single 'There Ain't No Good Chain Gang'.

"'I Would Like to See You Again' is the 57th album by American country singer Johnny Cash." -> (has) -> [Johnny Cash (American country singer and songwriter)]

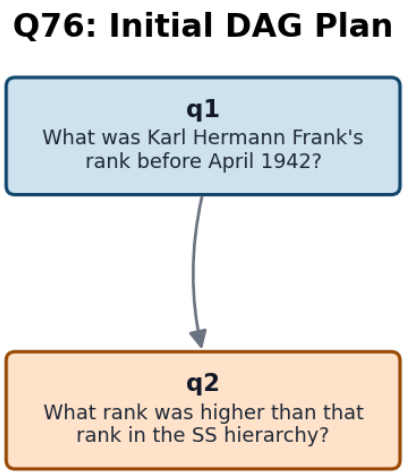
"'I Would Like to See You Again' is the 57th album by American country singer Johnny Cash." -> RELEASED_BY -> "'I Would Like to See You Again' was released on Columbia Records in 1978."
Logical Conclusions:
The album 'I Would Like to See You Again' was released …[truncated for report]

Q76 [bridge / hard] 5ab28c89554299340b525513

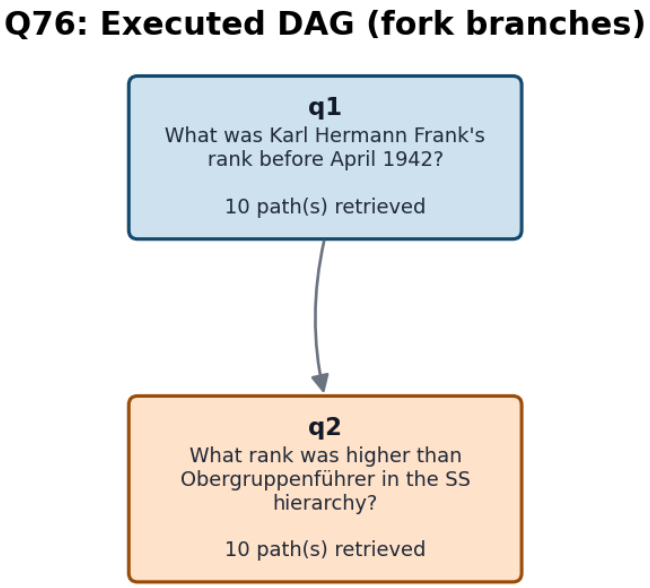
What rank was higher than Karl Hermann Frank's rank until April 1942?

Gold answer	Reichsführer-SS
LLM answer	Reichsführer-SS
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What was Karl Hermann Frank's rank before April 1942?	10 path(s)
q2	1	q1	What rank was higher than Obergruppenführer in the SS hierarchy?	10 path(s)

Final context fed to generation (1221 tokens)

```
Query plan:
- q1: What was Karl Hermann Frank's rank before April 1942?
- q2: What rank was higher than that rank in the SS hierarchy? (depends on: q1)

Level 1:

Subquery: What rank was higher than Obergruppenführer in the SS hierarchy?
[SS rank Obergruppenführer (Nazi paramilitary rank)] -> (participates in event: SS-Oberst-Gruppenführer was
above Obergruppenführer.) -> "SS-Oberst-Gruppenführer was above Obergruppenführer."

[SS rank Obergruppenführer (Nazi paramilitary rank)] -> (participates in event: Until April 1942,
Obergruppenführer was the highest commissioned SS rank.) -> "Until April 1942, Obergruppenführer was the
highest commissioned SS rank."

[SS rank Obergruppenführer (Nazi paramilitary rank)] -> (participates in event: The rank of
Obergruppenführer was adopted by the Schutzstaffel (SS) one year aft) -> "The rank of Obergruppenführer was
adopted by the Schutzstaffel (SS) one year after its creation."

[SS rank Obergruppenführer (Nazi paramilitary rank)] -> (participates in event: Obergruppenführer was a
Nazi Party paramilitary rank.) -> "Obergruppenführer was a Nazi Party paramilitary rank."

"Until April 1942, Obergruppenführer was the highest commissioned SS rank." -> SUPPORTS -> "Until April
1942, Obergruppenführer was inferior only to Reichsführer-SS."
Logical Conclusions:
Until April 1942, Obergruppenführer was inferior only to Reichsführer-SS and was the highest commissioned
SS rank.

"Until April 1942, Obergruppenführer was the highest commissioned SS rank." -> SUPPORTS ->
"Obergruppenführer translates to 'senior group leader'."
Logical Conclusions:
Obergruppenführer translates to 'senior group leader' and until April 1942, it was the highest commissioned
SS rank.

"Until April 1942, Obergruppenführer was the highest commissioned SS rank." -> HIGHEST_RANK_WAS ->
"Obergruppenführer was a Nazi Party paramilitary rank."
Logical Conclusions:
Until April 1942, Obergruppenführer was the highest commissioned SS rank.

[schutzstaffel (ss) (Paramilitary organization adopting the rank)] -> (participates in event: Until April
1942, Obergruppenführer was the highest commissioned SS rank.) -> "Until April 1942, Obergruppenführer was
the highest commissioned SS rank."

Level 0:

Subquery: What was Karl Hermann Frank's rank before April 1942?
"Karl Hermann Frank was an SS-'Obergruppenführer'." -> IS_EXEMPLIFIED_BY -> "Obergruppenführer was a Nazi
Party paramilitary rank."
Logical Conclusions:
Obergruppenführer was a Nazi Party paramilitary rank, and Karl Hermann Frank held the rank of
SS-Obergruppenführer.
```

"Karl Hermann Frank was an SS-'Obergruppenführer'." -> CORROBORATES -> "Karl Hermann Frank was a prominent Sudeten German Nazi official in Czechoslovakia prior to and during World War II."
Logical Conclusions:
Karl Hermann Frank's position as a Nazi official is associated with his rank in the SS.

"Karl Hermann Frank was an SS-'Obergruppenführer'." -> ASSOCIATED_WITH -> "Scouting was prohibited by German State Secretary Karl Hermann Frank during the Nazi occupation of Czechoslovakia on October 28, 1940."
Logical Conclusions:
Karl Hermann Frank, an SS-Obergruppenführer, prohibited scouting during the Nazi occupation of Czechoslovakia.

"Karl Hermann Frank was an SS-'Obergruppenführer'." -> (has) -> [Karl Hermann (German historical figure)]

[schutzstaffel (ss) (Paramilitary organization adopting the rank)] -> (participates in event: Until April 1942, Obergruppenführer was the highest commissioned SS rank.) -> "Until April 1942, Obergruppenführer was the highest commissioned SS rank."

[Karl Hermann Knoblauch (German physicist)] -> (participates in event: Karl Hermann Knoblauch was born on 11 April 1820.) -> "Karl Hermann Knoblauch was born on 11 April 1820."

[Karl Hermann Knoblauch (German physicist)] -> (participates in event: Karl Hermann Knoblauch was a German physicist.) -> "Karl Hermann Knoblauch was a German physicist."

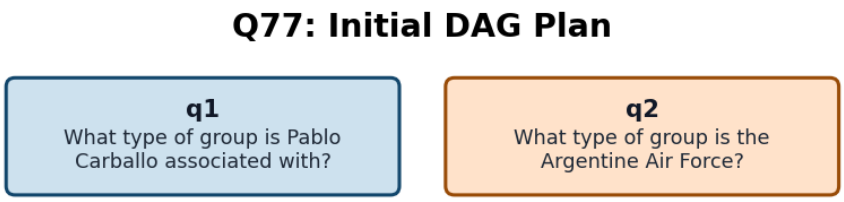
[Karl Hermann Knoblauch (German physicist)] -> (participates in event: Karl Hermann Knobl …[truncated for report]

Q77 [bridge / hard] 5a84eecd5542994c784dda8e

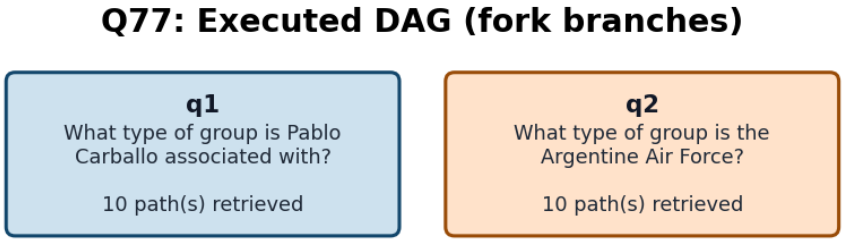
What type of group does Pablo Carballo and Argentine Air Force have in common?

Gold answer	military
LLM answer	Military air force
EM / F1	0.00 / 0.50
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What type of group is Pablo Carballo associated with?	10 path(s)
q2	0	—	What type of group is the Argentine Air Force?	10 path(s)

Final context fed to generation (1143 tokens)

```
Query plan:
- q1: What type of group is Pablo Carballo associated with?
- q2: What type of group is the Argentine Air Force?

Level 0:

Subquery: What type of group is Pablo Carballo associated with?
[pablo carballo (Retired Argentine Air Force commodore and war hero)] -> (participates in event: Commodore
Pablo Marcos Rafael Carballo is a retired member of the Argentine Air ) -> "Commodore Pablo Marcos Rafael
Carballo is a retired member of the Argentine Air Force."
```

[pablo carballo (Retired Argentine Air Force commodore and war hero)] -> (participates in event: Pablo Carballo participated in actions that led to the sinking of three Royal Na) -> "Pablo Carballo participated in actions that led to the sinking of three Royal Navy ships."

"Pablo Carballo participated in actions that led to the sinking of three Royal Navy ships." -> ASSOCIATED_WITH -> "Commodore Pablo Marcos Rafael Carballo is a retired member of the Argentine Air Force."
Logical Conclusions:
Commodore Pablo Marcos Rafael Carballo, a retired Argentine Air Force member, participated in actions that sank three Royal Navy ships.

[pablo carballo (Retired Argentine Air Force commodore and war hero)] -> (participates in event: Pablo Carballo fought in the 1982 Falklands War.) -> "Pablo Carballo fought in the 1982 Falklands War."

[pablo carballo (Retired Argentine Air Force commodore and war hero)] -> (participates in event: Pablo Carballo was awarded the Argentine Nation to the Heroic Valour in Combat C) -> "Pablo Carballo was awarded the Argentine Nation to the Heroic Valour in Combat Cross."

[pablo carballo (Retired Argentine Air Force commodore and war hero)] -> (participates in event: Pablo Carballo was awarded the Highest Distinction of the Argentine Air Force.) -> "Pablo Carballo was awarded the Highest Distinction of the Argentine Air Force."

"Pablo Carballo was born on 11 December 1947 in Buenos Aires." -> IS_SAME_PERSON -> "Commodore Pablo Marcos Rafael Carballo is a retired member of the Argentine Air Force."
Logical Conclusions:
Commodore Pablo Marcos Rafael Carballo, born on 11 December 1947 in Buenos Aires, is a retired member of the Argentine Air Force.

"Pablo Carballo participated in actions that led to the sinking of three Royal Navy ships." -> ASSOCIATED_WITH -> "Pablo Carballo is a retired member of the Argentine Air Force."
Logical Conclusions:
Pablo Carballo, a retired Argentine Air Force member, participated in actions that sank three Royal Navy ships.

"Pablo Carballo was born on 11 December 1947 in Buenos Aires." -> IDENTIFIES -> "Pablo Carballo is a retired member of the Argentine Air Force."
Logical Conclusions:
Pablo Carballo is a retired Argentine Air Force member who was born in Buenos Aires in 1947.

Subquery: What type of group is the Argentine Air Force?
"The Argentine Air Force is the national aviation branch of the Armed Forces of the Argentine Republic." -> (has) -> [Argentine Air Force (Military air force and its intelligence service)]

"The Argentine Air Force is the national aviation branch of the Armed Forces of the Argentine Republic." -> ALSO_KNOWN_AS -> "The Argentine Air Force is also known as the Fuerza Aérea Argentina or FAA."
Logical Conclusions:
The Argentine Air Force is also known as the Fuerza Aérea Argentina or FAA.

"The Argentine Air Force is the national aviation branch of the Armed Forces of the Argentine Republic." -> HAS_PROPERTY -> "In 2010, the Argentine Air Force had 6,900 civilian personnel."
Logical Conclusions:
In 2010, the Argentine Air Force had 6,900 civilian personnel.

"The Argentine Air Force is the national aviation branch of the Armed Forces of the Argentine Republic." -> HAS_PROPERTY -> "In 2010, the Argentine Air Force had 14,600 military personnel."
Logical Conclusions:
In 2010, the Argentine Air Force had 14,600 military personnel.

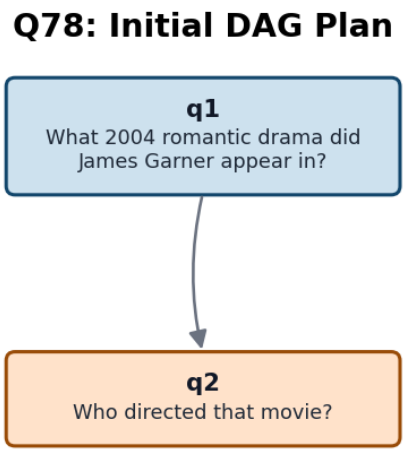
"The Argentine Air Force is the national aviation branch of the Armed Forces of the Argentine Republic." -> HAS_AGENCY -> "Servicio de Inteligencia de la Fuerza …[truncated for report]

Q78 [bridge / hard] 5ae056c755429945ae959319

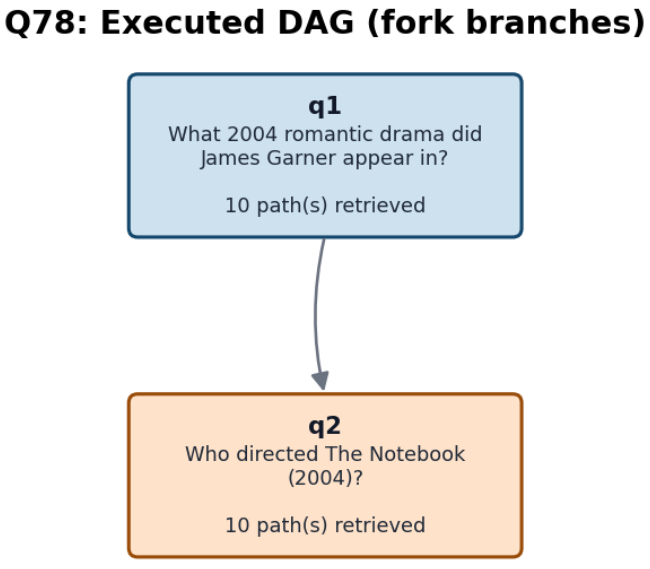
James Garner had a role in the 2004 romantic drama directed by whom?

Gold answer	Nick Cassavetes
LLM answer	Nick Cassavetes
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What 2004 romantic drama did James Garner appear in?	10 path(s)
q2	1	q1	Who directed The Notebook (2004)?	10 path(s)

Final context fed to generation (1257 tokens)

```
Query plan:
- q1: What 2004 romantic drama did James Garner appear in?
- q2: Who directed that movie? (depends on: q1)

Level 1:

Subquery: Who directed The Notebook (2004)?
"The Notebook is a 2004 American romantic drama film." -> DIRECTED_BY -> "The Notebook was directed by Nick Cassavetes."
Logical Conclusions:
The Notebook is a 2004 romantic drama directed by Nick Cassavetes.

"The Notebook is a 2004 American romantic drama film." -> (has ) -> [James Garner project (Film or TV project featuring James Garner)]

"The Notebook is a 2004 American romantic drama film." -> ADAPTED_FROM -> "The Notebook is based on the 1996 novel of the same name by Nicholas Sparks."
Logical Conclusions:
The Notebook is based on the 1996 novel by Nicholas Sparks.

"The Notebook is a 2004 American romantic drama film." -> RELATED_TO -> "Gena Rowlands is Nick Cassavetes's mother."
Logical Conclusions:
Gena Rowlands is Nick Cassavetes's mother.

"The Notebook is a 2004 American romantic drama film." -> STARS -> "The Notebook stars Ryan Gosling and Rachel McAdams as a young couple."
Logical Conclusions:
The Notebook stars Ryan Gosling and Rachel McAdams as a young couple.

"The Notebook is a 2004 American romantic drama film." -> RELATED_GENRE -> "Just Friends is a 2005 American romantic comedy film."
Logical Conclusions:
Both films are romantic movies released in the mid-2000s, with one being a comedy and the other a drama.

[night life (film) (1989 zombie comedy film)] -> (participates in event: Night Life was directed by David Acomba.) -> "Night Life was directed by David Acomba."

[Artists and directors (Music and film directors)] -> (participates in event: 'Ghetto Love' features Canadian artist Kardinal Offishall.) -> "'Ghetto Love' features Canadian artist Kardinal Offishall."

[Artists and directors (Music and film directors)] -> (participates in event: Other members of the production team for Puteri Gunung Ledang are music director) -> "Other members of the production team for Puteri Gunung Ledang are music director Roslan Aziz."

[national film board of canada (Organization Irwin served as Commissioner)] -> (participates in event: William Arthur Irwin served as the Commissioner of the National Film Board of Ca) -> "William Arthur Irwin served as the Commissioner of the National Film Board of Canada (NFB)."

Level 0:

Subquery: What 2004 romantic drama did James Garner appear in?
```

"James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams." -> (has) -> [rachel mcadams (Actress playing young couple in The Notebook)]

"James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams." -> CO-OCCURS_WITH -> "James Garner starred in 'Divine Secrets of the Ya-Ya Sisterhood' (2002)."
Logical Conclusions:
Both facts list films starring James Garner from the 2000s.

"James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams." -> RELATES_TO -> "The movie was filmed while Garner's series 'The Rockford Files' was on hiatus."
Logical Conclusions:
Fact 16 mentions a film shot during a hiatus of 'The Rockford Files', while fact 9 lists a film starring Garner from 2004.

"James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams." -> CO-OCCURS_WITH -> "James Garner portrayed Jim Egan in the television series '8 Simple Rules' (2003–2005)."
Logical Conclusions:
Both facts list roles played by James Garner, one in film and one in television.

"James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams." -> PART_OF -> "The following is the filmography for American actor James Garner."
Logical Conclusions:
Fact 10 introduces the filmography of James Garner, which includes the film listed in fact 9.

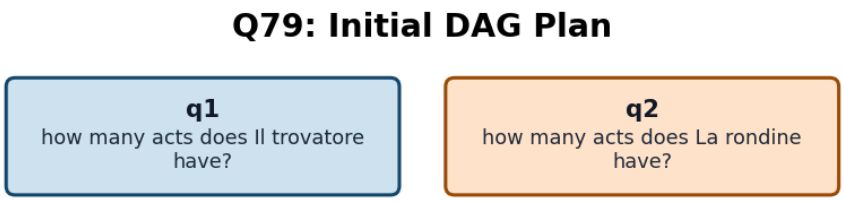
"James Garner starred in 'The Notebook' (2004) with Ryan Gosling and Rachel McAdams." -> CO-OCCURS_WITH -> "James Garner starred in 'Sunset' (1988) with Bruce Willis."
Logical Conclusions:
Both facts list films starring James Garner from different decad …[truncated for report]

Q79 [comparison / hard] 5a87136e5542991e771816b2

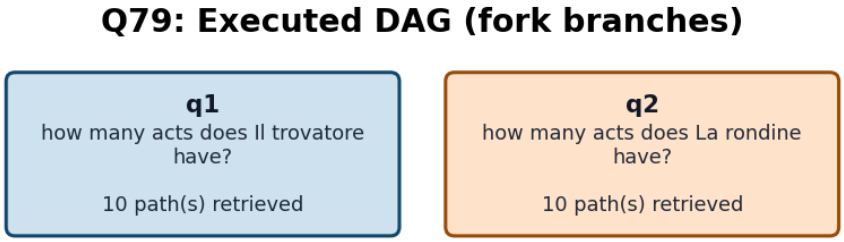
Does Il trovatore have less acts than La rondine

Gold answer	no
LLM answer	No
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	how many acts does Il trovatore have?	10 path(s)
q2	0	—	how many acts does La rondine have?	10 path(s)

Final context fed to generation (1162 tokens)

```
Query plan:
- q1: how many acts does Il trovatore have?
- q2: how many acts does La rondine have?

Level 0:

Subquery: how many acts does Il trovatore have?
"Il trovatore is an opera in four acts." -> (has ) -> [il trovatore (Opera by Giuseppe Verdi)]

"Il trovatore is an opera in four acts." -> HAS_LIBRETTO -> "Il trovatore has an Italian libretto largely
written by Salvadore Cammarano."
Logical Conclusions:
```

Il trovatore is an opera in four acts with an Italian libretto largely written by Salvadore Cammarano.

"Il trovatore is an opera in four acts." -> COMPOSED_BY -> "Il trovatore was composed by Giuseppe Verdi."
Logical Conclusions:
Il trovatore is an opera in four acts composed by Giuseppe Verdi.

"Il trovatore is an opera in four acts." -> HAS_LIBRETTIST -> "Salvatore Cammarano was the librettist of Verdi's 'Il trovatore'."
Logical Conclusions:
The four-act opera Il trovatore had its libretto written by Salvatore Cammarano.

"Il trovatore is an opera in four acts." -> HAS_LIBRETTO -> "The libretto of Il trovatore is based on the play 'El trovador' (1836) by Antonio García Gutiérrez."
Logical Conclusions:
The four-act opera Il trovatore has a libretto based on a play by Antonio García Gutiérrez.

"Il trovatore is an opera in four acts." -> ACTS_OF -> "'Il trovatore' was first performed at the Teatro Apollo, Rome on 19 January 1853."
Logical Conclusions:
'Il trovatore' was first performed on 19 January 1853 and is an opera in four acts.

[il trovatore (Opera by Giuseppe Verdi)] -> (participates in event: At least 83 recordings exist of the opera 'Il trovatore' as a whole.) -> "At least 83 recordings exist of the opera 'Il trovatore' as a whole."

[il trovatore (Opera by Giuseppe Verdi)] -> (participates in event: Of the 83 recordings, 45 are live audio recordings.) -> "Of the 83 recordings, 45 are live audio recordings."

[il trovatore (Opera by Giuseppe Verdi)] -> (participates in event: 'Il trovatore' was first performed at the Teatro Apollo, Rome on 19 January 1853) -> "'Il trovatore' was first performed at the Teatro Apollo, Rome on 19 January 1853."

Subquery: how many acts does La rondine have?
"La rondine is an opera in three acts." -> (has) -> [la rondine (Opera by Giacomo Puccini)]

"La rondine is an opera in three acts." -> PERFORMED_AT -> "La rondine was first performed at the Grand Théâtre de Monte Carlo in Monte Carlo."
Logical Conclusions:
La rondine is an opera in three acts that was first performed at the Grand Théâtre de Monte Carlo.

"La rondine is an opera in three acts." -> HAS_LIBRETTO -> "La rondine has an Italian libretto by Giuseppe Adami."
Logical Conclusions:
La rondine is an opera in three acts with an Italian libretto by Giuseppe Adami.

"La rondine is an opera in three acts." -> ACTS_OF -> "'La rondine' ('The Swallow') is an opera in three acts by Giacomo Puccini."
Logical Conclusions:
La rondine is an opera in three acts by Giacomo Puccini.

"La rondine is an opera in three acts." -> COMPOSED_BY -> "La rondine was composed by Giacomo Puccini."
Logical Conclusions:
La rondine is an opera in three acts that was composed by Giacomo Puccini.

"La rondine is an opera in three acts." -> PERFORMED_ON -> "La rondine was first performed on 27 March 1917."
Logical Conclusions:
La rondine is an opera in three acts that was first performed on 27 March 1917.

[la rondine (Opera by Giacomo Puccini)] -> (participates in event: La rondine was first performed at the Grand Théâtre de Monte Carlo in Monte Carl) -> "La rondine was first performed at the Grand Théâtre de Monte Carlo in Monte Carlo."

[la rondine (Opera by Giacomo Puccini)] -> (participates in event: Giuseppe Adami collaborated with Giacomo Puccini on the opera 'La rondine' (1917) -> "Giuseppe Adami collaborated with Giacomo Puccini on the opera 'La rondine' (1917)."

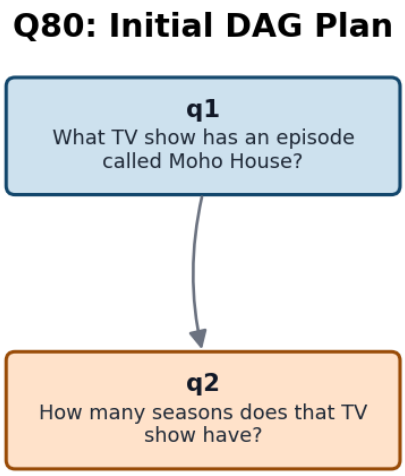
[la rondine (Opera by Giacomo Puccini)] -> (participates in event: Francesco Dominici created the role of Prunier in the original 1917 production o) -> "Francesco Dominici created the …[truncated for report]

Q80 [bridge / hard] 5a80c79155429938b61421da

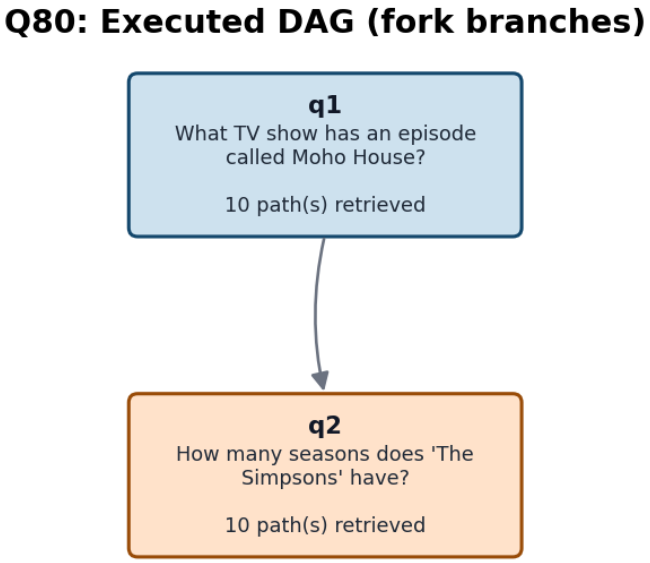
How many seasons has a popular tv show had, in which one of the episodes is called Moho House?

Gold answer	28
LLM answer	28
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What TV show has an episode called Moho House?	10 path(s)
q2	1	q1	How many seasons does 'The Simpsons' have?	10 path(s)

Final context fed to generation (1449 tokens)

```
Query plan:
- q1: What TV show has an episode called Moho House?
- q2: How many seasons does that TV show have? (depends on: q1)

Level 1:

Subquery: How many seasons does 'The Simpsons' have?
"The twenty-eighth season of 'The Simpsons' ended on May 21, 2017." -> FOLLOWS -> "On May 4, 2015, Fox
announced that 'The Simpsons' had been renewed for season 28."
Logical Conclusions:
Fact 39 states that season 28 ended on May 21, 2017, while fact 40 states that the renewal was announced on
May 4, 2015.

"The twenty-eighth season of 'The Simpsons' ended on May 21, 2017." -> CONTAINS_EPISODE -> "The first
episode of the twenty-eighth season of the animated television series 'The Simpsons' is the 617th episode
of the series overall."
Logical Conclusions:
Fact 1 refers to the first episode of season 28, and fact 11 provides the end date of that season.

"The twenty-eighth season of 'The Simpsons' ended on May 21, 2017." -> (has ) -> [the simpsons (American
animated television series)]

"The twenty-eighth season of 'The Simpsons' ended on May 21, 2017." -> FOLLOWS -> "The twenty-eighth season
of the animated television series 'The Simpsons' began airing on Fox in the United States on September 25,
2016."
Logical Conclusions:
The twenty-eighth season of 'The Simpsons' began airing on September 25, 2016 and ended on May 21, 2017.

"The twenty-eighth season of 'The Simpsons' ended on May 21, 2017." -> CONTAINS_EPISODE -> "'Moho House' is
the twenty first episode of the twenty-eighth season of the animated television series 'The Simpsons'."
Logical Conclusions:
Fact 7 places 'Moho House' in season 28, and fact 11 gives the end date of that season.

"The twenty-eighth season of 'The Simpsons' ended on May 21, 2017." -> CONTAINS_EPISODE -> "'Moho House' is
the 617th episode of 'The Simpsons' series overall."
Logical Conclusions:
Fact 4 identifies 'Moho House' as the 617th episode, and fact 11 gives the end date of season 28, which
contains this episode.

[the simpsons (American animated television series)] -> (participates in event: The first episode of the
twenty-eighth season of the animated television series ) -> "The first episode of the twenty-eighth season
of the animated television series 'The Simpsons' is the 617th episode of the series overall."

[the simpsons (American animated television series)] -> (participates in event: "Future-Drama" is the
fifteenth episode of The Simpsons' sixteenth season.) -> ""Future-Drama" is the fifteenth episode of The
Simpsons' sixteenth season."

[the simpsons (American animated television series)] -> (participates in event: 'Moho House' aired in the
United States on Fox on May 7, 2017.) -> "'Moho House' aired in the United States on Fox on May 7, 2017."

[the simpsons (American animated television series)] -> (participates in event: 'Girly Edition' originally
aired on the Fox network in the United States on Apri) -> "'Girly Edition' originally aired on the Fox
```

network in the United States on April 19, 1998."

Level 0:

Subquery: What TV show has an episode called Moho House?
"'Moho House' aired in the United States on Fox on May 7, 2017." -> BROADCAST -> "'Moho House' is the twenty first episode of the twenty-eighth season of the animated television series 'The Simpsons'."
Logical Conclusions:
'Moho House' aired on Fox on May 7, 2017.

[moho house (episode of The Simpsons)] -> (participates in event: 'Moho House' is the twenty first episode of the twenty-eighth season of the anim) -> "'Moho House' is the twenty first episode of the twenty-eighth season of the animated television series 'The Simpsons'."

"'Moho House' aired in the United States on Fox on May 7, 2017." -> BROADCAST -> "'Moho House' is the 617th episode of 'The Simpsons' series overall."
Logical Conclusions:
The 617th episode 'Moho House' aired on May 7, 2017.

"'Moho House' aired in the United States on Fox on May 7, 2017." -> (has) -> [moho house (episode of The Simpsons)]

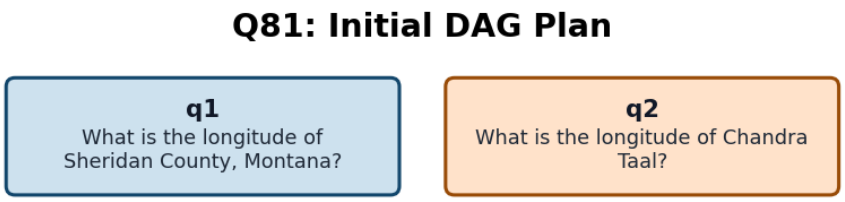
"'Moho House' aired in the United States on Fox on May 7, 2017 …[truncated for report]

Q81 [comparison / hard] 5a8841ee5542994846c1ce6b

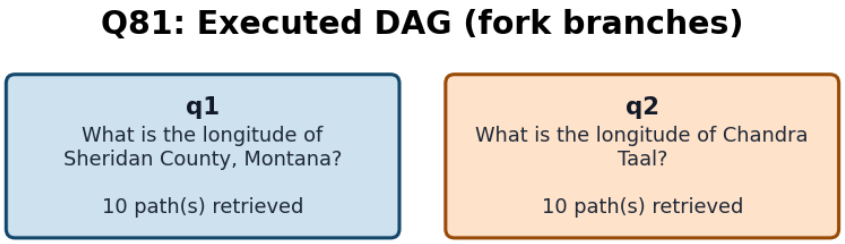
Which is farther west, Sheridan County, Montana or Chandra Taal?

Gold answer	Sheridan County
LLM answer	Sheridan County, Montana is farther west.
EM / F1	0.00 / 0.50
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the longitude of Sheridan County, Montana?	10 path(s)
q2	0	—	What is the longitude of Chandra Taal?	10 path(s)

Final context fed to generation (1217 tokens)

```
Query plan:
- q1: What is the longitude of Sheridan County, Montana?
- q2: What is the longitude of Chandra Taal?

Level 0:

Subquery: What is the longitude of Sheridan County, Montana?
"Sheridan County is a county located in the U.S. state of Montana." -> BOUNDED_BY -> "The northern boundary
of Sheridan County is the Canada–United States border south of Saskatchewan."
Logical Conclusions:
```

The northern boundary of Sheridan County is the Canada–United States border south of Saskatchewan.

"Sheridan County is a county located in the U.S. state of Montana." -> (has) -> [Sheridan County (County in Montana or Wyoming)]

"Sheridan County is a county located in the U.S. state of Montana." -> HAS_PROPERTY -> "As of the 2010 census, the population of Sheridan County was 3,384."
Logical Conclusions:
Sheridan County is a county in Montana with a population of 3,384 as of the 2010 census.

[Sheridan County (County in Montana or Wyoming)] -> (participates in event: The northern boundary of Sheridan County is the Canada–United States border sout) -> "The northern boundary of Sheridan County is the Canada–United States border south of Saskatchewan."

"Sheridan County is a county located in the U.S. state of Montana." -> HAS_SEAT -> "The county seat of Sheridan County is Plentywood."
Logical Conclusions:
The county seat of Sheridan County is Plentywood.

[Sheridan County (County in Montana or Wyoming)] -> (participates in event: The county seat of Sheridan County is Plentywood.) -> "The county seat of Sheridan County is Plentywood."

[Sheridan County (County in Montana or Wyoming)] -> (participates in event: The Sheridan County Courthouse was the first courthouse built in Sheridan County) -> "The Sheridan County Courthouse was the first courthouse built in Sheridan County."

[Sheridan County (County in Montana or Wyoming)] -> (participates in event: Sheridan County Airport is a county owned, public use airport in Sheridan County) -> "Sheridan County Airport is a county owned, public use airport in Sheridan County, Wyoming, United States."

[Sheridan County (County in Montana or Wyoming)] -> (participates in event: Leiter is an unincorporated community in eastern Sheridan County, Wyoming, Unite) -> "Leiter is an unincorporated community in eastern Sheridan County, Wyoming, United States."

[Sheridan, Wyoming (City and county seat of Sheridan County)] -> (participates in event: Sheridan County Airport is located two nautical miles southwest of the central b) -> "Sheridan County Airport is located two nautical miles southwest of the central business district of Sheridan, Wyoming."

Subquery: What is the longitude of Chandra Taal?
"The road from Kunzum Pass to Chandra Taal is about 8 km from Chandra Taal." -> (has) -> [Chandra Taal (crescent-shaped lake in Himachal Pradesh, India)]

"The road from Kunzum Pass to Chandra Taal is about 8 km from Chandra Taal." -> SUPPORTS -> "Chandra Taal is accessible on foot from Kunzum Pass from late May to early October."
Logical Conclusions:
Chandra Taal is accessible on foot from Kunzum Pass, and the road from Kunzum Pass to Chandra Taal is about 8 km from the lake.

[Chandra Taal (crescent-shaped lake in Himachal Pradesh, India)] -> (participates in event: The name of Chandra Taal originates from its crescent shape.) -> "The name of Chandra Taal originates from its crescent shape."

[Chandra Taal (crescent-shaped lake in Himachal Pradesh, India)] -> (participates in event: Chandra Taal is a popular destination for trekkers and campers.) -> "Chandra Taal is a popular destination for trekkers and campers."

[Chandra Taal (crescent-shaped lake in Himachal Pradesh, India)] -> (participates in event: The road from Kunzum Pass to Chandra Taal is accessible only on foot.) -> "The road from Kunzum Pass to Chandra Taal is accessible only on foot."

"The road from Kunzum Pass to Chandra Taal is about 8 km from Chandra Taal." -> (has) -> [kunzum pass (Mountain pass near Chandra Taal)]

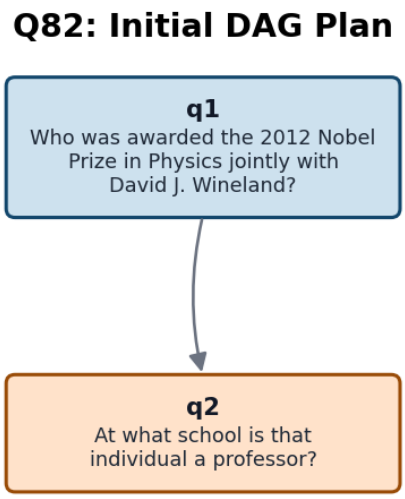
[Chandra Taal (crescent-shaped …[truncated for report]

Q82 [bridge / hard] 5ab9bb8355429970cfb8eb8b

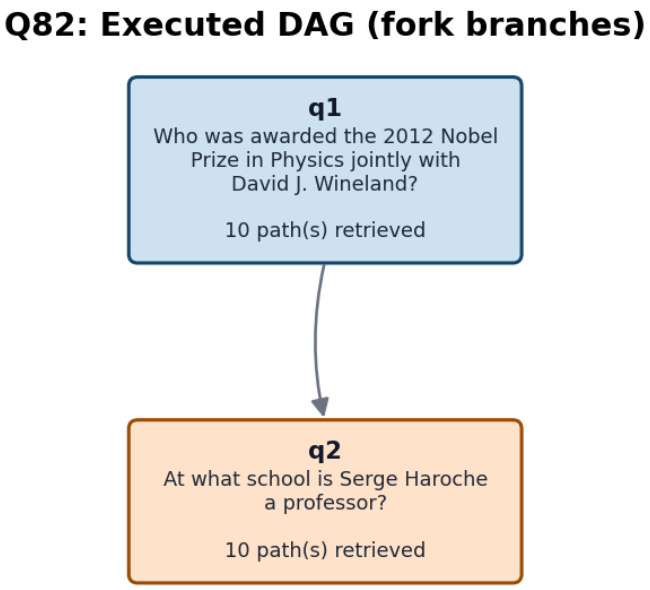
At what school is the individual who was awarded the 2012 Nobel Prize in Physics, jointly with David J. Wineland, a professor?

Gold answer	Collège de France
LLM answer	Collège de France
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who was awarded the 2012 Nobel Prize in Physics jointly with David J. Wineland?	10 path(s)
q2	1	q1	At what school is Serge Haroche a professor?	10 path(s)

Final context fed to generation (1324 tokens)

```
Query plan:
- q1: Who was awarded the 2012 Nobel Prize in Physics jointly with David J. Wineland?
- q2: At what school is that individual a professor? (depends on: q1)

Level 1:

Subquery: At what school is Serge Haroche a professor?
[serge haroche (French physicist and 2012 Nobel laureate)] -> (participates in event: Since 2001, Serge
Haroche is a Professor at the Collège de France.) -> "Since 2001, Serge Haroche is a Professor at the
Collège de France."

"Serge Haroche holds the Chair of Quantum Physics at the Collège de France." -> CHAIR_HELD_BY -> "Since
2001, Serge Haroche is a Professor at the Collège de France."
Logical Conclusions:
Since 2001, Serge Haroche is a Professor at the Collège de France and holds the Chair of Quantum Physics
there.

"Serge Haroche holds the Chair of Quantum Physics at the Collège de France." -> (has ) -> [serge haroche
(French physicist and 2012 Nobel laureate)]

"Serge Haroche holds the Chair of Quantum Physics at the Collège de France." -> RELATED_INFO -> "Serge
Haroche is a French physicist."
Logical Conclusions:
Both facts provide biographical details about Serge Haroche.

"Serge Haroche holds the Chair of Quantum Physics at the Collège de France." -> CORROBORATES -> "Serge
Haroche's research has been conducted under the direction of Claude Cohen-Tannoudji."
Logical Conclusions:
Both facts provide biographical information about Serge Haroche.

"Serge Haroche holds the Chair of Quantum Physics at the Collège de France." -> CORROBORATES -> "Serge
Haroche was awarded the 2012 Nobel Prize for Physics."
Logical Conclusions:
Both facts provide biographical information about Serge Haroche.

"Serge Haroche holds the Chair of Quantum Physics at the Collège de France." -> CORROBORATES -> "Serge
Haroche was awarded the 2012 Nobel Prize for Physics jointly with David J. Wineland."
Logical Conclusions:
Both facts provide biographical information about Serge Haroche.

"In 1971, Serge Haroche defended his doctoral thesis in physics at the University of Paris VI." ->
RELATED_INFO -> "Serge Haroche is a French physicist."
Logical Conclusions:
Both facts provide biographical details about Serge Haroche.

"In 1971, Serge Haroche defended his doctoral thesis in physics at the University of Paris VI." -> (has )
-> [university of paris vi (Institution where Haroche earned his doctorate)]

"In 1971, Serge Haroche defended his doctoral thesis in physics at the University of Paris VI." ->
RELATED_INFO -> "Serge Haroche's works developed laser spectroscopy."
```

Logical Conclusions:
Both facts provide biographical details about Serge Haroche.

Level 0:

Subquery: Who was awarded the 2012 Nobel Prize in Physics jointly with David J. Wineland?
"David Wineland was awarded the 2012 Nobel Prize in Physics jointly with Serge Haroche." -> CORROBORATES ->
"Serge Haroche and David J. Wineland were awarded the 2012 Nobel Prize for Physics for 'ground-breaking experimental methods that enable measuring and manipulation of individual quantum systems'."
Logical Conclusions:
Both facts mention that Serge Haroche and David J. Wineland were awarded the 2012 Nobel Prize for Physics.

"David Wineland was awarded the 2012 Nobel Prize in Physics jointly with Serge Haroche." -> CORROBORATES ->
"Serge Haroche was awarded the 2012 Nobel Prize for Physics jointly with David J. Wineland."
Logical Conclusions:
Both facts mention Serge Haroche and David J. Wineland sharing the 2012 Nobel Prize.

"David Wineland was awarded the 2012 Nobel Prize in Physics jointly with Serge Haroche." -> RELATED_INFO ->
"David Wineland's work has included advances in optics."
Logical Conclusions:
Both facts provide information about David Wineland.

"David Wineland was awarded the 2012 Nobel Prize in Physics jointly with Serge Haroche." -> SAME_AWARD ->
"David Jeffrey Wineland is an American Nobel-laureate physicist."
Logical Conclusions:
Both facts describe David Wineland's 2012 Nobel Prize.

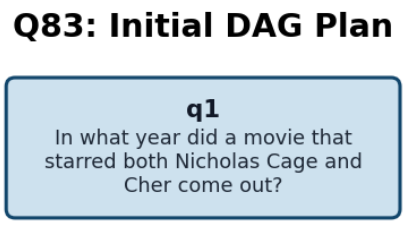
"David Wineland was awarded the 2012 Nobel Prize in Physics jointly wi …[truncated for report]

Q83 [bridge / hard] 5a88f7855542993b751ca8da

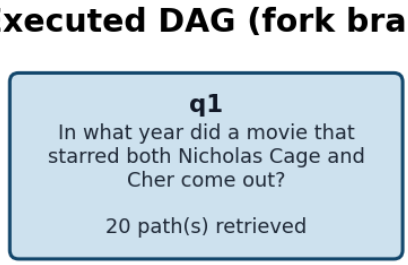
In what year did a movie that starred both Nicholas Cage and Cher come out?

Gold answer	1987
LLM answer	1987
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	In what year did a movie that starred both Nicholas Cage and Cher come out?	20 path(s)

Final context fed to generation (1423 tokens)

```
"During his early career, Cage starred in the film 'Fire Birds' (1990)." -> (has ) -> [Nicolas Cage
(American actor, director and producer)] -> (participates in event: Moonstruck is about a widowed
37-year-old Italian-American woman (Cher) who fall) -> "Moonstruck is about a widowed 37-year-old
Italian-American woman (Cher) who falls in love with her fiancé's (Danny Aiello) estranged, hot-tempered
younger brother (Nicolas Cage)." -> (has ) -> [cher (Actress and singer with filmography)] -> (participates
in event: Two of Cher's film appearances were as a cameo.) -> "Two of Cher's film appearances were as a
cameo."

[cher (Actress and singer with filmography)] -> (participates in event: Cher's filmography is the subject
of a section titled 'Cher filmography'.) -> "Cher's filmography is the subject of a section titled 'Cher
filmography'."
```

[cher (Actress and singer with filmography)] -> (participates in event: Cher has appeared in one starring theater role.) -> "Cher has appeared in one starring theater role."

[cher (Actress and singer with filmography)] -> (participates in event: 'Where Do You Go' was released for Cher's second album 'The Sonny Side of Cher'.) -> "'Where Do You Go' was released for Cher's second album 'The Sonny Side of Cher'."

[cher (Actress and singer with filmography)] -> (participates in event: Cher has starred in various international television commercials.) -> "Cher has starred in various international television commercials."

[cher (Actress and singer with filmography)] -> (participates in event: Throughout her acting career, Cher has mainly in comedy.) -> "Throughout her acting career, Cher has mainly in comedy."

"Moonstruck is about a widowed 37-year-old Italian-American woman (Cher) who falls in love with her fiancé's (Danny Aiello) estranged, hot-tempered younger brother (Nicolas Cage)." -> (has) -> [danny aiello (Actor playing fiancé in Moonstruck)]

"During his early career, Cage starred in the film 'Fire Birds' (1990)." -> STARS -> "Nicolas Kim Coppola is known professionally as Nicolas Cage." -> (has) -> [Nicolas Cage (American actor, director and producer)]
Logical Conclusions:
During his early career, Cage starred in the film Fire Birds (1990).

"Moonstruck is about a widowed 37-year-old Italian-American woman (Cher) who falls in love with her fiancé's (Danny Aiello) estranged, hot-tempered younger brother (Nicolas Cage)." -> (has) -> [moonstruck (1987 American romantic comedy film)] -> (participates in event: Moonstruck was directed by Norman Jewison.) -> "Moonstruck was directed by Norman Jewison."

[moonstruck (1987 American romantic comedy film)] -> (participates in event: Moonstruck was written by John Patrick Shanley.) -> "Moonstruck was written by John Patrick Shanley."

[moonstruck (1987 American romantic comedy film)] -> (participates in event: Vincent Gardenia and Olympia Dukakis play supporting roles in Moonstruck.) -> "Vincent Gardenia and Olympia Dukakis play supporting roles in Moonstruck."

[moonstruck (1987 American romantic comedy film)] -> (participates in event: Moonstruck is a 1987 American romantic comedy film.) -> "Moonstruck is a 1987 American romantic comedy film."

"Moonstruck is about a widowed 37-year-old Italian-American woman (Cher) who falls in love with her fiancé's (Danny Aiello) estranged, hot-tempered younger brother (Nicolas Cage)." -> SUPPORTS -> "Vincent Gardenia and Olympia Dukakis play supporting roles in Moonstruck."
Logical Conclusions:
The supporting actors in Moonstruck are part of the film's cast alongside the main characters.

"During his early career, Cage starred in the film 'Fire Birds' (1990)." -> EXEMPLIFIES -> "Nicolas Cage is an American actor." -> (has) -> [Nicolas Cage (American actor, director and producer)]
Logical Conclusions:
Nicolas Cage starred in 'Fire Birds' during his early career, exemplifying his work as an actor.

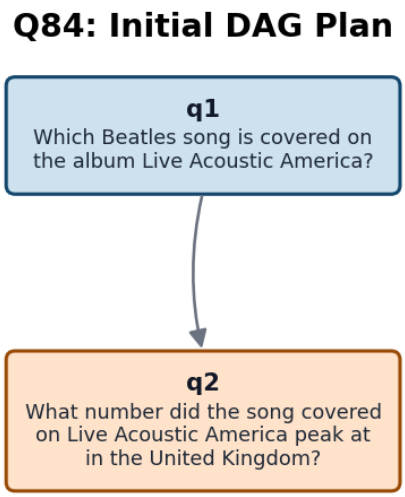
"Moonstruck is about a widowed 37-year-old Italian-American woman (Cher) who falls in love with her fiancé's …[truncated for report]

Q84 [bridge / hard] 5a8782f25542996e4f308818

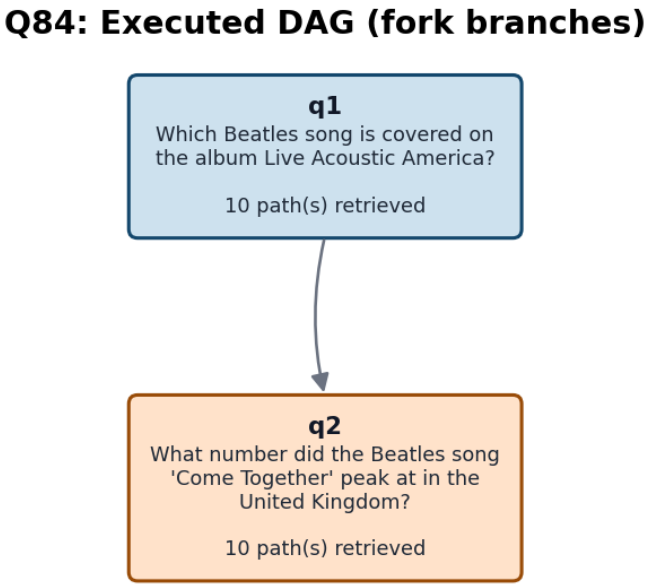
Live Acoustic America features a cover of a Beatles song that peaked at what number in the United Kingdom?

Gold answer	4
LLM answer	4
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which Beatles song is covered on the album Live Acoustic America?	10 path(s)
q2	1	q1	What number did the Beatles song 'Come Together' peak at in the United Kingdom?	10 path(s)

Final context fed to generation (1141 tokens)

```
Query plan:
- q1: Which Beatles song is covered on the album Live Acoustic America?
- q2: What number did the song covered on Live Acoustic America peak at in the United Kingdom? (depends on:
q1)

Level 1:

Subquery: What number did the Beatles song 'Come Together' peak at in the United Kingdom?
"'Come Together' peaked at No. 4 in the United Kingdom." -> CHARTED_FOR -> "'Come Together' is a song by
the Beatles."
Logical Conclusions:
'Come Together' peaked at No. 4 in the United Kingdom.

"'Come Together' peaked at No. 4 in the United Kingdom." -> (has ) -> [Classic Rock Songs (Songs by The
Beatles and The Who)]

"'Come Together' peaked at No. 4 in the United Kingdom." -> CONTRASTS_WITH -> "'Come Together' reached the
top of the charts in the United States."
Logical Conclusions:
'Come Together' peaked at No. 4 in the UK and reached the top of the charts in the US.

"'Come Together' is a song by the Beatles." -> ACHIEVED -> "'Come Together' reached the top of the charts
in the United States."
Logical Conclusions:
'Come Together' reached the top of the charts in the United States.

"'Come Together' is a song by the Beatles." -> APPEARS_ON -> "'Come Together' is the opening track on the
album 'Abbey Road'."
Logical Conclusions:
'Come Together' is the opening track on the album 'Abbey Road'.

"Forbes took 'Beatles' to the United Kingdom for the Eurovision Song Contest 1977." -> SUPPORTS ->
"'Beatles' won the Swedish Melodifestivalen 1977."
Logical Conclusions:
Fact 65 states Forbes took 'Beatles' to Eurovision, and fact 73 states it won Melodifestivalen.

"Forbes took 'Beatles' to the United Kingdom for the Eurovision Song Contest 1977." -> TOOK_PLACE_IN ->
"The contest took place in the United Kingdom."
Logical Conclusions:
The contest took place in the United Kingdom.

"Forbes took 'Beatles' to the United Kingdom for the Eurovision Song Contest 1977." -> (has ) -> [Stjórnin
and Heart 2 Heart (Icelandic Eurovision groups and songs)]

"Forbes took 'Beatles' to the United Kingdom for the Eurovision Song Contest 1977." -> PERFORMED_AT ->
"'Beatles' was performed by Swedish dansband Forbes."
Logical Conclusions:
Forbes took 'Beatles' to the United Kingdom for the Eurovision Song Contest 1977.

Level 0:
```

Subquery: Which Beatles song is covered on the album Live Acoustic America?
"The tour featured a pared-back version of The Beatles cover 'Come Together'." -> PART_OF -> "The tour featured pared-back versions of album tracks."
Logical Conclusions:
The classic Howard Jones acoustic tour featured pared-back versions of album tracks, including a pared-back version of The Beatles cover 'Come Together'.

"The tour featured a pared-back version of The Beatles cover 'Come Together'." -> PART_OF -> "The acoustic tour travelled the world."
Logical Conclusions:
The acoustic tour that travelled the world featured a pared-back version of The Beatles cover 'Come Together'.

"The tour featured a pared-back version of The Beatles cover 'Come Together'." -> FEATURED_IN -> "The tour accompanied the release of the piano-based album 'In the Running'."
Logical Conclusions:
The tour featured a pared-back version of The Beatles cover 'Come Together'.

"The tour featured a pared-back version of The Beatles cover 'Come Together'." -> SUPPORTS -> "The recording was to a sell out crowd."
Logical Conclusions:
The tour featuring a pared-back version of The Beatles cover 'Come Together' had a recording to a sell out crowd.

[live acoustic america (acoustic live recording by Howard Jones)] -> (participates in event: 'Live Acoustic America' is an acoustic live recording of classic Howard Jones so) -> "'Live Acoustic America' is an acoustic live recording of classic Howard Jones songs."

[live acoustic america (acoustic live recording by Howard Jones)] -> (participates in event: 'Live Acoustic America' was released in 1996.) -> "'Live Acoustic America' was released in 1996."

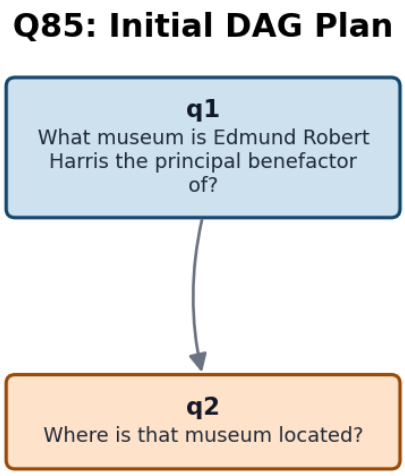
[live acoustic america (acoustic live recording by Howard Jones)] -> (participates in event: …[truncated for report]

Q85 [bridge / hard] 5ab207f65542993be8fa98b8

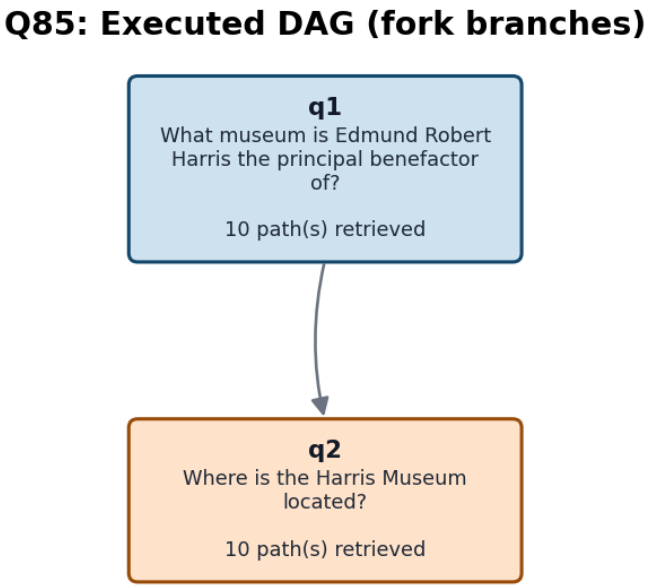
Edmund Robert Harris is the principal benefactor of a museum located where ?

Gold answer	Preston
LLM answer	Preston
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What museum is Edmund Robert Harris the principal benefactor of?	10 path(s)
q2	1	q1	Where is the Harris Museum located?	10 path(s)

Final context fed to generation (1257 tokens)

```
Query plan:
- q1: What museum is Edmund Robert Harris the principal benefactor of?
- q2: Where is that museum located? (depends on: q1)

Level 1:

Subquery: Where is the Harris Museum located?
"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> NAMED_AFTER -> "The Harris Museum, Art Gallery & Preston Free Public Library is a Grade I listed museum building in Preston."
Logical Conclusions:
The Harris Museum, named after its principal benefactor Edmund Robert Harris, is a Grade I listed building in Preston.

[Harris Museum (Grade I listed museum in Preston)] -> (participates in event: The Harris Museum, Art Gallery & Preston Free Public Library is a Grade I listed) -> "The Harris Museum, Art Gallery & Preston Free Public Library is a Grade I listed museum building in Preston."

[Harris Museum (Grade I listed museum in Preston)] -> (participates in event: Edmund Robert Harris was the principal benefactor of the Harris Museum.) -> "Edmund Robert Harris was the principal benefactor of the Harris Museum."

"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> SUPPORTS -> "Edmund Robert Harris was an English lawyer from Preston, Lancashire, UK."
Logical Conclusions:
Both facts describe Edmund Robert Harris, with one noting he was an English lawyer and the other that he was principal benefactor of the Harris Museum.

"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> SUPPORTS -> "Edmund Robert Harris was the principal benefactor of the Harris Orphanage."
Logical Conclusions:
Both facts describe Edmund Robert Harris as a principal benefactor, with one noting the Harris Orphanage and the other the Harris Museum.

"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> HAD_BENEFACTOR -> "Edmund Robert Harris was born c. 1804."
Logical Conclusions:
Edmund Robert Harris was the principal benefactor of the Harris Museum.

"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> SUPPORTS -> "Edmund Robert Harris was the principal benefactor of the Harris Institute or Art School."
Logical Conclusions:
Both facts describe Edmund Robert Harris as a principal benefactor, with one noting the Harris Institute or Art School and the other the Harris Museum.

"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> (has ) -> [Edmund Robert Harris (Principal benefactor of Harris Museum and schools)]

[Harris Museum (Grade I listed museum in Preston)] -> (participates in event: The Sports Museum was founded in nearby Cambridge, Massachusetts.) -> "The Sports Museum was founded in nearby Cambridge, Massachusetts."

[The Sports Museum (non-profit museum in TD Garden)] -> (participates in event: The Sports Museum is currently located in the TD Garden in Boston, Massachusetts) -> "The Sports Museum is currently located in
```

the TD Garden in Boston, Massachusetts."

Level 0:

Subquery: What museum is Edmund Robert Harris the principal benefactor of?
"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> SUPPORTS -> "Edmund Robert Harris was the principal benefactor of the Harris Institute or Art School."
Logical Conclusions:
Both facts describe Edmund Robert Harris as a principal benefactor, with one noting the Harris Institute or Art School and the other the Harris Museum.

"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> SUPPORTS -> "Edmund Robert Harris was the principal benefactor of the Harris Orphanage."
Logical Conclusions:
Both facts describe Edmund Robert Harris as a principal benefactor, with one noting the Harris Orphanage and the other the Harris Museum.

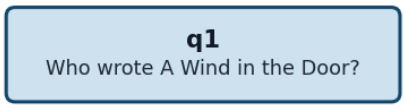
"Edmund Robert Harris was the principal benefactor of the Harris Museum." -> SUPPORTS -> "Edmund Robert Harris was an English lawyer from Preston, Lancashire, UK."
Logical Conclusions:
Both facts describe Edmund Robert Harris, with one noting he was an English lawyer and the other that he was principal benefactor o …[truncated for report]

Q86 [bridge / hard] 5ab870ee5542990e739ec8f6

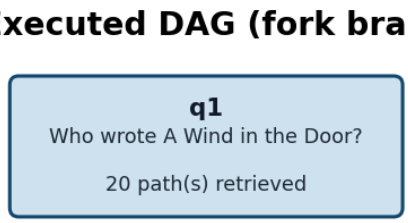
A Wind in the Door is part of a series written by who?

Gold answer	Madeleine L'Engle
LLM answer	Madeleine L'Engle
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who wrote A Wind in the Door?	20 path(s)

Final context fed to generation (997 tokens)

```
"A Wind in the Door was written by Madeleine L'Engle." -> AUTHOR_OF -> "A Wind in the Door is a young adult
science fantasy novel." -> SUPPORTS -> "The Time Quintet was written by Madeleine L'Engle."
Logical Conclusions:
Fact 25 identifies A Wind in the Door as a young adult science fantasy novel, and fact 26 states it was
written by Madeleine L'Engle.
Madeleine L'Engle wrote the Time Quintet and also wrote A Wind in the Door.

"A Wind in the Door was written by Madeleine L'Engle." -> SUPPORTS -> "The Time Quintet was written by
Madeleine L'Engle."
Logical Conclusions:
Madeleine L'Engle wrote the Time Quintet and also wrote A Wind in the Door.

"A Wind in the Door was written by Madeleine L'Engle." -> (has ) -> [madeleine l'engle (Author of A Wind in
the Door)] -> (participates in event: The Time Quintet was written by Madeleine L'Engle.) -> "The Time
Quintet was written by Madeleine L'Engle."
```

"The Time Quintet was written by Madeleine L'Engle." -> CONTAINS -> "A Wind in the Door is part of the Time Quintet."
Logical Conclusions:
The Time Quintet was written by Madeleine L'Engle and includes the novel 'A Wind in the Door'.

"A Wind in the Door was written by Madeleine L'Engle." -> SUPPORTS -> "A Wind in the Door is part of the Time Quintet." -> INCLUDES -> "A Wind in the Door is a young adult science fantasy novel."
Logical Conclusions:
A Wind in the Door is part of the Time Quintet and was written by Madeleine L'Engle.
Fact 25 identifies A Wind in the Door as a novel, and fact 28 states it is part of the Time Quintet.

"The Time Quintet was written by Madeleine L'Engle." -> SUPPORTS -> "A Wind in the Door is a companion book to 'A Wrinkle in Time'."
Logical Conclusions:
The novel 'A Wind in the Door' is a companion book to 'A Wrinkle in Time' and was written by Madeleine L'Engle.

"A Wind in the Door is a young adult science fantasy novel." -> RELATES_TO -> "Green Angel is a 2003 post-apocalyptic young adult novel."
Logical Conclusions:
Both A Wind in the Door and Green Angel are young adult novels, though of different subgenres.

[A Wind in the Door (Young adult science fantasy novel)] -> (participates in event: A Wind in the Door was written by Madeleine L'Engle.) -> "A Wind in the Door was written by Madeleine L'Engle."

"A Wind in the Door is a young adult science fantasy novel." -> (has) -> [A Wind in the Door (Young adult science fantasy novel)] -> (participates in event: The series of six novels was written by Compton Mackenzie.) -> "The series of six novels was written by Compton Mackenzie."

"A Wind in the Door is a young adult science fantasy novel." -> RELATES_TO -> "The Time Quintet is a fantasy/science fiction series of five young adult novels."
Logical Conclusions:
The Time Quintet is a fantasy/science fiction series, and A Wind in the Door is a young adult science fantasy novel.

"A Wind in the Door is part of the Time Quintet." -> (has) -> [A Wind in the Door (Young adult science fantasy novel)]

"The Time Quintet was written by Madeleine L'Engle." -> AUTHORED -> "The Time Quintet is a fantasy/science fiction series of five young adult novels."
Logical Conclusions:
The Time Quintet series was written by Madeleine L'Engle.

"A Wind in the Door is part of the Time Quintet." -> PART_OF -> "The Time Quintet is a fantasy/science fiction series of five young adult novels."
Logical Conclusions:
The Time Quintet series includes the novel 'A Wind in the Door'.

"A Wind in the Door is part of the Time Quintet." -> CONTAINS -> "A Wind in the Door is a companion book to 'A Wrinkle in Time'."
Logical Conclusions:
The novel 'A Wind in the Door' is a companion book to 'A Wrinkle in Time' and is also part of the Time Quintet series.

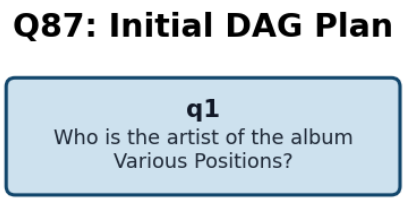
[A Wind in the Door (Young adult science fantasy novel)] -> (participates in event: A Wind in the Door is a companion book to 'A Wrinkle in Time'.) -> "A Wind in the Door is a companion book to 'A Wrinkle in Time'."

Q87 [bridge / hard] 5a8408a9554299123d8c21d3

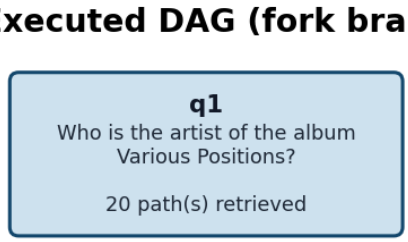
Which Canadian artist is credited with first coming up with the title of Various Positions?

Gold answer	Leonard Cohen
LLM answer	Leonard Cohen
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who is the artist of the album Various Positions?	20 path(s)

Final context fed to generation (1250 tokens)

```
"The album 'Various Positions' was released in 1985 in Europe and the United States." -> (has ) -> [Various Positions (Album and tour by Leonard Cohen)] -> (participates in event: The Various Positions Tour was a concert tour by Leonard Cohen.) -> "The Various Positions Tour was a concert tour by Leonard Cohen." -> (has ) -> [Leonard Cohen (Canadian singer-songwriter and poet)] -> (participates in event: Leonard Cohen's work explored personal relationships.) -> "Leonard Cohen's work explored personal relationships."

[Leonard Cohen (Canadian singer-songwriter and poet)] -> (participates in event: Leonard Cohen's work explored isolation.) -> "Leonard Cohen's work explored isolation."

[Leonard Cohen (Canadian singer-songwriter and poet)] -> (participates in event: Leonard Cohen was a musician.) -> "Leonard Cohen was a musician."

[Leonard Cohen (Canadian singer-songwriter and poet)] -> (participates in event: Leonard Cohen was inducted into the Rock and Roll Hall of Fame.) -> "Leonard Cohen was inducted into the Rock and Roll Hall of Fame."
```

[Leonard Cohen (Canadian singer-songwriter and poet)] -> (participates in event: The album 'Various Positions' was released in 1984 in Canada.) -> "The album 'Various Positions' was released in 1984 in Canada."

"The album 'Various Positions' was released in 1985 in Europe and the United States." -> PRECEDES -> "The Various Positions Tour was a concert tour by Leonard Cohen."
Logical Conclusions:
The album 'Various Positions' was released in 1985 in Europe and the United States and the Various Positions Tour was a concert tour by Leonard Cohen.

[Leonard Cohen (Canadian singer-songwriter and poet)] -> (participates in event: The Various Positions Tour supported Leonard Cohen's album 'Various Positions'.) -> "The Various Positions Tour supported Leonard Cohen's album 'Various Positions'."

"The Various Positions Tour was a concert tour by Leonard Cohen." -> SUPPORTS -> "The Various Positions Tour supported Leonard Cohen's album 'Various Positions'."
Logical Conclusions:
The Various Positions Tour supported Leonard Cohen's album 'Various Positions'.

"The Various Positions Tour was a concert tour by Leonard Cohen." -> IDENTIFIES -> "The film 'Various Positions' takes its title from an album by Leonard Cohen."
Logical Conclusions:
The film 'Various Positions' takes its title from an album by Leonard Cohen and the Various Positions Tour was a concert tour by Leonard Cohen.

"The Various Positions Tour was a concert tour by Leonard Cohen." -> FOLLOWS -> "The album 'Various Positions' was released in 1984 in Canada."
Logical Conclusions:
The album 'Various Positions' was released in 1984 in Canada and the Various Positions Tour was a concert tour by Leonard Cohen.

[Various Positions (Album and tour by Leonard Cohen)] -> (participates in event: The film 'Various Positions' takes its title from an album by Leonard Cohen.) -> "The film 'Various Positions' takes its title from an album by Leonard Cohen." -> (has) -> [various positions (film) (2002 film by Ori Kowarsky)] -> (participates in event: 'Various Positions' stars Carly Pope and Tygh Runyan.) -> "'Various Positions' stars Carly Pope and Tygh Runyan."

[various positions (film) (2002 film by Ori Kowarsky)] -> (participates in event: 'Various Positions' won the 2002 Prix de Montréal at the Montreal World Film Fes) -> "'Various Positions' won the 2002 Prix de Montréal at the Montreal World Film Festival."

"The film 'Various Positions' takes its title from an album by Leonard Cohen." -> ELABORATES -> "'Various Positions' was made by Vancouver, BC, lawyer and filmmaker Ori Kowarsky."
Logical Conclusions:
The film 'Various Positions' takes its title from an album by Leonard Cohen and was made by Vancouver, BC, lawyer and filmmaker Ori Kowarsky.

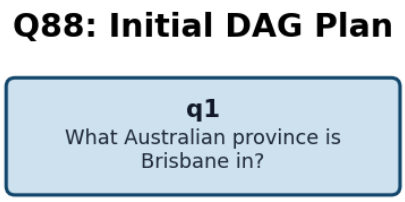
[various positions (film) (2002 film by Ori Kowarsky)] -> (participates in event: 'Various Positions' was made by Vancouver, BC, lawyer and filmmaker Ori Kowarsky) -> "'Various Positions' was made by Van
…[truncated for report]

Q88 [bridge / hard] 5a791ce955429974737f798e

Brett McClure competed in the pommel horse event at the 2001 Goodwill Games in what Australian province?

Gold answer	Queensland
LLM answer	Queensland
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What Australian province is Brisbane in?	20 path(s)

Final context fed to generation (1023 tokens)

```
[queensland (state hosting the games)] -> (participates in event: The 2001 Goodwill Games was held in
Brisbane, Queensland, Australia.) -> "The 2001 Goodwill Games was held in Brisbane, Queensland, Australia."
-> (has ) -> [brisbane (host city of the games)]

"The 2001 Goodwill Games was held in Brisbane, Queensland, Australia." -> (has ) -> [australia (Song by
Australian rock band Gyroscope)] -> (participates in event: Australia is a location.) -> "Australia is a
location." -> SAME_CATEGORY -> "Cyprus is a location."
Logical Conclusions:
Both Australia and Cyprus are locations where deer species are found.

"Australia is a location." -> SAME_CATEGORY -> "Lebanon is a location."
Logical Conclusions:
Both Australia and Lebanon are locations where deer species are found.
```

"Australia is a location." -> SAME_CATEGORY -> "Cape Verde is a location."
Logical Conclusions:
Both Australia and Cape Verde are locations where deer species are found.

"Australia is a location." -> SAME_CATEGORY -> "Algeria is a location."
Logical Conclusions:
Both Australia and Algeria are locations where deer species are found.

"Australia is a location." -> SAME_CATEGORY -> "New Zealand is a location."
Logical Conclusions:
Both Australia and New Zealand are locations where deer species are found.

"'Australia' is from the album 'Breed Obsession'." -> (has) -> [australia (Song by Australian rock band Gyroscope)]

"Australia is a location." -> SAME_CATEGORY -> "Réunion is a location."
Logical Conclusions:
Both Australia and Réunion are locations where deer species are found.

[australia (Song by Australian rock band Gyroscope)] -> (participates in event: Red deer have been introduced to other areas, including Australia.) -> "Red deer have been introduced to other areas, including Australia." -> INTRODUCED_TO -> "Red deer have been introduced to Canada."
Logical Conclusions:
Red deer have been introduced to Australia and Canada.

[australia (Song by Australian rock band Gyroscope)] -> (participates in event: 'Australia' is a song by Australian rock band Gyroscope.) -> "'Australia' is a song by Australian rock band Gyroscope." -> IDENTIFIES -> "One of the acoustic pieces was a live acoustic version of 'Australia'."
Logical Conclusions:
Fact 28 states 'Australia' was an acoustic piece on One Acoustic Night, and fact 29 identifies it as a song by Gyroscope.

"'Australia' is a song by Australian rock band Gyroscope." -> IS_FROM -> "'Australia' is from the album 'Breed Obsession'."
Logical Conclusions:
The song 'Australia' is from the album 'Breed Obsession'.

"Red deer have been introduced to other areas, including Australia." -> INTRODUCED_TO -> "Red deer have been introduced to other areas, including Peru."
Logical Conclusions:
Red deer have been introduced to Australia and Peru.

"'Australia' is a song by Australian rock band Gyroscope." -> RELEASED_ON -> "'Australia' was released to radio in May 2008."
Logical Conclusions:
The song 'Australia' was released to radio in May 2008.

"Red deer have been introduced to other areas, including Australia." -> INTRODUCED_TO -> "Red deer have been introduced to New Zealand."
Logical Conclusions:
Red deer have been introduced to Australia and New Zealand.

"Red deer have been introduced to other areas, including Australia." -> INTRODUCED_TO -> "Red deer have been introduced to other areas, including Uruguay."
Logical Conclusions:
Red deer have been introduced to Australia and Uruguay.

"Red deer have been introduced to other areas, including Australia." -> (has) -> [red deer (One of the largest deer species)] -> (participates in event: Red deer have been introduced to Australia.) -> "Red deer have been introduced to Australia."

"'Australia' is a song by Australian rock band Gyroscope." -> IS_SINGLE_FROM -> "'Australia' was released

as the third single from the album 'Breed Obsession'."
Logical Conclusions:
The song 'Australia' was released as the third single from its album.

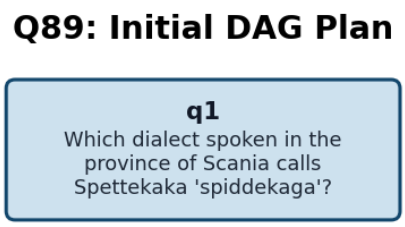
[red deer (One of the largest deer species …[truncated for report]

Q89 [bridge / hard] 5a70f0a75542994082a3e403

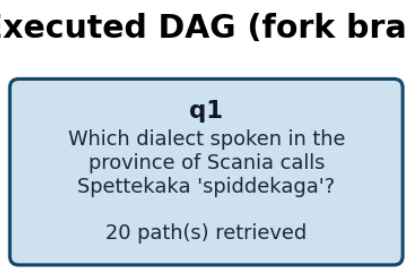
Which dialect spoken in the province of Scania calls Spettekaka, "spiddekaga"?

Gold answer	Scanian dialects
LLM answer	Scanian
EM / F1	0.00 / 0.67
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which dialect spoken in the province of Scania calls Spettekaka 'spiddekaga'?	20 path(s)

Final context fed to generation (804 tokens)

```
"Spettekaka is called 'spiddekaga' in native Scanian." -> CONTEXTUALIZES -> "Scanian is a closely related
group of South Swedish dialects." -> LOCATES -> "Scanian is spoken in the province of Scania in southern
Sweden."
Logical Conclusions:
Fact 5 gives the Scanian name for Spettekaka, and fact 9 defines Scanian as a group of South Swedish
dialects, linking the dialect to the dessert's name.
Scanian is a group of South Swedish dialects spoken in the province of Scania in southern Sweden.

"Scanian is a closely related group of South Swedish dialects." -> (has ) -> [scanian dialect (South
Swedish dialect group in Scania)] -> (participates in event: Scanian is spoken in the province of Scania in
southern Sweden.) -> "Scanian is spoken in the province of Scania in southern Sweden."
```

[spettekaka (local dessert of southern Sweden)] -> (participates in event: Spettekaka is called 'spiddekaga' in native Scanian.) -> "Spettekaka is called 'spiddekaga' in native Scanian." -> (has) -> [Scania (Province in southern Sweden)] -> (participates in event: Scanian is spoken in the province of Scania in southern Sweden.) -> "Scanian is spoken in the province of Scania in southern Sweden."

[scanian dialect (South Swedish dialect group in Scania)] -> (participates in event: Scanian is by most historical linguists considered to be an East Danish dialect) -> "Scanian is by most historical linguists considered to be an East Danish dialect group."

[scanian dialect (South Swedish dialect group in Scania)] -> (participates in event: Due to modern-era influence from Standard Swedish in the region, Scanian dialect) -> "Due to modern-era influence from Standard Swedish in the region, Scanian dialects have normally been treated as a South Swedish dialect group in Swedish dialect research."

[scanian dialect (South Swedish dialect group in Scania)] -> (participates in event: Scanian formed part of the old Scandinavian dialect continuum.) -> "Scanian formed part of the old Scandinavian dialect continuum."

[scanian dialect (South Swedish dialect group in Scania)] -> (participates in event: Some linguists, such as Elias Wessén, considered Old Scanian a separate language) -> "Some linguists, such as Elias Wessén, considered Old Scanian a separate language."

[scanian dialect (South Swedish dialect group in Scania)] -> (participates in event: Many of the early Scandinavian linguists, including Adolf Noreen and G. Sjöstedt) -> "Many of the early Scandinavian linguists, including Adolf Noreen and G. Sjöstedt, classified Scanian as 'South Scandinavian'."

"Scanian is spoken in the province of Scania in southern Sweden." -> LOCATES -> "Scanian is by most historical linguists considered to be an East Danish dialect group."
Logical Conclusions:
Scanian is spoken in Scania and is considered an East Danish dialect group.

"Scanian is spoken in the province of Scania in southern Sweden." -> LOCATES -> "Scanian formed part of the old Scandinavian dialect continuum."
Logical Conclusions:
Scanian is spoken in Scania and formed part of the old Scandinavian dialect continuum.

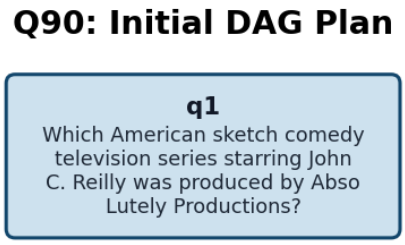
[Scania (Province in southern Sweden)] -> (participates in event: Spettekaka is also in Halland.) -> "Spettekaka is also in Halland." -> RELATES_TO -> "Spettekaka is called 'spiddekaga' in native Scanian."
Logical Conclusions:
Spettekaka is also in Halland and is called 'spiddekaga' in native Scanian.

Q90 [bridge / hard] 5a8ec02e5542995a26add501

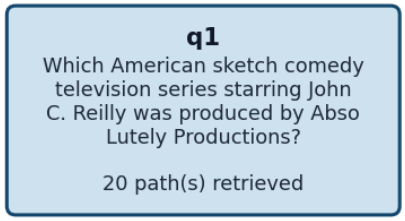
Which American sketch comedy television series starring John C. Reilly was produced by Abso Lutely Productions?

Gold answer	Check It Out!
LLM answer	Check It Out! with Dr. Steve Brule
EM / F1	**0.00 / 0.60**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which American sketch comedy television series starring John C. Reilly was produced by Abso Lutely Productions?	20 path(s)

Final context fed to generation (1227 tokens)

```
"Check It Out! with Dr. Steve Brule is an American sketch comedy television series." -> (has ) -> [check it out! with dr. steve brule (Sketch comedy series starring John C. Reilly)] -> (participates in event: Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Bru) -> "Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'." -> CO_PRODUCT -> "'555' was produced by Abso Lutely Productions."
Logical Conclusions:
Both facts are specific shows produced by Abso Lutely Productions.
```

"Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'." -> (has) -> [Abso Lutely Productions (Film and TV production company)] -> (participates in event: '555' was produced by Abso Lutely Productions.) -> "'555' was produced by Abso Lutely Productions."

"Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'." -> CO_PRODUCT -> "Hot Package was produced by Abso Lutely Productions."
Logical Conclusions:
Both facts are specific shows produced by Abso Lutely Productions.

"Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'." -> CO_PRODUCT -> "Abso Lutely Productions is known for producing 'Tom Goes to the Mayor'."
Logical Conclusions:
Both facts are specific shows produced by Abso Lutely Productions.

[Abso Lutely Productions (Film and TV production company)] -> (participates in event: Hot Package was produced by Abominable Pictures.) -> "Hot Package was produced by Abominable Pictures."

"Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'." -> EXEMPLIFIES -> "Abso Lutely Productions produces a variety of television and films."
Logical Conclusions:
Both facts describe the production activities of Abso Lutely Productions.

[Abso Lutely Productions (Film and TV production company)] -> (participates in event: Abso Lutely Productions is known for producing 'Tim and Eric Awesome Show, Great) -> "Abso Lutely Productions is known for producing 'Tim and Eric Awesome Show, Great Job!'."

[Abso Lutely Productions (Film and TV production company)] -> (participates in event: Abso Lutely Productions is known for producing 'Nathan For You'.) -> "Abso Lutely Productions is known for producing 'Nathan For You'."

[Abso Lutely Productions (Film and TV production company)] -> (participates in event: Abso Lutely Productions is known for producing 'Tom Goes to the Mayor'.) -> "Abso Lutely Productions is known for producing 'Tom Goes to the Mayor'."

[Abso Lutely Productions (Film and TV production company)] -> (participates in event: Abso Lutely Productions is a film and television production company.) -> "Abso Lutely Productions is a film and television production company."

"Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'." -> PRODUCED_BY -> "Abso Lutely Productions is a film and television production company."
Logical Conclusions:
Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'.

"Check It Out! with Dr. Steve Brule is an American sketch comedy television series." -> STARRED_IN -> "Check It Out! with Dr. Steve Brule stars John C. Reilly as Dr. Steve Brule." -> (has) -> [check it out! with dr. steve brule (Sketch comedy series starring John C. Reilly)]
Logical Conclusions:
Check It Out! with Dr. Steve Brule stars John C. Reilly as Dr. Steve Brule.

[john c. reilly (Writer and star of Bagboy)] -> (participates in event: Check It Out! with Dr. Steve Brule stars John C. Reilly as Dr. Steve Brule.) -> "Check It Out! with Dr. Steve Brule stars John C. Reilly as Dr. Steve Brule."

"Abso Lutely Productions is known for producing 'Nathan For You'." -> CO_PRODUCT -> "Abso Lutely Productions is known for producing 'Check It Out! with Dr. Steve Brule'."
Logical Conclusions:
Both facts are specific shows produced by Abso Lutely Productions.

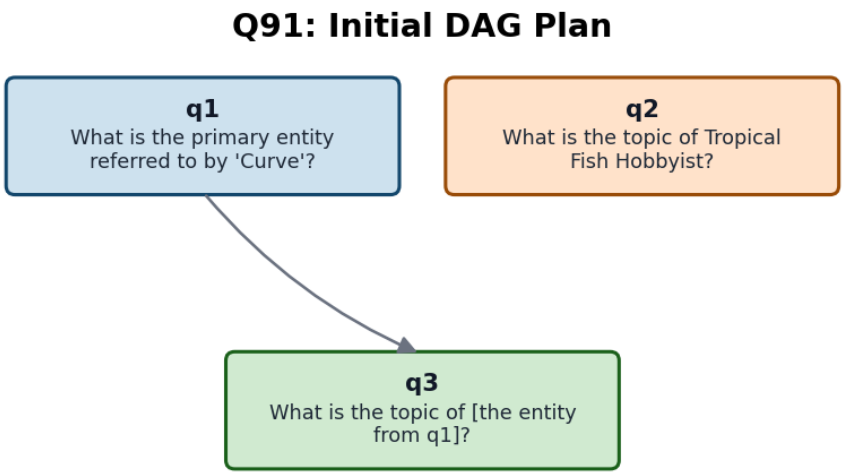
[check it out! with dr. steve brule (Sketch comedy series starring John C. Reilly)] -> (participat
…[truncated for report]

Q91 [comparison / hard] 5a8835245542994846c1ce27

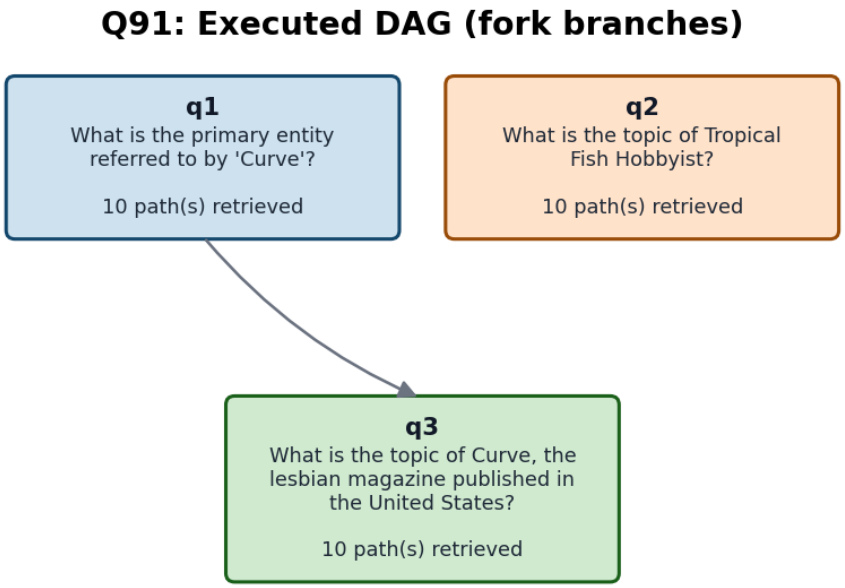
Do Tropical Fish Hobbyist and Curve cover the same topic?

Gold answer	no
LLM answer	No
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the primary entity referred to by 'Curve'?	10 path(s)
q2	0	—	What is the topic of Tropical Fish Hobbyist?	10 path(s)
q3	1	q1	What is the topic of Curve, the lesbian magazine published in the United States?	10 path(s)

Final context fed to generation (1317 tokens)

```
Query plan:
- q1: What is the primary entity referred to by 'Curve'?
- q2: What is the topic of Tropical Fish Hobbyist?
- q3: What is the topic of [the entity from q1]? (depends on: q1)

Level 1:

Subquery: What is the topic of Curve, the lesbian magazine published in the United States?
"Curve is a lesbian magazine published in the United States." -> INCLUDES -> "Curve covers social issues."
Logical Conclusions:
Curve, a lesbian magazine, covers social issues.

"Curve is a lesbian magazine published in the United States." -> INCLUDES -> "Curve has a website that
hosts an internet forum focusing on lesbian issues."
Logical Conclusions:
Curve has a website that hosts an internet forum focusing on lesbian issues.

"Curve is a lesbian magazine published in the United States." -> INCLUDES -> "Curve includes stories on
travel."
Logical Conclusions:
Curve, a lesbian magazine, includes stories on travel.

"Curve is a lesbian magazine published in the United States." -> INCLUDES -> "Curve includes celebrity
interviews."
Logical Conclusions:
Curve, a lesbian magazine, includes celebrity interviews.

"Curve is a lesbian magazine published in the United States." -> INCLUDES -> "Curve covers politics."
Logical Conclusions:
Curve, a lesbian magazine, covers politics.

[curve (Lesbian magazine published in US)] -> (participates in event: Curve is a lesbian magazine published
in the United States.) -> "Curve is a lesbian magazine published in the United States."

[curve (Lesbian magazine published in US)] -> (participates in event: Curve covers social issues.) ->
"Curve covers social issues."

[curve (Lesbian magazine published in US)] -> (participates in event: Curve has a website that hosts an
internet forum focusing on lesbian issues.) -> "Curve has a website that hosts an internet forum focusing
on lesbian issues."

Level 0:

Subquery: What is the primary entity referred to by 'Curve'?
[curve (Lesbian magazine published in US)] -> (participates in event: Curve is a lesbian magazine published
in the United States.) -> "Curve is a lesbian magazine published in the United States."

"Curve covers politics." -> PART_OF -> "Curve is a lesbian magazine published in the United States."
Logical Conclusions:
Curve, a lesbian magazine, covers politics.
```

[curve (Lesbian magazine published in US)] -> (participates in event: Curve includes celebrity interviews.) -> "Curve includes celebrity interviews."

[curve (Lesbian magazine published in US)] -> (participates in event: Curve covers social issues.) -> "Curve covers social issues."

[curve (Lesbian magazine published in US)] -> (participates in event: Curve includes stories on style.) -> "Curve includes stories on style."

"Curve covers politics." -> (has) -> [curve (Lesbian magazine published in US)]

"Curve covers politics." -> HAS_WEBSITE -> "Curve has a website that hosts an internet forum focusing on lesbian issues."
Logical Conclusions:
Curve covers politics and has a website that hosts an internet forum focusing on lesbian issues.

"Curve covers politics." -> COVERS -> "Curve covers social issues."
Logical Conclusions:
Curve covers politics and social issues.

"Curve covers politics." -> COVERS -> "Curve covers news."
Logical Conclusions:
Curve covers politics and news.

"Curve covers politics." -> COVERS -> "Curve includes stories on travel."
Logical Conclusions:
Curve covers politics and includes stories on travel.

Subquery: What is the topic of Tropical Fish Hobbyist?
"Tropical Fish Hobbyist Magazine is geared to hobbyist keepers of tropical fish." -> INCLUDES -> "Tropical Fish Hobbyist Magazine includes news and information on Species Reports."
Logical Conclusions:
Tropical Fish Hobbyist Magazine, geared to hobbyist keepers of tropical fish, includes news and information on Species Reports.

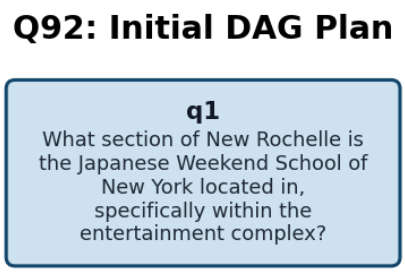
"Tropical Fish Hobbyist Magazine is geared to hobbyist keepers of tropical fish." -> IDENTIFIES -> "Tropical Fish Hobbyist Magazine is geared to hobbyist keepers of tropical fish."
Logical Conclusion …[truncated for report]

Q92 [bridge / hard] 5ae5415055429908b63265b7

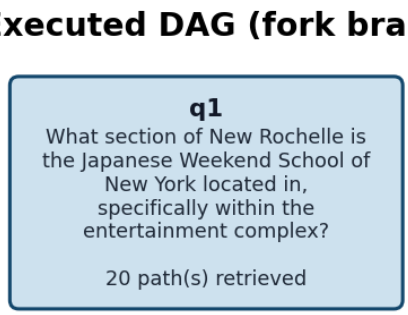
Japanese Weekend School of New York has offices in the entertainment complex in what section of New Rochelle?

Gold answer	Downtown
LLM answer	Downtown
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What section of New Rochelle is the Japanese Weekend School of New York located in, specifically within the entertainment complex?	20 path(s)

Final context fed to generation (1510 tokens)

```
[Japanese Weekend School of New York (Japanese supplementary school in New York area)] -> (participates in event: The Japanese Weekend School of New York has its offices in New Roc City in New R) -> "The Japanese Weekend School of New York has its offices in New Roc City in New Rochelle, New York." -> HOUSES -> "The Japanese Weekend School of New York (JWSNY) is a Japanese supplementary school in the New York City metropolitan area." -> IS_PART_OF -> "The JWSNY is one of the JEI's two weekend school systems."
Logical Conclusions:
The Japanese Weekend School of New York has its offices in New Roc City in New Rochelle.
```

The Japanese Weekend School of New York is a supplementary school and is one of the JEI's two weekend school systems.

[Japanese Weekend School of New York (Japanese supplementary school in New York area)] -> (participates in event: The JWSNY is one of the JEI's two weekend school systems.) -> "The JWSNY is one of the JEI's two weekend school systems."

[japanese educational institute of new york (Organization managing the Japanese school system)] -> (participates in event: The JWSNY is one of the JEI's two weekend school systems.) -> "The JWSNY is one of the JEI's two weekend school systems."

[Japanese Weekend School of New York (Japanese supplementary school in New York area)] -> (participates in event: The Japanese Weekend School of New York (JWSNY) is a Japanese supplementary scho) -> "The Japanese Weekend School of New York (JWSNY) is a Japanese supplementary school in the New York City metropolitan area."

[japanese educational institute of new york (Organization managing the Japanese school system)] -> (participates in event: JEI also operates two Japanese weekend school systems in the New York City area.) -> "JEI also operates two Japanese weekend school systems in the New York City area." -> INCLUDES -> "The JWSNY is one of the JEI's two weekend school systems."
Logical Conclusions:
JEI operates two weekend school systems, one of which is the JWSNY.

[japanese educational institute of new york (Organization managing the Japanese school system)] -> (participates in event: The JEI also operates two Japanese day schools in the New York area.) -> "The JEI also operates two Japanese day schools in the New York area." -> CORROBORATES -> "The JWSNY is one of the JEI's two weekend school systems."
Logical Conclusions:
The JEI operates both weekend and day schools in the New York area.

[New Roc City (Entertainment and residential complex)] -> (participates in event: New Roc City is located in the Downtown section of the city of New Rochelle.) -> "New Roc City is located in the Downtown section of the city of New Rochelle." -> (has) -> [New Rochelle (City in Westchester County, New York)] -> (participates in event: Roderick Kennedy, Jr. worked in partnership with the City of New Rochelle to cre) -> "Roderick Kennedy, Jr. worked in partnership with the City of New Rochelle to create the New Rochelle Walk of Fame." -> (has) -> [Roderick Kennedy, Jr. (Former resident who created and funded the Walk of Fame)] -> (participates in event: Channing was an All-American halfback for Princeton University.) -> "Channing was an All-American halfback for Princeton University."

[Roderick Kennedy, Jr. (Former resident who created and funded the Walk of Fame)] -> (participates in event: Channing later went into the mining business.) -> "Channing later went into the mining business."

[Roderick Kennedy, Jr. (Former resident who created and funded the Walk of Fame)] -> (participates in event: Roderick Kennedy, Jr. worked in partnership with the New Rochelle Business Impro) -> "Roderick Kennedy, Jr. worked in partnership with the New Rochelle Business Improvement District to create the New Rochelle Walk of Fame."

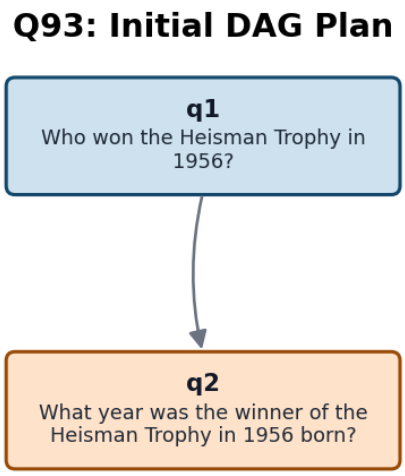
[New Roc City (Entertainment and residential complex)] -> (participates in event: New Roc City is an entertainment, retail and residential complex.) -> "New Roc City is an entertainment, retail and residential complex." …[truncated for report]

Q93 [bridge / hard] 5ae6fee3554299572ea54670

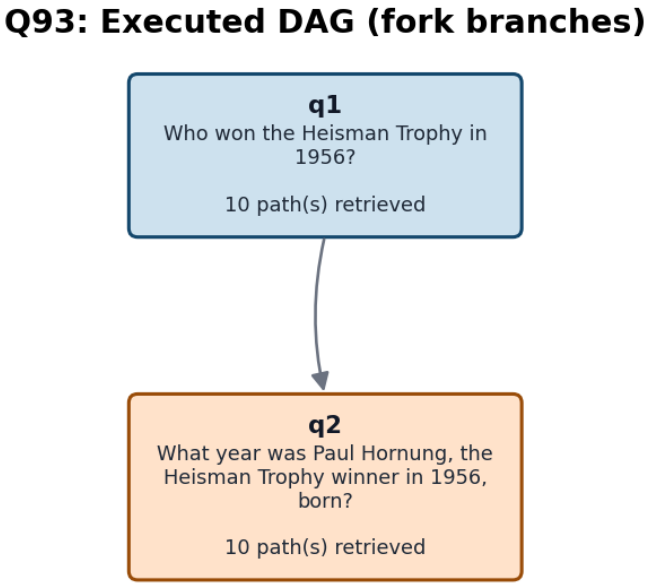
The player who defeated Johnny Majors for the Heisman Trophy in 1956 was born in what year?

Gold answer	1935
LLM answer	1935
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who won the Heisman Trophy in 1956?	10 path(s)
q2	1	q1	What year was Paul Hornung, the Heisman Trophy winner in 1956, born?	10 path(s)

Final context fed to generation (1470 tokens)

```
Query plan:
- q1: Who won the Heisman Trophy in 1956?
- q2: What year was the winner of the Heisman Trophy in 1956 born? (depends on: q1)

Level 1:

Subquery: What year was Paul Hornung, the Heisman Trophy winner in 1956, born?
"Paul Vernon Hornung was born on December 23, 1935." -> PRECEDES -> "Paul Hornung is the first in pro
football history to win the Heisman Trophy."
Logical Conclusions:
Paul Hornung's birth in 1935 is a biographical detail that relates to his later achievement of being the
first to win the Heisman Trophy.

"Paul Vernon Hornung was born on December 23, 1935." -> (has ) -> [paul hornung (Hall of Fame running back
and Heisman winner)]

"Paul Vernon Hornung was born on December 23, 1935." -> PRECEDES -> "Paul Hornung played for the Green Bay
Packers of the National Football League (NFL) from 1957 to 1966."
Logical Conclusions:
Paul Hornung's birth in 1935 is a biographical detail that relates to his later playing career for the
Green Bay Packers from 1957 to 1966.

"Paul Vernon Hornung was born on December 23, 1935." -> IDENTIFIES -> "Paul Hornung is nicknamed 'The
Golden Boy'."
Logical Conclusions:
Paul Hornung, born on December 23, 1935, is known by the nickname 'The Golden Boy'.

"Paul Vernon Hornung was born on December 23, 1935." -> PRECEDES -> "Paul Hornung is a Hall of Fame running
back."
Logical Conclusions:
Paul Hornung's birth in 1935 is a biographical detail that relates to his later status as a Hall of Fame
running back.

"Paul Vernon Hornung was born on December 23, 1935." -> PRECEDES -> "Paul Hornung was selected as the first
overall selection in the NFL Draft."
Logical Conclusions:
Paul Hornung's birth in 1935 preceded his selection as the first overall pick in the NFL Draft.

[paul hornung (Hall of Fame running back and Heisman winner)] -> (participates in event: Paul Hornung is
the first in pro football history to win the Heisman Trophy.) -> "Paul Hornung is the first in pro football
history to win the Heisman Trophy."

[paul hornung (Hall of Fame running back and Heisman winner)] -> (participates in event: John Terrill
Majors finished second to Paul Hornung in voting for the Heisman Tr) -> "John Terrill Majors finished
second to Paul Hornung in voting for the Heisman Trophy in 1956."

[paul hornung (Hall of Fame running back and Heisman winner)] -> (participates in event: Paul Hornung
played for the Green Bay Packers of the National Football League (N) -> "Paul Hornung played for the Green
Bay Packers of the National Football League (NFL) from 1957 to 1966."

[paul hornung (Hall of Fame running back and Heisman winner)] -> (participates in event: Paul Hornung won
the NFL most valuable player award.) -> "Paul Hornung won the NFL most valuable player award."
```

Level 0:

Subquery: Who won the Heisman Trophy in 1956?
"John Terrill Majors finished second to Paul Hornung in voting for the Heisman Trophy in 1956." -> (has)
-> [paul hornung (Hall of Fame running back and Heisman winner)]

"John Terrill Majors finished second to Paul Hornung in voting for the Heisman Trophy in 1956." -> SUPPORTS
-> "John Terrill Majors won the Southeastern Conference Most Valuable Player award in 1956."
Logical Conclusions:
John Terrill Majors finished second in Heisman voting in 1956 and won SEC MVP that same year.

"John Terrill Majors finished second to Paul Hornung in voting for the Heisman Trophy in 1956." ->
CORROBORATES -> "John Terrill Majors was an All-American in 1956."
Logical Conclusions:
John Majors was an All-American in 1956 and finished second in Heisman voting that same year.

"John Terrill Majors finished second to Paul Hornung in voting for the Heisman Trophy in 1956." -> SUPPORTS
-> "John Terrill Majors is a former American football player and coach."
Logical Conclusions:
John Terrill Majors, a former player and coach, finished second in Heisman voting in 1956.

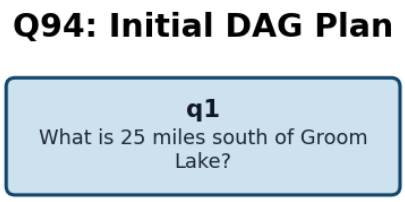
"John Terrill Majors finished second to Paul Hornung in voting for the Heisman Trophy in 1956." -> SUPPORTS
-> "John Terrill Majors was …[truncated for report]

Q94 [bridge / hard] 5adfff0755429925eb1afbce

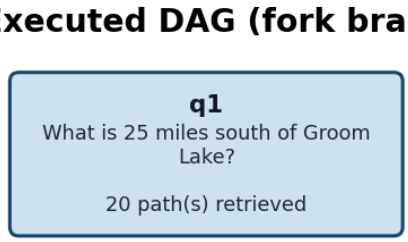
What is 25 miles south of Groom Lake?

Gold answer	Rachel, Nevada
LLM answer	Rachel, Nevada
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is 25 miles south of Groom Lake?	20 path(s)

Final context fed to generation (1517 tokens)

```
[Groom Lake (salt flat and restricted military facility in Nevada)] -> (participates in event: Groom Lake is approximately 3 miles from east to west at its widest point.) -> "Groom Lake is approximately 3 miles from east to west at its widest point." -> DIMENSION_OF -> "Groom Lake is 25 miles south of Rachel, Nevada." -> USED_FOR -> "Groom Lake is used for runways of the Nellis Bombing Range Test Site airport (KXTA)."
Logical Conclusions:
Groom Lake is 25 miles south of Rachel, Nevada, and is approximately 3 miles from east to west at its widest point.
Groom Lake is 25 miles south of Rachel, Nevada and is used for runways of the Nellis Bombing Range Test Site airport (KXTA).

"Groom Lake is 25 miles south of Rachel, Nevada." -> (has ) -> [rachel, nevada (Closest habitation to Area 51)] -> (participates in event: Rachel enjoys a modest celebrity among UFO hunters.) -> "Rachel enjoys a modest celebrity among UFO hunters."
```

"Groom Lake is 25 miles south of Rachel, Nevada." -> LOCATED_NORTH_OF -> "Rachel is the closest habitation to the Nellis Air Force Range and Area 51."
Logical Conclusions:
Rachel is the closest habitation to the Nellis Air Force Range and Area 51, and Groom Lake is 25 miles south of Rachel.

[rachel, nevada (Closest habitation to Area 51)] -> (participates in event: Rachel enjoys a modest celebrity among aviation enthusiasts.) -> "Rachel enjoys a modest celebrity among aviation enthusiasts."

"Groom Lake is 25 miles south of Rachel, Nevada." -> LOCATED_NORTH_OF -> "Groom Lake is a salt flat in Nevada."
Logical Conclusions:
Groom Lake is located 25 miles south of Rachel, Nevada.

"Groom Lake is 25 miles south of Rachel, Nevada." -> LOCATED_NORTH_OF -> "Groom Dry Lake is the home of the Area 51 facility."
Logical Conclusions:
Groom Dry Lake is the home of the Area 51 facility, and Groom Lake is 25 miles south of Rachel, Nevada.

[rachel, nevada (Closest habitation to Area 51)] -> (participates in event: North of Rachel is the Quinn Canyon Range.) -> "North of Rachel is the Quinn Canyon Range."

"Groom Lake is 25 miles south of Rachel, Nevada." -> LOCATED_SOUTHEAST_OF -> "Tonopah is located northwest of Groom Dry Lake."
Logical Conclusions:
Tonopah is located northwest of Groom Dry Lake, and Groom Lake is 25 miles south of Rachel, Nevada.

[rachel, nevada (Closest habitation to Area 51)] -> (participates in event: Rachel is the closest habitation to the Nellis Air Force Range and Area 51.) -> "Rachel is the closest habitation to the Nellis Air Force Range and Area 51."

[Groom Lake (salt flat and restricted military facility in Nevada)] -> (participates in event: Groom Lake is located immediately north of the Area 51 USAF installation.) -> "Groom Lake is located immediately north of the Area 51 USAF installation." -> LOCATED_NEAR -> "Groom Lake is 25 miles south of Rachel, Nevada."
Logical Conclusions:
Groom Lake is 25 miles south of Rachel, Nevada and is located immediately north of the Area 51 USAF installation.

[rachel, nevada (Closest habitation to Area 51)] -> (participates in event: As of the 2010 census, Rachel had a population of 54.) -> "As of the 2010 census, Rachel had a population of 54."

[rachel, nevada (Closest habitation to Area 51)] -> (participates in event: Rachel is a census-designated place (CDP) in Lincoln County, Nevada, United Stat) -> "Rachel is a census-designated place (CDP) in Lincoln County, Nevada, United States."

"Groom Lake is approximately 3.7 miles from north to south." -> (has) -> [Groom Lake (salt flat and restricted military facility in Nevada)]

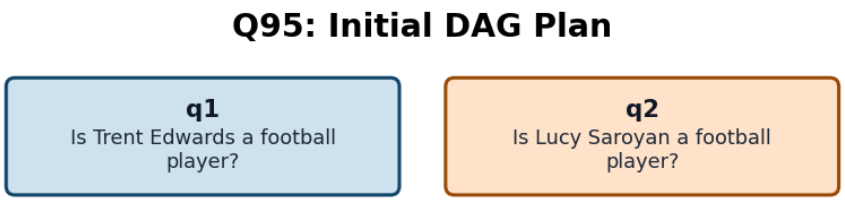
[Groom Lake (salt flat and restricted military facility in Nevada)] -> (participates in event: Groom Lake is located within the namesake Groom Lake Valley portion of the Tonop) -> "Groom Lake is located within the namesake Groom Lake Valley portion of the Tonopah Basin." -> LOCATED_NEAR -> "Groom Lake is 25 miles south of Rachel, Nevada."
Logical Conclusions:
Groom Lake is located within the namesake Groom Lake Valley portion of th …[truncated for report]

Q95 [comparison / hard] 5a8482e55542997175ce1edc

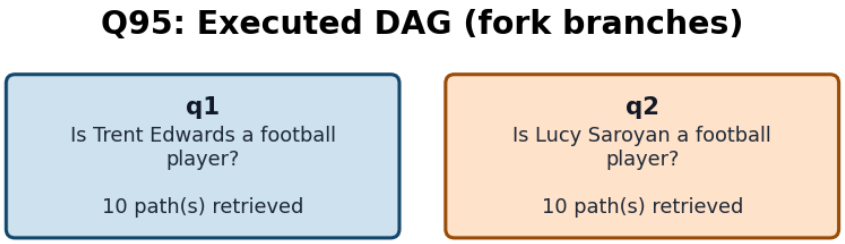
Are Trent Edwards and Lucy Saroyan both football players?

Gold answer	no
LLM answer	No, Trent Edwards is a football player, but Lucy Saroyan is not.
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Is Trent Edwards a football player?	10 path(s)
q2	0	—	Is Lucy Saroyan a football player?	10 path(s)

Final context fed to generation (953 tokens)

```
Query plan:
- q1: Is Trent Edwards a football player?
- q2: Is Lucy Saroyan a football player?

Level 0:

Subquery: Is Trent Edwards a football player?
"Trent Addison Edwards is a former American football quarterback." -> EDUCATED_AT -> "Trent Edwards played
college football at Stanford."
Logical Conclusions:
Trent Edwards, a former American football quarterback, played college football at Stanford.
```

"Trent Addison Edwards is a former American football quarterback." -> IDENTIFIES -> "Trent Edwards was drafted by the Buffalo Bills in the third round of the 2007 NFL Draft."
Logical Conclusions:
Trent Addison Edwards, a former American football quarterback, was drafted by the Buffalo Bills in the third round of the 2007 NFL Draft.

"Trent Addison Edwards is a former American football quarterback." -> (has) -> [trent edwards (former American football quarterback)]

"Trent Edwards played college football at Stanford." -> (has) -> [trent edwards (former American football quarterback)]

"Trent Edwards played college football at Stanford." -> PRECEDES -> "Trent Edwards was drafted by the Buffalo Bills in the third round of the 2007 NFL Draft."
Logical Conclusions:
After playing at Stanford, Trent Edwards was drafted by the Buffalo Bills in the third round of the 2007 NFL Draft.

"Trent Addison Edwards is a former American football quarterback." -> IDENTIFIES -> "Trent Addison Edwards was born on October 30, 1983."
Logical Conclusions:
Trent Addison Edwards, born October 30, 1983, is a former American football quarterback.

"Trent Edwards played college football at Stanford." -> (has) -> [Stanford University (Aldrich's PhD alma mater)]

Subquery: Is Lucy Saroyan a football player?
[lucy saroyan (American actress and photographer)] -> (participates in event: Lucy Saroyan was an American actress and photographer.) -> "Lucy Saroyan was an American actress and photographer."

[lucy saroyan (American actress and photographer)] -> (participates in event: Lucy Saroyan was born on January 17, 1946.) -> "Lucy Saroyan was born on January 17, 1946."

"Lucy Saroyan was born on January 17, 1946." -> CONTEXTUALIZES -> "Lucy Saroyan was an American actress and photographer."
Logical Conclusions:
Lucy Saroyan's birth date is a biographical detail for her career as an actress and photographer.

[lucy saroyan (American actress and photographer)] -> (participates in event: Lucy Saroyan died on April 11, 2003.) -> "Lucy Saroyan died on April 11, 2003."

"Lucy Saroyan was born on January 17, 1946." -> BOUNDS -> "Lucy Saroyan died on April 11, 2003."
Logical Conclusions:
Lucy Saroyan's birth and death dates define her lifespan.

"Les Joueurs de football is also referred to as Football Players." -> SUPPORTS -> "Unlike Serie A and Premier League, third-party ownership is allowed in Portugal."
Logical Conclusions:
Fact 5 mentions Albert Gleizes' painting 'Football Players', and fact 67 states 'Les Joueurs de football' is also referred to as 'Football Players'.

"Les Joueurs de football is also referred to as Football Players." -> IDENTIFIES -> "Les Joueurs de football is a 1912-13 painting."
Logical Conclusions:
Les Joueurs de football, also referred to as Football Players, is a 1912-13 painting.

"He represented SFU at the 2005 East West Bowl held in Waterloo, Ontario." -> IDENTIFIES -> "2006 was his first year playing professional football."
Logical Conclusions:
He represented SFU at the 2005 East West Bowl held in Waterloo, Ontario, and 2006 was his first year playing professional football.

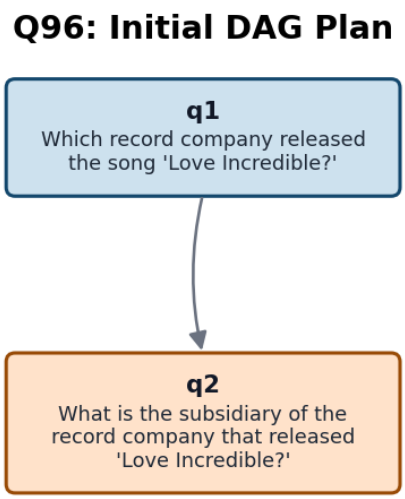
"He represented SFU at the 2005 East West Bowl held in Waterloo, Ontario." -> REPRESENTED_BY -> "Valli played for the Simon Fraser University football team, the Simon Fraser Clan."
Logical Conclusions:
Valli represented SFU at the 2005 East West Bowl while playing for the Clan.

Q96 [bridge / hard] 5a8db348554299068b959dc2

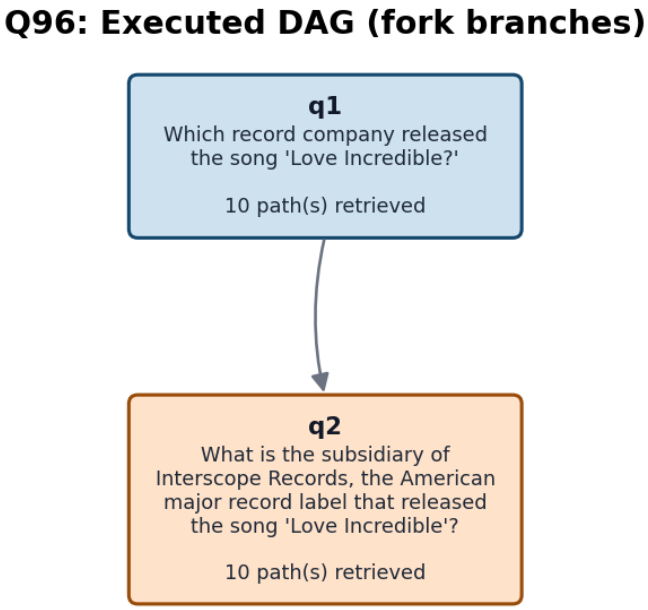
What is the subsidiary of the record company who released "Love Incredible?"

Gold answer	Vivendi S.A.
LLM answer	Mad Love
EM / F1	**0.00 / 0.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Which record company released the song 'Love Incredible?'	10 path(s)
q2	1	q1	What is the subsidiary of Interscope Records, the American major record label that released the song 'Love Incredible'?	10 path(s)

Final context fed to generation (1146 tokens)

```
Query plan:
- q1: Which record company released the song 'Love Incredible?'
- q2: What is the subsidiary of the record company that released 'Love Incredible?' (depends on: q1)

Level 1:

Subquery: What is the subsidiary of Interscope Records, the American major record label that released the song 'Love Incredible'?
[Interscope Records (American major record label)] -> (participates in event: "Love Incredible" was released through Interscope and Mad Love.) -> ""Love Incredible" was released through Interscope and Mad Love."

""Love Incredible" was released through Interscope and Mad Love." -> SUPPORTS -> ""Love Incredible" was released on 17 February 2017."
Logical Conclusions:
Both facts provide details about the release of 'Love Incredible', with one specifying the date and the other the record label.

""Love Incredible" was released through Interscope and Mad Love." -> (has ) -> [South African Music Entities (Record labels and song from South Africa)]

""Love Incredible" was released through Interscope and Mad Love." -> RELEASED -> ""Love Incredible" is a song by Norwegian DJ and record producer Cashmere Cat."
Logical Conclusions:
'Love Incredible' was released through Interscope and Mad Love.

""Love Incredible" was released through Interscope and Mad Love." -> COMPLEMENTS -> ""Love Incredible" features vocals by Cuban-American singer Camila Cabello."
Logical Conclusions:
Fact 10 gives the label for 'Love Incredible', while fact 16 notes it features Camila Cabello.

""Love Incredible" was released through Interscope and Mad Love." -> SUPPORTS -> ""Love Incredible" was released as the third single from Cashmere Cat's debut album '9'."
Logical Conclusions:
Fact 10 states 'Love Incredible' was released through Interscope and Mad Love, and fact 14 says it was the third single from Cashmere Cat's album.

[Interscope Records (American major record label)] -> (participates in event: Interscope Records is an imprint of Interscope Geffen A&M Records.) -> "Interscope Records is an imprint of Interscope Geffen A&M Records."

[Interscope Records (American major record label)] -> (participates in event: The parent company of Interscope Records is Universal Music Group.) -> "The parent company of Interscope Records is Universal Music Group."

[Interscope Records (American major record label)] -> (participates in event: Interscope Records is an American major record label.) -> "Interscope Records is an American major record label."
```

Level 0:

Subquery: Which record company released the song 'Love Incredible?'
""Love Incredible" was released through Interscope and Mad Love." -> (has) -> [Interscope Records (American major record label)]

""Love Incredible" was released through Interscope and Mad Love." -> SUPPORTS -> ""Love Incredible" was released on 17 February 2017."
Logical Conclusions:
Both facts provide details about the release of 'Love Incredible', with one specifying the date and the other the record label.

""Love Incredible" was released through Interscope and Mad Love." -> RELEASED -> ""Love Incredible" is a song by Norwegian DJ and record producer Cashmere Cat."
Logical Conclusions:
'Love Incredible' was released through Interscope and Mad Love.

""Love Incredible" was released through Interscope and Mad Love." -> (has) -> [South African Music Entities (Record labels and song from South Africa)]

""Love Incredible" was released through Interscope and Mad Love." -> COMPLEMENTS -> ""Love Incredible" features vocals by Cuban-American singer Camila Cabello."
Logical Conclusions:
Fact 10 gives the label for 'Love Incredible', while fact 16 notes it features Camila Cabello.

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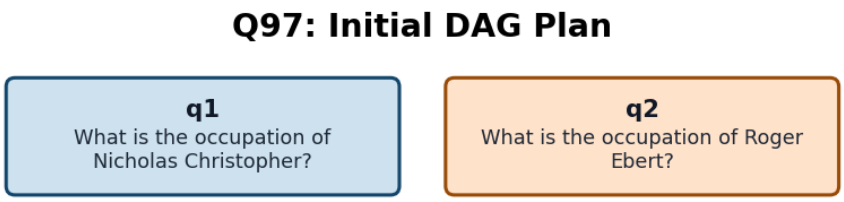
[American Record Corp …[truncated for report]

Q97 [comparison / hard] 5a844fd35542990548d0b274

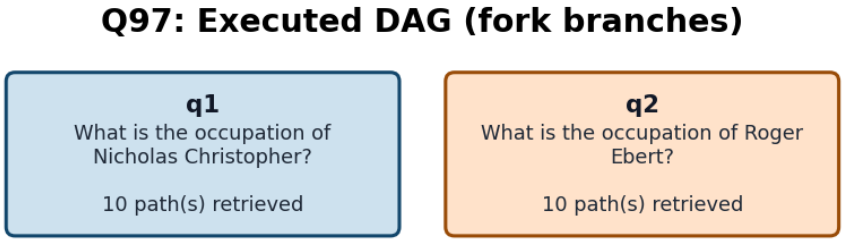
What occupation did Nicholas Christopher and Roger Ebert share?

Gold answer	author
LLM answer	novelist, poet, and critic, film critic and historian
EM / F1	0.00 / 0.00
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What is the occupation of Nicholas Christopher?	10 path(s)
q2	0	—	What is the occupation of Roger Ebert?	10 path(s)

Final context fed to generation (1320 tokens)

```
Query plan:
- q1: What is the occupation of Nicholas Christopher?
- q2: What is the occupation of Roger Ebert?

Level 0:

Subquery: What is the occupation of Nicholas Christopher?
"Nicholas Christopher is an American novelist, poet and critic." -> SUPPORTS -> "Nicholas Christopher's
sixteen books include six novels, eight volumes of poetry, a critical study of film noir, and a novel for
children."
Logical Conclusions:
Nicholas Christopher's sixteen books include six novels, eight volumes of poetry, a critical study of film
```

noir, and a novel for children, and he is an American novelist, poet and critic.

"Nicholas Christopher is an American novelist, poet and critic." -> SUPPORTS -> "Nicholas Christopher is the author of sixteen books."
Logical Conclusions:
Nicholas Christopher is the author of sixteen books and is an American novelist, poet and critic.

"Nicholas Christopher is an American novelist, poet and critic." -> (has) -> [nicholas christopher (American novelist, poet and critic)]

[Nicholas Brothers (Tap-dance duo starring in MGM musicals)] -> (participates in event: Fayard Antonio Nicholas was an American choreographer, dancer and actor.) -> "Fayard Antonio Nicholas was an American choreographer, dancer and actor."

[Nicholas Brothers (Tap-dance duo starring in MGM musicals)] -> (participates in event: The Nicholas Brothers starred in the MGM musicals 'An All-Colored Vaudeville Sho) -> "The Nicholas Brothers starred in the MGM musicals 'An All-Colored Vaudeville Show' (1935), 'Stormy Weather' (1943), 'The Pirate' (1948), and 'Hard Four' (2007)."

[Nicholas Brothers (Tap-dance duo starring in MGM musicals)] -> (participates in event: The Nicholas brothers also starred in the 20th Century-Fox musicals 'Down Argent) -> "The Nicholas brothers also starred in the 20th Century-Fox musicals 'Down Argentine Way' (1940), 'Sun Valley Serenade' (1941), and 'Orchestra Wives' (1942)."

[Nicholas Brothers (Tap-dance duo starring in MGM musicals)] -> (participates in event: Fayard Nicholas and his younger brother Harold Nicholas made up the Nicholas Bro) -> "Fayard Nicholas and his younger brother Harold Nicholas made up the Nicholas Brothers tap-dance duo."

[Nicholas Brothers (Tap-dance duo starring in MGM musicals)] -> (participates in event: Fayard Antonio Nicholas was born on October 20, 1914.) -> "Fayard Antonio Nicholas was born on October 20, 1914."

[Nicholas Brothers (Tap-dance duo starring in MGM musicals)] -> (participates in event: Fayard Antonio Nicholas died on January 24, 2006.) -> "Fayard Antonio Nicholas died on January 24, 2006."

[Christopher I of Denmark (King of Denmark 1252-1259)] -> (participates in event: Christopher I of Denmark was also known as 'Christoffer I'.) -> "Christopher I of Denmark was also known as 'Christoffer I'."

Subquery: What is the occupation of Roger Ebert?
"Roger Ebert was an American film critic and historian, journalist, screenwriter, and author." -> SUPPORTS -> "Roger Ebert was a film critic for the Chicago Sun-Times from 1967 until his death in 2013."
Logical Conclusions:
Roger Ebert was an American film critic and historian who worked for the Chicago Sun-Times from 1967 until his death.

[Roger Ebert (Film critic and television host)] -> (participates in event: Roger Ebert was an American film critic and historian, journalist, screenwriter,) -> "Roger Ebert was an American film critic and historian, journalist, screenwriter, and author."

"Roger Ebert was an American film critic and historian, journalist, screenwriter, and author." -> NATIVE_OF -> "Roger Ebert was a native of the adjoining town of Urbana, Illinois."
Logical Conclusions:
Roger Ebert, an American film critic, was a native of Urbana, Illinois.

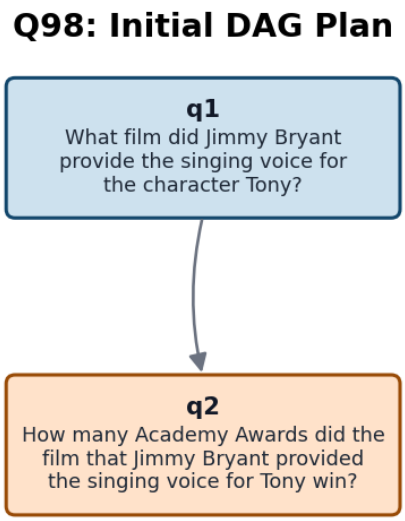
"Roger Ebert was an American film critic and historian, journalist, screenwriter, and author." -> WON -> "In 1975, Ebert became the first film critic to win the Pulitzer Prize for Criticism."
Logical Conclusions:
Roger Ebert, an American film critic, became the first film crit …[truncated for report]

Q98 [bridge / hard] 5addffff5542992200553bdd

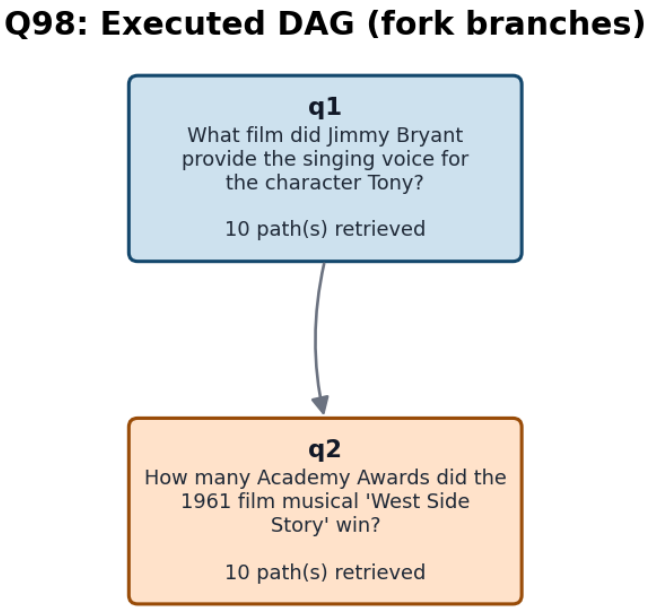
How many Academy Awards did the film, in which Jimmy Bryant provided the singing voice for the character Tony, win ?

Gold answer	10
LLM answer	10
EM / F1	**1.00 / 1.00**
Fallback mode	no

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	What film did Jimmy Bryant provide the singing voice for the character Tony?	10 path(s)
q2	1	q1	How many Academy Awards did the 1961 film musical 'West Side Story' win?	10 path(s)

Final context fed to generation (1245 tokens)

```
Query plan:
- q1: What film did Jimmy Bryant provide the singing voice for the character Tony?
- q2: How many Academy Awards did the film that Jimmy Bryant provided the singing voice for Tony win?
(depends on: q1)

Level 1:

Subquery: How many Academy Awards did the 1961 film musical 'West Side Story' win?
"West Side Story was nominated for 11 Academy Awards." -> NOMINATED -> "West Side Story is a 1961 American
musical romantic drama film."
Logical Conclusions:
West Side Story was nominated for 11 Academy Awards.

"West Side Story was nominated for 11 Academy Awards." -> (has ) -> [west side story (film) (1961 musical
romantic drama film)]

"West Side Story was nominated for 11 Academy Awards." -> LED_TO -> "West Side Story won 10 Academy Awards,
including Best Picture."
Logical Conclusions:
West Side Story won 10 Academy Awards, including Best Picture.

"West Side Story was nominated for 11 Academy Awards." -> SUPPORTS -> "West Side Story won a special award
for Jerome Robbins."
Logical Conclusions:
West Side Story was nominated for 11 Academy Awards and won a special award for Jerome Robbins.

"West Side Story was nominated for 11 Academy Awards." -> (has ) -> [academy honorary award (special award
for career achievements)]

[west side story (film) (1961 musical romantic drama film)] -> (participates in event: West Side Story won
a special award for Jerome Robbins.) -> "West Side Story won a special award for Jerome Robbins."

[west side story (film) (1961 musical romantic drama film)] -> (participates in event: West Side Story
became the record holder for the most wins for a movie musical.) -> "West Side Story became the record
holder for the most wins for a movie musical."

[west side story (film) (1961 musical romantic drama film)] -> (participates in event: West Side Story
received high praise from critics and viewers.) -> "West Side Story received high praise from critics and
viewers."

[west side story (film) (1961 musical romantic drama film)] -> (participates in event: The Broadway musical
'West Side Story' was inspired by William Shakespeare's pla) -> "The Broadway musical 'West Side Story' was
inspired by William Shakespeare's play 'Romeo and Juliet'."

Level 0:

Subquery: What film did Jimmy Bryant provide the singing voice for the character Tony?
"James Howard Bryant is most well known for providing the singing voice of Tony in the 1961 film musical
'West Side Story'." -> VOICE_FOR -> "Tony was played onscreen by Richard Beymer in 'West Side Story'."
```

Logical Conclusions:
Tony was played onscreen by Richard Beymer in 'West Side Story'.

"James Howard Bryant is most well known for providing the singing voice of Tony in the 1961 film musical 'West Side Story'." -> (has) -> [west side story (film) (1961 musical romantic drama film)]

[Jimmy Bryant (Singer and musician)] -> (participates in event: James Howard Bryant is most well known for providing the singing voice of Tony i) -> "James Howard Bryant is most well known for providing the singing voice of Tony in the 1961 film musical 'West Side Story'."

"James Howard Bryant is most well known for providing the singing voice of Tony in the 1961 film musical 'West Side Story'." -> SIMILAR_TO -> "James Howard Bryant sang for James Fox in the 1967 film musical 'Thoroughly Modern Millie'."
Logical Conclusions:
James Howard Bryant sang for James Fox in the 1967 film musical 'Thoroughly Modern Millie'.

"James Howard Bryant is most well known for providing the singing voice of Tony in the 1961 film musical 'West Side Story'." -> CONTRASTS_WITH -> "James Howard Bryant received no screen credit for his singing in 'West Side Story'."
Logical Conclusions:
James Howard Bryant received no screen credit for his singing in 'West Side Story'.

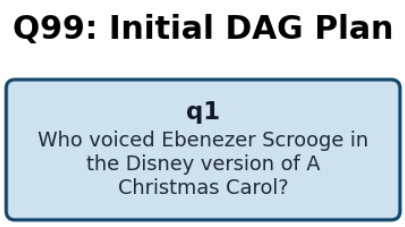
"James Howard Bryant is most well known for providing the singing voice of Tony in the 1961 film musical 'West Side Story'." -> SUPPORTS -> "James Howard Bryant stated that Richard Beymer was 'a nice guy, and every time he did an interview he …[truncated for report]

Q99 [bridge / hard] 5ac00f795542996f0d89cb16

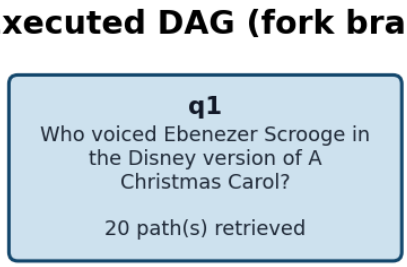
The Disney version of Ebenezer Scrooge was voiced by whom?

Gold answer	Alan Young
LLM answer	Alan Young
EM / F1	**1.00 / 1.00**
Fallback mode	single-node plan — delegated straight to the wrapped engine (no DAG structure)

Initial DAG Plan



Executed DAG (all fork branches)



Branch detail

Branch	Level	Parents	Refined query	Result
q1	0	—	Who voiced Ebenezer Scrooge in the Disney version of A Christmas Carol?	20 path(s)

Final context fed to generation (1528 tokens)

```
"Alan Young first voiced Scrooge McDuck in the Academy Award-nominated short film 'Mickey's Christmas
Carol' (1983)." -> APPEARS_IN -> "The cartoon stars Scrooge McDuck as Ebenezer Scrooge." -> (has ) ->
[scrooge mcduck (Disney character playing Ebenezer Scrooge)] -> (participates in event: Alan Young was the
voice of Disney's Scrooge McDuck for over thirty years.) -> "Alan Young was the voice of Disney's Scrooge
McDuck for over thirty years."
Logical Conclusions:
The cartoon 'Mickey's Christmas Carol' stars Scrooge McDuck as Ebenezer Scrooge, a role first voiced by
Alan Young in that same film.

[ebenezer scrooge (Miserly protagonist of A Christmas Carol)] -> (participates in event: The cartoon stars
Scrooge McDuck as Ebenezer Scrooge.) -> "The cartoon stars Scrooge McDuck as Ebenezer Scrooge."
```

"Alan Young first voiced Scrooge McDuck in the Academy Award-nominated short film 'Mickey's Christmas Carol' (1983)." -> (has) -> [scrooge mcduck (Disney character playing Ebenezer Scrooge)]

[scrooge mcduck (Disney character playing Ebenezer Scrooge)] -> (participates in event: In Act I, Scene 5 of 'A Christmas Carol', the Ghost of Christmas Past takes Scro) -> "In Act I, Scene 5 of 'A Christmas Carol', the Ghost of Christmas Past takes Scrooge to revisit his youthful days in Fezziwig's world."

[scrooge mcduck (Disney character playing Ebenezer Scrooge)] -> (participates in event: Scrooge apprenticed under Fezziwig.) -> "Scrooge apprenticed under Fezziwig."

"Alan Young first voiced Scrooge McDuck in the Academy Award-nominated short film 'Mickey's Christmas Carol' (1983)." -> STARTED_ROLE -> "Alan Young was the voice of Disney's Scrooge McDuck for over thirty years."
Logical Conclusions:
Alan Young's voice role as Scrooge McDuck began with 'Mickey's Christmas Carol'.

[scrooge mcduck (Disney character playing Ebenezer Scrooge)] -> (participates in event: Scrooge is the very antithesis of the person he worked for as a young man.) -> "Scrooge is the very antithesis of the person he worked for as a young man."

[scrooge mcduck (Disney character playing Ebenezer Scrooge)] -> (participates in event: Alan Young voiced Scrooge McDuck in various other films, TV series and video gam) -> "Alan Young voiced Scrooge McDuck in various other films, TV series and video games until his death."

"Alan Young was the voice of Disney's Scrooge McDuck for over thirty years." -> (has) -> [alan young (actor voicing Scrooge McDuck)] -> (participates in event: The television version of 'The Alan Young Show' gained Alan Young two Emmy Award) -> "The television version of 'The Alan Young Show' gained Alan Young two Emmy Awards."

"The cartoon stars Scrooge McDuck as Ebenezer Scrooge." -> PART_OF -> "The Disney characters were primarily from the Mickey Mouse universe, 'Robin Hood', and 'The Adventures of Ichabod and Mr. Toad'."
Logical Conclusions:
The Disney characters in 'Mickey's Christmas Carol' are primarily from the Mickey Mouse universe, and the cartoon stars Scrooge McDuck as Ebenezer Scrooge.

[Mickey's Christmas Carol (1983 Disney animated featurette)] -> (participates in event: Alan Young first voiced Scrooge McDuck in the Academy Award-nominated short film) -> "Alan Young first voiced Scrooge McDuck in the Academy Award-nominated short film 'Mickey's Christmas Carol' (1983)."

[ebenezer scrooge (Miserly protagonist of A Christmas Carol)] -> (participates in event: Seymour Hicks appears as Ebenezer Scrooge in the 1935 film.) -> "Seymour Hicks appears as Ebenezer Scrooge in the 1935 film."

[ebenezer scrooge (Miserly protagonist of A Christmas Carol)] -> (participates in event: Jack Palance gives a performance as Ebenezer Scrooge in the 1998 film 'Ebenezer') -> "Jack Palance gives a performance as Ebenezer Scrooge in the 1998 film 'Ebenezer'."

"Scrooge apprenticed under Fezziwig." -> (has) -> [mr. fezziwig (character from A Christmas Carol)] -> (participates in event: Mr. Fezziwig has a large Welsh wig.) -> "Mr. Fezziwig has a large Welsh wig."
…[truncated for report]